

Masters Thesis

ME506 – An automated mini lab platform using gravitational and centrifugal flow

School of Mechanical & Manufacturing Engineering

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Nomenclature

BSA	Bovine Serum Albumin
CAD	Computer-Aided Design
DNA	Deoxyribonucleic Acid
E.Coli	Escherichia Coli
LoaD	Lab-on-a-Disc
LFA	Lateral Flow Assay
µm	Micrometer
µL	Microlitre
µTAS	Micro Total Analysis System
PCR	Polymerase Chain Reaction
q-PCR	Real-Time Quantitative PCR
SLA	Stereolithography
CT	Threshold Cycles
vLoaD	vertical-Lab-on-a-Disc

Abstract

Microfluidics is a field of science and technology that involves systems which process and manipulate fluids on the micro scale. The required amount of fluid and the properties associated with, make microfluidic devices desirable for performing medical tests. They serve as an alternative to traditional laboratory testing, that can save time and money. Their most desirable use case is point of care testing, where a blood or DNA sample can be taken from a patient and a test can be performed there to aid medical practitioners in administering aid. A common form of these devices are lab-on-a-chips, these devices aim to perform all required laboratory tasks on a single device the size of a credit card. An issue with these devices is that they often required bulky machinery, as they need external forces applied to operate which increases overall total size. A proposed alternative to them are Lab-on-a-Disc devices, these aim for a similar goal of encapsulating all required laboratory tasks onto a single disc the size of a common CD. The usage of centrifugal forces allows for the actuation of valves, centrifuging of liquids, pumping of fluids, all with a simple commonly available motor. This makes these devices, cheaper and far more compact the Lab-on-a-chips. The fatal flaw with these devices is that centrifugal forces can only move fluids radially outward, leading to unidirectional flow. Much research has gone into solutions for this problem, capillary valves, siphon valves, dissolvable films, heat activated solutions. However, the solutions develop into two other flaws. They either become too complex or too expensive to operate. Diminishing the goal of simple cost effective point of care testing. Furthermore, these discs often have underutilised surface area, due to unidirectional nature of centrifugal flow. This project is the design of a microfluidic disc, aiming to solve these flaws, it is a vertically mounted Lab-on-a-Disc. This allows the usage of gravitational flow in conjunction with centrifugal flow, to allow for bidirectional flow. The primary aim of this project is to create a disc capable of performing “Non-Magnetic Silica bead based DNA extraction” as a test to show the capabilities of the microfluidic device. This project was performed in collaborations with the RAPID institute and the PhD student “Nicklas Rondot”. It carries on from Nicklas’s initial research, starting with conceptional designs created by himself and past assistants. This thesis serves as a summary to all research and work performed towards the ME506 project. It outlines the project problem itself, the researched literature, the theory behind the design, the methods applied to prototyping the design, an analysis of the final vLoaD disc and full details of the testing required to achieve a successful DNA Isolation assay.

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4 Introduction

4.1 Medical Diagnostics

In the modern medical world, diagnostic testing is vital for understanding how to best treat a patient. They allow doctors to see beyond physical symptoms of their patients and make better decisions. It is fair to say that diagnostics have become the cornerstone of the current generation of medicine.

Diagnostics can come in various forms such as, X-rays, MRIs, ultrasounds, mammographs, EKGs, blood panels and genetic screenings to. Genetic screenings are particularly interesting as they allow doctors to understand more about the individual they are treating, allowing for tailored personalized medicine. This is a form of medical practice in which the doctor can analyse a person's DNA, essentially their code in terms of software engineering, from this they can predict the risk of certain diseases based on genetics or diagnose existing symptoms linked to specific genetic mutations.

There are plenty of diseases that individuals can be at a higher risk of based on their genetics, such as breast/ovarian cancer being linked to the BRCA1/BRCA2 mutation, and there are other diseases that are specifically caused by an existing mutation like cystic fibrosis. In early years children may present some physical symptoms, but doctors would not be able to properly diagnose the child if diagnostics tests were unavailable.

A key issue with these tests is they often have to be conducted in a laboratory, something that not every hospital has, especially in impoverished parts of the world. Even when sending samples to a laboratory is an option it is still expensive in terms of time and economics. As such, point of care testing has become an aspiration for the medical world. The ability to take a DNA sample at the patients side and provide a result from the same location.

4.2 Microfluidics

Microfluidics is the science of manipulating meniscus fluid volumes, using micrometre-sized channels. They were first used for their abilities to take small amounts of samples and reagents to carry out analytical tasks where high sensitivity was required.

They do not require the same quantity of sample nor reagents that a traditional laboratory testing would require and they also do not require a laboratory itself. This makes them perfect for the goal of achieving point of care testing.

Additionally, when fluids are “shrunk down” to this scale they exhibit some interesting properties, they typically only have laminar flow, which allow for more precise control, and due to their small volume they can be heated and cooled in a fraction of the time, when compared to a traditional laboratory test, allowing for chemical reactions to occur faster.

They already exist in everyday life, a good amount of the population have used microfluidic devices without even realising, a good example is the modern pregnancy test. These devices take a small amount of urine and mix it with a preloaded reagent to indicate a positive or negative result. Another more recent example is the at home covid test kits, they require some more sample preparation but they are also considered as microfluidic devices.

Both of these examples are defined as lateral flow assays (LFAs). They are capable of performing a single simple task, to produce a positive or negative test result. For most testing, more complex steps are required, where valves, pumping and multi-step processes occur, LFAs can not achieve this goal. A device that could accomplish this is referred to as a “Lab-on-a-chip”.

4.3 Lab-on-a-Chip

Lab-on-a-chip devices aim to shrink all the functionality of a laboratory into a microfluidic device the size of a credit card. Typically, they are made from glass, silicon or specialized plastics, they are designed in a similar fashion to micro circuits on computer chips where channels, pumps, valves and chambers are all etched into the material.

An example would be the Abbott i-Stat, this is a lab-on-a-chip used in ambulances to perform a blood screen from a patient's blood sample, allowing paramedics to make informed decisions before arriving at the hospital [1]. They can only run a limited set of tests per cartridge, such as one for checking troponin for a possible heart attack and another to check organ functionality. However, they are still incredibly useful.

The main inhibitor with Lab-on-a-Chip devices is they require external forces to operate. So, while the device itself can be the size of a credit card it needs to be inserted into a much larger device in order to pump or heat fluids. A proposed solution to this is Lab-on-a-disc, these devices while larger than a credit-card, use centrifugal microfluidics to achieve their operations, allowing for more compact devices.

4.4 Lab-on-a-Disc

Lab-on-a-Disc (LoaD) are a distant relative to Lab-on-a-chip, utilizing centrifugal microfluidics instead. This subfield of microfluidics uses a rotating microfluidic device, during operation the disc has its rotational speed increased or decreased to control the centrifugal force applied to fluid inside the disc. These devices were originally inspired by CD players, hence the name Lab-on-a-disc.

During rotation there are three main forces generated, Centrifugal force, Euler force and Coriolis force. LoaD devices use these forces to transport fluids within their disc, controlling valves, pumping fluids, actuating siphon and even changing flow direction with a Coriolis switch. These devices can achieve all of this functionality via a simple standard rotary motor, this makes them much more affordable and compact as opposed to Lab-on-a-Chip devices.

In addition these devices also have the upper hand in complex assays due to their centrifugal nature. They can manipulate samples in a similar way to traditional laboratory centrifugal, for example the separation of red blood cells from clear plasma.

Over the years many approaches have been taken in this field, most resolving towards the issue that the discs become too large or too complex. These discs generally follow the approach of starting with liquids towards the centre of the disc and using centrifugal forces to push the liquid to the outside of the disc. Using this centrifugal flow to route the liquids through various chambers and channels. However, as these forces act evenly around the disc, this leads to all the liquids present on the disc moving at once.

This is ideal for where parallel testing is desired, where many identical assay can be run at the same time on a single disc. This is not ideal for when complex routing is desired as centrifugal force will only transport liquid outwards, making transporting fluids bidirectionally more difficult than for a Lab-on-a-chip .

Various techniques have been researched to overcome this issue, capillary valves, siphon valves, dissolvable films, heat activated solutions but they all become either too complex or expensive to operate, straying from the key goal of simple cost effective point of care testing. In addition, they rarely solve the issue that for a single test, the majority of the disc surface area is underutilized.

4.5 Vertical Lab-on-a-Disc

The proposed solution to this problem is the key focus of this project, a vertically mounted “lab on a disc”. This approach allows for the usage of gravitational and centrifugal flow.

In theory this would allow for movement to the outside of a disc using centrifugal flow and then a return to the inside of the disc using gravitational flow. Allowing for the “chaining” of sequential operations, moving liquids to the outside of the disc and back inside, although this will still require some of the valving techniques previously mentioned. This would all lead towards a disc that can remain compact by being able to utilize the full surface area of the disc, as opposed to only a “slice” of the disc as seen previously.

For this project the capability of this disc will be tested via “Non-Magnetic Silica Bead DNA Isolation”, as this displays aptitude of the device for point of care testing. This was chosen as it is a complex multi-step assay that can be contrasted to traditional laboratory protocol using a CT shift on a q-PCR.

This project was originally created by “Nicklas Rondot” from the RAPID institute at DCU Glasnevin, concept designs were created by himself and prior assistants. This project was done in collaboration with Nicklas with the aim to create a masters final year project as well as a published engineering journal paper.

This project builds upon these concept designs, to create a finalized disc adept for practical usage. The testing “vLoaD” rig (vertical lab-on-a-disc) is pre-developed, however it will require additional improvements both to hardware and software to reach all of this project’s objectives.

The main objective statement of this project is: “The development of a microfluidic lab-on-a-disc that uses gravitational and centrifugal flow to perform non-magnetic silica bead based DNA extraction”

4.6 Engineering Contributions

The engineering contributions of this project are directly linked to the research questions posed in **section 5.28**. The overarching goal of this project was to develop a new type of microfluidic system and test it appropriately (see **section 7.1** for the project mission), as such the engineer contributions link directly to this.

1: Development of a Microfluidic disc that utilizes Gravitational and Centrifugal forces

Associated Research Questions: RQ1, RQ2

A plethora of microfluidic discs were designed, tested and developed as part of this project. Demonstrating the challenges and advancements that are associated with developing a LoaD that utilizes gravitational flow. The design calculations performed (see **section 8.2**) display the phenomenon associated with using gravitational flow in conjunction with centrifugal forces.

The prototype designs (see **section 9.4**) show the iterative approach taken to the development of the vLoaD disc, showing the shortcomings that needed to be overcome for the progression of this project. Ultimately, a fully functional disc was developed, using gravity driven flow to actuate storage chambers, with centrifugal forces to perform mixing, actuate burst valves and control separation between product and waste via a Coriolis switch.

This process directly answers research questions one and two, by displaying how gravitational and centrifugal flow can be used in conjugation as well as the operational challenges associated with doing so. Future researchers can read and analyse this thesis to understand how to develop their own vLoaD devices for different chemical assays. The vLoaD should hopefully develop into its own branch of microfluidic devices due to the work performed in this project acting as a stepping stone for further development.

2: Assessment of the feasibility for 3D printing microfluidic discs

Associated Research Questions: RQ3, RQ4

For this project the chosen fabrication method was SLA 3D printing. It is a method that is seeing further usage with each year for the development of microfluidic devices, but it is still not the most popular choice. Questions still arise surrounding the biocompatibility of standard resins and the tolerances producible by commercial 3D printers. The analysis segment of this thesis (**section 11**) investigates

both of these aspects of 3D printed microfluidic discs and presents the findings so other researchers can take notes of the capabilities and drawbacks associated with this fabrication method.

3: Creating a fully automated mini-lab system

Associated Research Questions: RQ4

As part of the research statement of this project the device was to be fully automated. Initially the device was only controlled via a basic Arduino code. As an enhancement to the rig for this project, an application (see **section 9.1.3**) was developed to make controlling the device easier and automation a feature. In order to automate the device, data files are developed within the application, acting as instructional recipes with each step laid out. These files are capable of being stored as JSON files. They can be loaded into the application at a later time and ran directly from it, allowing for the automation of assays on the device.

4: Testing of the System via a complex multi-step assay

Associated Research Questions: RQ4

To fully test the performance of the developed vLoaD device, DNA isolation assays were performed as they are a complex multi-step assay with strong implications for point of care testing. The assays were performed using lysed E.Coli cells and compared to a control performed using traditional laboratory protocol (see **section 9.7**).

For comparison, a real time quantitative Polymerase Chain Reaction (q-PCR) was run with both the vLoaD DNA and the control lab DNA. The q-PCR takes the DNA samples provided and amplifies the amount of DNA present. A CT shift was performed, this measures the change in cycle threshold value between the vLoaD DNA and the control, allowing for comparison of the amount of nucleic acid present within the samples.

This allows the assessment of the vLoaD capabilities in a format appropriate of the molecular diagnostic goals that the project is targeted towards. Informing future research of what can be achieved with the vLoaD device.

5 Literature Review

5.1 Origins of Microfluidics

Microfluidics is a field of science and technology that involves systems that process and manipulate fluids on the micro scale, -9 to -18 SI of litres [2]. They were first used for their abilities to take small amounts of samples and reagents to carry out analytical tasks where high sensitivity was required.

The Inkjet printer is heralded as the first microfluidic device. The mechanics behind the invention were originally prescribed by Lord Rayliegh in 1879 [3]. He discovered that a falling continuous stream of fluid breaks up into discrete droplets to minimize their surface area and energy. This became known as the Plateau-Rayleigh instability or the Rayleigh instability.

At Stanford University in 1965, Richard Sweet developed the first working continuous inkjet printer, prior to mass commercialization [4]. His device took advantage of the Rayleigh instability to force ink out of a vibrating nozzle with a diameter of only 35 micrometres. The droplets passed through a uniform electric field which controlled the direction of the droplets based on their electric charge prior to making contact with the target piece of paper. These charges could be recorded with respect to time and plotted on an oscillograph. This device is said to be the first microfluidic device, this went on to lead to advancements seen today in both commercial inkjet printing as well as expansions in photolithography techniques by IBM [5].

Photolithography is a common method of fabrication in microelectronics as it is used to create semiconductors which formulate the base of many microscopic electronic circuits. This technology was originally seen as the path for fabricating microfluidic components, some of the earliest microfluidic devices were created using silicon and glass. Although in the modern day these materials have been overtaken by various forms of plastics. As it is more apt than silicon for the purposes of microfluidic devices.

Research continued at Stanford university which lead to devices capable of gas-phase chromatography [6]. This is a microanalytical method used in analytical chemistry to separate and analyse compounds capable of vaporization without breaking down [7]. This device was manufactured using photolithography and etching to form a combination of microelectronics and microfluidic systems. It was a highly sensitive device that was able to analyse the composition of gases inputted into the system. This device was seen as the first “micro-Total Analysis System”, prior to the actual definition but it still fulfils the criteria.

5.2 Physics of Microfluidics

In Microfluidic systems, fluids behave differently due to the nature of physics at the microscale. Reynolds Number [8] is a dimensionless number used to describe the ratio between inertial and viscous forces in a fluidic system.

$$Re = \frac{\rho v L}{\mu}$$

For this equation, ρ represents fluid density, μ is dynamic viscosity of the fluid, v is the flow velocity and lastly L is the "characteristic linear dimension of the system". This often ends up being the volume or surface area, however it depends on the system, for example for a pipe used to transport fluid this would be the diameter. This equation shows that the Reynolds number (Re) and characteristic dimension are proportional to one another, if the dimensions decrease as they do in a microfluidic system so too does the Reynolds number. If Re is below 2000, the system displays a laminar flow regime. This makes the flow of the system highly predictable as opposed to turbulent flow. This laminar flow condition (Re < 2000) is the case for most microfluidic systems, giving them predictable flow characteristics. The Péclet Number [9] is another dimensionless number; it describes the ratio between advective to diffusive transport of molecules in a fluid. Advective transport is one in which the substance is moved by bulk motion of a fluid, properties are conserved such as energy. An example is how silt can be carried by waters in rivers. Diffusion is the gradual movement of a body as opposed to the bulk movement. It operates based on gradients where molecules move from areas of higher concentration to lower concentration, "spreading out" or "defusing". An example is how fresh air replaces used air in the lungs when humans breathe due to the new air having a higher oxygen concentration.

$$Pe = \frac{vL}{D}$$

Here again L is the "characteristic dimension", D is the mass diffusion coefficient and v is the local flow velocity. Just as with the Reynold number, reducing the dimensions of the system to the micro scale leads to the reduction of the Péclet Number. The reduction in this number shows that microfluidic systems have diffusion-dominated transport. The Reynolds number already displays that there is no turbulence flow, so there is no "mixing from" the fluid motion, so transport is dominated by molecules as opposed to flow. Mixing methods commonly depends on turbulence or chaotic advection [10]. Making this a key design consideration for Microfluidic systems that depend on mixing of different samples/reagents. Implementation of augmented channels designed specifically to mix two different fluids is a frequent design goal for microfluidic systems that has been approached in a variety of forms over years of research. Such as augmented tesla structures [11] or taking advantage of three-

dimensional structures to introduce turbulent flow into a system by manipulating the Reynolds number [12].

Surface tension is the property of a liquid to resist external forces; it acts like a film that pulls molecules of the liquid inward and minimizes surface area. It is clearly exhibited by liquid droplets; they form into spheres due to surface tension. The area of droplet microfluidics focuses heavily on this fluid property [13].

Reaction times refers to the time interval between the introduction of a stimulus and the occurrence of a response. For chemical reactions, this refers to how long it takes for the interaction between two chemicals to occur. At the microscale, a shorter diffusion time is present for all molecules due to the reduction in characteristic dimensions [14]. This is vitally important for larger molecules with low coefficients of diffusion, such as DNA. Additionally, due to the micro scale, smaller amounts of reagent and sample that are required for chemical testing, microfluidics are preferred for certain diagnostics.

Capillary forces refer to forces that occur for liquids in small dimensional areas, cohesion and adhesive forces occur between the liquid and the surrounding container. An example of the property is liquid being drawn through a straw. On the microscale seen in microfluidic devices these capillary forces are dominant with respect to gravity, meaning they can resist gravitational forces and allow fluids to travel in desired directions. Since they can be controlled via channel dimensions and are purely passive, these forces can be used to create relatively cheap microfluidic devices such as paper analytical devices (PADs) [15].

5.3 Micro-Total Analysis Systems

Ciba was a chemical company based in Switzerland, in the 90s a team of researchers in their Central Analytical Research department set out to create a novel concept for chemical sensing. A miniaturized total chemical analysis system [16]. The goal of this was to create a system which periodically transforms chemical information into electronic information. Allowing for the detection of chemicals in sample with rapidly changing compositions such as chemical reaction mixtures. With the key aim of creating a device capable of operating extremely close to the place of measurement, defining it as a "Micro-Total Analysis System". The Micro-Total Analysis System was a truly novel yet revolutionary idea, with the concepts still echoing to the modern day through research such as the "Lab-on-a-chip". The definition is labelled as a system that is capable of carrying out all functions required for analysis while maintaining a form that allows for usage near the location of sample/measurement. This concept engaged interest from other fields of science, defence being a key one. With the end of the cold war, the Defence Advanced Research Projects Agency (DARPA) of the US Department of Defence became engaged with microfluidic technology [2]. Seeing it as a form of biodefence, a way to combat chemical and biological weaponry.

5.4 Manufacturing Methods for Microfluidic Devices

As mentioned previously, Silicon was the material used originally to construct microfluidic devices, as photolithography methods were first used [17]. Over time this shifted towards Glass as its properties became requirements for microfluidic devices [18], such as how silicon is opaque to visible and ultraviolet light. This makes the material unsuitable for popular microscopy methods.

However, both of these materials were pre-existing, chosen for their convenience, they were not advancements specific to the field of microfluidics [19]. They were brittle materials with non-trivial bonding protocols, and ultimately they were too expensive for the aims of microfluidic devices. The microfluidic device field was aspiring to be one in which fabrication was both cheap and accessible, something that was not possible with the current materials and methods. Leaving a research gap for advancements to be made.

5.5 PDMS

Poly(dimethylsiloxane), better known as PDMS was created by a team at Harvard University, specifically with the key aim of developing a polymer for fabricating microfluidic devices [20]. They identified the issues with previous techniques, such as cost and incompatibility with rapid prototyping.

PDMS is an optically transparent, gas- and vapour-permeable elastomer, that does not require a clean room to be fabricated. Their techniques were based on soft lithography [21], [22], [23], this is a suite of nonphotolithographic methods for replicating a pattern. This allowed for the usage of computer-aided design (CAD) programs to design the channels for their microfluidic device.

With their methods it was possible to fabricate features with sub 0.1 micrometre dimensions easily [24]. It is an incredible material that can tune its hydrophobic surface properties to be more hydrophilic instead [25], [26] and in certain cases can be irreversibly bonded to glass, plastic, other materials or itself. Its most important feature is the elasticity of PDMS makes it possible to create valves and actuators.

PDMS was used to fabricate elastomer based valves that are capable of being actuated due to their far greater young's modulus as opposed to the silicon based devices seen previously [27]. These valves were actuated by pneumatics and allowed for active devices to be densely arranged. Allowing for independent control that could not be seen in the first microfluidic devices [28] [29] that used electrokinetic flow control [30]. The valve created here from PDMS would go on to become known as the quake valve, which was greatly influential to the field of microfluidics [27].

Over the years various fabrication methods have been developed for the material and it is still a popular choice for microfluidic devices [31] [32]. It is easy to fabricate, has good biocompatibility and rapid iteration with modern methods, such as 3D printed moulds. However, the material is not suitable for scaled production, leaving the microfluidics industry to continue searching for an ideal material, which will be required when microfluidic devices leave the research sphere and become an everyday technology.

5.6 PMMA

Polymethyl Methacrylate (PMMA) is a thermoplastic polymer that is commonly known as acrylic. It is a glass like material in appearance having a transparent nature with high optical clarity. It is a rigid material with good biocompatibility, and it is easy to fabricate devices from. It first saw usage in the early 1990s where its optical clarity made it a clear winner over silicon devices, prior to the introduction of PDMS soft lithography. DNA analysis [33] chips were one of the large areas that PMMA first saw usage within. It has always been popular in the microfluidics space as the "industrial" material, one that is easy to scale manufacturing with. This is due to how PMMA is less fragile when compared to other materials used in microfluidic devices and it does not require expensive manufacturing

techniques. Hot embossing, injection moulding, laser ablation, reactive ion etching and deep UV lithography are all manufacturing techniques that can be used with PMMA.

Laser micromachining is a fabrication process that uses a laser beam to remove material, making it a subtractive manufacturing method. This method has been studied for the creation of microchannels within many different thermoplastic materials, however PMMA stands out amongst the rest [34]. A study found it to be the most stable polymer structure for laser processing, seeing little to no change in colouration. Although it results do indicate that thermal degradation occurs, similar to unzipping, the inverse process to polymerization.

PMMA has also seen usage within laminate manufacturing, this is a process in which individual layers are formed then stacked together to create a microfluidic device [35]. Research has shown that this technique can make rapid prototypes for microfluidic devices with low investment costs [36]. With this process up to 19 layers can be bonded at a time allowing for incredible intricate designs.

5.7 PADs

PADs (paper analytical devices), these microfluidic devices are the ones that the public are possibly most familiar, with a common example being a modern pregnancy test. They are a low-cost disposable microfluidic devices, that have seen great commercial success. Initial devices were extremely rudimentary focusing on simple lateral flow assays, with no complex pathing. A breakthrough was made in 2007 [15] where it was found that chromatography paper can be treated with chemicals to creative resistive areas allowing for more complex assay design.

These devices see the highest usage in the molecular diagnostics field, with the most recent surge in popularity being tests for the COVID-19 virus [37]. The pandemic saw huge investments into molecular diagnostic testing, with PADs alongside it. As these devices can be ultra-low cost allowing for usage in deprived countries where medical equipment standards are lower.

5.8 SLA 3D Printing

Stereolithography is the process in which two dimensional layers are stacked to created three dimensional objects, commonly referred to as SLA 3D printing. This technique is used in modern day "resin" printers. These printers use liquid photopolymer resin that cures into a solid when exposed to UV light, a screen in the base of the printer projects images of UV light that cure specific layers of resin. A "slicer" program is used with a CAD model to separate a model along the z-axis into images which when stacked together recreate the 3D model. These images are fed into the printer to create the object one layer at a time.

Compared to soft-lithography which needs to be carried out within a cleanroom area, it is far simpler and cheaper. Since the patents on Stereolithography printing expired in 2014, the market has exploded with various brands offering SLA printers. Making the devices affordable and an ideal budget manufacturing method [38].

For Microfluidic devices, SLA printing is able to create some models in time periods as low as 1 hour at resolutions less than one micrometre. Although for most commercial machines this is not the case, the level of detail is still exceptional. Rapid-prototyping is ideal with these devices due to their fast print speed and low-cost, it has been show that devices can be made for less than a dollar [39].

There is still one additional key factor that makes these devices so desirable, they do not require specialised facilities. The desktop nature of these machines allows them to be setup easily, with no need for a cleanroom or other such specialised area of operations.

For safety a fume hood should be used due to the toxic resin fumes. For researchers and students, SLA printing is as easy as selecting the file, print, wash and cure, the devices are suitable for workshops or laboratories.

Creation of microfluidic devices, using open channels has been explored with these printers. These are easier to print than closed channels and rely on using an adhesive tape to seal the 3D printed channels [39]. The issue with printing enclosed channels is removing the uncured resin from the devices post-fabrication. A crucial element to consider when speculating on using SLA printing for a microfluidic device is the resin material. Microfluidic devices are exposed to a wide variety of chemicals based on the desired usage of the device, as such Biocompatibility is a critical design concern for the manufacturing method. A study performed with various resins on the culturing of HeLa cells, found that the resin used can drastically alter the cell vitality [38].

Recent advancements in resin chemistry have lead towards low-toxicity resins that are biocompatible [40], such as polyethylene glycol diacrylate (PEGDA)- based resins which are suitable for short-term

cell culturing. In addition to the formation of new resin, research into resin coating has progressed. As SLA resin printed objects can have new surface layers applied through various treatment method which support device operations [41]. Coating resin-based microfluidic devices with natural bio-adhesive compounds is a promise future for obtaining biocompatibility with these devices [42].

SLA printers have shown great promise as a fabrication method suitable for microfluidic devices. Various researchers have tested the capabilities and limitations of various printers on the market. The main constraints for these devices are the resins material properties and the physical resolution achievable by the printer [43]. However, which each year these devices are becoming increasingly precise with resolutions starting to come down into the 50 micrometre range.

5.9 Valves in Microfluidics

A valve is a mechanical device that allows for the control of fluid flow within a system, acting as a gate/door that can open or close. For microfluidic devices they are vital for controlling the flow of fluids, they can be used for separating fluids from one another, sequencing operations or timing transportation operations. There are many valid use cases for valves in microfluidic systems, aptly named microvalves.

They can be mostly divided between two distinct categories, active and passive microvalves. From here each group can be further subdivided but generally the two categories do encompass most definitions [44]. As discussed previously the origins of microfluidic devices are entangled with electronic devices, more specifically microelectromechanical systems (MEMS) [6]. Active microvalves are often constructed using MEMS-based technologies, the first active microvalve was a solenoid plunger that was actuated using electromagnetic forces [6].

Magnetic [45], [46], electric [47], electrochemical phase change [48], rheological materials [49], paraffin [50], sol-gel [51], ferrofluids [52], thermal [53] and piezoelectric [54] are all example methods of actuation for active microvalves.

Passive microvalves are ones that do not require an actuation method, they can still contain moving mechanical parts however they open via forward pressure, acting similarly to a diode in an electronic system. They come in many forms, flaps [55], membranes [56], spherical balls [57] or mobile structures [58] and for non-mechanical valves, nozzles [59], diffusers [60], Tesla [61], Burst [62] and Hydrophobic [63]. This style of valve commonly sits at the inlet or outlets of a reciprocal displacement micropump acting as a part of the system.

As discussed, there is a huge variety in the valving techniques that can be used for microfluidic devices, making choice an important design factor that should be carefully considered. From the research performed here burst and hydrophobic valves both stood out as passive systems that can be cheaply produced.

5.10 Rotational Forces

When linked to a rotating reference frame, objects experience several relative forces. The equation of motion of such an object can be prescribed as the combination of three forces, Centrifugal, Coriolis and Euler. A rotating object experiences all three of these forces while rotational motion is applied. Understanding these three forces is vital to using rotational motion to transport fluids in microfluidic systems.

Centrifugal force is an inertial force based in Newtonian mechanics, it appears on all rotating objects, it stems perpendicular from the axis of rotation of the object. For a rotating circular platform, it will appear as force that drives all objects on the platform from the centre towards the outer radius of the circle.

$$F_{Centri} = -m\omega \, x \, (\omega \, x \, r')$$

Equation for centrifugal force, where m = mass, ω = angular velocity, r' = positional vector

The Coriolis forces act as a deflection, should the frame of reference (previous described as the centre of a circular platform) rotate clockwise, the Coriolis force will act to the left, should the rotation be counterclockwise, then likewise the force will act towards the right. It is a deflection force. The force is proportional to both the mass of the object and the cross-product between angular velocity & the object's relative velocity.

$$F_{Coriolis} = -2m(\omega \, x \, v')$$

Equation for Coriolis force, where m = mass, ω = angular velocity, v' = velocity vector

The Euler force occurs due to changes in the rotating frames angular velocity, it is a tangential force that "pushes" or "pulls" the object. An example would be during Olympic hammer toss as the hammer and athlete rotate together at the start up, the hammer experiences the Coriolis force which acts tangential to the rotational path, making it want to "escape" from the athlete's grip, requiring extra torque to continue holding onto it. The athlete often increases rotational velocity prior to release to purposefully increase Euler force to assist the hammer's trajectory as desired.

$$F_{Euler} = -m(\frac{d\omega}{dt} \, x \, r)$$

Equation for Coriolis force, where m = mass, ω = angular velocity, v' = velocity vector

5.11 Microfluidic Pumps

At the microscale, fluids do not move on their own due to viscous and capillary forces. Due to this flow control is required, which can come in active or passive systems [64].

Pumps are classified as active flow control systems. They are devices capable of moving fluid from one place to another through adding energy to the fluid, which induces flow or increases fluid pressure. They come in many forms each using different mechanisms to imbue energy into a fluid.

5.12 Gravitational Flow

Gravity-Driven flow is a form of passive flow control, as it relies on natural effects as opposed to the forces generated by pumps. Gravity is a rather simple and effective technique, with the main pitfall being the requirement for mass [64]. As microfluidic systems deal with small sample volume, generally they also have small mass, which makes gravity a relatively weak force. However, there has been proven success for devices that use this as their flow control method [65], [66], [67].

A restricting limitation to this application is that these systems can only generate continuous flow [68], meaning that flow is constant and there are no start-stop cycles. For certain systems dynamic flow or pulsatile flow is required, in which gravity-driven flow will not meet those standards.

5.13 Capillary Action Flow

Capillary forces are often used in microfluidic systems to allow for autonomous liquid movement without the need of any external energy. They are another example of passive flow control. Capillary action is dependent on the surface properties of the microfluidic channel, the internal capillary dimensions, the fluids type and viscosity. Adjusting these parameters allows for control over the capillary flow and thus the fluids velocity.

There are many examples of different applications of capillary action flow in microfluidic systems [69], [70], [71], [72]. When designed correctly they are a potent passive flow control system. However, their design is incredibly difficult to engineer due to the list of parameters required to tune. Satisfying all design conditions is arduous and the manufacturing techniques necessary to develop the capillaries is similarly obtuse.

5.14 Centrifugal Flow

For Lab-on-a-Disc (LoaD) devices (**section 5.23**), pumping can be handled by the centrifugal forces already present. Through rapidly rotating the disc, centrifugal force can pump fluids in proportion to the applied angular velocity. However, this is only capable of moving fluids outward radially, meaning the pumping action only has a singular direction. Additionally, the degree of pumping corresponds to the radial distance from the centre which can be a limiter to some designs.

Experimentation of combining pneumatic forces with unique geometry has found success in achieving bidirectional flow for centrifugal devices [73]. Meaning that while unidirectional flow is the limitation of centrifugal flow on its own, combination with other types of flow can allow for new opportunities.

5.15 Centrifugal Microfluidics

Centrifugal Microfluidics is a subfield of microfluidics that relies on centripetal forces to transport fluids. These microfluidic systems often come in the form of CDs, or compact discs, they centrifugally induce pressure by rotating the disc. These systems rely on varying the rotational speed of the disc as well as the geometry and layout of channels to control the transportation of fluid within the system [74].

Centrifugal forces are used to pump the fluid but can also be used to actuate valves such as burst or capillary valves. In addition to the centrifugal force induced via rotation there is also Euler and Coriolis forces, both of which must be considered when designing system geometry. The Coriolis force is particularly useful for switching operations as it can be harnessed to create a “Coriolis Switch”.

The flow rate of aqueous solutions in centrifugal microfluidic has been contrasted to the simple centrifuge theory [75]. This was achieved by using two reservoirs connected with a microfluidic channel, this setup was constructed multiple times each experimenting with different geometries and materials. The simple centrifuge theory states that the average velocity of a liquid is given as:

$$U = \frac{D_h^2 \rho \omega^2 \bar{r} \Delta r}{32 \mu L}$$

Thus, Volumetric Flow Rate is given by:

$$Q = UA$$

D_h is the hydraulic diameter of the channel, ρ is the density of the liquid, ω is the angular velocity of the CD, r bar is the average distance of the liquid within the channel from the centre of the CD, delta r

is the radial extent of the fluid and μ is the viscosity of the fluid and L is the length of the liquid in the capillary channel.

During testing it was found that the simple centrifuge theory has a 18.5% coefficient of variation when compared to experimental results. There was an absence of systematic deviation which thus validated the theory for describing flow induced by centripetal forces. This proved the centrifugal like action can be achieved within a microfluidic device. Allowing for assay operation that usually require a centrifuge such as DNA Isolation to be achieved on a microfluidic device.

Thus, these systems have shown great success in cell analysis due to centrifugal actuation being independent of fluid properties such as pH, viscosity and conductivity [76]. As such centrifugal microfluidics has expanded into the Lab-on-a-Chip branch of microfluidics. Where they have become known as “Lab-on-a-Disc” devices.

5.16 Capillary Valves

Capillary valves are a form of passive valve that rely on capillary forces to open and close. On a fundamental level these are forces exhibited by liquids within small areas. In a more technical sense they are a form of surface tension that occurs at a liquid-air interface of a microfluidic channel [69]. Dependant on whether the surface of the channel is hydrophobic or hydrophilic this generated pressure will be either negative or positive. As such the key factors are the liquid’s surface tensions, the channel dimensions, and the contact angles the liquid makes with the microchannel surfaces.

A common example of usage for microfluidic devices is burst valves. Which can be designed to use negative pressure to keep the liquid enclosed within the channel until additional pressure is applied, “bursting” the capillary seal and allowing liquid flow to occur. This concept is incredibly powerful as it allows for allow for actuation of liquid without mechanical components, if enough pressure can be generated through passive forces, such as gravitational, or centrifugal force.

5.17 Hydrophobic Valves

Hydrophobic valves use a hydrophobic surface to prevent fluid flow [74]. These valves are based on the pressure drop within a channel, using the Hagen-Poiseuille equation for their design. For these valves to open, centripetal forces are exerted on the liquid to overcome the pressure to move past the hydrophobic surface area. While similar to capillary valves there are slight difference between the two which make one preferable over the other dependant on the system.

5.18 Siphon Valves

Siphon valves are a form of valve that can be used to move liquid inward radially in a centrifugal microfluidic device. They work by spinning a disc at a low speed which allows for capillary forces to prime a siphon, then the valve can be actuated by sequentially increasing and decreasing spin speed [77]. These are powerful valves, but require a lot of space to execute properly which is a valuable resource for LoaDs that aim to be as compact as possible. A major issue associated with these valves is reliability [78]. These valves are reliant on capillary force to advance or retard the flow, as capillary forces are sensitive to a liquids surface energy which can change with temperature. Thus, it is possible during the operation of microfluidic devices, that these valves can fail due to increases in temperature, for example LAMP devices [79].

Kinahan et al [80], utilized Centrifugo-pneumatic siphon valves for a LoaD device that processed whole blood on a centrifugal microfluidic disc. For their specific purposes, these valves were a better option than common capillary burst valves. These siphon valves have low-pass flow-control elements which remain closed at fast centrifugal speed and then open at a geometrically defined threshold spin rate. When linked with capillary burst valves they allow for sequential release of liquids.

5.19 Fishbone valves

Fishbone valves are a form of trapping fluids by inducing a vacuum effect. They are alternative to capillary valves, that were designed to overcome the problem of capillary valves losing function when holding protein solutions [81]. Fishbone valves solve this by creating a superhydrophobic surface, which is done by creating a series of nanosized protrusions, the design is based on the Lotus leaf. In practice, these valves appear like capillary valves that have square like protrusions on either side. Their superhydrophobic nature make them useful beyond just protein applications, they can be used to improve capillary designs where it is not possible to reduce channel diameters for some specified design constraint.

5.20 Burst Valves

Burst Valves is a term that is used to refer to Capillary valves that use Centrifugal pumping to open. Thus, they are commonly seen in Lab-on-a-Disc devices. These are a passive valving technique that can be actuated using the centrifugal forces that are already in place for a Lab-on-a-Disc system. [82]

These valves are designed so that at a specific frequency centrifugal force overcomes capillary force allowing liquid to flow through the valve. During the design process these “burst” frequencies can be tailored and adjusted through various parameters, which can allow for synchronous or asynchronous operations. Making these valves desirable for complex assays [83], [84], [85] [86]. Some attempts have been made to control this burst frequency like a programmable variable [87].

As there is a desire to design around the parameters of these valves, several mathematical models for describing their characteristics have been developed. These models show that the operation of these valves depends greatly on the physical design of the valves, the material used to construct as well as the fluid being transported, as all these variables influence the burst frequency of the valve. This made calculating the exact burst frequency a challenging task. Both Hydrophobic and hydrophilic materials [88] have been used to create these valves however complex models address hydrophilic valves only [89],[90].

5.21 Coriolis Switch

A Coriolis Switch is a component that achieves transversal flow control within a centrifugal microfluidic device. It is based around the Coriolis force, a pseudo force that occurs when observing an object under centrifugal force. It appears as a slight deflection when viewed from the rotate frame of reference, in the case of a LoaD device it appears as though the fluid is skewing left or right dependent on the rotational direction. A y-shaped junction can be built to harness this effect, taking a fluid from one input and moving it to one of two different outputs dependent on the rotational direction of the disc.

Research has found that channel geometry, radial position and angular velocity of the disc are all key factors for designing around this force [91]. Of these factors, channel geometry is a key design concern. This force is considered “weak” so in order for it to sufficiently “move” a liquid into a desirable position a channel of suitable length is required. To ensure that the liquids tangential position is skewed before reaching its radial destination, “the end of the switch”.

These switches are useful for flow sequencing and are ideal for operations such as separating product and waste from one another within a complex assay.

5.22 Micromixers

Rapid mixing is vital in most microfluidic systems in order to perform biochemical analysis, sequencing or synthesis of nucleic acid and/or drug delivery [92]. Importantly they are necessary for Lab-on-a-Chip platforms to perform complex chemical reactions. Micromixers can be separated into two distinct categories, active and passive mixers.

Passive micromixers require no external energy and come in an array of forms, such as lamination, injection, chaotic advection and droplet. Active micromixers use external forces to generate disturbances required for mixing. They are available in a wide variety of types, pressure disturbance, electrohydrodynamic, electrokinetic, acoustic and thermal. These mixers are often more complicated and require more advanced fabrication processes. Making integrating them into microfluidic system challenging and costly when contrasted to their passive alternatives.

When considering the combination of two different fluids, stirring may be the first mechanical motion that comes to mind, placing both liquid into a contained area and mixing the two via large fluid motion. This is known as turbulent mixing, where turbulent flow is induced into the combined liquid through external mechanical motion. The key word here is turbulent flow, something that normally does not occur in microfluidic devices due to the low Reynold number present (**section 5.2**).

Microfluidic devices require the usage of other methods such as chaotic advection, this is the concept of splitting, stretching, folding and breaking up fluid streams, which is highly effective at a low Reynolds number making this method more plausible [10]. Advection only takes place parallel to the flow direction, limiting usage for transversal mixing, creating a design constraint that requires geometric design modification or external driving forces to address it.

The approaches taken to solve mixing requirement vary dependant on the Reynolds number present, in cases with extremely low Reynolds numbers ($Re < 10$), the flow can be described as strictly laminar and mixing can occur exclusively via molecular diffusion. Both of these outlined methods are examples of passive micromixers, using diffusion or chaotic advection to achieve mixing action. However, it should be noted that purely diffusive based mixing is a very inefficient approach to mixing.

For larger Reynolds numbers (approximately between 2 -> 100), modifications to channel geometry can be utilized to artificially induce flow through inertia effects [93], [94], [95]. Design channels with 90 degrees bends that “fold” the liquid mixture onto itself to create a lamination based micromixer. Similarly, Chaotic advection-based micromixers can be created through usage of intersecting channels [96], [97], [98]. Dependant on the fluid and Reynolds number present there is a diverse range of passive approaches to improving fluid mixing.

5.22.1Micromixers in centrifugal microfluidics

In the field of centrifugal microfluidics, the forces present have been investigated thoroughly for their abilities to enhance mixing operations [99]. Euler forces generated by periodically adjusting the centrifugal speed which can provoke fluid mixing [100]. Coriolis force can also be used to influence mixing with unique channels [101].

Taking advantage of the rotational motion through a moving magnetic field with corresponding magnetic beads to improve mixing conditions has also occurred, which can be optimal for certain assays which already use these components such as magnetic bead based DNA isolation [102]. Novel mechanical approaches have also been researched such as a disc that moves also a rotational shaft with the disc to agitate liquids by pressing up against a film, creating chaotic advection inside the disc [103].

Notably due to its independence from changes in disc velocity the Coriolis force is suitable for mixing application in wide variety of processes, such as standard laboratory centrifuges [104].

5.23 Lab on a Disc

Lab-on-a-disc (LoaD) systems are centrifugal microfluidics devices designed for the purpose of delivering point-of-care (POC) diagnostics that are affordable and fast, when compared to traditional laboratory operation, such as molecular diagnostics [105].

They centre around the idyllic goal of “Sample-in-answer-out” [105]. These systems aim at being light-weight and compact to allow for practical field usage, hoping to remove the need to send samples away to a laboratory to have biological testing performed by a trained professional.

The process of sending samples to a laboratory for testing is both costly in terms of monetary value and time for critical applications [106]. Ideally these systems are able to be used at the required point of care by operators with minimal training, allowing for the insertion of the sample onto a platform preloaded with the supplementary required chemicals for bioassay testing [106], to deliver results within an hour [107].

Centrifugal flow is a key factor in allowing these LoaD systems to be so lightweight and easy to operate. However, valving is an essential technique to achieve centrifugal microfluidic analysis, all systems require them to properly retain fluids until the analysis step where they are required.

Many approaches have been taken to valving technology on the LoaD platform, burst valves, capillary valves, gravitational flow [108], heat activated wax deposition systems and dissolvable films, to name a few.

These platforms are able to automate various laboratory task such as mixing of reagents or metering sample fluids, allowing for a sequence of analysis stages on a single disc [109]. A key usage of the LoaD system is for molecular diagnostics, using the platform to gather answers based on a blood or saliva sample.

This application has seen a great degree of usage across various fields, methicillin-resistant Staphylococcus aureus (MRSA) [110], respiratory tract infections [111], respiratory tract disease pathogens [112], cancer therapy [113], detection of foodborne pathogens [114], and notably detection of SARS-CoV-2 infections [115]. LoaD system have also proven themselves to be viable in extreme environments where typical point of care testing is not available [116].

5.24 LoaD Case Study: Flipping

The work of *A. Gholizadeh et al* presents a unique approach to achieving more compact sequential operations on LoaD devices [117]. They used microfluidic chips inserted into a disc platform, acting like cartridges which can be removed and “flipped”, changing the relative direction of centrifugal forces present within the chip, while leaving the disc platform itself unchanged.

Their paper’s novel approach was studied as part of research towards this project to investigate any shortcomings or discoveries that were made, using this gathered information as a stepping stone.

The chips consisted of a well, metering chamber, waste chamber, vent channel and storage chamber. The goal was to meter the liquid from the well to the metering chamber using centrifugal flow, any excess was sent to the waste chamber, the chip was then flipped allowing the liquid to flow from the metering chamber to the storage chamber. The liquid chosen for testing was a diluted dye solution, and the chips themselves were sealed using a sealing tape.

Tests were performed in two distinct configurations. For the first the device consisted of a circular disc platform with 3 equidistant spacings around the disc for the insertion of microfluidic chips. Three identical chips were placed into the disc, and two layers of poly(methyl methacrylate) (PMMA) was used to “sandwich” the chips and disc together. This is referred to as the “chip-on-a-disc” configuration.

The second configuration utilized a vertical arrangement, placing four chips upright into a 3D printed support affixed into the swinging-bucket rotor of a commercial lab centrifuge. This is referred to as the “chip-off-a-disc” configuration.

Measurements took place primarily using the “chip-on-a-disc” however for very high thresholds the “chip-off-a-disc” was required. This was due to the “chip-off-a-disc” rig having a far greater maximum speed than the “chip-on-a-disc” rig.

An instability at the liquid interface was observed during the “flipping” process. Where fluid trapped at the bottom of the chambers moves away upon flipping, if the centrifugal acceleration was above a certain threshold.

This is similar to the Rayleigh-Taylor instability [118], which prescribes an instability that occurs between two fluids of different densities, in this example the dye solution and air. The centrifugal forces create an inertial effect that destabilizes the liquid interface, while capillary forces stabilize it.

This paper goes in depth on the control parameters of this threshold value linking it to the chamber’s dimensions at the level where the dye and air fluids interface. This instability is a key design parameter to be considered for LoaD systems where reorientation of samples occurs.

5.25 LoaD Case Study: Mechanically Programmed Valving Technology

A wide variety of valving techniques have been tested on the LoaD platform, generally these valves need to be relatively lightweight so they can be fabricated for each disposable microfluidic chip, meaning that they cannot be reused, which drastically reduces the scope on available valve technologies. Pinch valves are widely used in microfluidic fields due to the efficiency and stability, studies performed by Xiang et al. (2018) outlines an approach to incorporating them into a centrifugal microfluidic device [87]. It proposes a pinch valve mechanism that is composed of a microfluidic channel that is sandwiches between a membrane and the disc itself. A spring-loaded plunger applies a “pinching” force to the membrane to the keep the valve normally closed. When experiencing centrifugal force, the weight below the plunger slides across, allowing the plunger to rest into a groove, this in turn lessens the pinch pressure against the membrane enough to allow fluid to flow. Due to the spring loaded nature of the platform, when the centrifugal forces become less than that of the platform spring it returns to its original position, sliding back across, in turn raising the surface beneath the plunger which then reapplies the “pinching” force.

Experimenting with this spring force, the weight beneath and groove position allows for adjustment of the centrifugal force required to open the valve. Centrifugal forces on the disc are varied by the rotational speed of the disc itself, due to the valve acting like a burst valve under these conditions the speed required the open the valve was coined as “burst frequency”. Precise control over this burst frequency opens the door to sequential operations, by having various valves positioned around the disc with different frequencies the order of operations can be mechanically programmed. Having the first operation linked to the lowest burst frequency in the range and the last operation to the highest.

When testing a single chip with this pinch valve approach the theoretical calculations for the burst frequency closely matched the experimental measured frequency across various spring stiffnesses, plunger-groove distances and platform weights. With the strongest correlations occurring for the varying of platform weight.

Issues began to arise when they increased the total number of valves across the disc, with the burst frequency straying further from the theoretical values with the addition of each valve. The normal compressing force from a single plunger to valve chip was speculated to be the cause of error as it was measured using a force sensitive resistor.

Initial attempts at using a flow switch were not effective at low speeds [119]. The stopping of the disc and high rotational frequencies were also found to cause issues as it could prematurely trigger other operational units within the system such as the siphon and capillary valves [120], thus adding more

limitations onto the disc design, meaning the pinch valve is not a one size fits all solution. There was some success with creating a flow switch with this pinch valve design, choosing to use two different burst frequencies to control the flow switch, as opposed to the traditional approach of changing rotational direction.

While displaying itself as an effective and relatively low-cost solution to sequential operations on centrifugal microfluidic devices, the pinch valve still requires more research to improve its accuracy and reliability.

5.26 DNA Terminology

5.26.1 Molecular diagnostics why they are important and such, point of care testing

Molecular diagnostics are tests that analyse the molecular structure of a patient, such as DNA to determine a person's medical health, for example, if they have a certain diseases or conditions. A molecular diagnostic that the public will be familiar with is the COVID-19 home tests. In which a small sample from the source patient is taken and tested to see if they were infected. In modern cancer research they have become vital for detection and targeting therapy treatments.

5.26.2 Assays and DNA

An assay is an investigative procedure used to assess or measure the presence, functional activity or amount of a targeted entity. The entity can be a drug, biochemical substance, chemical element, compound or a specific cell.

Deoxyribonucleic acid, better known as DNA, is a polymer that carries genetic instructions for the growth, development, reproduction and function of all known organisms. It is a nucleic acid which is one of the four major types of macromolecules required to for all known forms of life. It is a vital component for the field of Genetics. As DNA is the source from which all known forms of life inherit traits, such as Human receiving distinct characteristics of their appearance.

DNA acts as an identifier code for each human making it vital for diagnosing diseases, creating personalized medicine and researching many medical ailments such as cancer. In 1990 the Human Genome Project commenced, it was an international project with the aim of mapping and sequencing all the genes of the human genome. It was publicly funded by the National Institute of health and the US Department of energy, with a cost of 3 billion US dollars [121]. It was fundamental to modern understandings of the human genome.

5.26.3 DNA Sequencing

DNA is composed of four base nucleotides, adenine, thymine, cytosine and guanine. DNA sequencing is the process of determining the sequence of these bases. By sequencing these bases the genetic information of the data can be read, it is the process of data extraction from DNA. It was vital to Human Genome project, which in turn caused a push for the advancement of microfluidic technology.

5.26.4 DNA Isolation

DNA isolation is the process of separating DNA from a cell so it may be studied. Cells are taken from a sample, and Lysis is used to break up all the components. This DNA sample can then be bonded to some material such as beads. Wash Buffers are utilized to wash all components away bar the DNA itself. An elution buffer is added to the extract the DNA from the beads, this solution can then be used for PCR.

5.26.5 PCR

Polymerase Chain Reaction is a technique used to copy a segment of DNA, it acts like a “photocopier”. This technique allows a tiny amount of DNA to be amplified to a scale that it can be studied in detail, as such it is vital to molecular diagnostics.

5.27 DNA Isolation on LoaD Devices

Due to their usage of centrifugal force, LoaD systems are innately well-suited to cell analysis tasks. For molecular diagnostic purpose a key task that has been studied in depth is DNA Isolation. Over years of research many teams have taken different approaches to achieving DNA Isolation on a LoaD device.

A comparative study into Microfluidic systems for extraction of nucleic acids via human samples reveals some interesting insights [122]. Typically for centrifugal systems, magnetic beads are utilised. This assay requires design considerations towards the magnetic forces required for the extraction process. The first step of any DNA isolation process is lysing cells. LoaD devices have opted to do this on disk and off disk. For certain devices it is not possible to perform lysis on disk without major modifications, making it part of off disk pre-treatment more feasible.

This is not to say that on-disc cell lysing is impossible. *Kim et al* [123], achieved this goal through usage of beads and mechanical action. Rotating the platform at high speed in oscillating directions for upwards of 5 minutes to achieve the desired interaction. Proving that it is possible, however they only achieved 65% of the efficiency of a traditional off-disk lysis protocol, meaning integration of on-disk lysing may be not worth the resources it consumes for certain devices.

Kido et al [124], took a magnetic based approach to cell lysis. Creating a microfluidic platform that has an oscillating magnetic field, and ferromagnetic discs inserted into the LoaD device. During operation these discs move creating shear forces that ensured lysis occurs in even the most difficult cell types. Again, proving the on-disc lysis is possible but it becomes an element that the whole device must be designed around.

Lee et al [86], created a LoaD platform capable of performing molecular diagnostics from a whole blood sample all contained on the disc platform. This was achieved through laser actuated valves and centrifugal force. They used a capillary valve to separate blood from plasma, then used the plasma to isolate a DNA sample, using the proper reagents to perform molecular detection. In this study they performed a microbead-based suspension assay, as opposed to magnetic microparticles. Interestingly this study researches the effects of spin speed on the beads. Spin speeds that were too great caused the beads to sediment; this is an interesting design concern. Optimal spin conditions did allow for the return of beads to an optimal suspension state.

5.28 Learning Gaps & Research Questions

Research shows that microfluidics are an ever-evolving research sector with continuing advancements. Many different groups across the world have come up with novel solutions for addressing various tasks, valving, pumping, mixing etc. These devices have shown their successes in performing different molecular diagnostic assays. Although single-layer LoaD devices that rely on purely passive mechanisms show a trend where moving inwards radially is either difficult or requires a large amount of disc space. The most common passive methods for inward radial movement as well as their trade-offs are highlighted in table 5.28.1.

Table 5.28.1: Radial Inward Movement Methods and their trade-offs

Method	Trade-Off
Pneumatic Pumping	Requires a large air compression chamber that takes up large area of disc surface
Capillary Action	A weak force relative to centrifugal force so often requires the disc to stop spinning entirely or spin very slowly.
Vacuum Suction	Uses differential pressure, also requires large reservoirs to create necessary pressure which uses disc space

For a single-layer purely passive LoaD device, there is no true method to inward radial movement that is easy and does not require crucial disc real estate. The vLoaD proposes the usage of gravity-driven flow to overcome both limitations. Using gravitational flow to pump fluids, in conjunction with capillary valves to allow for sequential liquid operations. This approach will minimize the required disc space, and it is easy to design for control, as it is dependent on geometry. This is achieved by rotated the disc such that the plane of rotation is parallel to Earth's gravity, hence the "vertical" naming.

The aim of this thesis is to document and analyse the entirety of the vLoaD project from initial design to final assay testing. Presenting a new approach to LoaD devices that should allow future development to the field. To achieve this the following research questions are proposed:

RQ1: Fluid Mechanics

How do gravitational and centrifugal flow influence microfluidic device operations in a vertical rotating frame specifically metering, valving and fluid flow?

RQ2: Microfluidic Valving

What are the **operational limits** of passive capillary valves within the vLoaD architecture, and can they maintain hydraulic sealing against the gravitational pressure?

RQ3: Fabrication & Feasibility

Is 3D printing a viable fabrication method for vLoaD devices, given the require surface finish and dimensional tolerance requirements of a microfluidic device dependent on capillary action?

RQ4: System Performance Validation

Can the proposed vLoaD platform autonomously perform a multi-step assay such as DNA Isolation with comparable performance to traditional benchtop laboratory methods?

6 Ethical and Sustainability Considerations

Good Ethical and Sustainability practices are one and the same with good engineering practices. Engineers are one of the roles in society where ethical and sustainability practices are of the utmost importance, Engineers have a responsibility to conduct their research and work with consideration to ethics and with the best sustainable practices employed. The Engineers Ireland Code of Ethics states *"Members must act at all times with fairness, courtesy and good faith toward all persons with whom the member has professional dealings, and in accordance with the public good".* In research, it can be easy to solely focus on the end goal and not consider the means to which the goal is achieved, waiving aside proper ethical and sustainability practices. However, that is not the way in which an engineer should conduct themselves. Thus, for the vLoaD project where relevant, good ethical and sustainable practices were always considered, shaping the course of the project itself.

6.1 Ethical Concerns

6.1.1 Stakeholder Impact

Doctors, Nurses, Paramedics, and Patients are the main beneficiaries of the vLoaD project. As a device capable of performing assays that would traditionally require a laboratory, the vLoaD should allow for future improvements in point of care testing.

This will allow Doctors, Nurses and Paramedics to receive diagnostic information, in turn patients will receive a high quality of personalized medical treatment. While the vLoaD so far has only been used for a DNA isolation assay, it is an architecture that can be further researched and expanded for usage of other assays, for example the separation of plasma from whole blood could be applied to the vLoaD.

Since the vLoaD aims to explore a new platform for future devices it has additional beneficiaries', Scientific researchers / medical device engineers. They can take the research applied within this paper to construct their on vLoaD devices for different chemical assays, applying similar techniques for utilizing gravitational flow to adapt existing LoaD devices to the vLoaD platform or to create new microfluidic devices. Additionally, as a DNA isolation device the vLoaD could be used by chemists/biologists to extract DNA from cells.

6.1.2 Transparency and Public Relations

Public acceptance is an often overlooked part of innovation, it is of no benefit to create a medical marvel if the public do not trust the technology, as adoption will never arise from mistrust. An example that is relevant to the vLoaD is Theranos.

Theranos was a medical company that promised a breakthrough in health technology. Rapid and accurate blood test from a tiny amount of blood using compact automated devices. Their technology was guaranteed to work for years according to statements from the company. However, when questioned by third parties the company had no evidence to back up their claims it had all been smoke and mirrors. The company collapsed from there due to various legal and economic issues, but the point remains even if the company stayed afloat the public would never place their trust in any product Theranos created going forward.

For the vLoaD, it is important to specify exactly what the device can achieve and any drawback associated with it. A notable flaw is while the device was designed to perform DNA isolation assays it is not able to do so from a whole blood cell. It is not a magic machine that can take a single drop of blood and isolate DNA wholly. It is a smaller part of a subsystem, realistically a device for separating blood and lysing cells would need to be used first, then the vLoaD, then some machine afterwards would take the DNA and produce the actual genetic diagnostic read out. It is not an all in one point-of-care machine, it is part of a family of machine to achieve such a goal.

6.1.3 DNA Source and Usage

The main ethical concern for this project was the DNA Isolation process. As a piece of genetic information DNA handling is a sensitive topic. This problem first arose during the risk assessment process where initial plans incorporated the student using their own DNA for the DNA isolation process, to minimize risk of infection or contamination of a foreign body.

Human DNA contains information about health, ancestry, familiar connections and genetic risk factors. Therefore, the collection, storage, usage and manipulation of this “data” is an extreme ethical concern. Even though it would be the students own genetic information, should it be mishandled or used to uncover some health risk as a result there is a case of uncertain liability. Who is responsible for these shortcomings, the student or DCU for allowing the student to perform the testing comes into concern.

Using a lab-safe DNA sample, is a far more ethical approach to this problem, which avoids unnecessary concerns, the extraction process is the same regardless of the sample used. Thus, a decision was made to use lab-safe E. Coli samples for the DNA isolation assay portion of the project.

6.1.4 3D Printing, Chemical Exposure

During the fabrication process where SLA 3D printing was utilized. The student was going to need to handle 3D printer resin, a material which is highly toxic to humans. It produces fumes which can cause respiratory issues, with skin contact it can cause irritation and allergic reactions, and long term exposure can affect the nervous system. The health and safety of the operator was an extreme ethical concern for the fabrication element of this project. The 3D printer and resin always remained under a fume hood to prevent exposure to Volatile Organic Compounds (VOCs) fumes during printing.

Prior to the student commencing any independent fabrication it was necessary to construct a risk assessment form (see **Appendix 3**) and for the student to receive training from an PhD Engineer experienced with the equipment. The student was informed on PPE, proper operational procedures and resources should exposure occur such as eye wash stations. PPE was worn at all times when handling resin during the 3D printing process. When discs were removed from the printer they were handled with extreme caution and were not removed from the fume hood until cleaned and cured. As curing seals excess resin within the disc preventing exposure.

6.1.5 Laboratory Conduct and Safety

For the DNA preparation and vLoaD operation with the DNA isolation assay, laboratory work was required. When conducting work in a laboratory environment, safety of the operators as well as other individuals using the laboratory space is critically important. The risk assessment form, see **Appendix 3**, was fully completed and approved prior to commencement of any operations in the laboratory.

Chemical Usage:

The main chemicals used in the vLoaD project are the wash buffers and elution from the QIAGEN mini-DNA kit. These chemicals were stored together in their appropriate box, within the bacterial cell culture lab. When required for testing they were transported from the cell culture lab to the vLoaD rig lab, in which they were stored underneath a benchtop fume hood. When in use, each bottle was opened only for pipetting then resealed immediately.

Biohazards:

The vLoaD requires a lysed E. Coli DNA sample to be loaded onto it, and it has a DNA-elution solution as its end-product, both of these are biological materials which be seen as a potential hazard. The lysed DNA sample was prepared alongside Abrar Abdeslam, a PhD researcher experienced with handling the biological materials and chemicals involved. The sample was then stored inside a labelled collection tube inside the fridge of the bacterial cell culture lab. For vLoaD assay test it was transported to the vLoaD rig lab carefully and kept in Nicklas Rondot's storage box. Once testing was completed it was transported back to the fridge within the bacterial cell culture laboratory. The DNA-elution solution was the product of the vLoaD assays. They were stored in labelled collection tubes, within the same fridge storage box. Each sample is labelled in accordance with the test iteration, to make it easy to trace records of each sample.

Additionally, the disposables of the project are contaminated with these biological samples marking them as biohazards. The two main disposable products are the pipette tips and the vLoaD discs themselves once they are used. The tips were disposed of in two ways, a sharps bin or a hazardous waste bag. The sharps bin was always kept on the vLoaD workbench, being sealed when not being used and reopened during operations. The hazardous waste bags were used with an appropriate holder, a fresh bag was used at the start of each day and then sealed once testing was completed. The bags were then moved to the appropriate disposal point. For the vLoaD discs themselves, due to their size they were only disposed of within these biohazard waste bags.

Laboratory equipment:

For the preparation and testing of the vLoaD DNA assays, four main pieces of laboratory equipment were required, a centrifuge, a mini vortex mixer, the NanoDrop 2000 and a q-PCR machine.

For the centrifuge a standard bench top centrifuge was used. This was used to centrifuge samples, which was important for DNA sample preparation and on-bench control preparation. The student was trained to ensure to use the centrifuge in a safe manner and to also not damage the equipment inhibiting the research of others. When placing sample tubes into the centrifuge there is one golden rule, there must be a counterweight. Any time a sample was centrifuged it was ensured that a second sample or even empty tube was in the opposite positions. Once samples were placed within the rotor, they were secured using the rotor clamp, the lid was closed securely and then the centrifuge was used.

For the mini-vortex mixer, the student was mindful to hold collection tubes firmly and to place them against the vortex mixer appropriately, to ensure that the tubes could not be launched, causing damage or exposure to biohazards.

For the NanoDrop 2000, the student received training from Nicklas Rondot, on how to use the equipment appropriately. Ensuring to be gentle when using the lid mechanism, to avoid damage to the sensor or sample platform. Additionally, the machine was cleaned thoroughly in-between samples and once its usage was completed to leave the machine ready for its next user and to avoid cross-contamination.

For the q-PCR machines, the Rotor-gene Q or the AriaMX, the student did not use these machines as they did not have the appropriate training. Operations were carried out by either Nicklas Rondot or Abrar Abdelsalam.

6.1.6 vLoaD Rig

The vLoaD rig itself acts like a centrifuge, spinning its discs at high angular velocity, this posed two safety concerns. The disc could come loose and the hit the operator or someone else within the vicinity, or if a leak occurred within the disc to the external environment, it was possible for adverse chemical exposure to occur. These two problems were tackled with a single solution, a Perspex shield. This was affixed in front of the vLoaD to avoid the disc being able to launch freely and to trap chemicals inside should a leak occur. Reducing the chance of hazards to the operator and surrounding personnel.

For Basler camera usage, the IR sensor would not operate correctly with the Perspex shield due to its reflectivity and how it requires being in extremely proximity with the discs surface. In addition to this the strobe produced extremely bright light which was uncomfortable for all surrounding personal while posing a risk of causing an epileptic attack. To combat these issues, a wall of black Perspex was built around the rig, so the Perspex shield could be removed for water-dye tests and the camera could function appropriately. For DNA tests however the Perspex shield stayed in place.

6.1.7 Reliability of Device

As a medical tool, it is important that functionality can be guaranteed, a medical device cannot fail unexpectedly as it could lead to severe medical implications. The vLoaD itself is a framework for other applications not solely a DNA isolation device. However, as it is presented in this way in the thesis it should be discussed as such when it comes to reliability. In the analysis and the conclusions of this thesis the vLoaD devices abilities to perform a DNA isolation assay were critiqued harshly due to this. To present the findings and results in an honest format to avoid misleading readers about the success rate of the device. Additionally, the outputted DNA was unbiasedly compared to a laboratory on-bench control, to truthfully display the capabilities of the device.

6.1.8 Ethics of Research

To maintain the standards of practice in accordance with Dublin City University and Engineers Ireland, ethical and sustainable practices were implemented throughout the entirety of the vLoaD project. Respecting ethical concerns when they were flagged and acknowledging the responsibility of a research engineer. All testing throughout this project was conducted in the safe manners possible with the resources available. Design decisions such as using E. Coli as opposed to Human DNA were made to minimize ethical risks. The student received training on all equipment required or was under supervision when required. At the front of this thesis a plagiarism declaration can be found, signed by the author of this thesis, this is to state that all work produced for this thesis is of the authors own hand or is reference appropriately. As a representative of DCU it was my responsibility to do this to uphold their academic standards as well as my own. All testing for this project was done in ways to avoid bias such that all findings are true and not falsified in any way. All graphs and figures present are unaltered, all metrics are their true values, no data has been edited or constructed in anyways to lead readers astray. For the analysis of results, a critical point of view was presented to avoid looking at the project favourably or introducing any bias that obscures the truth of the projects findings.

6.1.9 Artificial Intelligence

Dublin City University recognises that Artificial intelligence can be a positive impact on research, however users must act in an ethical manner when using AI. Prior to the usage of any forms of AI with respect to this project, DCU's stance on AI was thoroughly researched.

(https://www.dcu.ie/ocoo/dcu-position-statement-use-artificial-intelligence-tools)

For this project usage of AI was used in an ethical manner where no compromising data was provided to any AI tools, to avoid the loss of privacy or ownership on data. Google Gemini was the sole AI tool used as it is one of DCU's approved AI tools. When used with a DCU account Gemini does not use any prompts or chats to train its model.

Google Gemini was primarily used as a search tool, providing prompts to find research papers or other relevant sources as opposed to taking answers directly from the AI. In cases where information was not available via traditional means for the purposes of developing this project and AI had to be employed, the information was taken while exercising my engineering intuition and employing critical thinking. This ensured that AI did not replace proper engineering judgement. Where information provided by Google Gemini was utilized in this project, an appropriate reference is provided which state that Google Gemini was used.

6.1.10 Summary of Design decisions made based on Ethical Concern

The ethical discussions and considerations throughout the project have resulted in several distinct design choices that were made to upkeep good ethical practices, they are as follows;

- Integration of Perspex shield at the front of the rig to protect operators
- Usage of E. Coli for DNA isolation as opposed to a Human DNA sample
- Sealing of disc with PSA, to minimize chance of leakage for samples or reagents.
- Choice to use q-PCR, as the testing metric as it does not examine the DNA itself, only the quantity of DNA present.

6.2 Sustainability Concerns

Every engineering project involves working with consumables, materials, chemicals, electricity, etc. Sustainability is a vital consideration when using such resources, as it wasteful and harmful to unnecessarily use that which is not needed.

6.2.1 3D Printing

3D printing is an area in modern times where engineers need to consider sustainability. This technology has made rapid prototyping extremely convenient. Which such convenient sometimes proper design processes are abandoned and prototypes are printed repeatedly, haphazardly. This can lead to unnecessary waste of materials and is an unsustainable practice.

For this project it was ensured for each design iteration to maintain proper design processes, taking time to evaluate the design before using the 3D printer. This ensured minimal wastage. Additional care was taken, as the discs are a single-use devices, to ensure optimal testing conditions were always maintained as opposed to conducting tests inadequately.

6.2.2 Reagent Usage

For the DNA Isolation assays performed on the vLoaD rig, the usage of reagents is required. These come in the form of the QIAGEN mini-DNA kit, with two wash buffers and one elution buffer. While required for DNA isolation, these chemicals are expensive and are consumable. To avoid unnecessary wastage, all prototype testing was done using water-dye solutions. Additionally, at some points throughout testing major design changes were required, such as the introduction of the S-Chamber. Water-dye testing preceded any DNA testing when a change such as this occurred. This was at the expense of “wasting” a disc for water testing, however it was deemed more appropriate to “waste” a disc as opposed to wasting reagents. It was a hard decision to make as wastage at all is unsustainable, but it was a forced trade-off between the two.

7 Project Planning

7.1 Project Mission

The main goal of this project is to design and test a new automated Lab-on-a-Disc system which uses both gravitational and centrifugal forces to pump fluids and actuate valves, with the aim of performing a DNA isolation assay on-disc to validate the concept. To achieve this there were three components, the rig, the disc, and the operational software. The rig was already designed and constructed by the PhD associate for basic testing, but further adjustment was a possibility dependent on the assay chosen. The disc was the main design challenge of the project, proof of concept testing was required to choose a general approach then the disc went through phases of prototyping to refine this approach. As a fully automated device a piece of software was required to control the system, aiming for the software to be able to automatically run and control the disc through the required assay steps.

The reasoning behind this project is to prove that gravitational forces can be used to move fluids inward on a LoaD device, opening the door to more compact assays in the future. While only a concept, further research and incorporation of already known technologies such as dissolvable films or siphon valves, could enable the vLoaD to be even more compact making it a possible solution to practical point-of-care testing within the future.

7.2 Project Aims

At the beginning of the project, a list of aims was constructed, these would inform decision making for the project;

- Create a Microfluidic Disc that can use Gravitational Flow
- Use Gravitational Flow in conjunction with Centrifugal Flow
- Have the disc perform a DNA Isolation Assay
- Fully automated the developed platform

7.3 Project Objectives

From this initial list of aims, an array of objectives was devised for the project, which went on to inform the project scope (**Section 7.4**) and schedule (**Section 7.5**);

- Perform a through literature review of microfluidics and LoaD devices, to ensure that the concept is novel as well as to become familiar with the theoretical concepts of microfluidics. This is seen in **Section 5**.
- Apply this research to performing theoretical design calculations, ensuring that the design is following appropriate design rules and concepts. Minimizing 3D printing wastage due to half-baked designs. This is seen in **Section 8.2**
- Formulate an understanding of the materials used to construct the vLoaD discs and processes required. This is seen in **Section 8.1**
- Test the pre-existing designs concepts, drafted by the PhD associate, understand each approach and decide on which one should progress further in the design process. This is seen in **Section 9.3**
- Prototype the chosen design, progressing the design through various stages to meet the design requirements, adapting the design as necessary when changes are made in the testing assay. This is seen in **Section 9.4**
- Update the vLoaD testing rig as required for the DNA Isolation assay operation. This is seen in **Section 9.1.1** and **9.1.2**.
- Develop a software program that can control the vLoaD rig throughout all the assay steps automatically. This is seen in **Section 9.1.3**
- Perform a DNA Isolation Assay on vLoaD (**Section 10**) and using traditional benchtop approach, compare the two using appropriate testing metrics. This is seen in **Section 11**
- Meticulously analyse the final design to identify shortcoming and opportunities for further improvement with supplementary research. This is seen in **Section 12**

7.4 Project Scope

The goal of this project is the design of the vLoaD system during the student's final year, due to time constraints and other tasks required for the master's degree, time is a critical resource for this project. As such a Work Breakdown Structure was applied to this project to control the scope. This is a hierarchical breakdown of the tasks required to complete this project. Each of which with their own deliverables that when summed formulate the projects goals. This structure was a dynamic aspect of the project that adapted alongside the scope and scheduling requirements. This Work Breakdown Structure has three key areas;

- Work Package 1: Research and Planning, this was all the preparation work prior to commencing any lab work. It incorporates the literature review, theory analysis and project planning.
- Work Package 2: Proof of Concept, this is where the pre-existing designs were tested and a formulation of understanding for both the materials and fabrication methods was formed.
- Work Package 3: Design and Test, after selection of concept, prototype testing occurred until a testing design was reached. DNA isolation was performed on the testing design and the results were analysed. The software program was developed in the background alongside these tasks.

The Work Breakdown Structure diagram can be seen in **Appendix 1.**

7.4.1 Changes in Scope

The main change in scope that occurred in this project was the change from magnetic bead-based DNA Isolation to non-magnetic silica bead based. This change was made at the end of semester one. The design change was believed to be for the benefit of the system as it would lead to a more automated design. Magnetic bead-based DNA isolation required to activation and deactivation of magnetic forces during operation which would have required a code-controllable electromagnet or a mechanism to move magnets in and out of place. Thus, non-magnetic beads were chosen as the easier process to automate. During testing, issues occurred with the non-magnetic beads and magnetic beads were trialled instead, they showed more promising results at the trade-off of losing true automation.

7.5 Project Resources

This masters project was carried out as a joint research endeavour between the student and the RAPID institute. Most works were carried out independently by the student or via assistance of the PhD associate Nicklas Rondot.

7.5.1 Human Resources

The key stakeholders to this project are the individuals who contributed to the development of this project. Aiding at various stages to provide their skills to the student to assist them in project completion.

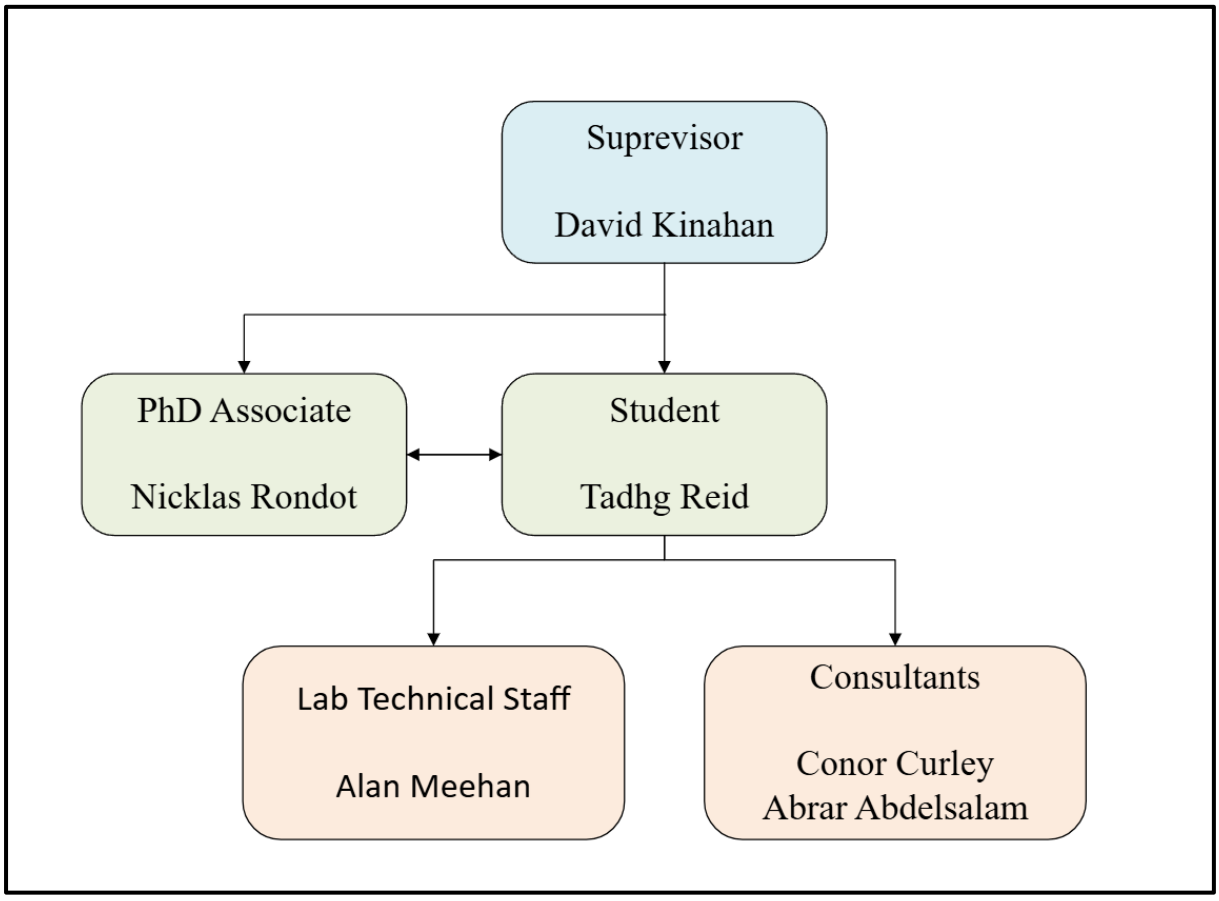


Figure 7.5.1.1: Diagram of Project Stakeholders

7.5.2 Financial Resources

The budget for this project was handled between DCU School of Mechanical Engineering and the RAPID institute. The majority of resources were already available at the RAPID institute, such as the chemical required for the DNA Isolation assay and the 3D printer setup used for fabrication. The only resource that was needed on the budget of this project, SLA Resin.

3D printer Resin was the main fabrication material for this project, as such it was needed to buy additional resin during the project to keep up with consumption. This was done through the School of Mechanical Engineering, 5 bottles of ANYCUBIC High Clear Resin were purchased directly from their EU website for the cost of 140€

7.5.3 Technical Resources

For this project the key technical resources were access to online publishments and software's provided by DCU. Publishments are available through DCU that cannot be ordinarily viewed such as Springer Nature, Elsevier and ScienceDirect. SolidWorks was the sole CAD software used for this project and accessing it would have been impossible without the license provided by DCU.

As the project was carried out in the RAPID institute all the lab facilities and equipment required for this project were provided by them. This included all fabrication and testing equipment, such as the 3D printers, hotplates, Basler camera and more.

7.6 Governance

7.6.1 Project Scheduling

Time management was possibly the most important skill for the proper execution of this project. As this project started in October 2025 and needed to be finished by March 2026, time was a finite resource. From the project outside proper time keeping was maintained to its upmost. This was done via Gantt Charts, these charts made using Google Sheets as it was possible to share the files with Nicklas Rondot, to always maintain communication. Two charts were created, one for each semester, and the tasks are aligned with the work breakdown structure. They can be seen in **Appendix 2**.

7.6.2 Hazard Assessment

Before any practical work was conducted a risk assessment form was created in collaboration with Nicklas Rondot, David Kinahan and Alan Meehan. This form outlines all the risks associated with this project, with a heavy focus on the chemicals required in fabrication and testing, each potential risk has had safety measures considered and implemented. The full Risk Assessment is available in **Appendix 3.**

7.6.3 Project Risk Mitigation

Potential shortcomings for the project were investigated in early phases by conducting a Failure Mode and Effect Analysis. Each potential risk was identified and given a severity score. A potential cause for each was researched and given an occurrence score. Controls were selected and a detection score was formed. The product of these three scores formed the RPN, risk priority number. As this was done on a scale of 1 to 5, the highest potential RPN was 125. Actions were recommended and implemented, then the three scores were reevaluated aiming to minimize RPNs throughout. It was determined that any RPN above 20 would not be acceptable, as this would mean there was a good chance two of the three criteria were at or near the maximum. The figure below shows a scoring matrix where detection is equal to 5 (at its worst) with severity and frequency varying. This was a dynamic document updated throughout the design process, the final version of the FMEA is available in **Appendix 4**.

Table 7.6.3.1: Risk Mitigation Scoring Metric

			Severity				
			Very Low	Low	Medium	High	Very High
			1	2	3	4	5
Frequency	Very Low	1	5	10	15	20	25
	Low	2	10	20	30	40	50
	Medium	3	15	30	45	60	75
	High	4	20	40	60	80	100
	Very High	5	25	50	75	100	125

7.7 Engineers Ireland Criteria

As the Integrated Masters programme at DCU is an accredited degree by Engineers Ireland, there are learning outcomes designated by them. They are as follows, with responses attached explaining how this project meets them;

LO1 – Knowledge and Understanding: This project required a deep understanding of microfluidics and the theory behind them to properly design the vLoaD discs. Research was performed into the design behind microfluidic devices and similar devices were studied to understand where other LoaDs succeeded and failed. This knowledge was vital to executing the project.

LO2 – Problem Analysis: The vLoaD project targets a previously unaddressed design issue with LoaD devices (the inability to move fluids inward passively) and developed methods to solve the problem.

LO3 – Design: This project proposes a solution that addresses the complex problem in a format that is validated by thorough design calculations and real-world testing.

LO4 – Investigation: This project investigates previous research towards its goal in depth. Identifying similar microfluidic devices and models developed, applying this knowledge to the project.

LO5 – Professional and Ethical Responsibilities: Ethical Responsibilities were identified and good practices were maintained for the entirety of the project. Both the people working on the project and the environment were considered during major decisions.

LO6 – Teamwork and Lifelong Learning: This project involved teamwork alongside fellow researchers to develop a project. It required good communication and teamwork skills to see fruition. This project incorporated skills from both electronic and mechanical disciplines that the student experienced throughout university as well as the learning of new skills, such as an understanding of chemical testing.

LO7 – Communication: This project required in depth collaboration between the student, the PhD associate and the project supervisor at all times, to ensure that a collective vision was maintained between parties. Meetings and design discussions occurred on a frequent basis where the student had to present findings in a clear and explanatory manner.

LO8 – Engineering Management: Due to the limited time available for the execution of this project and the requirement for the student to complete other external tasks for their master's degree, good time management skills were vital to the project. There was also a need for the student to understand when tunnel visioning would be the determent of the project, and the scope was reduced several times to ensure project completion. Choosing alternative approaches where possible for the sake of efficiently utilising the time available.

8 Theory

To display good design principles where possible theoretical microfluidic calculations were used to select parameters for the disc design, such as the width of capillary channels. This segment of the thesis will cover these calculations in depth, the decision-making process and the theoretical fundamentals behind each calculation. These calculations were performed in a dynamic excel design workbook, as such for any numerical values provided these are the values used for the testing designs. Calculations are presented based on the order they were performed in, during the methodology segment there will be references to these calculations and how they augmented the design between prototype iterations. For each segment the papers studied will be referenced and the background material described.

8.1 Material study

8.1.1 Water Testing on Hydrophobic and Hydrophilic Films

Currently at this stage in the project there were two concept designs, both of which required a hydrophobic and/or hydrophilic film to cover one or two of the disc's faces. As such it was required to perform tests on these films prior to usage on the discs themselves. Measuring their contact angles with a sample of water dye solution was key to understanding each film's properties. The contact angle indicates how well a liquid spreads on its surface, an angle below 90 degrees is typically that of a hydrophilic film whereas above 90 is indicative of a hydrophobic film. Table 8.1.1.1 shows the measured contact angle for each of the films. Due to the nature of how angles are measured for all contact angle measurements made on the right there is a converted value indicative of if they were measured on the left. A small amount of food colouring was mixed with the water to make it visible.

Table 8.1.1.1 Measured Contact Angle values for each film used

Hydrophobic	Left	Right	Converted right
IMG258	131.5	59.04	120.96
IMG261	124.11	61.78	118.22
IMG262	123.18	60.95	119.05
MEAN	122.84		
Hydrophilic	Left	Right	Converted right
IMG266	23.65	152.51	27.49
IMG272	19.94	157.5	22.5
IMG276	25.1	152.84	27.16
MEAN	24.307		

8.1.2 Channel Characterisation using Films

Sample discs (Figure 8.1.2.1) were constructed using chambers of varying lengths and widths to understand the effects of chamber dimensions in conjunction with each type of film. For each channel there was a chamber at either end, in these tests' liquid was placed into the chamber on the outer edge of the disc. This was done with the chamber in the 180 degree position, so liquid would be against the walls of the chamber, the disc was then rotated 180 degrees so that gravity would move the liquid towards the exit of the chamber. The time taken for the liquid to fully flow from one chamber to another through the channel was recorded. Analysing the resulting times from these tests for each channel dimension. All channel's dimensions were selected so each one has the same cross sectional area, 0.5 mm^2.



Figure 8.1.2.1: Testing Disc with various channel dimensions

Plot of these results are available in **Appendix 5**, the results were sorted so depth decreases while width increases, this is important to note as with the increase in width there is also an increase in surface area contact between the liquid and film. This process was done twice, once for the hydrophobic film, once for the hydrophilic film.

For the hydrophilic film testing it can be observed that as the channel width increased, the time taken for the liquid to travel remained mostly the same. For the hydrophobic film it was too difficult to see much of a correlation, as such a sample of the data was taken and it was seen that as width increased so too did liquid "movement speed".

8.1.3 Water Testing on Resin

During the process of perform calculations for the capillary forces it became necessary to consider the contact angle that the fluid made with the 3D printed disc. As during flow, the liquid will contact the 3D printed channel as well as the chosen film.

Table 8.1.3.1 Measured Contact Angle values for Resin

Resin	Left	Right	Converted right
IMG547	142.85	40.06	130.06
IMG548	138.27	40.36	130.36
IMG551	138.37	39.52	129.52
IMG552	134.06	38.48	128.48
IMG553	141.95	41.31	131.31
IMG554	135.66	43.33	133.33
IMG555	137.22	42.66	132.65
MEAN	134.578		

From this analysis, it can be observed that the disc is hydrophobic so for the liquid to flow easily a hydrophilic film will need to be applied as the top layer.

8.1.4 Ethanol Testing on Hydrophilic Film And Resin

The final part of this project will be to perform DNA Isolation analysis, to approximate the fluid used for this operation 96% ethanol was used. The contact angle was measured against the hydrophilic film and the 3D printed resin disc. A small amount of food colouring was mixed with the ethanol to make it visible.

Table 8.1.4.1 Measured Contact Angle values for Ethanol, Hydrophilic Film

Ethanol	Left	Right	Converted right
IMG580	11.93	163.46	16.54
IMG592	12.99	165.96	14.04
IMG594	12.32	167.77	12.23
MEAN	13.3417		

Table 8.1.4.2 Measured Contact Angle values for Ethanol, Resin

Ethanol	Left	Right	Converted right
IMG641	27.47	151.83	28.17
IMG642	20.91	163.83	16.17
IMG643	24.78	160.07	19.93
MEAN	22.905		

8.2 Design Calculations, Testing Design 1

This segment covers the full design calculations used to design the first main testing design, “Testing Design 1”. Some of these calculations were initially performed for prototype designs, this segment only covers the numerical values used for Testing Design 1.

8.2.1 Storage Chambers

The storage chambers located on the outer radius of the disc for storing the samples and reagents are a critical element of the angled chamber design. These chambers must be the correct volume to store enough liquid for the DNA Isolation process. Due to the nature of these chambers, they must be a volume larger than the amount of liquid to store the liquid properly during operation. See Figure 8.2.1.1.

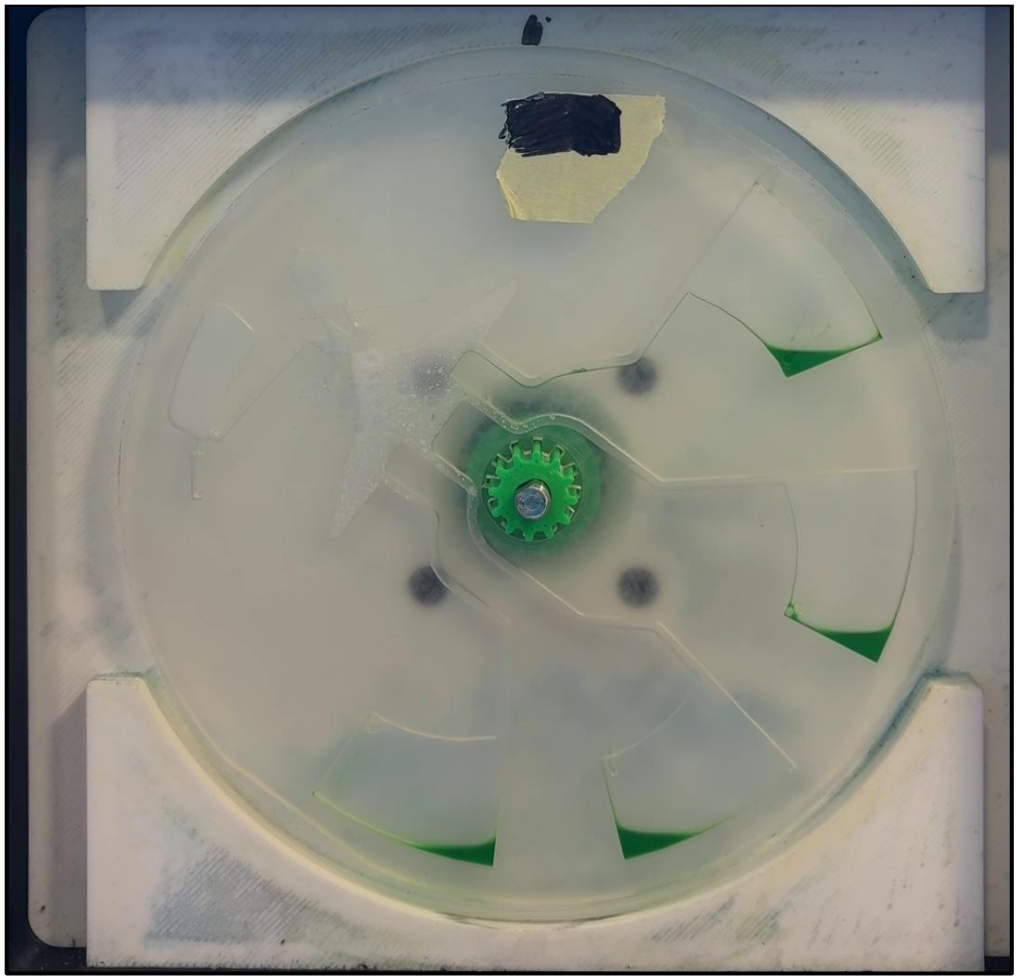


Figure 8.2.1.1: Example of how fluid is stored in vLoaD

The areas of each one of these chambers was approximated using the formula for a circular sector area, see eq. 1. This was done twice, first to get the approximate area of the entire chamber, then a second time to get the area at the exit.

$$A = \frac{1}{2}(R^2 - r^2)\theta \quad (1)$$

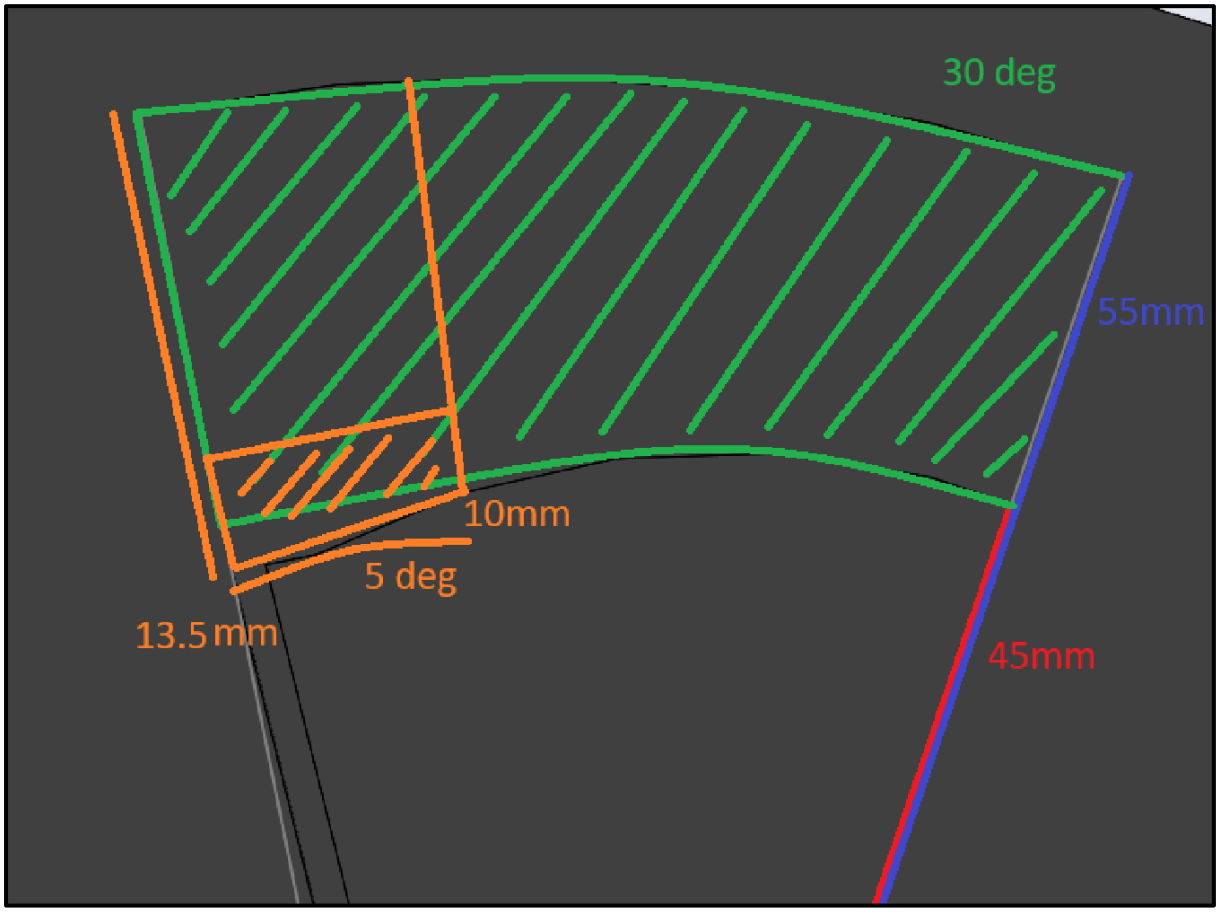


Figure 8.2.1.2: Area Segmentation for Storage Chamber

Table 8.2.1.1: Measurements of Storage Chamber

Depth	2	mm
Disc Diameter	120	mm
Disc Radius	60	mm
Distance from Disc edge	5	mm
Radial Length Short	10	mm
Radial Length Long	13.5	mm
Chamber Angle (Width)	30	deg
Chamber Angle Exit	5	deg
Outer Radius	55	mm
Inner Radius	45	mm

Table 8.2.1.2: Volume Calculations of Storage Chamber

Area (Green)	261.7994	mm^2
Extra Segment (Orange)	13.20996	mm^2
Total Area	275.0093	mm^2
90% Accuracy	247.5084	mm^2
Volume	495.0168	mm^3

A 90% value was taken of this area as it is an approximation at it would be better for the chamber to be oversized. The required amount of fluid was 50 µL for each sample/reagent, thus each chamber must have a volume of at least 50 cubic millimetres.

This is where the trick of the angled channel design becomes relevant. Through experimentation on the concept design, it was found that the first chamber could reliably hold 40 µL during operations, with a volume of 495 mm^3. This metric was used to scale the chambers so they could hold 50 µL. It was found that the desired chambers would need a volume of at least 618.75 mm^3. The inner radius of the chamber was adjusted until the desired volume was obtained. Once this design was printed, it was tested to the limits and found it could hold up to 250 µL. Although with the design of the mixing chamber and planned protocol if the volume was greater than 50 µL, backflow would occur into the storage chambers.

Table 8.2.1.3: Desired Volume Calculations of Storage Chamber

Desired Volume	618.75	mm^3
Outer Radius	55	mm
Chamber Angle	30	deg
Inner Radius	40	mm
Area (Green)	373.0643	mm^2
Extra Segment (Orange)	13.20996	mm^2
Total Area	386.2742	mm^2
90% Accuracy	347.6468	mm^2
Volume	695.2936	mm^3

Another key element of the storage chamber design was the spacing of the storage chambers apart from one another. This was to ensure that individual actuation of each chamber was possible. This was proven through experimentation of concept designs to be four chambers, as there needed to be 50 degrees of spacing between the exits. This was one of the key limitations of the design but for the current assay, four chambers was an acceptable amount.

8.2.2 Storage Chamber Leakage Prevention

In prototype version 11, a design update was made to the storage chambers. This involved the creation of a notch within the chamber which left behind a pocket for the liquid to sit in during spin cycles. This aided liquid retention in the chambers during operations. To ensure this area would be large enough to store the liquids volume, a calculation was performed like as seen in **Section 8.2.1.**

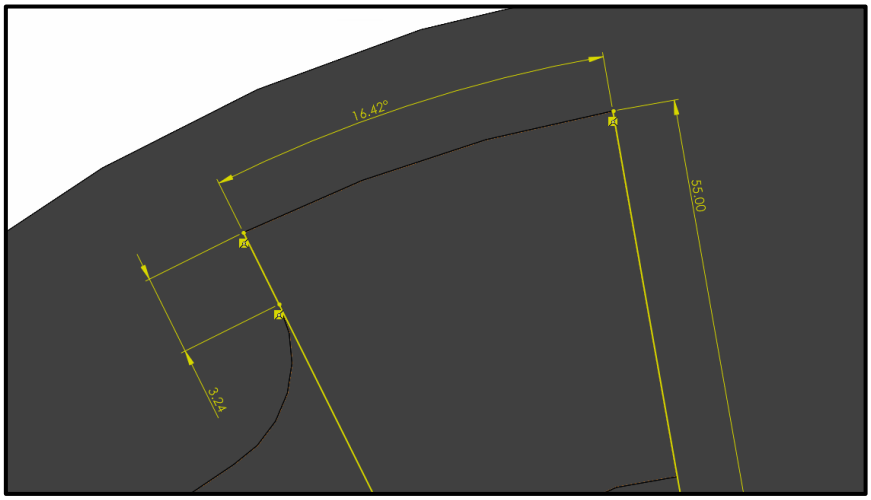


Figure 8.2.2.1: Area Segmentation for Pocket

Table 8.2.2.1: Measurements of Pocket

Depth	2	mm
Chamber Angle (Width)	16.42	deg
Outer Radius	55	mm
Radial Length	3.24	mm
Inner Radius	51.76	mm

Table 8.2.2.2: Volume Calculations of Pocket

Area (Green)	49.5649	mm^2
Volume	99.1298	mm^3

This volume is approximately double the intended fluid volume, leaving an adequate margin of safety.

8.2.3 Burst Valve Analysis

The mixing chamber of the vLoaD disc has a Burst Valve at its exit. In original design concepts this valve let fluid flow out of the mixing chamber towards the waste chamber. A burst valve is a type of capillary valve that opens once a centrifugal force with spin speed greater than a specified threshold is applied. The frequency required to achieve this spin speed is named the “Burst” frequency, hence the naming of the valve.

For the vLoaD design it was required to develop a Burst Valve with a known burst frequency, such that it was known what speed that platform can be operated at without opening the valve. This will allow centrifugal force to be applied for other operations such as mixing within the mixing chamber, without dumping fluid from it to waste accidentally.

Research began with T. H. G. Thio et al [82], this paper full describes and analyses the theoretical equations used to develop burst valves. It was instrumental to the design analysis I performed for my own design.

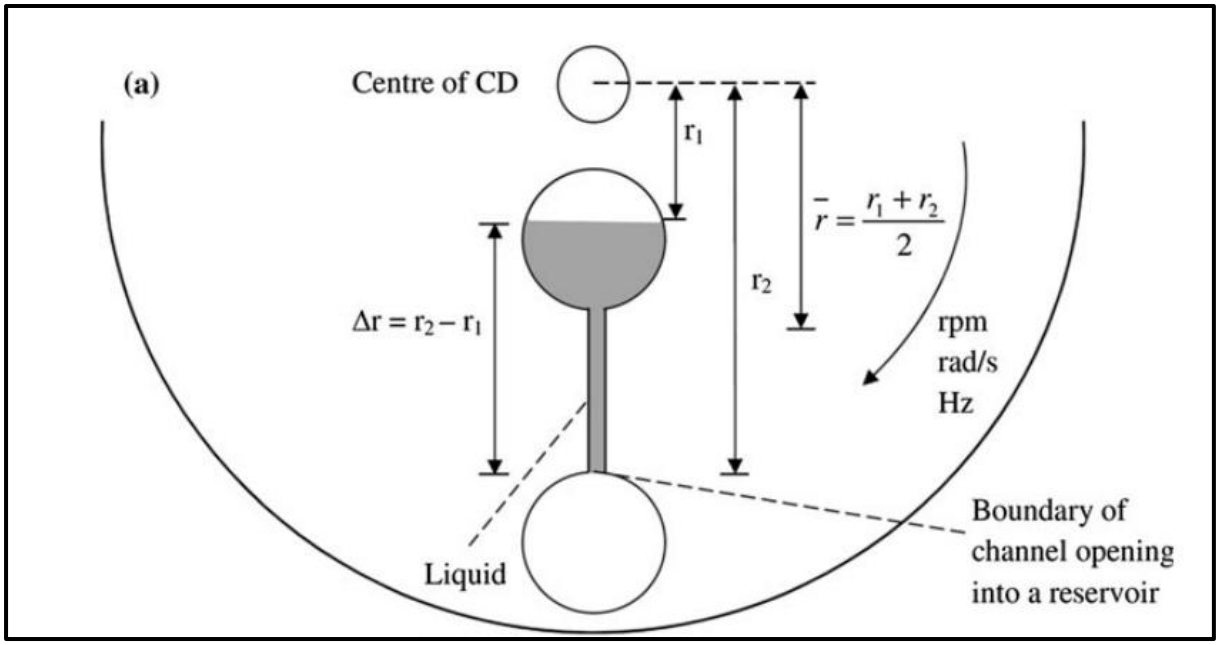


Figure 8.2.3.1: Diagram of Burst Valve on CD, [82]

As the burst valve operates based upon the balance between centrifugal and capillary forces, both forces need to be analysed separately.

Centrifugal forces are a force generated due to rotational motion. As the disc rotates, the centrifugal force experienced by liquids is proportional to the disc’s angular velocity. The centrifugal pressure for fluids on a microfluidic disc can be described as.

$$P_{Centrifugal} = \rho\omega^2 \Delta r \bar{r} \quad (2)$$

Where, ρ is density, ω is the disc's angular velocity, Δr is the change in radial position and $\bar{r}$ is the average radial position.

For Capillary pressure the analysis is more complex. Capillary Flow can be divided into three stages. The first stage: flow has stopped within the channel and the liquid is at rest near the opening with a concave meniscus. The meniscus is the surface of the fluid at the liquid to air barrier. The second stage: The liquid is still at rest, but the meniscus becomes convex, as it bulges outwards towards the opening. The third stage: the liquid has now expanded past the threshold of the opening and begins to flow into the next area.

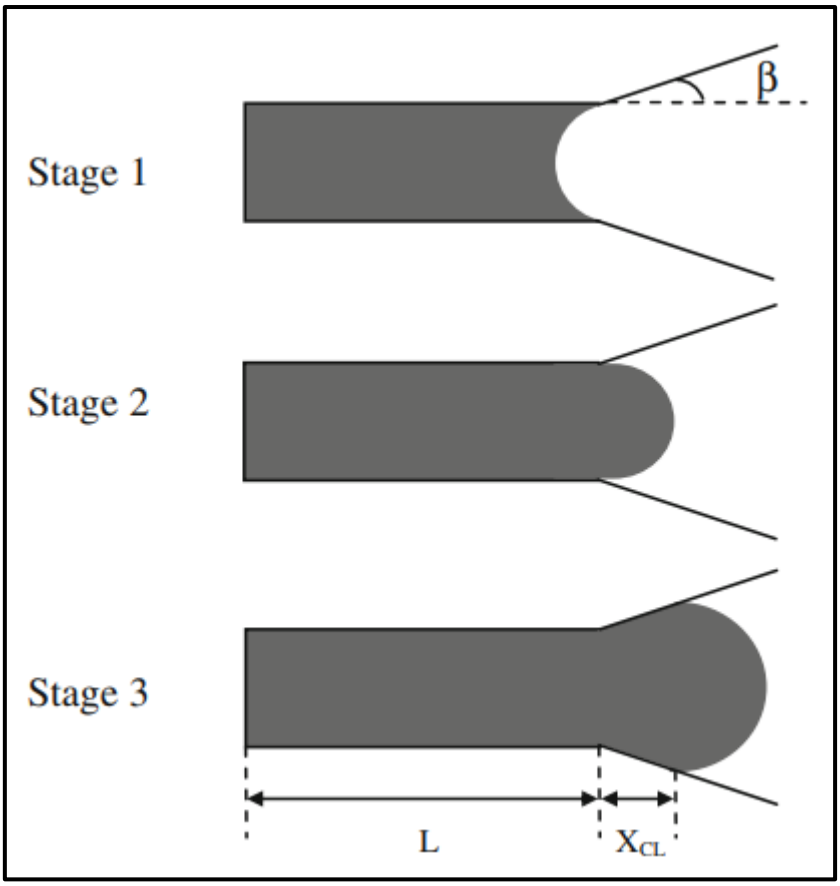


Figure 8.2.3.2: Diagram of Capillary Flow, [82]

At its core, capillary pressure can be described as change of total interfacial energy of a solid-liquid-air system, U_T with respect to the injected liquid volume, V;

$$P_{Capillary} = -\frac{dU_T}{dV} \quad (3)$$

$$U_T = A_{sl}\gamma_{sl} + A_{sa}\gamma_{sa} + A_{la}\gamma_{la}$$

Where A is for area, γ is for surface energy, the subscripts s, l and a, each represent respectively, solid, liquid and air. For the microfluidic system being studied, this can be reduced to eq 4. Where θ_c is the contact angle between the liquid and solid.

$$P_{cap} = \frac{dA_{sl}}{dV}\cos\theta_c\,\gamma_{la} \quad (4)$$

For a circular cross-section area channel, the difference in area between solid and liquids with respect to the difference in volume can be described as.

$$\frac{dA_{sl}}{dV} = \frac{4}{D}$$

This will not work for a rectangular channel such as the one of the vLoaD. Therefore, a conversion must be made from circular equations to rectangular, for this the hydraulic diameter must be calculated. This is the equivalent diameter if the channel was circular.

$$D_h = \frac{2wh}{w+h} \quad (5)$$

This can be merged with the previous equations to give one final equation to describe the capillary pressure in a rectangular channel;

$$P_{cap} = \frac{4\cos\theta_c\,\gamma_{la}}{D_h} \quad (6)$$

For a rectangular channel the capillary pressure can also be approximated as;

$$P_{cap} = 2\cos\theta_c\,\gamma_{la}\,(\frac{1}{h} + \frac{1}{w}) \quad (7)$$

If centrifugal pressure is equivalent or greater than this calculated capillary pressure, capillary flow will begin. The work by T. H. G. Thio *et al* [82], goes further into analysing each stage of capillary flow, however these equations only work for hydrophilic valves.

An issue with the vLoaD design is it is a multi-material composition. Meaning there is hydrophobic and hydrophilic surfaces in use on the same channels. Following the methods outlined in this paper would be extremely difficult and lead to inaccurate results. Meaning that only the 1D model can be taken from this. To counteract this an average contact angle was taken from the resin and the film, taking ¼

of the film contact angle and ¾ of the resin, to represent 1 wall being film and the other 3 walls being resin. While the model was simple it did approximate well to the testing observations.

To calculate the burst frequency, first the capillary pressure and the gravitational pressure were calculated, the gravitational pressure was subtracted from the capillary pressure, then this pressure was used to calculate the required Centrifugal Pressure and associated burst frequency in RPM.

NOTE: R1 is measured for approximately where the fluid sits in the mixing chamber, R2 is measured for the exit of the burst valve.

$$P_{Grav} = \rho g \Delta h \quad (8)$$

$$rpm = \sqrt{\frac{ABS(P_{cap} - P_{Grav})}{\rho \Delta r \bar{r}}} \left(\frac{30}{\pi}\right) \quad (9)$$

Table 8.2.3.1: Calculation of max Gravitational Pressure

g	9.81	m/s
Δh	5.43	m
Fluid Density (ρ)	997	Kg/m^3
Gravitational Pressure	53.1085	Pa

Table 8.2.3.2: Calculation of max Capillary Pressure

Contact Angle (Film)	24.3	Deg
Contact Angle (Resin)	135	Deg
Average Contact Angle	1.87317	rad
Surface Energy of Water in Air	0.072	J/m^2
Channel Width	0.1	Mm
Channel Depth	0.5	Mm
Capillary Pressure	-514.58	Pa

Table 8.2.3.3: Calculation of Burst Frequency

Capillary Pressure	-514.58	Pa
Gravitational Pressure	53.1085	Pa
Fluid Density (ρ)	997	Kg/m^3
Δr	5.43	mm
$\bar{r}$	30.785	mm
rpm	502.491	

The calculated burst rpm is achievable by the vLoaD rig and closely matches experimental results, in practice values were closer to 550 or 600 rpm. There is a possibility that beads will increase the required burst RPM as they can pack against the capillary entrance during operation. The maximum centrifugal pressure generatable by the vLoaD rig was calculated in Table 8.2.3.4.

Table 8.2.3.4: Calculation of max Centrifugal Pressure

Angular Velocity (ω)	1200	rpm
r1	28.07	mm
r2	33.5	mm
r bar	30.785	mm
delta r	5.43	mm
Pressure	2631.80	Pa

The calculated max centrifugal pressure should allow for the valve to operate in the case of bead packing. Although the effects of bead packing can only truly be found during operation.

8.2.4 Coriolis Switch

When the decision was made to use silica-bead based DNA extraction, a Coriolis switch was introduced to the design. The original design inspiration for this Coriolis switch came from the work by *Mark et al* [125]. This paper was studied and the design of the Coriolis switch replicated as the base for the implementation on the vLoaD. Notably, the Coriolis switch implement was radially aligned with the disc, some research into non-radial positioned Coriolis switch was performed [126], however this approach seemed unnecessarily complex for the application of the vLoaD.

For the Coriolis switch used the main design calculation required was switch length. A Coriolis switch operates off of the Coriolis force, which is a pseudo force that is observed when an object moves within a frame of reference that itself is rotating. In plain English, it is the observable force of the fluid relative to the disc. When observed it appears to be a curving or deflection motion, hence how it is used for a switch. Guiding the liquid left or right dependent on spin direction to move it into one of two channels at a Y-channel split.

The Coriolis force is a relatively weak force as opposed to the centrifugal and gravitational flow occurring. Thus, a Y-channel needs to be designed to be long enough so that the Coriolis force can guide the fluid towards the desired exit prior to the end of the switch. If this is not achieved, then there will not be proper control over the fluid, and it will not go entirely into desired channel. Some quantity of the fluid will end up in both channels. The work by *Brenner et al* [119] studies in detail transversal flow due to Coriolis force and served as the template for the calculations performed here.

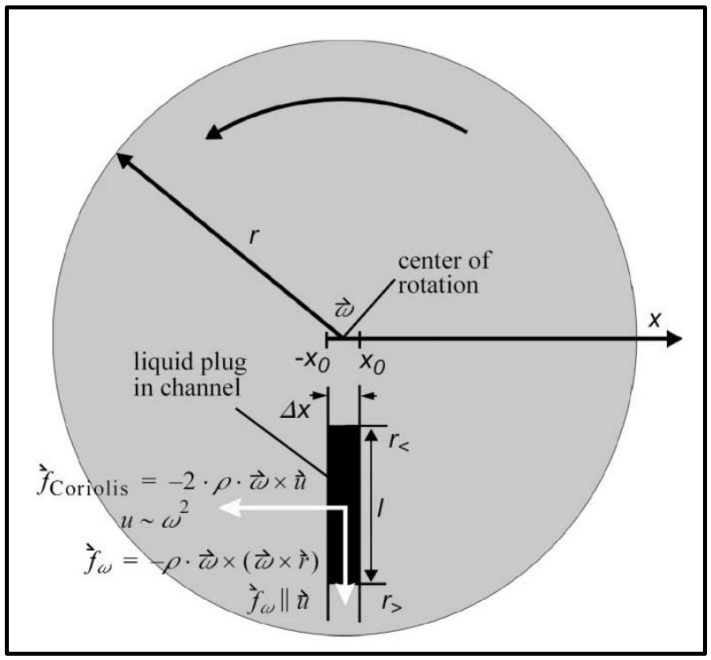


Figure 8.2.4.1: Diagram to prescribe Coriolis force and flow, taken from [119]

First the width of the channel is defined as;

$$\Delta x = 2x_0 \ (10)$$

Such that dependent on the direction of flow the fluid will travel ± X distance. This research paper proposes an equation to describe the maximum velocity within the centre of the channel.

$$u_{max} = \frac{\rho \omega^2 \bar{r}}{2\eta} x_0^2 \ (11)$$

The gravitational force is relatively small when compared to centrifugal, so it is not accounted here. As seen when performing calculations in **section 8.2.3**.

Table 8.2.4.1: Velocity Calculations for Coriolis Switch

Angular Velocity (ω)	125.664	rad/s
r1	33	mm
r2	33.5	mm
r bar (Mean radial position)	33.25	mm
Density (ρ)	997	Kg/m^3
Viscosity (η) (Water @ 20 C)	1.002	mPa.s
Tangential Distance (x)	3	mm
Maximum Flow Velocity	2.3510	m/s

This velocity will act as the radial velocity or y-axis velocity, it will now be necessary to find the tangential velocity, or x-axis. This will be the "velocity" induced by the Coriolis force. The Coriolis force itself is described by the following equation;

$$\vec{f}_{Coriolis} = -2\rho\vec{\omega} \times \vec{u} \ (12)$$

The tangential (perpendicular) acceleration due to the Coriolis force can be described as:

$$a_{Coriolis} = 2\vec{\omega} \times \vec{u} \ (13)$$

Then using the following formula to find travel time for the given distance;

$$x = \frac{1}{2}at^2$$

$$x = \frac{1}{2}(2\vec{\omega} \times \vec{u})t^2$$

$$t = \sqrt{\frac{x}{\omega * u}} \ (14)$$

This works based on the assumption that tangential velocity is equal to 0 at the start of motion. This is assumed as the liquid is being pumped out a capillary where there is only outward radial movement.

This time can then be used to find the required length;

$$L = v * t \quad (15)$$

Table 8.2.4.2: Distance Calculations for Coriolis Switch

Angular Velocity (ω)	125.664	rad/s
Tangential Distance (x)	3	mm
Maximum Flow Velocity	2.3510	m/s
Time	3.18662	ms
Length (L)	7.49172	mm

In practical experimentation (see **section 9.4**), it was discovered that this did not work as expected, this was likely due to inaccuracies in the theoretical calculation or the neglect of relatively weak gravitational force. Through iterations the Coriolis length was increased to have a margin of safety on the theoretical calculation, 9mm to be precise.

8.2.5 Volume of Product and Waste Chambers

The product and waste chambers are the final destinations for fluids on this device; their only requirement is to have enough volume to store the sample/reagents once DNA isolation is completed. The total amount of fluid was summed, and it was noted that the chambers would need to be at least this volume, ideally larger to account for potential air pockets. Specifically, the waste chamber would need to be larger for its sealing functionality, as a vacuum needs to be formed see **section 9.4.3**.

Table 8.2.5.1: Volume Calculations for Waste Chamber

Sample	50	µL
Wash Buffer 1	50	µL
Wash Buffer 2	50	µL
Total Volume	150	µL

Table 8.2.5.2: Volume Calculations for Product Chamber

Elution	50	µL
Total Volume	50	µL

8.2.6 Ventilation of Coriolis Switch, Product and Waste Chambers

To prevent the generation of vacuums that would inhibit operations, vents were placed in the Coriolis Switch, Product Chamber and Waste Chamber. The inclusion of these vents were inspired via the work by *Mark et al* [125]. The main design challenge here was to create a capillary that would allow air to flow as ventilation without allowing liquid outward which would be a hazard risk when handling the chemical during the DNA Isolation assay.

The same calculation was performed for each vent, calculating capillary pressure and maximum gravitational pressure, i.e. pressure due to gravity when the vent is at its maximum height. From where adjustments were made to the channel dimensions such that capillary pressure is greater than gravitational pressure.

$$P_{Capillary} = 2\gamma_{la} cos\theta_c (\frac{1}{w} + \frac{1}{h}) \quad (16)$$

$$P_{Gravity} = \rho g h \quad (17)$$

Table 8.2.6.1: Syntax for Pressure Calculations

Capillary	
γ_{la}	Surface Energy, liquid to air
θ_c	Contact Angle, Mean Average
w	Width of channel
h	Depth (height) of channel
Gravity	
ρ	Fluid Density
g	9.81, acceleration due to gravity
h	Height, from base of vLoaD to vent entrance

Table 8.2.6.2: Centrifugal Pressure Calculations for ALL vents

Surface Energy (Water)	0.072	J/m^2
Contact Angle	24.3	deg
Channel Width	0.1	mm
Channel Depth	2	mm
Centrifugal Pressure	1378.042	Pa

To ascertain the height, a measurement was taken from the base of the vLoaD rig to the centre of the disc, then for each vent the radial distance from the centre to the opening of the vent was measured in SolidWorks, these two values were surmised to obtain the height at which gravitational pressure is its strongest.

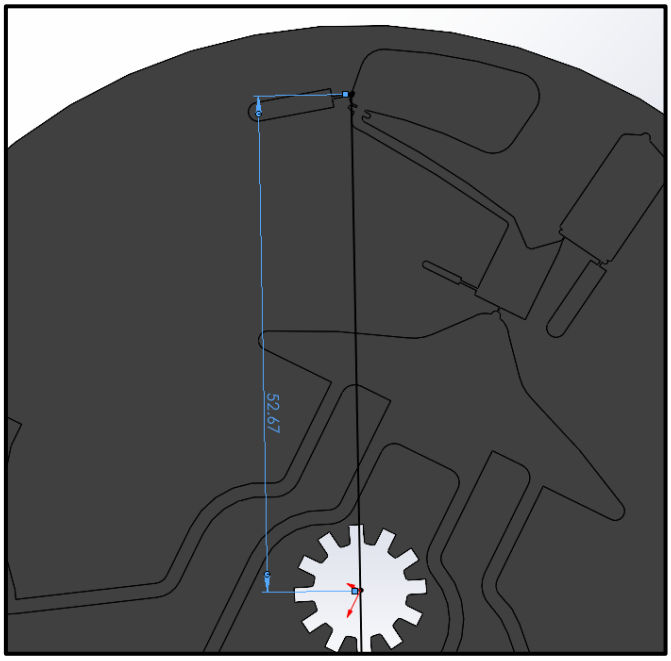


Figure 8.2.6.1: Example Radial Measurement for Vent, Waste Chamber

Table 8.2.6.3: Gravitational Pressure Calculations for Waste Chamber

Density (Water)	997	Kg/m^3
Disc Radius	60	mm
Base to Edge of Disc	16.71	mm
Vent Radial Distance	52.67	mm
Height	129.38	mm
Gravitational Pressure	1265.41	Pa

Table 8.2.6.4: Gravitational Pressure Calculations for Product Chamber

Density (Water)	997	Kg/m^3
Disc Radius	60	mm
Base to Edge of Disc	16.71	mm
Vent Radial Distance	43.24	mm
Height	11.95	mm
Gravitational Pressure	1173.179	Pa

Table 8.2.6.5: Gravitational Pressure Calculations for Coriolis Switch Chamber

Density (Water)	997	Kg/m^3
Disc Radius	60	mm
Base to Edge of Disc	16.71	mm
Vent Radial Distance	34.25	mm
Height	110.96	mm
Gravitational Pressure	1085.252	Pa

It was found that using a vent entrance width of 0.1mm, worked for all vent conditions. Thus, for consistency this entrance width was used for all three vents.

8.2.7 Ethanol Calculations

To see if the same design is applicable to a vLoaD disc that uses samples/reagents for DNA Isolation as opposed to water, the design calculations were recomputed substituting water's surface energy and contact angle for Ethanol 96%. Table 8.2.7.1 features the values that were used to alter that calculations for Ethanol 96%.

Table 8.2.7.1: Property values for Ethanol 96%

Density	789	Kg/m^3
Surface Energy	0.02239	J/m^2
Contact Angle (Film)	14.2	Deg
Contact Angle (Resin)	22.905	Deg
Contact Angle (MEAN)	20.72875	Deg

8.2.7.1 Burst Valve

Table 8.2.7.2: Ethanol 96% Calculations for Burst Valve

Centrifugal Pressure	2082.743	Pa
Gravitational Pressure	42.0286	Pa
Capillary Pressure	502.575	Pa
Burst RPM	564.287	

Burst RPM has increased for Ethanol as opposed to water, but it is still an achievable frequency by the vLoaD rig. The planned operational frequency of 1200 rpm far exceeds this calculated value.

8.2.7.2 Coriolis Switch

Table 8.2.7.3: Ethanol 96% Calculations for Coriolis Switch

Ethanol 96% Viscosity	1.2	mPa.s
Maximum Flow Velocity	1.4384	m/s
Required Length	5.85989	mm

The Coriolis Switch requires less length to operate than with water, therefore the same design vLoaD disc should work without any modifications.

8.2.7.3 Capillary Vents

Table 8.2.7.4: Ethanol 96% Calculations for Capillary Vents

Capillary Pressure	439.753	Pa
Max Grav Pressure Waste Vent	1001.413	Pa
Max Grav Pressure Product Vent	928.4238	Pa
Max Grav Pressure Coriolis Vent	858.8404	Pa

The capillary vents for the waste and product chamber in theory should not seal properly for Ethanol 96%. With the current 3D printer it is not possible to get accurate chamber dimensions at this scale, 0.1mm is already approaching the limit of accuracy, see **section 11.1**. So, the vents cannot be modified at this time to improve sealing capabilities.

8.2.8 Mixing Forces, Reynolds Number and Peclet Number

The plan for “mixing” the fluids on the vLoaD platform is to place them within the mixing chamber and hold the disc in such a position that the liquid makes maximum contact with the silica beads. For the mixing process to be successful there is two key fluid properties to analyse, the Reynolds and Peclet numbers.

8.2.8.1 Reynolds Number

The Reynolds number represents the balance between inertial forces and viscous forces. For the vLoaD it is expected that the number will be low as it is a microfluidic system. This will indicate that flow within the system is Laminar.

This number must be investigated to ensure "jetting" does not occur, this is when the fluid enters the mixing chamber with too high of a moment leading it to go straight from inlet to outlet, making positioning the fluid with the beads difficult, it also increases the chance of fluid over powering the burst valves seal and entering the Coriolis switch prematurely. Ideally this Reynolds number is low enough such that the fluid will spread towards the beads. The Reynolds number is calculated using the partial format for the loading phase, this is in accordance with Darcy's Law.

Darcy's Law is a fluid mechanic equation that describes flow through a porous medium such as beads used on the vLoaD.

Darcy's Law requires the "Superficial" flow rate, which is the rate at which the liquid would flow if the beads were not there. To keep the model simple the flow was considered as the gravity driven flow from the storage chamber to the mixing chamber.

$$R_{e_p} = \frac{\rho \,.\, Q \,.\, d_p}{\mu \,.\, (1 - \varepsilon) \,.\, W \,.\, H} \quad (18)$$

Hagan-Poiseuille flow equation was used, replacing the radius for the hydraulic diameter of a rectangular channel, this is due to the lack of flow equations for a rectangular channel, so this approximation is required.

$$\Delta P = \rho \,.\, g \,.\, \Delta h \; (19)$$

$$D_h = \frac{2wh}{w + h} \; (5)$$

$$Q = \frac{\pi \,.\, \Delta P \,.\, (\frac{D_H}{2})^4}{8\mu L} \quad (20)$$

Table 8.2.8.1: Flow Rate Calculation due to Gravity from Storage to Mixing, for E. Coli

Fluid Density	1000	Kg/m^3
Gravity	9.81	m/s/s
Difference in height	44.29	mm
Pressure Differential	434.4849	Pa
Channel Width	1	mm
Channel Depth	2	mm
Hydraulic Diameter	0.001333333	m
Channel Length	34.91	Mm
Fluid Viscosity	0.003	Pa . s
Flow Rate	3.21809 x10^-7	m^3/s

Table 8.2.8.2: Flow Rate Calculation due to Gravity from Storage to Mixing, for Wash Buffer 1

Fluid Density	990	Kg/m^3
Difference in height	57.55	mm
Pressure Differential	558.919845	Pa
Channel Length	48.72	Mm
Fluid Viscosity	0.003	Pa . s
Flow Rate	2.96631 x10^-7	m^3/s

Table 8.2.8.3: Flow Rate Calculation due to Gravity from Storage to Mixing, for Wash Buffer 2

Fluid Density	900	Kg/m^3
Difference in height	60.61	mm
Pressure Differential	535.12569	Pa
Channel Length	53.17	Mm
Fluid Viscosity	0.0023	Pa . s
Flow Rate	3.39435 x10^-7	m^3/s

Table 8.2.8.4: Flow Rate Calculation due to Gravity from Storage to Mixing, for Elution

Fluid Density	1000	Kg/m^3
Gravity	9.81	m/s/s
Difference in height	52.79	mm
Pressure Differential	517.87	Pa
Channel Length	41.46	Mm
Fluid Viscosity	0.001	Pa . s
Flow Rate	9.6892 x10^-7	m^3/s

Table 8.2.8.5: Reynolds Number Calculation for E. Coli

Flow Rate (Q), due to Gravity	3.21809 x10^-7	m^3/s	Table 8.2.8.1
Fluid Density (ρ)	1000	Kg/m^3	For Ethanol?
Fluid Viscosity (μ)	0.003	Pa . s	For Ethanol?
Silica Bead Diameter (d_p)	181	µm	Mean Average
Void Fraction (ε)	40%		Assumed
Chamber Width (Flat Edge of Semi-Circle), (W)	40	mm	
Chamber Depth (H)	2	mm	
Partial Reynolds Number	0.4045		

Table 8.2.8.6: Reynolds Number Calculation for Wash Buffer 1

Flow Rate (Q), due to Gravity	2.96631 x10^-7	m^3/s	Table 8.2.8.1
Fluid Density (ρ)	990	Kg/m^3	
Fluid Viscosity (μ)	0.003	Pa . s	
Partial Reynolds Number	0.3691		

Table 8.2.8.7: Reynolds Number Calculation for Wash Buffer 2

Flow Rate (Q), due to Gravity	3.39435 x10^-7	m^3/s	Table 8.2.8.1
Fluid Density (ρ)	900	Kg/m^3	
Fluid Viscosity (μ)	0.0023	Pa . s	
Partial Reynolds Number	0.5009		

Table 8.2.8.8: Reynolds Number Calculation for Elution

Flow Rate (Q), due to Gravity	9.6892 x10^-7	m^3/s	Table 8.2.8.1
Fluid Density (ρ)	1000	Kg/m^3	
Fluid Viscosity (μ)	0.001	Pa . s	
Partial Reynolds Number	3.653621537		

Note: For the calculations seen above, the fluid density and fluid viscosities had to be approximated by Google Gemini based on current knowledge of the fluids and their compositions, since for each values were not available. [127]

Examining the calculated Reynolds numbers in accordance with Darcy's Law, E. Coli sample, wash buffer 1 and wash buffer 2 all have values less than 0 meaning that the fluid will move like a "thick liquid", this is good for maximizing contact with the beads. The Elution mixture has a number greater than 1 meaning that the fluid might create dead zones in between each bead, however this is acceptable due to the role of elution to remove DNA from the beads.

8.2.8.2 Peclet Number

The Peclet number represents the rate at which particles move through a fluid, it is the ratio between advection and diffusion. Advection is the physical flow of the fluid, as it moves downstream. Diffusion is the movement that occurs as the fluid spreads out. This is covered in full detail in the literature review (**section 5.2**). This number can be calculated to analyse the flow occurring within the mixing chamber.

$$P_e = \frac{U \cdot d_p}{D_{diff}} \quad (21)$$

$$U \approx \frac{Q}{A \cdot \varepsilon} \quad (22)$$

$$A = \frac{\pi r^2}{2} \quad (23)$$

U is the interstitial velocity, the speed at which the fluid moves inside the bead mixture. As applied to the Reynolds number calculation, the only flow considered is the rate at which the fluid enters the mixing chamber, as the disc is stationary when liquid is moving meaning this is no additional physical flow of the fluid. For the mixing chamber, the area was approximated as that of a semi-circle. D_{diff} is the diffusion coefficient this is the measure of how fast the specific molecule naturally spreads out in the fluid-bead mixture. For the E. Coli sample, this coefficient was for genomic DNA, for wash buffer 1, the coefficient is for medium proteins, for wash buffer 2 it was guanidine salts, and for the elution mixture the coefficient is for genomic DNA.

Table 8.2.8.9: Peclet Number Calculation for E. Coli

Radius of Mixing Chamber	20	mm
Area of Mixing Chamber (A)	6.28 x10^-4	m^2
Flow Rate (Q)	3.21809 x10^-7	m^3/s
Void Fraction (ε)	40%	
Velocity (U)	5.12 x10^-4	m/s
Silica Bead Diameter (d_p)	181	µm
Diffusion coefficient	5E-13	m^2/s
Peclet Number	463518.57	

Table 8.2.8.10: Peclet Number Calculation for Wash Buffer 1

Flow Rate (Q)	2.96631 x10^-7	m^3/s
Velocity (U)	4.72 x10^-4	m/s
Silica Bead Diameter (d_p)	181	µm
Diffusion coefficient	2E-11	m^2/s
Peclet Number	10681	

Table 8.2.8.11: Peclet Number Calculation for Wash Buffer 2

Flow Rate (Q)	3.39435 x10^-7	m^3/s
Velocity (U)	5.40 x10^-4	m/s
Silica Bead Diameter (d_p)	181	µm
Diffusion coefficient	6E-10	m^2/s
Peclet Number	407.42	

Table 8.2.8.12: Peclet Number Calculation for Elution

Flow Rate (Q)	9.6892 x10^-7	m^3/s
Velocity (U)	1.54 x10^-4	m/s
Silica Bead Diameter (d_p)	181	µm
Diffusion coefficient	2E-12	m^2/s
Peclet Number	348895	

Note: For the calculations seen above, the diffusion coefficient had to be approximated by Google Gemini based on current knowledge based on current knowledge of the target particles and the fluids they are immersed in. [127]

For E. Coli and Elution, the Peclet numbers are very large, reinforcing the protocol decision to hold the disc stationary during diffusion times. For wash buffer 1 & 2, the lower Peclet numbers indicate that the fluid will not need to remain in contact for as long, if speeding up the DNA isolation process is desired.

8.3 Design Calculations, Testing Design 2-1

In the final stages of the project a critical error was spotted within the design, a major overhaul was required. This segment covers the new design calculations required for this design and any updates to previous calculations. The new area of the design is the “S-Chamber” this sits between the prior mixing chamber and the Coriolis switch, the beads now reside in this area, and liquids are pumped through them. All calculations are done for ethanol 96%.

Table 8.3.1: Property values for Ethanol 96%

Density	789	Kg/m^3
Surface Energy	0.02239	J/m^2
Contact Angle (Film)	14.2	Deg
Contact Angle (Resin)	22.905	Deg
Contact Angle (MEAN)	20.72875	Deg

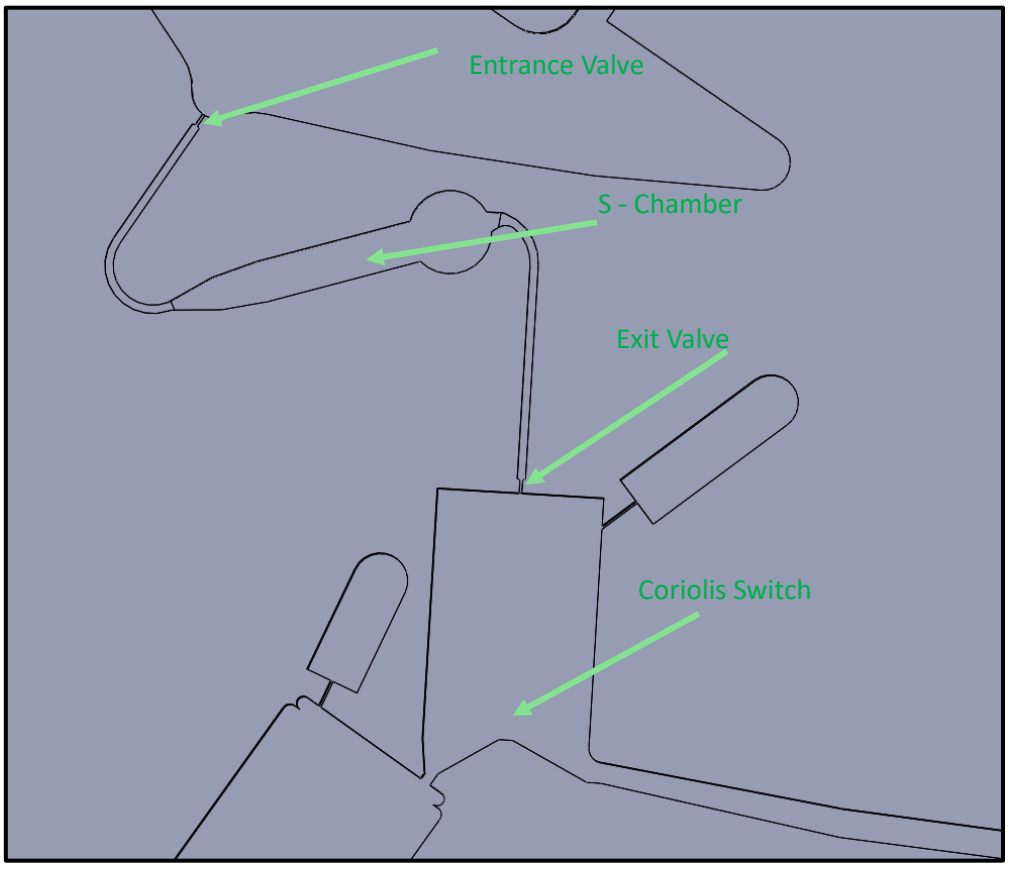


Figure 8.3.1: Image of vLoaD Revamp, highlighting design calculation relevant areas

8.3.1 Burst Valve at entry to S-Chamber

Burst Valve calculations were performed using the same method as seen in **section 8.2.3**. Adjusting radius values and capillary values to account for the new design changes.

Table 8.3.1.1: Calculation of max Gravitational Pressure

g	9.81	m/s
Δh	3	m
Fluid Density (ρ)	789	Kg/m^3
Gravitational Pressure	23.22	Pa

Table 8.3.1.2: Calculation of max Capillary Pressure

Contact Angle (Film)	14.2	Deg
Contact Angle (Resin)	22.905	Deg
Average Contact Angle	0.36178	rad
Surface Energy of Ethanol 96%	0.02239	J/m^2
Channel Width	0.1	mm
Channel Depth	0.5	mm
Capillary Pressure	502.575	Pa

Table 8.3.1.3: Calculation of Burst Frequency

Capillary Pressure	502.575	Pa
Gravitational Pressure	23.22	Pa
Δr	3	mm
$\bar{r}$	22.5	mm
rpm	905.96	

Table 8.3.1.4: Calculation of max Centrifugal Pressure

Angular Velocity	1200	rpm
r1	21	mm
r2	24	mm
r bar	22.5	mm
delta r	3	mm
Pressure	841	Pa

8.3.2 Burst Valve at exit of S-Chamber

Burst Valve calculations were performed using the same method as seen in **section 8.2.3**. Adjusting radius and capillary dimension values to account for the new design changes.

Table 8.3.2.1: Calculation of max Gravitational Pressure

g	9.81	m/s
Δh	9	mm
Fluid Density (ρ)	789	Kg/m^3
Gravitational Pressure	69.66081	Pa

Table 8.3.2.2: Calculation of max Capillary Pressure

Contact Angle (Film)	14.2	Deg
Contact Angle (Resin)	22.905	Deg
Average Contact Angle	0.36178	rad
Surface Energy of Ethanol 96%	0.02239	J/m^2
Channel Width	0.1	Mm
Channel Depth	1	Mm
Capillary Pressure	502.575	Pa

Table 8.3.2.3: Calculation of Burst Frequency

Capillary Pressure	502.575	Pa
Gravitational Pressure	69.66081	Pa
Δr	9	mm
$\bar{r}$	28.5	mm
rpm	441.662	

Table 8.3.2.4: Calculation of max Centrifugal Pressure

Angular Velocity (ω)	1200	rpm
r1	24	mm
r2	33	mm
r bar	28.5	mm
delta r	9	mm
Pressure	3195.833	Pa

The burst rpm, is well below the operational rpm of 1200, this exit valve should not inhibit the liquids entry toward the Coriolis switch

8.3.3 Centrifugal Lift

A design concern for the new "S-Chamber" was if the fluid would more correctly around the bend and through the chamber as desired. This is due to the requirement for the fluid to move radially inward to clear the second bend. To confirm this the Centrifugal Hydrostatic Pressure equation was used.

$$\Delta P = \frac{1}{2}\rho\omega^2(R_{Crest}^2 - R_{Liquid}^2) \quad (24)$$

Where R_{Crest} represents the highest point in the S-Chamber and R_{Liquid} represents the point of liquid level prior to chamber entry.

Table 8.3.3.1: Calculation of Centrifugal Hydrostatic Pressure

Angular Velocity (ω)	1200	rpm
Fluid Density (ρ)	789	Kg/m^3
R Crest	23.19	mm
R Liquid	23.5	mm
Delta Pressure	-90.1679	Pa

This pressure is a negative value which indicates that the liquid will not move over the bend itself, towards the exit of the S-Chamber. This means that other forces must be relied upon for the S-Chamber to operate as intended, it must be viewed as a siphon valve. Therefore, the capillary pressure of this chamber was calculated to see if it is enough to wick the liquid up the slope and towards the exit.

Table 8.3.3.2: Calculation of Chamber Capillary Pressure

Contact Angle (Film)	14.2	Deg
Surface Energy of Ethanol 96%	0.02239	J/m^2
Channel Width	1.5	mm
Channel Depth	1	mm
Capillary Pressure	72.3529	Pa

Table 8.3.3.3: Calculation of max Gravitational Pressure

g	9.81	m/s
Δh	7.58	mm
Fluid Density (ρ)	789	Kg/m^3
Gravitational Pressure	58.67	Pa

It was assumed that the liquid will take the path of least resistance and thus only the film contact angle was consulted. The gravitational pressure was calculated as it in theory would work against the fluid as it attempts to "climb" the slope. As the capillary pressure is greater than the gravitational pressure, the liquid should wick itself up the slope. The S-Chamber can now be deemed as siphon-like valve. Once the liquid passes beyond the peak the capillary narrows ensuring flow towards the exit.

8.3.4 Bead Capillary Force

The previous segment did not consider the silica beads that fill the S-Chamber, the fluid needs to be pumped through these beads. To see if this was possible, Darcy's Law was consulted.

First the permeability of the bead mixture was calculated using the Kozeny-Carman equation;

$$\kappa = \frac{d_p^2 \varepsilon^3}{150(1 - \varepsilon)^2} \quad (25)$$

Table 8.3.4.1: Calculation of Permeability

Porosity (ε)	0.40	
Mean Bead Diameter (d_p)	181	µm
Permeability (κ)	3.88E-11	m^2

Now using Darcy's Law to find the pressure required to achieve an acceptable flow rate, 0.001 m/s was chosen as it is a suitable value for a microfluidic system. The fluid viscosity of the DNA sample was used as it is the "highest" meaning it would be the hardest fluid to pump. If the pressure for this fluid was achievable then all other fluids for the vLoaD DNA isolation assay should work.

$$\Delta P = \frac{\mu L v}{\kappa} \quad (26)$$

Table 8.3.4.2: Calculation of Pressure Required to pump through beads

Permeability (κ)	3.88E-11	m^2
Channel Length (L)	28.17	mm
Target Velocity (v)	0.001	m/s
Fluid Viscosity (µ)	0.003	Pa.s
Delta Pressure	2176.53	Pa

See Table 8.3.2.4, this pressure is achievable on the vLoaD system. Therefore, fluids should be able to flow through the beads in the S-Chamber.

8.3.5 Coriolis Switch

The Coriolis Switch design remained the same as previous versions, however radial positions of the disc affect the maximum flow velocity, so the calculations were run again.

Table 8.3.5.1: Calculations for Coriolis Switch

Angular Velocity (ω)	125.664	rad/s
r1	24	mm
r2	33	mm
r bar	28.5	mm
Density (ρ)	789	Kg/m^3
Viscosity (η) Ethanol 96%	1.2	mPa.s
Tangential Distance (x)	3	mm
Maximum Flow Velocity	1.3316	m/s
Time	4.234	ms
Length (L)	5.638	mm

The current length of the Coriolis switch is greater than this calculated value, meaning no design changes are required, the switch design is adequate.

8.4 Design Calculations, Testing Design 2-2

These calculations use similar formula as seen in **section 8.2** and **section 8.3**, full calculations can be found in **Appendix 6**.

8.5 Design Calculations, Testing Design 3

For the new chamber introduced the design calculations were as follows:

Table 8.5.1.1: Calculation of max Gravitational Pressure

g	9.81	m/s
Δh	0.5	mm
Fluid Density (ρ)	789	Kg/m^3
Gravitational Pressure	3.87	Pa

Table 8.5.1.2: Calculation of Capillary Pressures

Contact Angle (Film)	14.2	Deg
Contact Angle (Resin)	22.905	Deg
Average Contact Angle	0.36178	rad
Surface Energy of Ethanol 96%	0.02239	J/m^2
Channel Width	0.3	mm
Channel Depth	0.5	mm
Capillary Pressure	223.367	Pa

Table 8.5.1.3: Calculation of Inlet Centrifugal Pressure

Angular Velocity (ω)	1200	rpm
r1	25	mm
r2	25.5	mm
r bar	25.25	mm
delta r	0.5	mm
Pressure	157.29	Pa
Burst Frequency	1417	RPM

Table 8.5.1.4: Calculation of Outlet Centrifugal Pressure

Angular Velocity (ω)	1200	rpm
r1	32.5	mm
r2	33	mm
r bar	32.75	mm
delta r	0.5	mm
Pressure	204.022	Pa
Burst Frequency	1244	RPM

9 Methodology

9.1 Rig Development (Background)

9.1.1 vLoaD Rig

The rig of the vLoaD device is the piece that is used to operate the disc and run assays. It is comprised of a 3D printed housing, an acrylic shield, a stepper motor (Nema 17), an Arduino UNO and a CNC relay power shield. Its construct as basic as its main intended function are to mount the vLoaD disc to the motor and then operate said motor. The design did have a backing cover, however this was not attached during the project to leave the CNC board reset switch accessible, the cover was printed and attached later, for future demonstrations such as the Final Year Expo.



Figure 9.1.1.1: Photo of vLoaD Rig Standalone

Construction of this rig and the scripting of the base Arduino code were both performed by Nicklas Rondot, as the progenitor of the vLoaD project. The Arduino code was structured in such a format that the serial monitor acts as the communicator between operator and device. The two key functions SpinToAngle and MoveToAngle can both be sent to the Arduino via commands from the serial monitor.

The first command “SpinToAngle” takes the desired speed, run time and end position. It spins the disc at the desired speed for the requested amount of seconds and finish its movement at the specified position. This was used to generate centrifugal forces on the disc, which opened the burst valve and moved liquid away from the mixing chamber.

The second command “MoveToAngle” receives a movement speed and angular position, it then moves the disc in the indicated direction at the desired speed to the specified point. This was used to change the angular positions of the disc, which was vital to actuating different storage chambers and the position of the mixing chamber for diffusion to occur.

A third command “Persuade” was added by the student during testing. This command moves the disc from side to side in an oscillating manner. Used to vibrate the disc, agitating fluid and helping remove blockages. It takes as inputs the speed for which it will move at, the desired movement range, for example 90 degrees is +45 and -45 degrees from the current position, and the run count, how many times it will oscillate.

Full Arduino Code is available in **Appendix 14**.

9.1.2 Basler Camera

The Basler Camera refers to the “acA2040-90uc” made by Basler, this is a digital global shutter camera, that can synchronise with a stroboscope for the capturing of videos at very fast frame rates. To capture video on vLoaD rig during operation this camera was employed. Using an IR sensor and an Arduino UNO to send a trigger signal once a rising edge was detected from the sensor. This trigger signal was then sent to the stroboscope and Basler Camera actuating both at once. A black sticker was applied to the vLoaD disc itself so the sensor would flag once per rotation. This will make it so the Basler Camera will take a photo once per rotation.



Figure 9.1.2.1: Photo of vLoaD rig with Basler Camera

The stroboscope does emit extreme light during operation, hence why black acrylic was placed around the rig, to reduce light exposure for the user, as seen in figure 9.1.2.1.

Pylon Viewer is the Basler software used to operate the camera. This was used to capture screenshots and videos of the vLoaD in operation and to save it to a PC. Additionally, this software is used to tweak camera parameters such as exposure time and the activation of trigger mode.

The Arduino code was programmed to also send a trigger signal if enough time passed, such that the Basler will take photos of the vLoaD during “slower” operations, such as the actuation of storage chambers, or when diffusion is occurring. When the sensor was being uncooperative this time was dropped all the way down to 50 milliseconds, the periodic time of the disc when spinning at 1200 rpm. Which allowed for the capturing of video still however the disc would not appear to be “locked” in positions. This was not ideal, but it was the only way to capture video under certain lighting conditions.

All videos were saved and catalogued in accordance with the design/test they link to. The were also uploaded to a shared google drive with Nicklas Rondot for future research.

9.1.3 Software Development

It was desired to make an application to interface with the Arduino used to control the vLoaD. There were two main design objectives:

- Create a way for users to control the vLoaD manually in a fashion easier than using the serial monitor
- Create a way to write automated programs for the vLoaD

These two design objectives be known as manual mode and program mode. The application was created as a Windows Presentation Foundation (WPF) desktop application. This format was chosen as the student had some experience with the framework. The main limitation of this approach was the application only ran on windows devices.

This framework operates on a XAML code script which forms the front-end of the application and a C# code script for the back end. The front end controls all visuals of the application, the back-end covers the code scripts required to make the application operate. Example: In the XAML there will be a button placed called “Button1”, there will be a linked click function called “Button1_click”, when the button is clicked by the user the code in the C# file under the name “Button1_click” is executed. Both of these files as well as associated functions can be found in **Appendix 15**.

The application window is divided into 3 areas, the connectivity bar, the mode tab (Manual or Program) and the communication area. The connectivity bar is used to select which computer port the Arduino is plugged into and to connect to it. These buttons connect to serial functions in the C# code. The mode tab was created using a “TabControl” item in XAML, this allows for the selection between a manual window and a program window, see figures 9.1.3.1 and 9.1.3.3. The communication area is present in both modes, it contains a command history and alert centre. The command history captures the commands sent to the Arduino and allows for them to copied so a user can take note of their steps. The alert centre, prints messages received back from the Arduino via serial communications, this was useful for debugging.

The manual mode tab features three segments, one for each vLoaD user command, Angle adjustment, Spin disc and Vibrate disc. Each of these areas have sliders with text boxes to control command variables, with their own respective send buttons. As the text boxes are updated or the sliders adjusted, the C# code tracks values and sends a serial message to the Arduino in a similar manner to how a user operating the serial monitor would. Each command has a visualization on the right, this represents the vLoaD disc itself, each of which is useful for visualizing what the command will do. The angular position shows where the disc will move to. The spin disc shows where the stop position will

be. The vibrate disc shows the range of oscillations, this was the most difficult to implement as .cs functions were written to halve and inverse numerical values for correct visualization purposes.



Figure 9.1.3.1: Visual of Manual Mode in the application

The program mode can be broken into two sections, the control buttons and the program step grid. The control buttons are located at the top and are used to add/remove steps, load/save program files, paste commands (copied from the command history) and run the loaded program. Programs are saved as .JSON files.

The program step grid has 6 columns, the first 4 are used for defining the CMD and its parameters, then there is a delay and a comment column. Each column has a tool tip which can be seen when hovered over, these are useful for users writing programs.

The delay begins once the command is sent to the Arduino so this must be accounted for when scripting. An example would be if the disc is spun for 20 seconds and then wait 10 seconds for the next command, the delay must be 30 seconds. As steps are added to the grid the window expands to

accommodate. An example program, which was used for test 4 (**Appendix 11**) can be seen in figure 9.1.3.4.

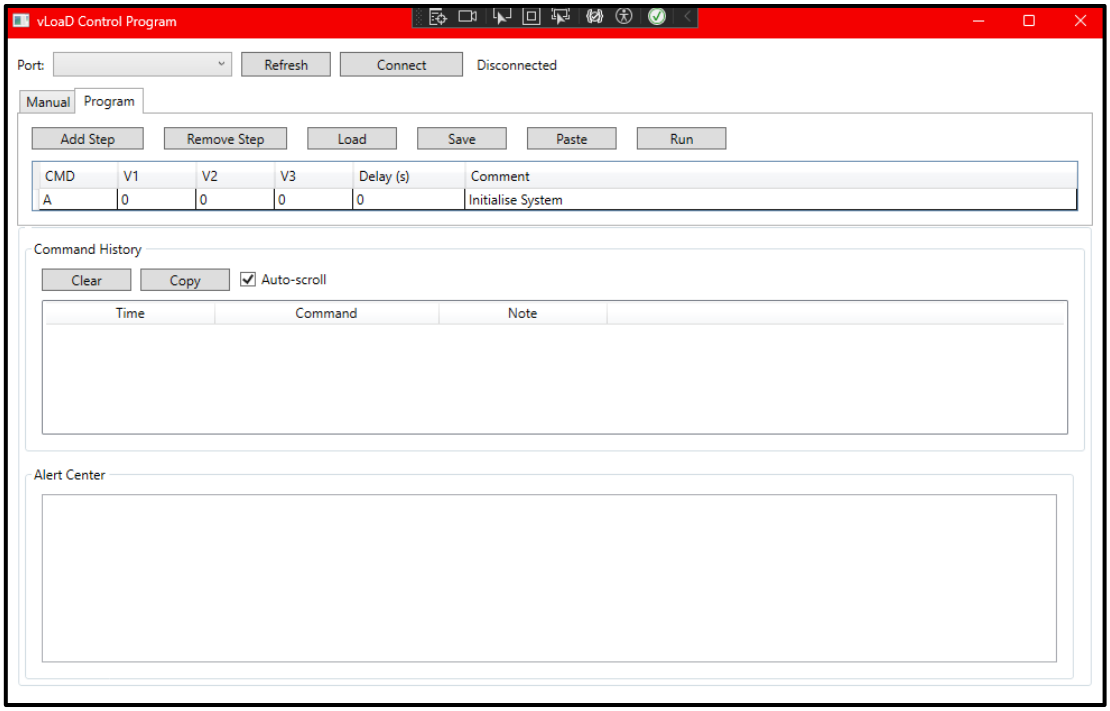


Figure 9.1.3.2: Visual of Program Mode in the application

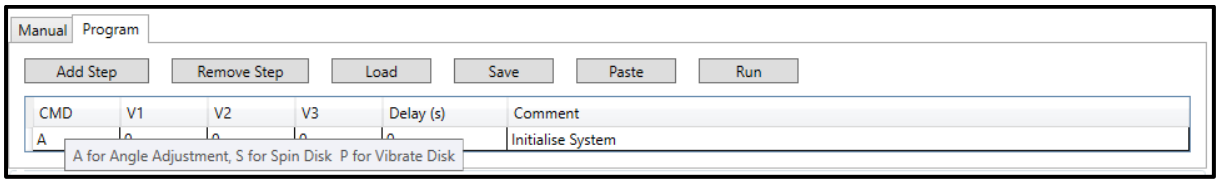
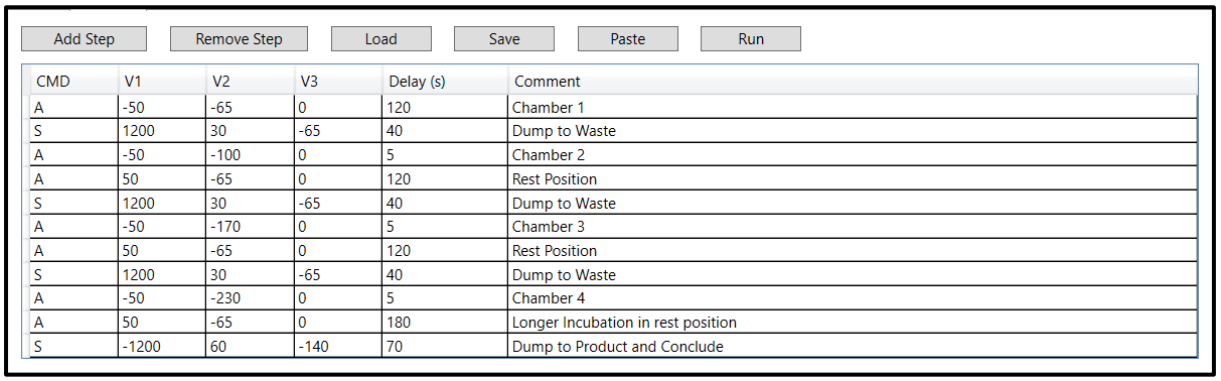


Figure 9.1.3.3: Example of Program Mode Tool Tip



CMD	V1	V2	V3	Delay (s)	Comment
A	-50	-65	0	120	Chamber 1
S	1200	30	-65	40	Dump to Waste
A	-50	-100	0	5	Chamber 2
A	50	-65	0	120	Rest Position
S	1200	30	-65	40	Dump to Waste
A	-50	-170	0	5	Chamber 3
A	50	-65	0	120	Rest Position
S	1200	30	-65	40	Dump to Waste
A	-50	-230	0	5	Chamber 4
A	50	-65	0	180	Longer Incubation in rest position
S	-1200	60	-140	70	Dump to Product and Conclude

Figure 9.1.3.4: Example Program, in application

9.2 Fabrication Method

Fabrication of each vLoaD disk can be separated into 3 distinct areas, 3D Printing, sealing and filling. For 3D printing an Elegoo Saturn 4 Ultra was used for the fabrication of the first discs, from version 10 (**Appendix 9**) onwards the Elegoo Saturn 4 Ultra 16K was used instead. This 16K version was initially reserved for testing designs, so as the vLoaD became further refined the swap occurred.

The CAD models were developed in SolidWorks, then they were exported as an STL file to a software called Chitubox, this sliced the STL files and prepared them for usage on the 3D printer, here positioning and orientation with respect to the print bed could be controlled. Then files were exported to a secondary software called UVTools which further cleaned up files and adjusted parameters such as exposure time for more optimal printing. From UVTools files were saved to a USB drive that was inserted into the printer.

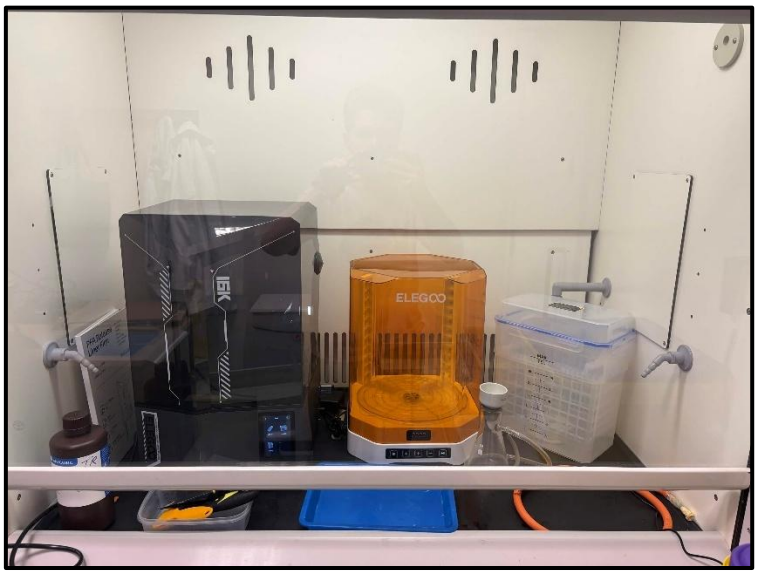


Figure 9.2.1: Image of 3D printer setup

All steps of 3D printing operations took place under a laboratory fume hood while wearing a lab coat and gloves to protect the operator.

Once printing was completed the entire print bed was removed from the printer. This was then held over a tray and a scraper was used to remove the disc, being careful to not allow excess resin to drip onto surfaces. Once removed the disc had excess resin removed using tissue.

The disc was then placed into an IPA Wash Tank then washed for 2 minutes and 30 seconds. Once removed the disc was dried once more and placed under a UV-Cure Station, this was run for 15 minutes to cure the disc. This strengthened, stabilised and hardened parts.

NOTE: During final version which were used for DNA Isolation assays the curing operation was changed, 10 minutes, an additional wash with water occurred, the disk was submerged under water for 5 minutes and then an additional cure occurred for 10 minutes. This was done to further remove resin from the disc which in theory should enhance its biocompatibility.

The fully finished disc was then sealed with a hydrophilic film. 219158 by Adhesives Research Ireland was used for this. A Graphtec Cutter Plotter was used to cut a circular disc out of the film sample, this was done via Graphtec Studio 2, a software that took designed files and sent instructions to the cutter plotter to cut the desired shape out of the film. The film was removed from its protective backing and applied to the disc, then the disc was pressed onto a hotplate set to 100 degrees Celsius which activated the films adhesive. The disc was pressed on the hotplate using a metal weight. Once removed from the hot plate after approximately 10 seconds the disc was pressed against a workbench with the same weight to prevent warping. Lastly a hole is poked into the mixing chamber using a pin, then a modified Pipette tip is used to load glass beads into the mixing chamber. Approximately 0.2 grams were used, then the hole was sealed using a PSA material.

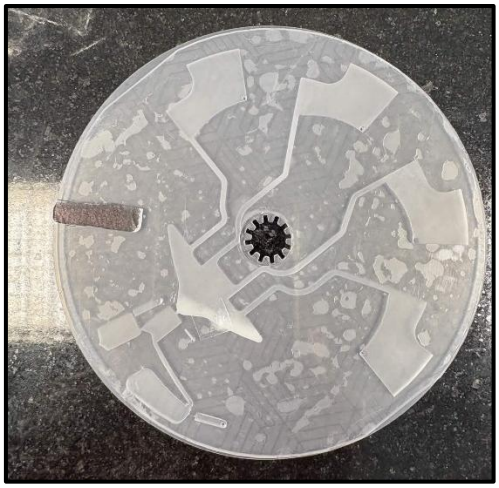


Figure 9.2.2: Photo of a finished disk with beads inserted

9.3 Proof of Concept Testing

There were two pre-developed concept designs for this project, the first design uses tentacle based burst valves, the second uses angled channels to control gravitational flow through geometric design. Each Disc was 3D printed using an SLA printer, washed then cured prior to attaching appropriate films on the disc's face(s). For these two disc designs, the 3D model files were not available as such they were recreated based on already constructed physical discs found in the lab.

9.3.1 Tentacle Disc

This disc uses a uniquely shaped burst valve, that can be described as an S like shape that loops back on itself, it is reminiscent of a tentacle found on an octopus hence where its name was derived from. This disc takes a dual layered approach where the top layer, displayed in figure 9.3.1.1, contains the main tentacle chambers where liquid is moved and stored, the bottom layer contains small chambers for transfer between the tentacles, see figure 9.3.1.2 where bottom layer is displayed. Using gravitational flow liquid is moved through the tentacle shape until it reaches a small hole that connects to the lower layer, marked with a blue circle in figure 9.3.1.1. Centrifugal forces are used to pump the liquid from the top layer into the bottom layer and subsequently into the top layer again. This allows for transfer between chambers, but the liquid cannot flow back on its own.

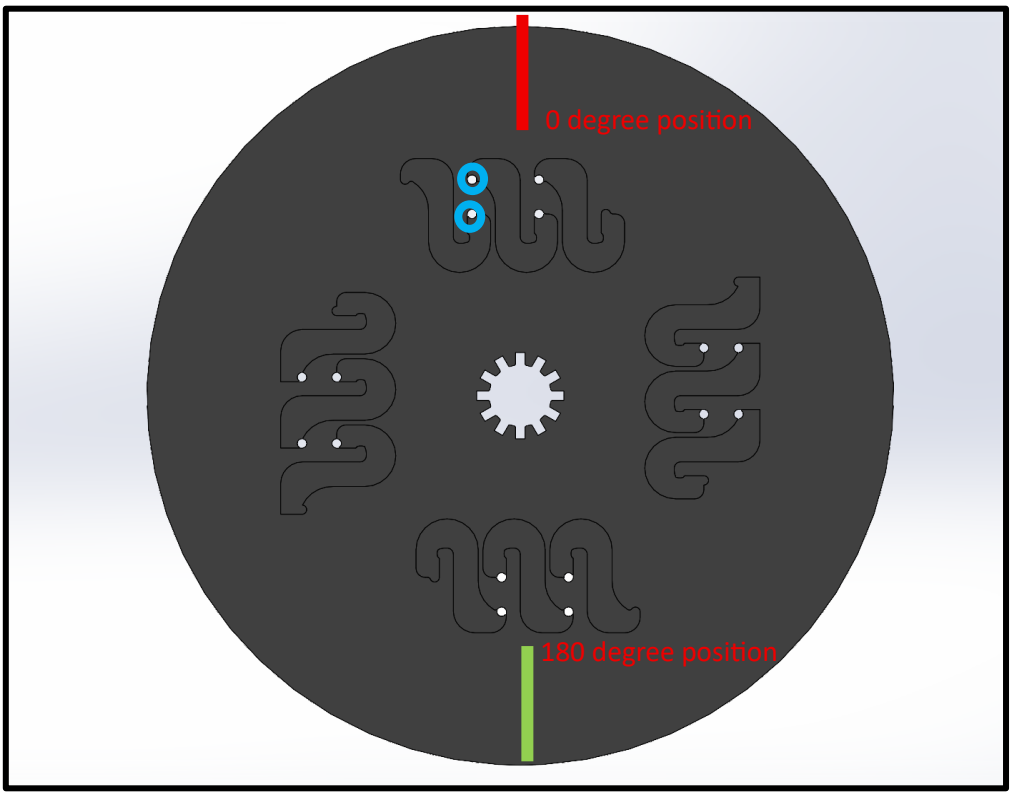


Figure 9.3.1.1: SolidWorks Model of Tentacle Disc Design, Top Layer

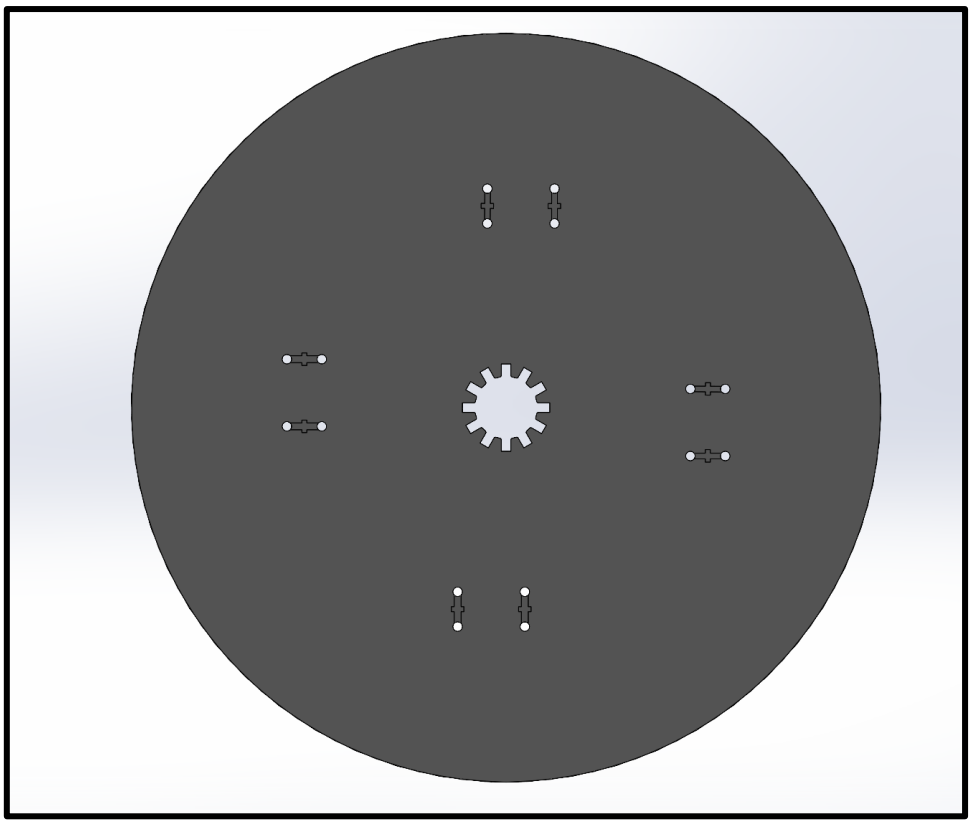


Figure 9.3.1.2: SolidWorks Model of Tentacle Disc Design, Bottom Layer

This disc uses a hydrophilic film on its top layer, with a hydrophobic layer on the bottom. The combination of capillary forces and the hydrophobic films resistance to liquid movement, makes it so the liquid cannot flow back into the previous chamber.

However, it can flow back into the channel between the two. A stroboscope and Basler camera were paired together to capture footage of the movement, using the “pylon viewer” software. This allowed for the capture of images during the centrifugal motion, to create a series of frames depicting the motion. It was observed that at the 180 degree position, spinning either clockwise or counterclockwise the liquid transferred successfully between the two chambers. When in the 0 degree position, the liquid will flow out of the chamber into the next prior to returning into the channel between with clockwise rotations. For counterclockwise rotations the liquid will transfer between chambers and remain within the second, not flowing back into the channel. The reason for this is due to the difference in direction of tangential accelerations, for clockwise rotations the liquid clings to the side wall allowing for gravitational flow to return it to the channel after centrifugal rotation. Whereas for counterclockwise rotations the liquid is moved tangentially in the opposite direction where gravitational flow pulls it further into the tentacle chamber. **Appendix 7** features “comic-strip” depictions of the captured videos for each test mentioned above.

Initially the disc failed to perform as intended, with liquid becoming trapped inside of the tentacle chambers, originally it was believed that this was due to air trapped within the chambers. After testing it was discovered that it was an error with the type of hydrophobic film being used. The final iteration of this disc was constructed using a different film and the disc worked as desired.

One issue noticed with this design was the need to puncture holes in the film to allow for air to be released. Otherwise, it could inhibit the flow of liquid within the chambers when their volume was reduced. This issue was flagged to be a limiting design constraint, as it could require the tentacles to have a larger than desired volume, conflicting with the key design goal of prioritizing compactness.

This disc is solely constructed of valves as they were the key testing element. If a proper design was created for this tentacle valve concept it would be similar to the disc seen in figure 9.3.1.3, where the valves would be used in series, to act as sequencers. This approach removes the requirement of precise burst frequencies as seen in the work by Xiang et al. (2018) [87]. However, these tentacle valves in series would take up a large portion of area on the disc. Miniaturizing of the tenacle chambers themselves does not seem possible with the chance of this causing air blockages in the system. These limitations affect this design's ability to be compact.

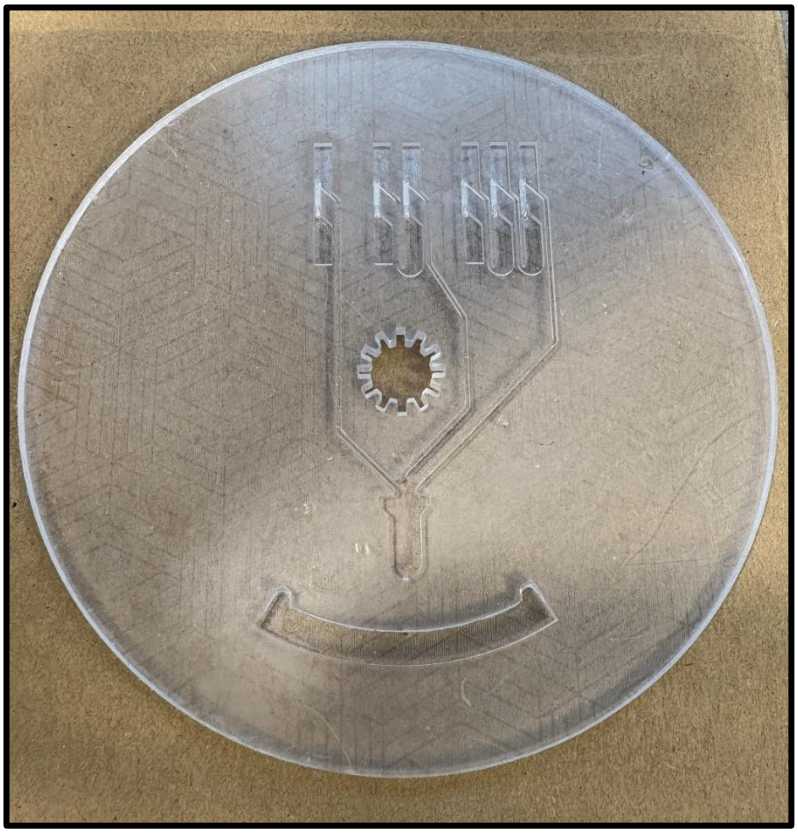


Figure 9.3.1.3: Image of sample disc, to display how the tentacle approach may look

9.3.2 Angle Channel Disc

This disc utilizes four chambers placed around the edge of the disc to store liquids, as the disc is rotated on the platform, storage chambers are brought sequentially to the 0 degree position. Each chamber is designed so that liquid will only flow towards the mixing chamber when the exit of the chamber is in this position. See figure 9.3.2.1 below where the 0 degree position, mixing chamber and waste chamber are detailed. The current design has no valve between the mixing and waste chamber.

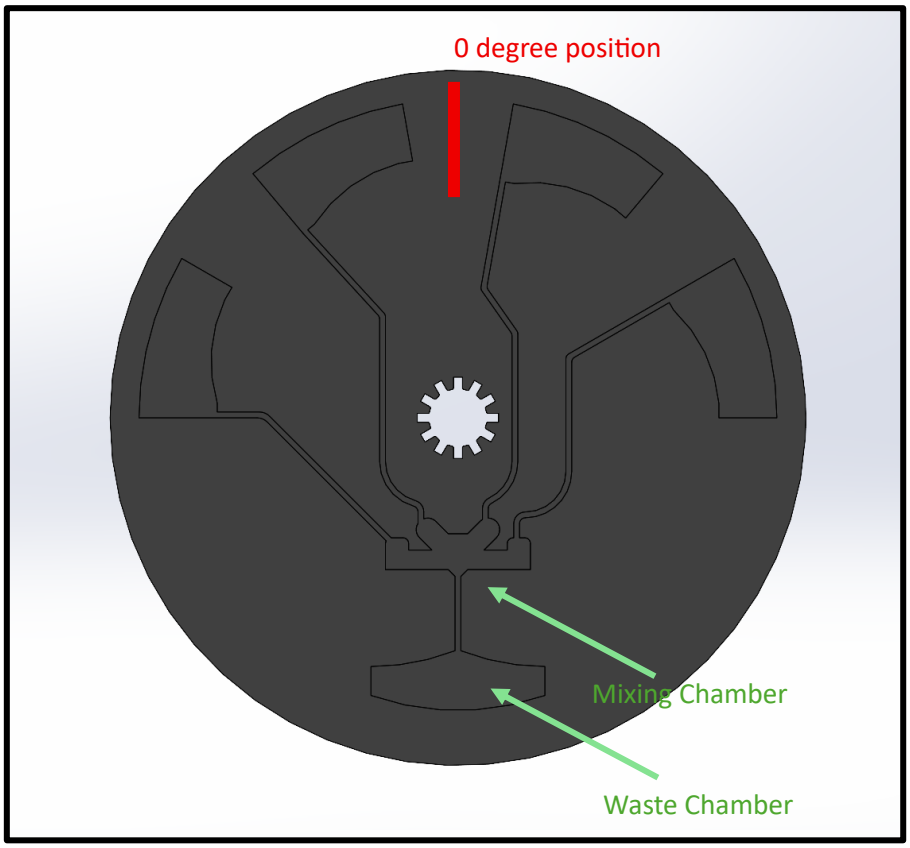


Figure 9.3.2.1: SolidWorks Model of Angled Channel Disc Design

During testing of this disc, it was confirmed that clockwise rotations would not cause liquid to leave the chambers because centrifugal force would cause the liquids to cling to the chamber walls. This means in a more mature design the disc could be rotated in this direction to apply Coriolis forces to the mixing chamber. This also presents the opportunity for a flow switch or burst valve in between the mixing and waste chambers. These tests used a relatively small amount of liquids versus what will be required for the testing designs, around 20 microliters where some of the reagents are planned to use between 50 and 200 microliters. Should this design be taken forward, this will need to be a design consideration. In **Appendix 8** a “comic strip” can be seen depicting the step-by-step actions of testing taken using multiple dye solutions of varying colours to show each chamber distinctly from one another.

9.3.3 Design Decisions Matrix

The key design goals of the vLoaD disc are as follows:

- Compactness
- Reliability
- Simplicity

These factors were determined based on the overall goals of the project having a disc with minimal instrumentation and a lightweight nature that can be used easily by operators with basic training. As such these key design goals were vital aspects for the decision making process and were used in the decisions matrices seen below. Each of these properties were weighted based on their relative importance to the project. The decision matrix shown below operates on a scale of 1 to 5, where 5 is the best score. Therefore, the design with highest total is deemed the better of the two.

Table 9.3.3.1: Decision Matrix for Concept Designs

	Weighting	Tentacle	Angled Channels
Compactness	5	2	3
Reliability	4	4	3
Simplicity	2	2	5
Weighted Total		30	37

The angled channel design ranked higher given the weight totals. It passed the tentacle disc in both simplicity and compactness; these were both linked to how this design handles the sample/chemical chambers. The tentacle disc did edge out beyond the angled channel disc in the reliability category, this is due to the ability for the angled disc to be rotated in the wrong direction or be over rotated, either of which will result in the order of sequential operations being disrupted.

9.4 Prototype Designs (IP)

After the initial proof of concept testing concluded, and the angled channel design was selected, iterative prototype design begun. This spanned for 14 versions of the disc, intermediate revisions are covered briefly in **Appendix 9**. This segment only covers the notable versions that show how the designs evolved over time. Notably, prototypes start from version two, as version one was the concept design

9.4.1 Version 2

Differences from previous version:

- Introduction of burst valve from mixing chamber to waste chamber
- Augmentation of mixing chamber, introduction of “side wings” to prevent backflow into storage chambers

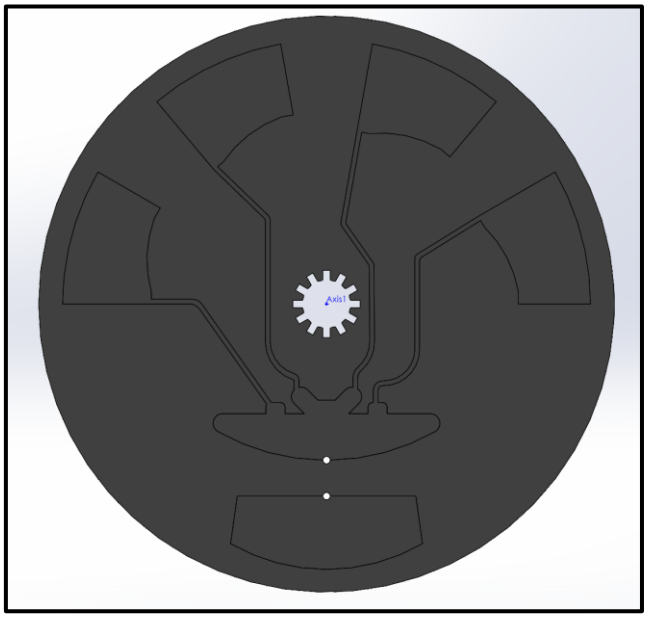


Figure 9.4.1.1: Image of Prototype Design V2

Testing Results:

Successes	Shortcomings
• Backflow was minimized thanks to the introduction of the side wings • The burst valve worked as intended	• Issues with actuation of fluids into the mixing chamber, it is possible for liquid to flow across the outlet towards the other liquid outlets leaving no fluid in the chamber

9.4.2 Version 5

Differences from previous version:

- Adjustments of mixing chamber design to “fix” centrifugal flow

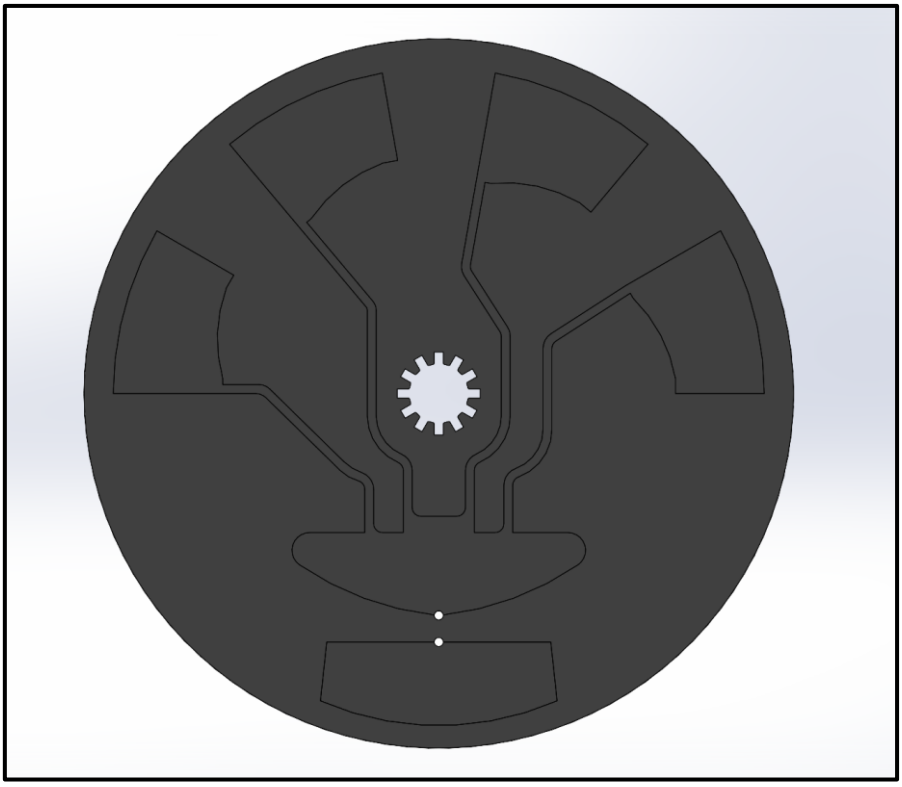


Figure 9.4.2.1: Image of Prototype Design V5

Testing Results:

Successes	Shortcomings
• Burst Valve works as intended • Fully resolve all issues relating to getting fluid from the storage chambers into the mixing chamber	• Here is where the change in scope occurred, making it necessary to make changes to adapt the design for non-magnetic silica beads.

9.4.3 Version 6

Differences from previous version:

- Introduction of Coriolis Switch
- Introduction of Product and Waste chambers after Coriolis Switch

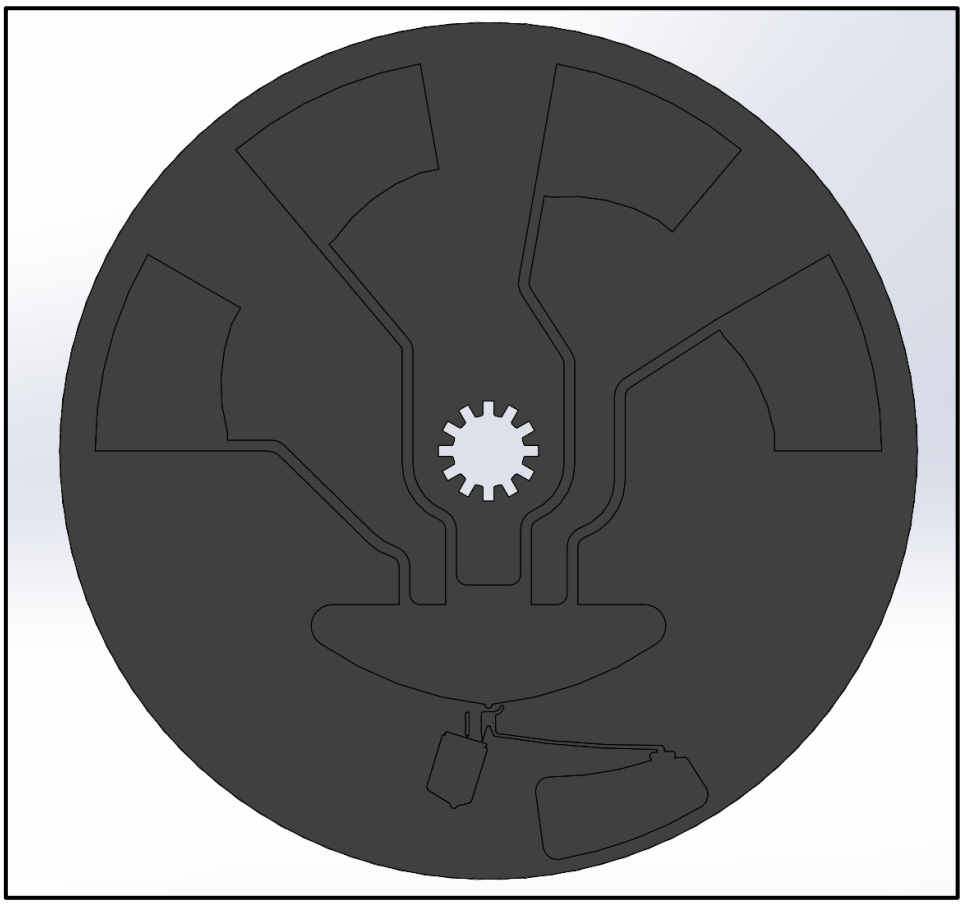


Figure 9.4.3.1: Image of Prototype Design V6

Testing Results:

Successes	Shortcomings
• Burst Valve worked to pump fluid into the Coriolis switch	• Coriolis switch and waste chamber did not work as intended, thought due to a vacuum being created and the directions of flow being reversed

9.4.4 Version 13

Differences from previous version:

- Final Coriolis Switch design calculations occurred
- Increase of length of Coriolis switch
- Adjustment of separator within the Coriolis switch

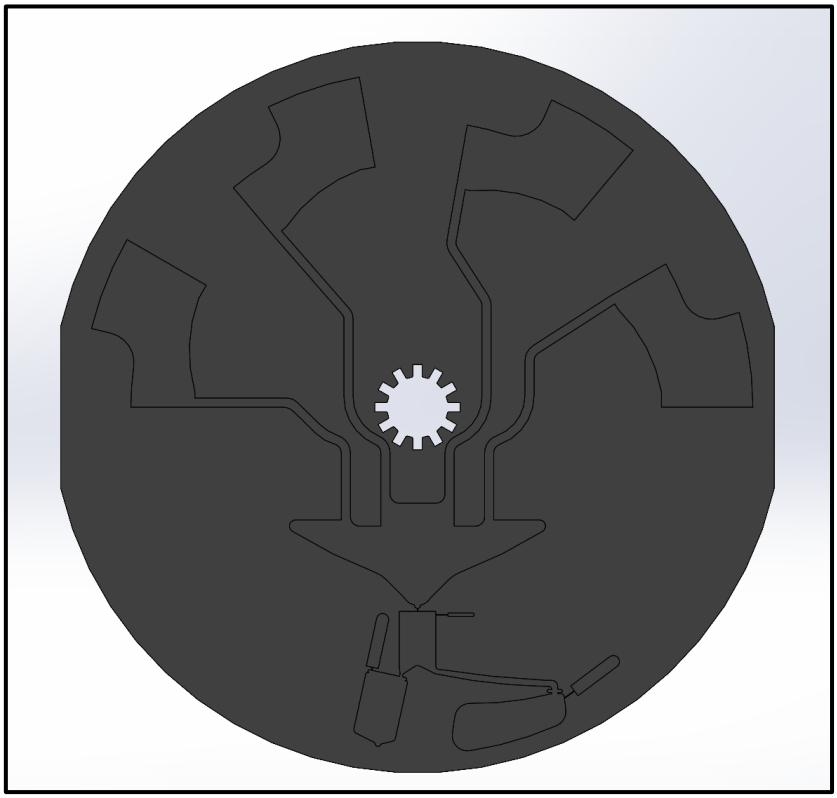


Figure 9.4.4.1: Image of Prototype Design V13

Testing Results:

Successes	Shortcomings
• Coriolis Switch worked as desired • Product and Waste chambers work as desired • Full completion of prototyping for these elements	• No new shortcomings noticed at this stage • Possibly suboptimal for DNA Isolation

This disc was the first prototype for Coriolis switch testing to work fully as desired. **Appendix 12** features a comic strip showing how this disc operates.

9.4.5 Version 14

Differences from previous version:

- Major redesign of mixing chamber, attempt to create a design that uses centrifugal flow to pump fluid through clogged beads as opposed to using diffusive mixing, should in theory improve DNA Isolation assay performance

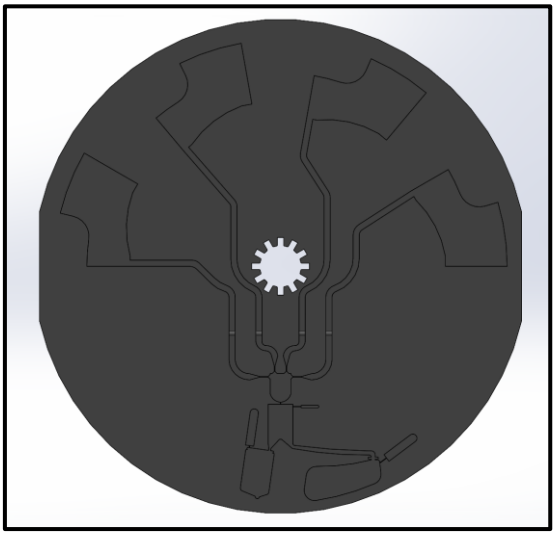


Figure 9.4.5.1: Image of Prototype Design V14

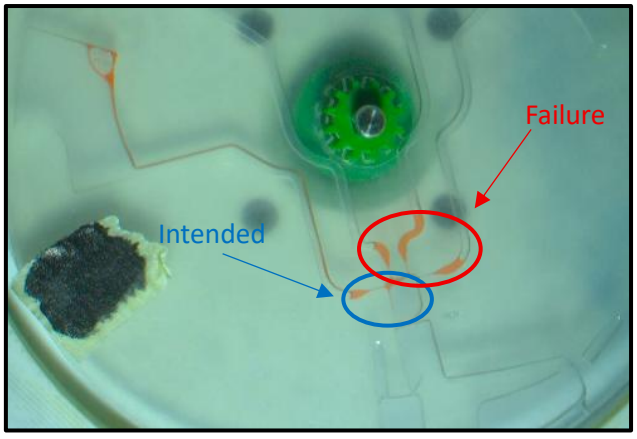


Figure 9.4.5.2: Image of Prototype Design V14, Failure During Operation

Testing Results:

Successes	Shortcomings
• Beads remained within the newly designed chamber	• Fluid flow into other storage outlets as opposed to into the desired mixing chamber, total failure, see figure 9.4.5.2

9.5 Testing Designs (IP)

The testing designs are the versions of the vLoaD discs that were used for performing DNA Isolation assays, this segment of the thesis highlights these designs as well as the tests they were used on. The protocol used for each one of these designs is covered in **Appendix 10**.

9.5.1 Testing Design 1 (vLoaD v13)

This design was prototype version 13, see **section 10**. This was used for tests one through eight when using non-magnetic silica beads. The design was used again for tests 11 through 13 with magnetic silica-beads. When used for silica beads a new attachment was designed for the rig to allow for the usage of magnets, see figure 9.5.1.1.

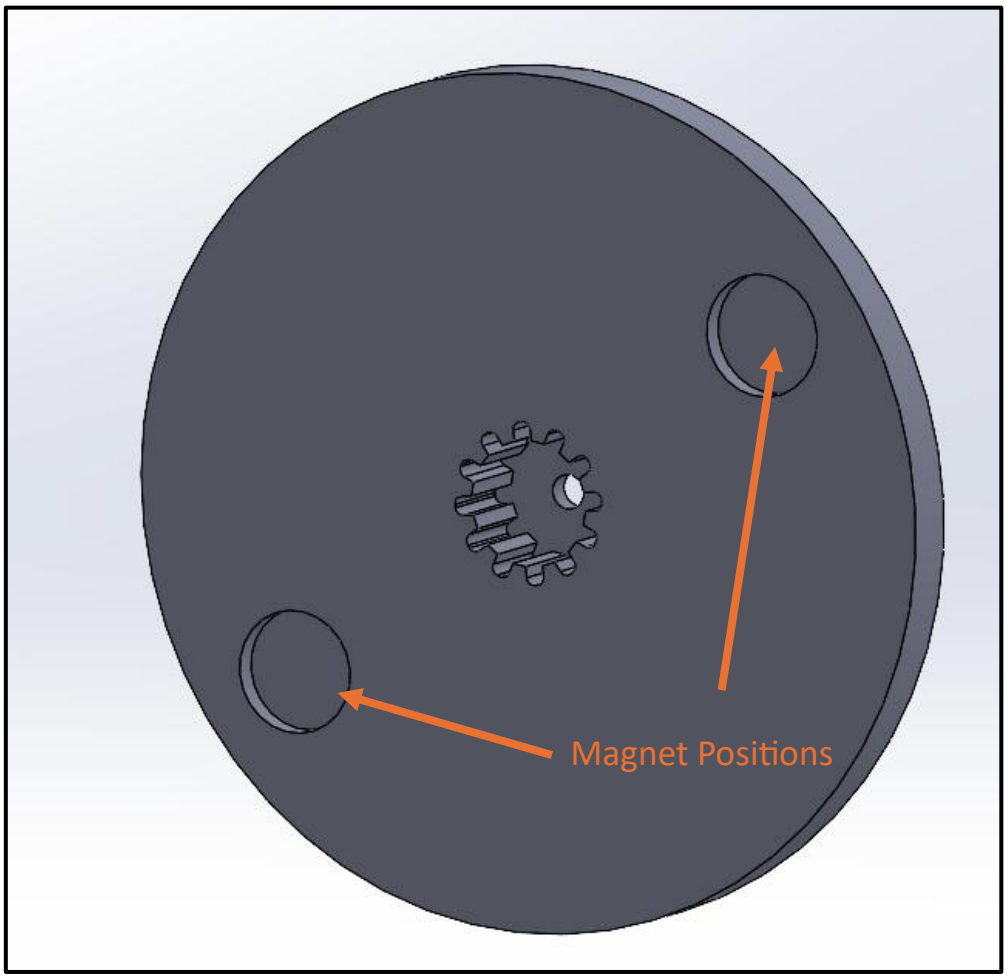


Figure 9.5.1.1: SolidWorks screenshot of vLoaD Magnet Holder

9.5.2 Testing Design 2-1

For Test One through Seven the previous design seen in version 13 was considered to be the final form of the vLoaD disc. A critical design error was spotted, the beads were not mixing properly with the DNA sample, preventing DNA capture from occurring. This disc, TD2 was used for test eight

The new area of the design is the “S-Chamber” which sits between the prior mixing chamber and the Coriolis switch, the beads now reside in this area, and liquids are pumped through them. Attempts were made (Test Seven and Version 14) to augment the mixing chamber to pump fluid through the beads but neither were successful. The major flaw was trying to keep the beads in position atop the burst valve as opposed to within one of the mixing chambers wings. Hence why the new additional chamber was added. The previous “mixing” chamber was required to remain as the wings added with liquid metering and prevent flowback. However, it was scaled back and repositions to allow room for the new S-Chamber.

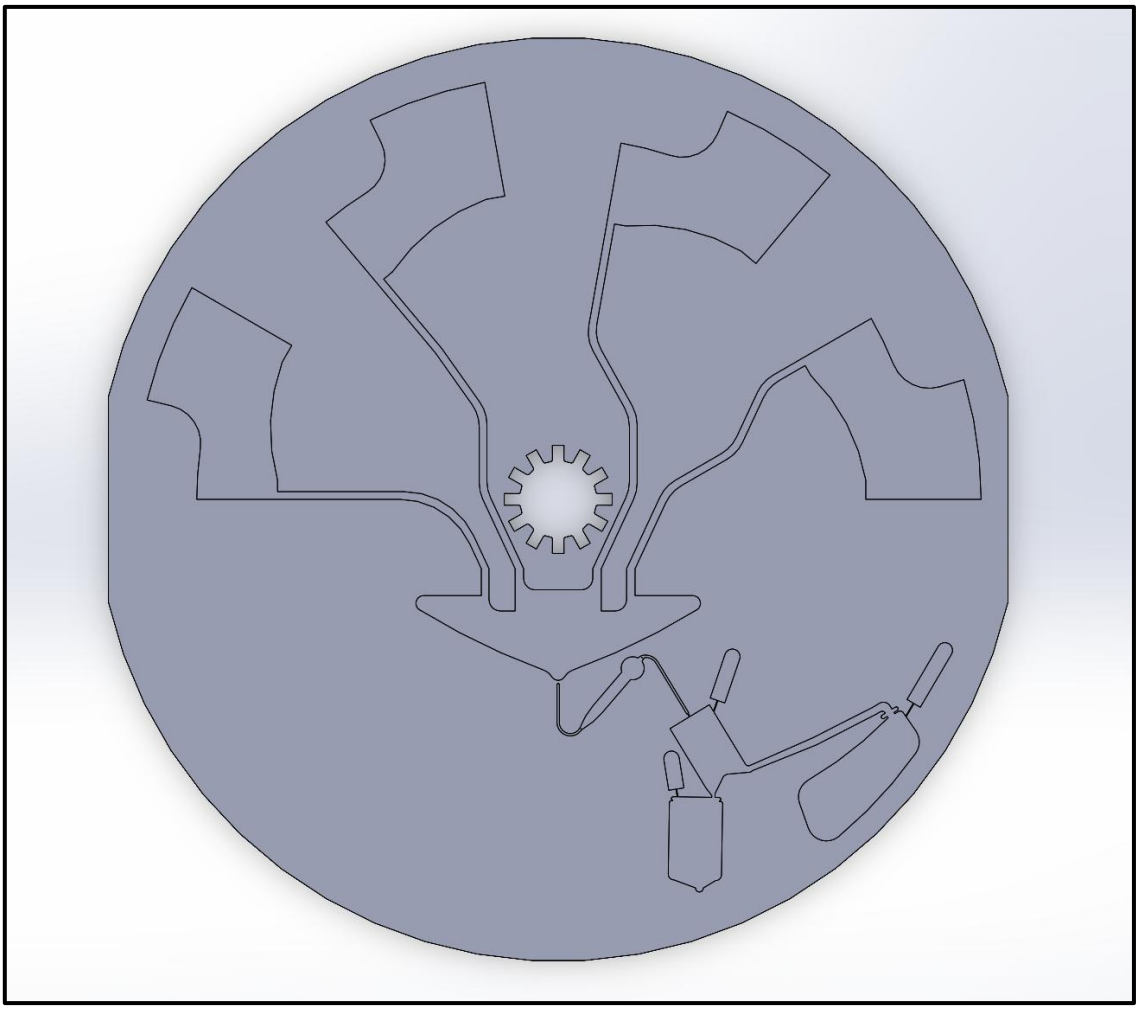


Figure 9.5.2.1: SolidWorks screenshot of vLoaD Testing Design 2-1

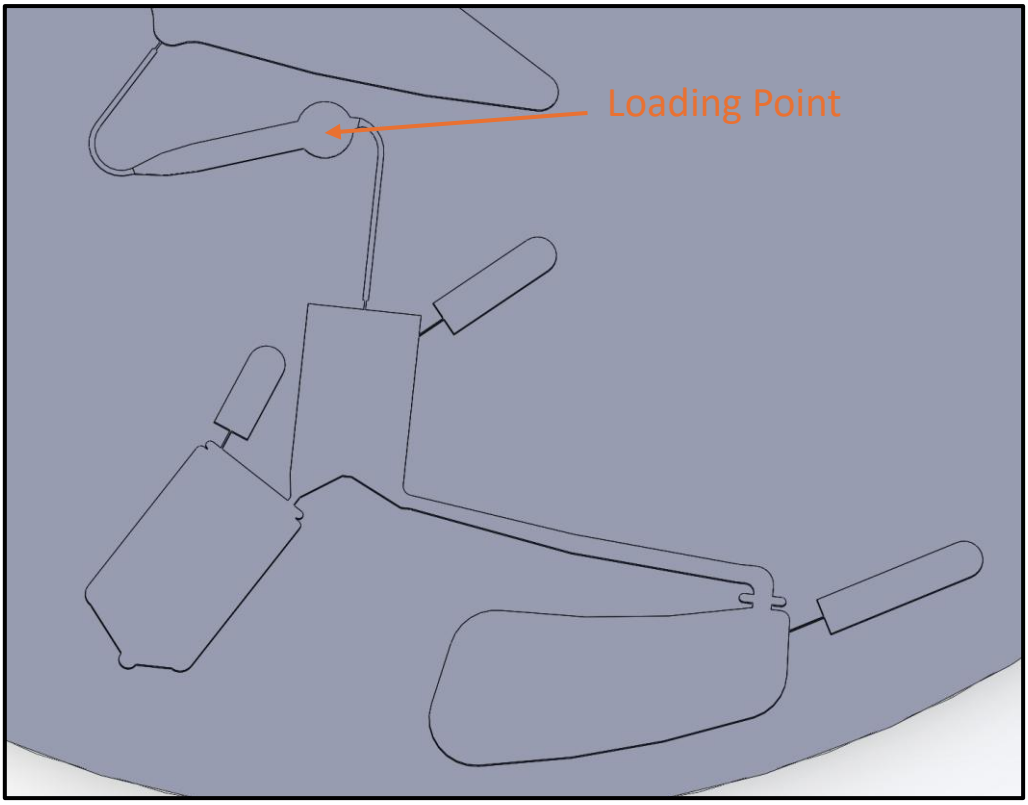


Figure 9.5.2.2: SolidWorks screenshot of vLoaD Testing Design 2 - 1, Focused on Lower Half

It was proven that the forces were sufficient for the liquid to pump around the "S-Chamber" curves via theoretical design calculations, see **Section 8.3.4.** It was also proven that the S-Chamber will act like a siphon valve, see **Section 8.3.3**. This will result in major changes in the vLoaD protocol as to send a liquid from the mixing chamber to the Coriolis switch, the disc will need to be spun to break past the first burst valve, then stopped to allow the Coriolis switch to work, then spun again to break past the second burst valve. The vertical nature of the vLoaD was taken advantage of here, while for the theoretical design calculations the worst-case scenario was assumed where the capillary force would need to work against gravitational force, the layout of the DNA isolation assay allows for the two forces to work in conjunction. One important design consideration was the desired clogging effect. The beads must clog the S-Chamber to form an obstruction to fluid flow; this makes it that samples/reagents must flow through the beads. The entrance and exit channel widths are 0.3mm to aid with the "clogging" of beads which ensure that the fluid cannot flow around the beads, this is because the ratio of channel width to bead diameter is so small that beads are almost guaranteed to form a wall/bridge. The burst valves use a width of 0.1mm so beads cannot escape the channel itself. The mid-section of the chamber is larger to allow for easier bead loading, with an area designated for loading, see the circular cut out in figure 9.5.2.2. The larger area makes it easier to get the pipette tip into the disc.

9.5.3 Testing Design 2-2

This disc design is a minor revision of the previous testing disc; it was created after the unsuccessful test eight. The previous disc failed to operate as desired due to the capillaries at the entrance and exit of the S-Chamber.

The overall disc maintains most of the same design, changes were not made to the storage chambers, “mixing” chamber or the Coriolis switch. Aside from reducing the wing size on the “mixing” chamber to ensure that the entrance to the S-Chamber is the further radial distance for centrifugal forces to apply correctly.

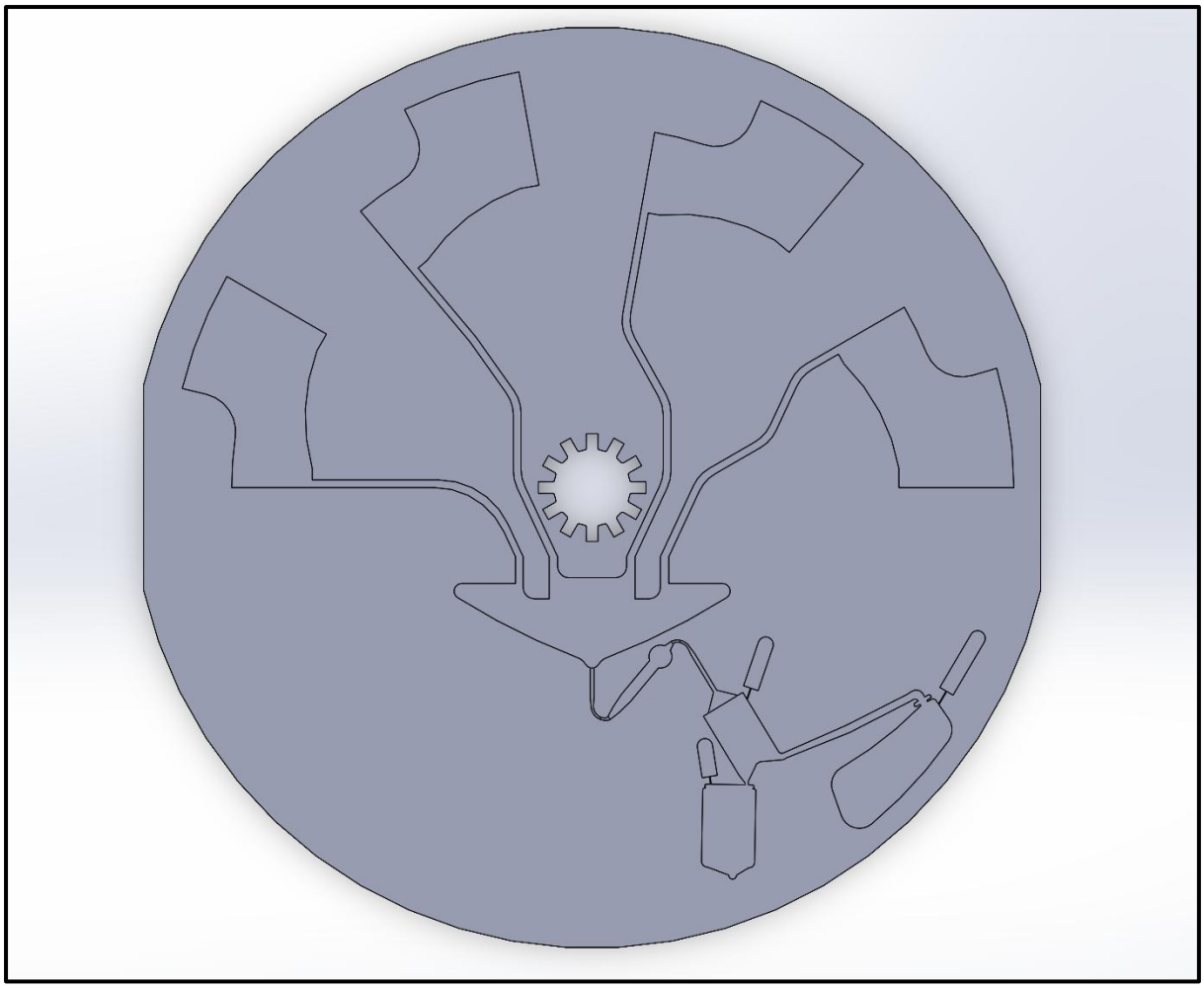


Figure 9.5.3.1: SolidWorks screenshot of vLoaD Testing Design 2-2

For the S-Chamber the main design changes made were at the capillaries of the disc. The burst valves at the entrance and exit of the S-Chamber have been removed. The dimensions of the entrance and exit have been widened to 0.5mm to improve fluid flow. The 0.3mm width has been preserved within the curves that lead into and out of the bead storage area, to maintain “clogging” keeping the bead locked here. At the exit of the S-Chamber a wide chamber has been implemented to improve fluid flow towards the Coriolis switch, as the second burst valve was deemed unnecessary.

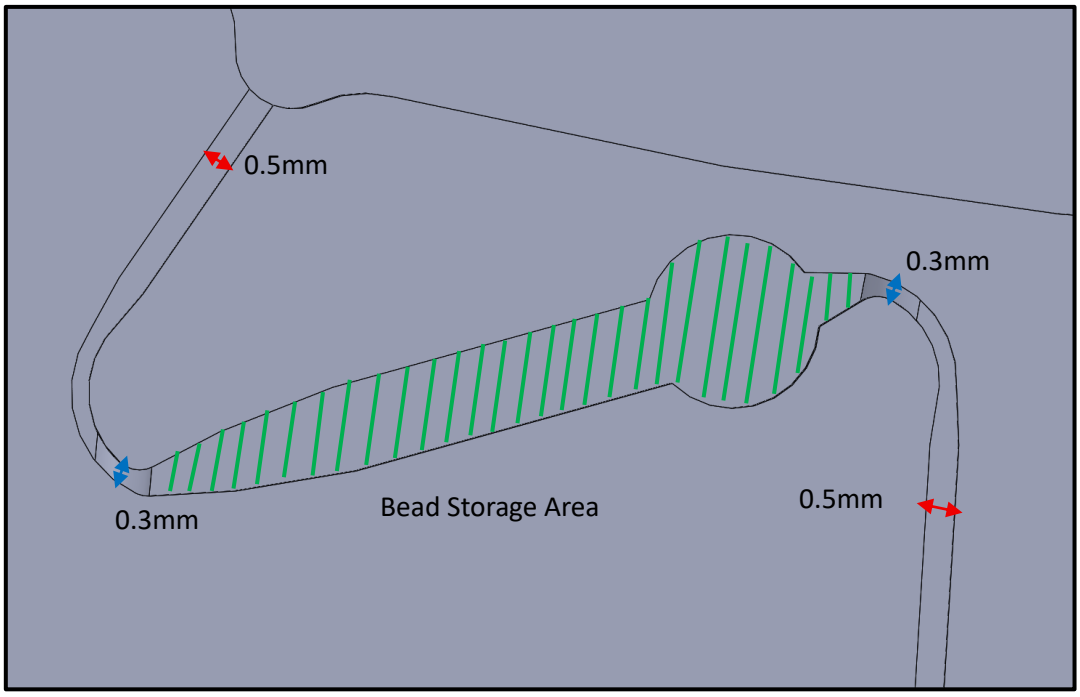


Figure 9.5.3.2: SolidWorks screenshot of vLoaD Testing Design 2 - 2, Focused on S-Chamber

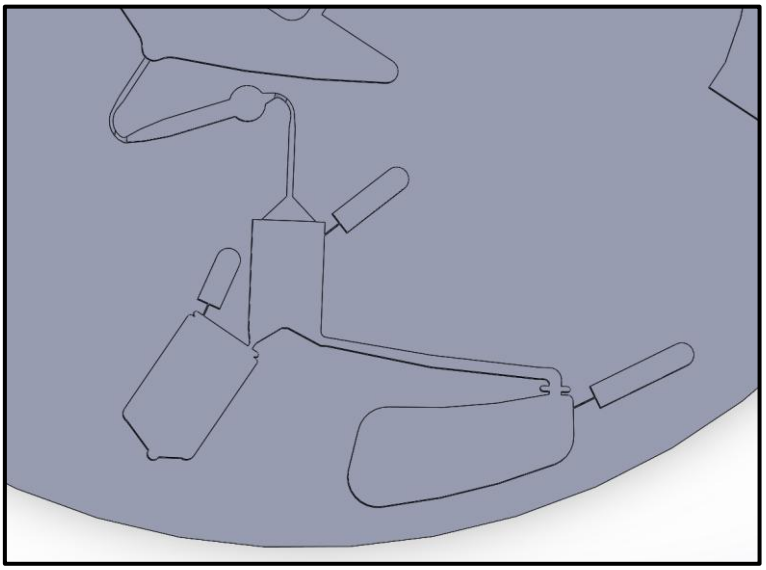


Figure 9.5.3.3: SolidWorks screenshot of vLoaD Testing Design 2 - 2, Focused on Lower Disc Half

9.5.4 Testing Design 3

When the siphon valve failed to operate as desired it was proposed to replace it with a straight chamber that would be packed with a column of silica beads. This was used for test ten (**section 10.6**)

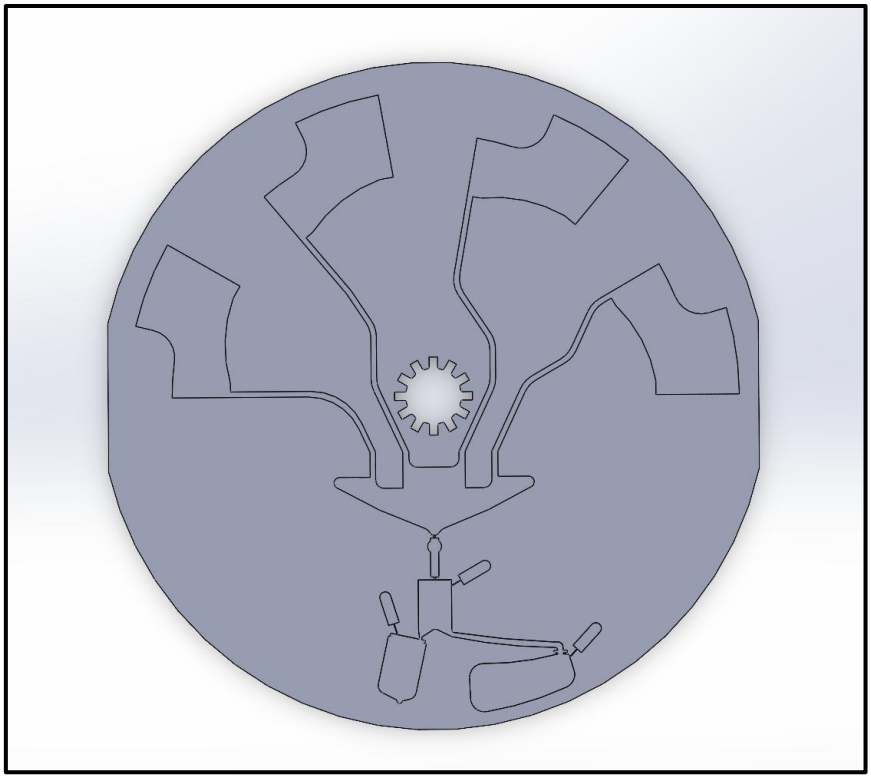


Figure 9.5.4.1: SolidWorks screenshot of vLoaD Testing Design 3

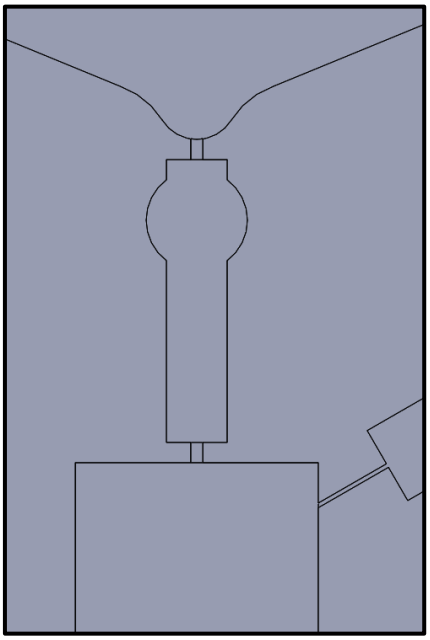


Figure 9.5.4.2: SolidWorks screenshot of vLoaD Testing Design 3, focus on new bead chamber

9.6 DNA Preparation protocol

9.6.1 First Protocol

For the test of the vLoaD device, a DNA Isolation assay was performed on-disc. This segment covers the preparation of DNA sample prior to usage on vLoaD, the off-disc work. The on-bench protocol is covered in the following section.

- First lab-safe E. Coli cells were cultured by Abrar Abdalsalam
- 1mL of E. Coli Cells were taken and added to a microcentrifuge tube
- 180 µL of Buffer ATL and 20µL of Proteinase K are added
- Mix via vortex
- Incubate at 56 degrees Celsius, for approximately 1 hour
- Add 200 µL of Buffer AL, vortex to mix
- Incubate this mixture at 70 degrees Celsius for 10 minutes
- Centrifuge briefly
- Add 200 µL of Ethanol 96%, vortex for 1.5 seconds then Centrifuge briefly

This mixture is the sample used on the vLoaD disc it is a mixture of lysed E. Coli cells that is appropriate for the DNA Isolation protocol. The Cells, ATL and Proteinase K make a volume of 1200 µL, this volume was split at this point and some of it was maintained as a control for possible later usage.

Therefore, the mixture used on disc only 50 µL of Buffer AL and 50 µL of Ethanol 96% were added, to make a total volume of 400 µL. This mixture was kept in the fridge in the Bacterial Cell Culture lab and taken out only when required to take a sample for the disc testing. NOTE: This mixture will be referred to as the cell sample going forward for vLoaD and on-bench isolation.

9.6.2 Second Protocol

This cell sample was used for test one to Four, for test five onwards a different sample was used with the following protocol, as per Abrar Abdelsalam's recommendation.

- First lab-safe E. Coli cells were cultured by Abrar Abdalsalam
- 1mL of E. Coli Cells were taken and added to a microcentrifuge tube
- 180 µL of Buffer ATL and 20µL of Proteinase K are added
- Mix via vortex
- Incubate at 56 degrees Celsius, for approximately 1 hour
- Pipette 200 µL of this mixture to a new 2ml collection tube
- Add 200 µL of Buffer AL, vortex to mix
- Incubate this mixture at 70 degrees Celsius for 10 minutes
- Centrifuge briefly
- Add 200 µL of Ethanol 96%, vortex for 1.5 seconds then Centrifuge briefly

9.7 On-Bench Control Sample

9.7.1 First Sample

To compare the vLoaD performance to traditional laboratory benchtop DNA Isolation, a sample was fully prepared using the following protocol;

- Pipette 50 µL of Cell Sample and add to a QIAamp Mini spin column in a 2ml collection tube
- Centrifuge at 8000 RPM for 1 min, discard the flow-through and collection tube.
- Place the QIAamp Mini spin column in a new 2ml collection tube and add 50 µL of Buffer AW1
- Centrifuge at 8000 RPM for 1 min, discard the flow-through and collection tube.
- Place the QIAamp Mini spin column in a new 2ml collection tube and add 50 µL of Buffer AW2
- Centrifuge at 13,300 RPM for 1 min, discard the flow-through and collection tube.
- Place the QIAamp Mini spin column in a new 2ml collection tube and add 100 µL of Buffer AE
- Incubate at room temperature for 1 minute
- Centrifuge at 8000 RPM for 1 min, discard the flow-through and collection tube.

This protocol was modified to match the volumes used on the vLoaD, this was the comparison on testing 4.

9.7.2 Second Sample

However, after discussion this was deemed to not be a good control as it modified the Qiagen Protocol by adjusting the volumes. For the following tests a new control was produced using a different protocol;

- Pipette the full 600 µL of Cell Sample and add to a QIAamp Mini spin column in a 2ml collection tube. This included lysed cells, buffer AL and ethanol 96%
- Centrifuge at 8000 RPM for 1 min, discard the flow-through and collection tube.
- Place the QIAamp Mini spin column in a new 2ml collection tube and add 500 µL of Buffer AW1
- Centrifuge at 8000 RPM for 1 min, discard the flow-through and collection tube.
- Place the QIAamp Mini spin column in a new 2ml collection tube and add 500 µL of Buffer AW2
- Centrifuge at 13,300 RPM for 3 min, discard the flow-through and collection tube.
- Place the QIAamp Mini spin column in a new 2ml collection tube and add 200 µL of Buffer AE
- Incubate at room temperature for 1 minute
- Centrifuge at 8000 RPM for 1 min, discard the flow-through and collection tube.

This new sample was used as the control for Test Five onwards.

9.8 DNA Testing

Testing of DNA samples was done in two forms for this project. Nanodrop and q-PCR. Once the DNA Isolation assay on disc concluded, the DNA + Elution mixture was removed from the product chamber via pipette. This mixture was then used in testing apparatus to gather results.

9.8.1 NanoDrop, Microvolume Spectrophotometer

The first test performed was NanoDrop, this is a microvolume spectrophotometer produced by ThermoFisher Scientific. This device takes a 1 microliter calibration liquid and compares it to the sample volume of sample. For the usage case in this project, the device is calibrated on the Elution buffer, QIAGEN AE then a droplet of the DNA + Elution mixture was measured on the machine. The device is connected to a laptop that runs a companion software. In between measurements the sensor is cleaned with lint-free tissue and ethanol.



Figure 9.8.1.1: Image of NanoDrop Machine

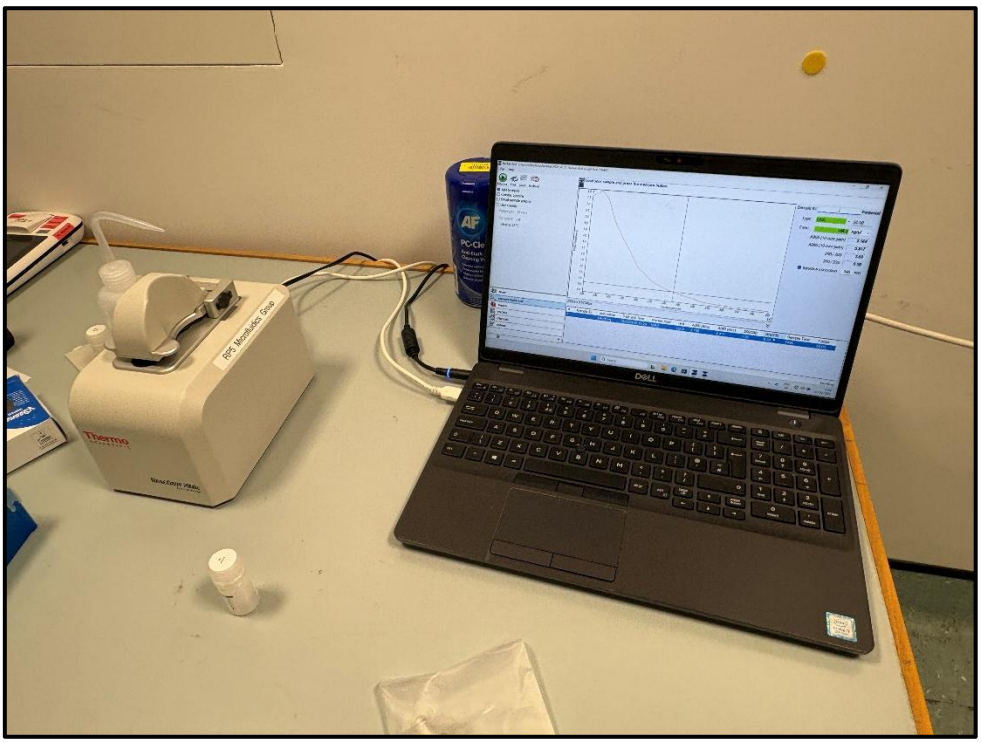


Figure 9.8.1.2: Image of NanoDrop Setup

This application displays the results from the device, providing a graph between 10mm Absorbance and Wavelength. This is a standard output graph for a spectrophotometer and can be read to analyse the measured material. The software also output other results metrics, Nucleic Acid Count, A260 (ABS), A280 (ABS), 260/280 and 260/230. Nucleic Acid Count is the sample count for number of nanograms per microlitre. A260 (ABS) is the count of sample absorbed at 260 nanometres, at this wavelength high absorbance indicates the presence of DNA or RNA. A280 is similar, it is for 280 nanometres, at this wavelength high absorbance indicates the presence of aromatic amino acids.

These two measurements indicate the presence of their respective measurand. 260/280 and 260/230 are purity checks for the sample. 260/280 can be utilised to determine the purity of the nucleic acid, a value of approximately 1.8 [128] indicates pure DNA. 260/230 is used to determine if there is an organic contaminant or salts present, a value from 2.0 to 2.2 [128] is indicative of a pure DNA sample, as the Isolation protocol uses QIAGEN wash buffer AW1 which has guanidine salts, this can determine if the wash buffer is present in the sample. Which would be an inhibitor for the q-PCR process.

9.8.2 Q-PCR

Real time quantitative PCR (q-PCR) is a molecular biology technique that amplifies and quantifies DNA in real-time. For the purposes of the vLoaD it can take a small amount of product, amplify the DNA present and measure it so it can be compared to an on-bench laboratory sample using a CT shift. This is the difference in cycle threshold between two q-PCR runs, which measures relative differences in the amount of DNA present. The q-PCR was first run for Test Four using the Qiagen Rotor-Gene Q. For these q-PCR tests in addition to the DNA sample, a negative control was also run. This was prepared in the same way as the DNA sample, however it does not have any sample included.



Figure 9.9.2.1: Image of Rotor-Gene Q

The tubes were prepared as follows:

- Add 5µL of PowerUp™ SYBR™ Green Master Mix (2X) to each well
- Add 0.5µL of each primer (Forward and Reverse) from 10µM solution to each well
- Add 4µL of Nuclease-Free water
- Add 1µL of DNA Sample

Table 9.8.2.1: Cycle times for Rotor-Gene Q

Step	Temperature	Time	Cycles
UDG Activation	50˚C	2 mins	1
Dual-Lock Polymerase Activation	95˚C	2 mins	1
Denature	95˚C	15 sec	40
Anneal/Extend	60˚C	1 min	40
Melt Curve	Auto	Auto	1

After the q-PCR for test four showed zero amplification, the machine was swapped to the AriaMX.



Figure 9.9.2.2: Image of AriaMX

For this test the tubes were prepared as follows:

- Add 10µL of SYBR™ Green Master Mix (2X) to each well
- Add 0.8µL of each primer (Forward and Reverse) from 10µM solution to each well
- Add 1µL of DNA Sample
- Add enough Nuclease-Free water for a total volume of 20 µL

Table 9.8.2.1: Cycle times for AriaMX

Step	Temperature	Time
Initial Denature	95˚C	2 mins
Denature (x40)	95˚C	15 sec
Anneal	57˚C	30 sec
Extend	72˚C	30 sec
Melt Curve	60-95˚C	0.3˚C/s ramp

10 Results

This results section is comprised of subsection, each of which corresponds to one test of the vLoaD device, each one documents what occurred during testing as well as the results derived. Which design was used for what test is covered in **section 9.5**. This segment only features tests with significant results additional testing is in **Appendix 11**.

10.1 Test Five

10.1.1 vLoaD Test

During testing it was noted that the elution buffer turned a yellowish colour. This was researched and it is likely due to uncured resin on the surface of the vLoaD disc, as SLA resin contains chemicals called photo-initiators which are naturally yellow, so these could have leached into the elution to cause the colour shift.

10.1.2 DNA Test

10.1.2.1 NanoDrop

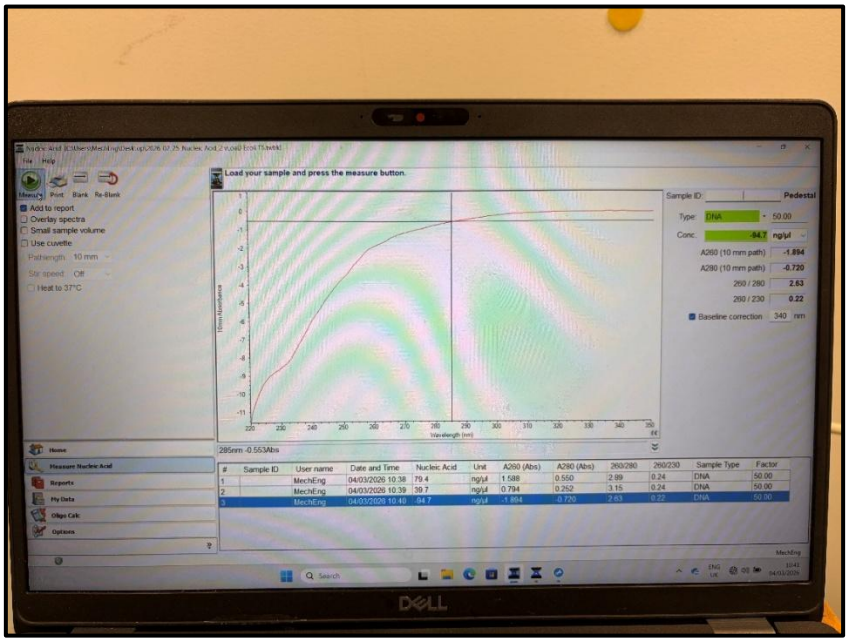


Figure 10.1.1: Image of NanoDrop Graph Test Five

Table 10.1.1: NanoDrop Measurement Results for Test Five

Measurement	ng/µL	A260	A280	260/280	260/230
1	79.4	1.588	0.55	2.89	0.24
2	39.7	0.794	0.252	3.15	0.24
3	-94.7	-1.894	-0.72	2.63	0.22

This DNA sample has a very low amount of A260 and neither of the two ratios are in the expected ranges, indicating an that DNA is not present within the sample. The third measurement is likely an error and thus was ignored for analysis. The extreme results are likely due to the contaminated elution.

10.1.2.2 q-PCR

q-PCR was performed using the ARIAMX machine. No amplification for the vLoaD sample, control sample was loaded and did show amplification, indicating the machine performed properly. See Figure 10.5.2, where the on-bench control sample was the yellow plot. This validates the control DNA sample.

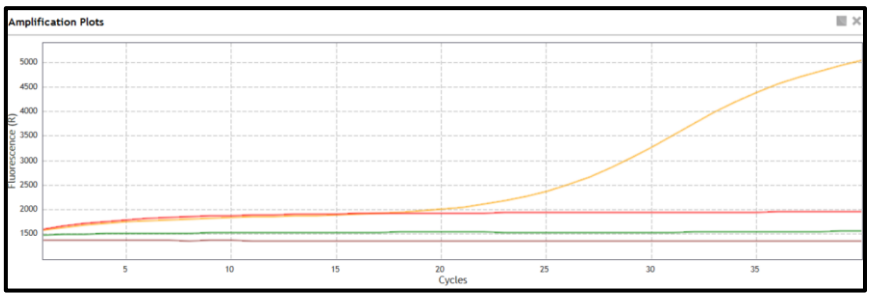


Figure 10.1.2: q-PCR Amplification Plot for Test Five

10.2 Test Six

10.2.1vLoaD Test

Disc preparation was adjusted, as the previous test it was believe excess uncured resin is an issue with biocompatibility. Disc was washed thoroughly with high-pressure. Then the disc was fully submerged in water for 5 minutes prior to curing for 10 minutes. It was expected that the water would remove oxygen pockets that prevent curing.

Additionally, this disc was coated in BSA, Bovine Serum Albumin, this acts as a physical buffer to prevent biocompatibility issues due to the toxic nature of the resin material that the vLoaD disc is fabricated from.

10.2.2 DNA Test

10.2.2.1 NanoDrop

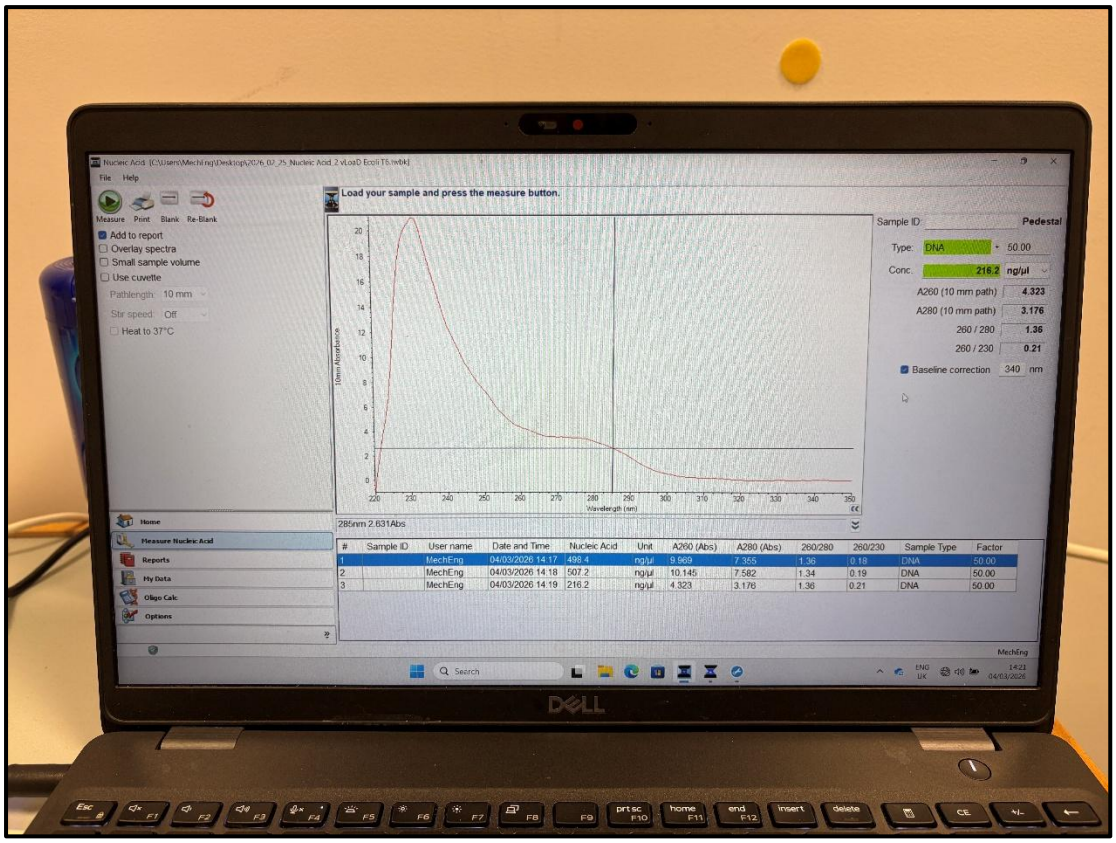


Figure 10.2.1: Image of NanoDrop Graph Test Six

Table 10.2.1: NanoDrop Measurement Results for Test Six

Measurement	ng/µL	A260	A280	260/280	260/230
1	498.4	9.969	7.355	1.36	0.18
2	507.2	10.145	7.582	1.34	0.19
3	216.2	4.323	3.176	1.36	0.21

Readings are extremely high, this is likely due to the BSA throwing off the count, as a protein it can appear similarly to the sensor. The ratios indicate that DNA is likely not present in the sample.

10.2.3 On Bench Control

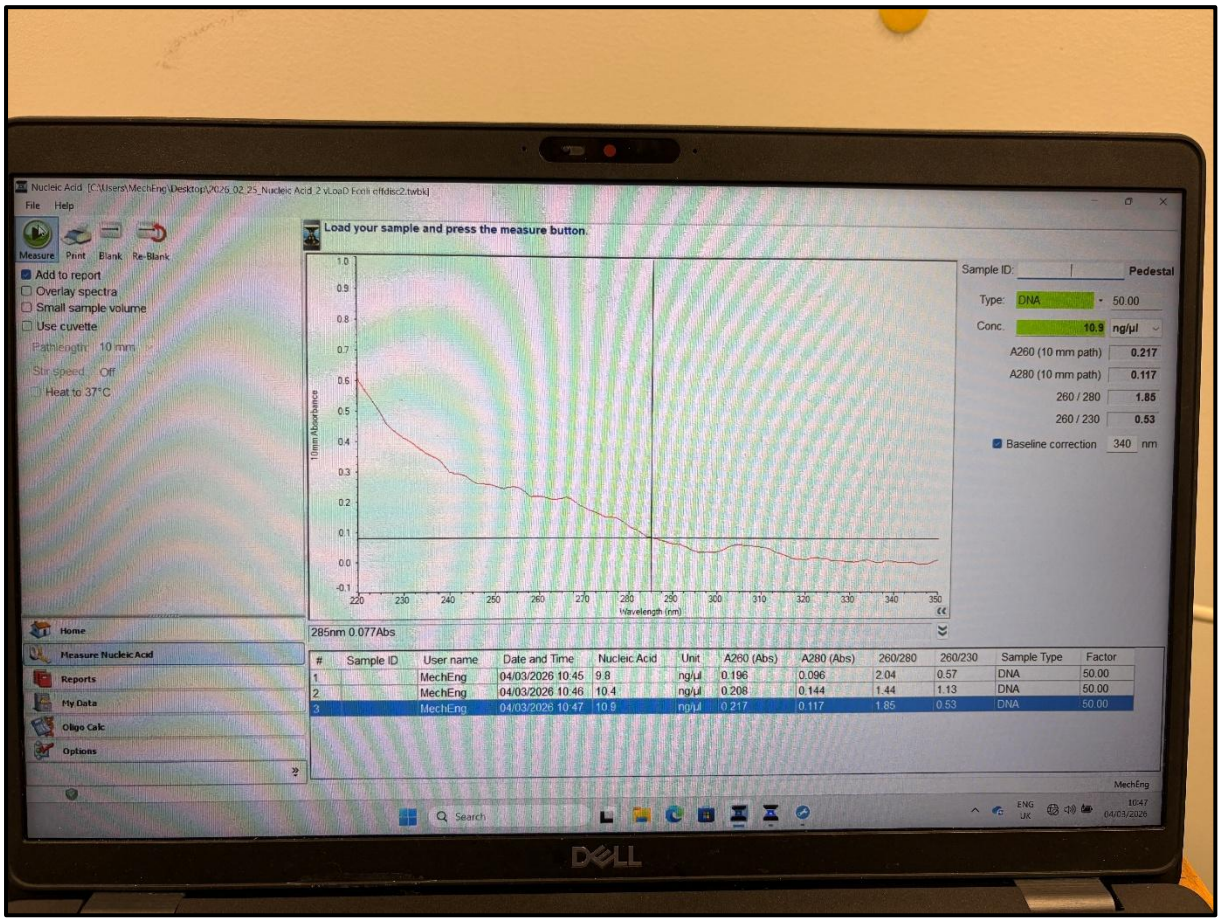


Figure 10.2.2: Image of NanoDrop Graph Murder On Bench Control

Table 10.2.2: NanoDrop Measurement Results for On Bench Control

Measurement	ng/µL	A260	A280	260/280	260/230
1	9.8	0.196	0.096	2.04	0.57
2	10.4	0.208	0.144	1.44	1.13
3	10.9	0.217	0.117	1.85	0.53

It is interesting to see that a proven control DNA sample still shows ratio readings that are not as expected, particularly the 260/230 ratio. Bringing the accuracy of the NanoDrop into question.

10.2.4 Biocompatibility Check DNA (Murder DNA)

A sample of the on-bench control was placed onto a vLoaD disc and left there for 30 minutes, to investigate the biocompatibility of the vLoaD discs. This DNA was expected to have cells removed due to the toxic nature of resin, hence why it was called the “Murder” DNA sample.

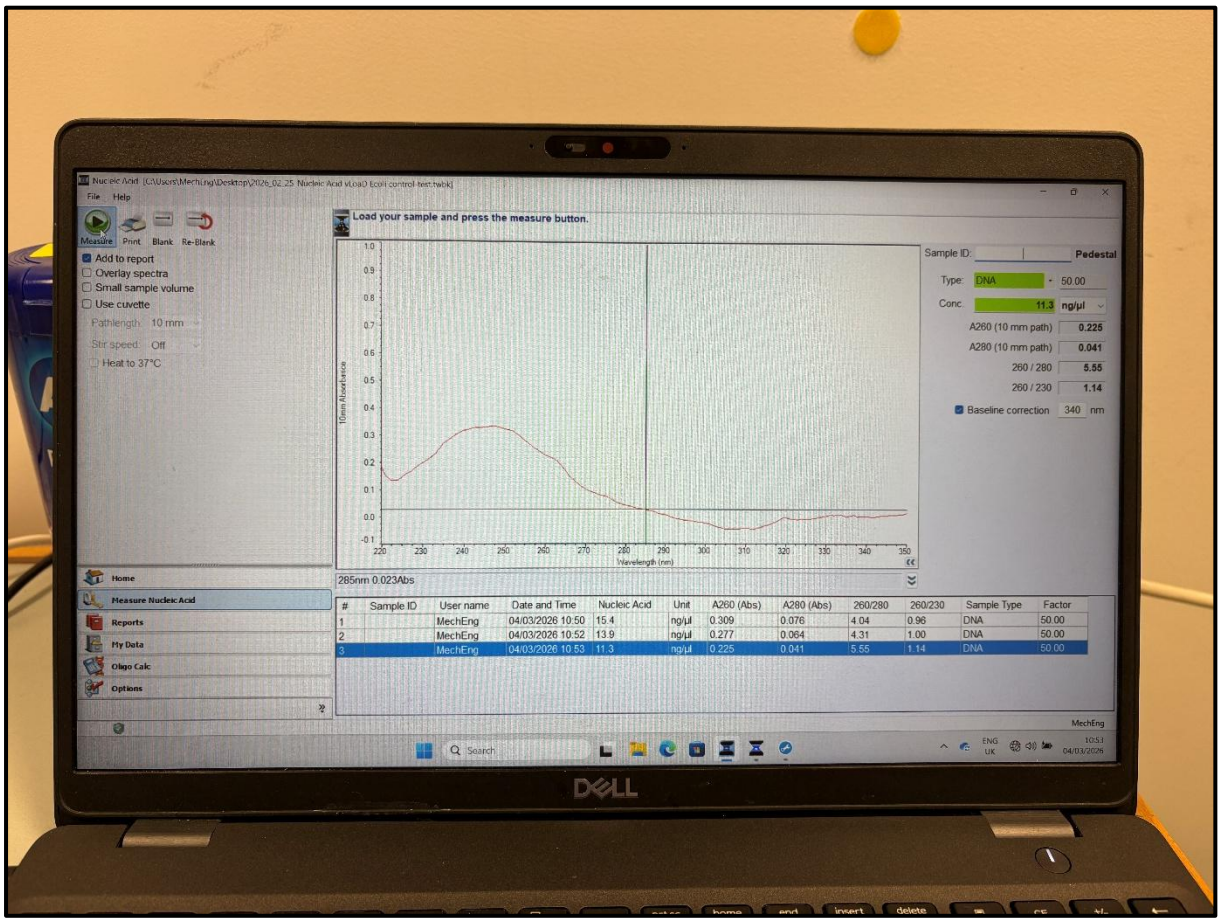


Figure 10.2.3: Image of NanoDrop Graph Murder DNA

Table 10.2.3: NanoDrop Measurement Results for Murder DNA

Measurement	ng/µL	A260	A280	260/280	260/230
1	15.4	0.309	0.076	4.04	0.96
2	13.9	0.277	0.064	4.31	1.00
3	11.3	0.225	0.041	5.55	1.14

Contrasting the “Murder” DNA sample to the on-bench control, the sample has an increase in nucleic acid count, indicating that the sample has been contaminated due to its time on disc. Additionally the ratios have increased values, possibly leading to the reduction of DNA present.

10.2.5 Q-PCR

This q-PCR test was prepared using the protocol for the AriaMX machine.

As part of this q-PCR testing, in addition to the on-bench control and vLoaD samples, the previously mentioned “Murder” DNA sample was also tested to examine biocompatibility.

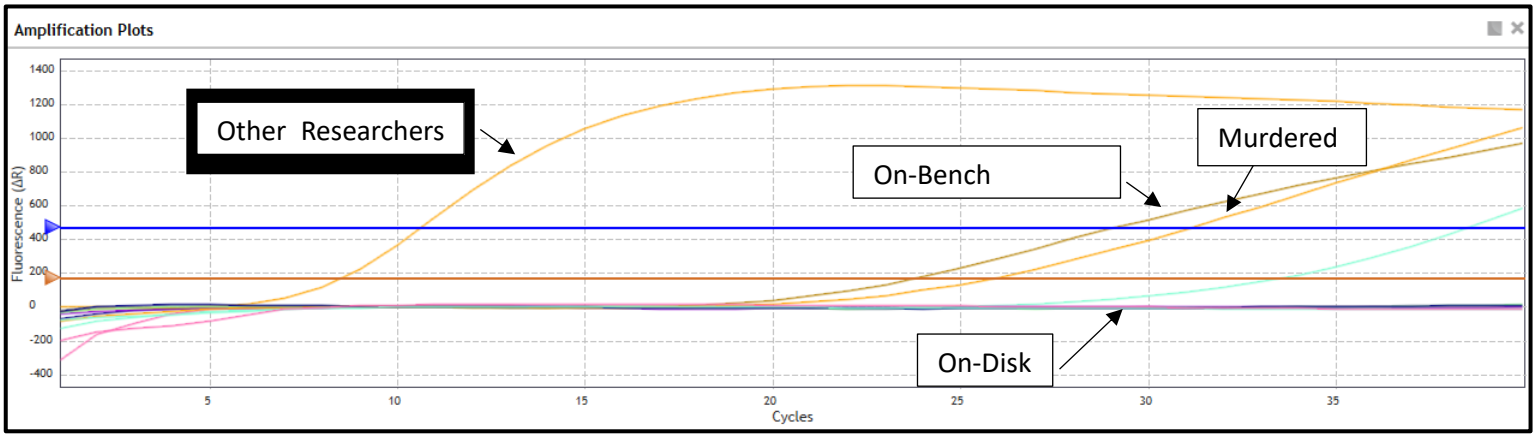


Figure 10.2.4: Image of q-PCR Results,

These q-PCR results validate the control DNA sample, show that the on-disc sample has no DNA present and that while biocompatibility will inhibit sample quality it will not lead to zero DNA being present. These results indicate that the vLoaD is still unsuccessful and that biocompatibility is not the source of error.

10.3 Test Seven

It was recognized during close examination of Test Six that the liquid was not mixing properly, more beads were added and placed them on top of the burst valve to use centrifugal forces to pump the liquid through the beads, changing the mixing action from being purely diffusive to using centrifugal forces.

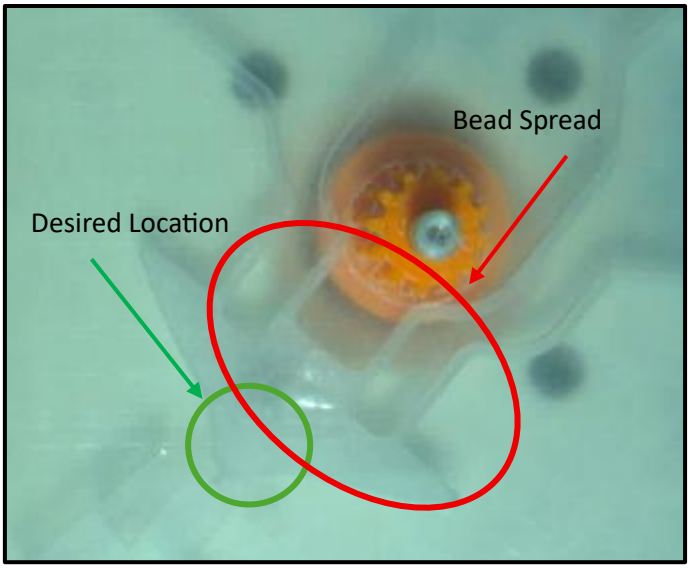


Figure 10.3.1: Image of Mixing Chamber, During Test Seven

The major error was silica beads were not kept in the desired location. Something new was required to overcome this.

10.4 Test Eight

This was the first test using testing design 2-1. This test was unsuccessful as liquid became trapped within the S-Chamber. The reasoning for this was thought to be due to hydrostatic pressure not being enough.

10.5 Test Nine

This test used the second revision of the S-Chamber, testing design 2-2. Still the same issue occurred where liquid became trapped within the S-Chamber, at this point it was determined that the motor on the vLoaD rig could not rotate fast enough for this siphon design. To verify this the same siphon design was ran on a traditional LoaD device at 1800 RPM. The design performed better however there was still a considerable amount of lost (trapped) fluid.

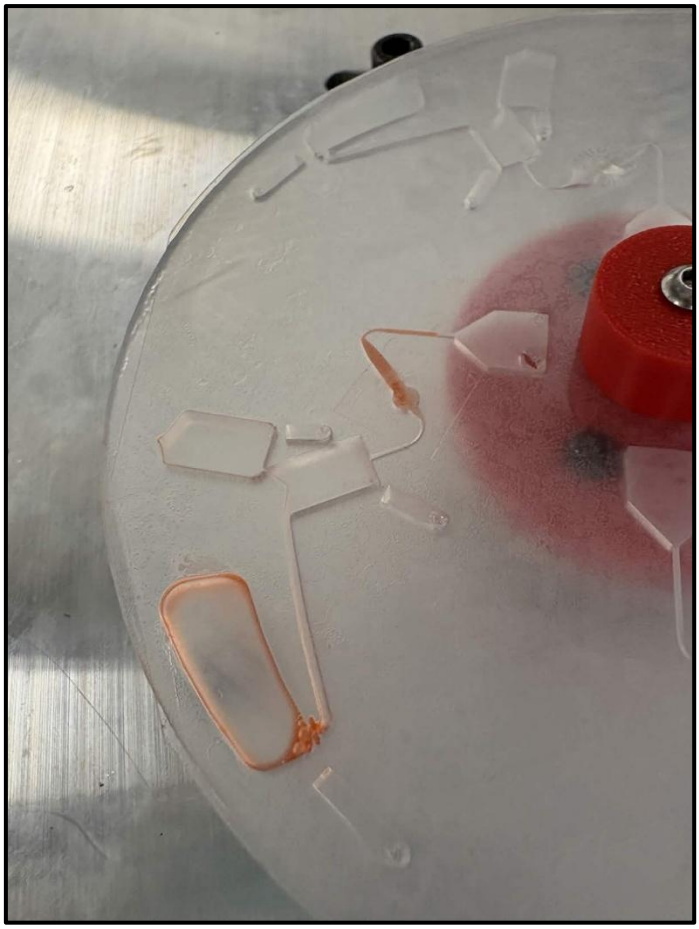


Figure 10.5.1: Image of Siphon Valve, after test on traditional LoaD platform

10.6 Test Ten

Testing Design 3, Liquid managed to travel through the newly designed bead chamber, despite the design calculations indication that the burst frequency was outside of the motor range. However, the Coriolis switch did not work as intended despite having the same positioning as testing design 1. This inhibited any positive results to come from this test as the product chamber was contaminated with Wash Buffer 1.

10.7 Test Eleven

10.7.1 vLoaD Test

This was the first magnetic bead based assay run using testing design 1. The assay seems to show promising results based on visuals; it could be seen that liquids were able to make contact with beads. The agitate mode was used on the vLoaD rig during diffusion times to aid the mixing process. However, this was only done for wash buffer 1 onwards.

10.7.2 DNA Test

10.7.2.1 NanoDrop

The NanoDrop results indicate a likelihood that contaminates have reached the final product. No BSA was used on this disc, as previously it was determined that it should not be required to get some form of DNA.

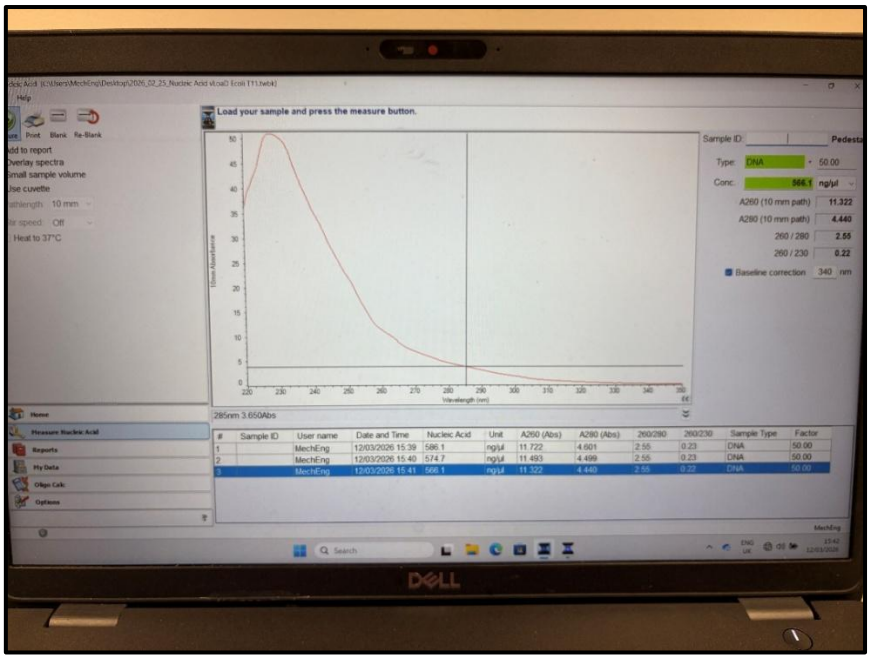


Figure 10.7.1: Image of NanoDrop Graph Test Eleven

Table 10.7.1: NanoDrop Measurement Results for Test Eleven

Measurement	ng/µL	A260	A280	260/280	260/230
1	586.1	11.722	4.601	2.55	0.23
2	574.7	11.493	4.499	2.55	0.23
3	566.1	11.322	4.440	2.55	0.22

10.7.2.2 q-PCR

The q-PCR was performed on the AriaMX, it showed no signs of amplification. It was determined that the vLoaD's agitation mode should be used for all diffusion processes during the assay, to improve the fluids contact with the beads. It was theorized that either the DNA did not bond with the beads or the contaminates inhibited the q-PCR process.

10.8 Test Twelve

10.8.1vLoaD Test

This test was conducted in the same format as the previous, however this time the disk was vibrated consistently for all of the fluid mixing processes.

10.8.2DNA Test

10.8.2.1 NanoDrop

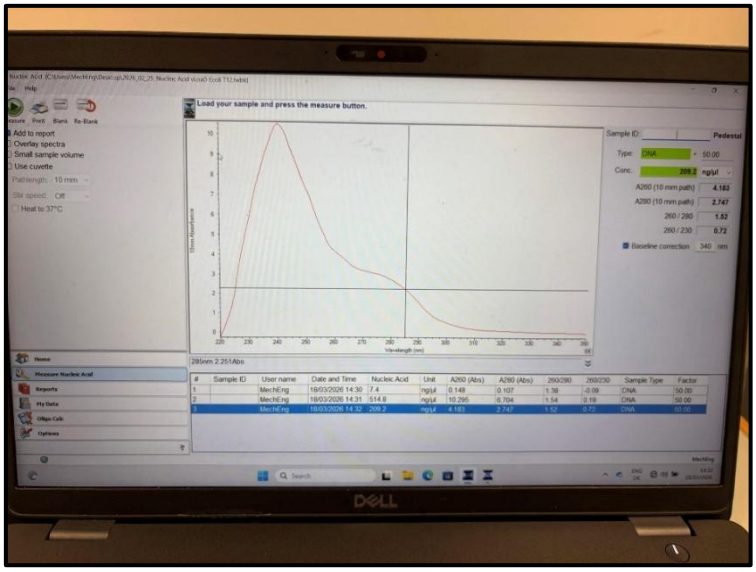


Figure 10.8.1: Image of NanoDrop Graph Test Twelve

Table 10.8.1: NanoDrop Measurement Results for Test Twelve

Measurement	ng/µL	A260	A280	260/280	260/230
1	7.4	0.148	0.107	1.38	-0.09
2	514.8	10.295	6.704	1.54	0.19
3	209.2	4.183	2.747	1.52	0.72

10.8.2.2 q-PCR

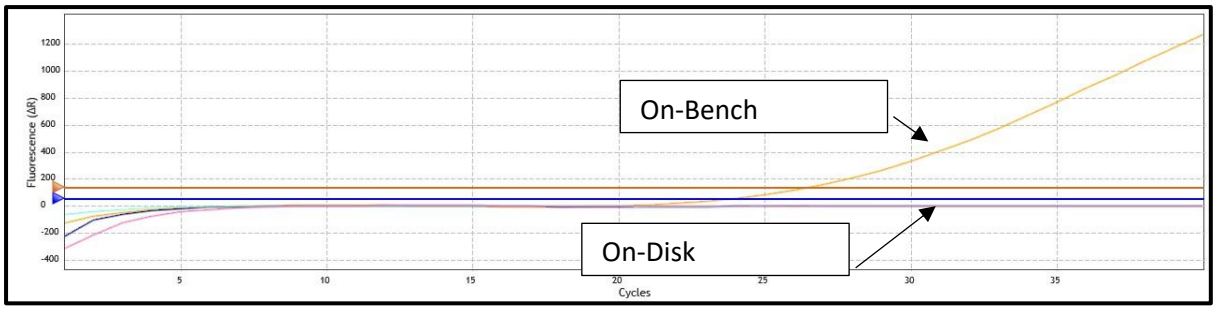


Figure 10.8.2: q-PCR results for Test Twelve

The q-PCR shows zero amplification for this test sample. At this point it was theorized that there could be an issue with the DNA sample itself

10.9 Final Test

10.9.1vLoaD Test

To try determine if the issue laid with the DNA sample itself, the sample was swapped out for the on-bench control, as this is a mixture of DNA + Elution that was proven to work. This sample was diluted to 50% with nuclease-free water.

10.9.2 DNA Test

10.9.2.1 NanoDrop

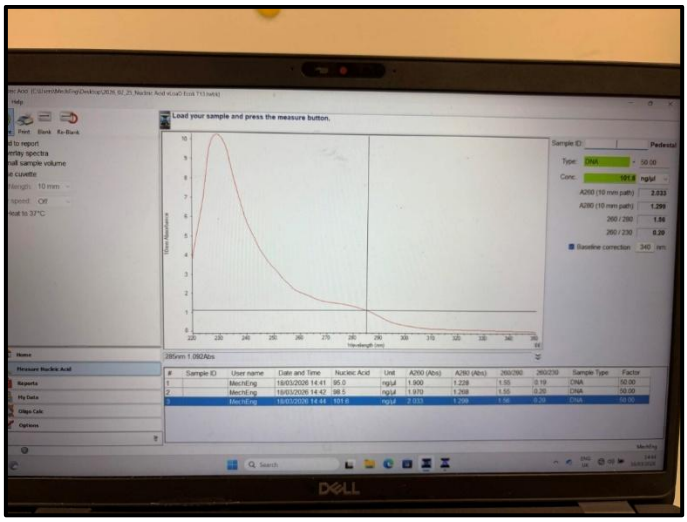


Figure 10.9.1: Image of NanoDrop Graph Final Test

Table 10.9.1: NanoDrop Measurement Results for Final Test

Measurement	ng/µL	A260	A280	260/280	260/230
1	95.0	1.900	1.228	1.55	0.19
2	98.5	1.970	1.268	1.55	0.20
3	101.6	2.033	1.299	1.56	0.20

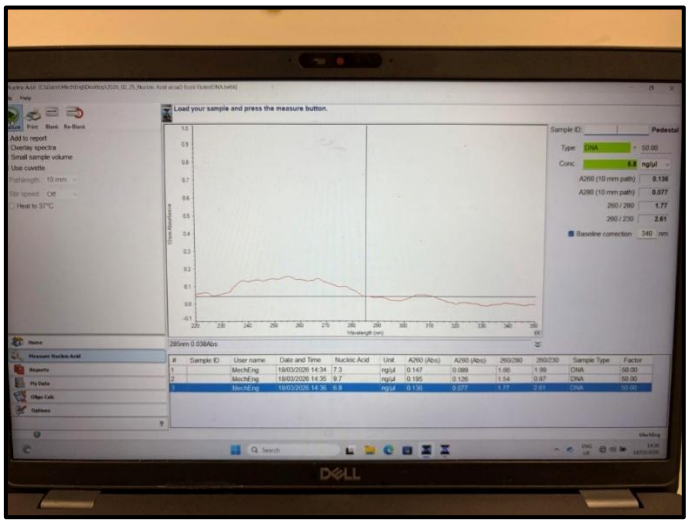


Figure 10.9.2: Image of NanoDrop Graph diluted DNA Mixture

Table 10.9.2: NanoDrop Measurement Results for diluted DNA Mixture

Measurement	ng/µL	A260	A280	260/280	260/230
1	7.3	0.147	0.089	1.66	1.99
2	9.7	0.195	0.126	1.54	0.87
3	6.8	0.136	0.077	1.77	2.61

10.9.2.2 q-PCR

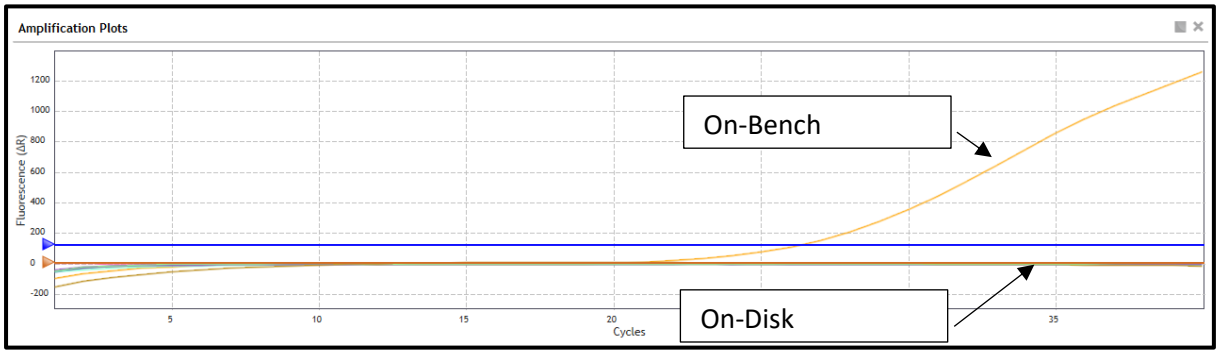


Figure 10.9.3: q-PCR results for Final Test

The PCR shows zero-amplification for the vLoaD sample. The fact that the amplification is at zero exactly, points to contamination being an issue.

11 Discussion

To quantify the success of the vLoaD there were three main areas examined, the accuracy of dimensions for the 3D printed disc, the biocompatibility of the produced discs, these both relate to research question 3 and the successful completion of DNA isolation assay with comparison to a control DNA sample, this ties into research question 4.

11.1 3D Printer / Fabrication Analysis

For the vLoaD Testing Design 1, a session was booked at the NRF to use the KEYENCE digital microscope. This device could capture high resolution images of the microfluidic disc's features, measure aspects of the design, and create topography style depth maps. These images are available in **Appendix 13**. Table 11.1.1 displays a list of measured features as well as their modelled SolidWorks counterparts, with accuracy percentages assigned.

Table 11.1.1: Feature Measurements with SolidWorks comparison

Feature	Measurements (µm)	SolidWorks (µm)	Percentage Error
Burst Valve Width	146.83	100	46.83
Fishbone Entrance	993.17	1000	0.683
Fishbone Midsection	2165.69	2500	13.3724
Fishbone Exit	1154.81	1000	15.481
Coriolis Length	8881.99	9000	1.311222

Overall the parts for the 3D printer were not as accurate as desired. The variance noticed in burst frequency from theoretical calculations is likely correlated to the inaccuracy of the valve dimensions from the 3D printer. Notably some features did not appear on the printed disc, the waste chamber vent and product chamber vents were both "filled in", see figure 15.13.6 and 15.13.7. Analysing the topography map of the product chamber vent shows some depth around the vent but not the expected 2mm of the SolidWorks design.

Notably, there was a persistent print defect, an inclusion at the base of the waste chamber, see figure 15.13.8. This defect persisted through multiple prints, attempts were made to remedy the flaw by moving the waste chamber further away from the outer edge. However, this still could not fix the error, showing the inconsistency associated with SLA printed parts and the pitfalls associated with this fabrication method. This inclusion had to be filled using a cyanoacrylate adhesive.

During testing it was noticed that when low centrifugal forces were applied, well below the burst pressure, the burst valve would still allow some amount of fluid to past. This was speculated to be due to some porosity of the disc allowing fluids to pass when pressed against the surface with centrifugal

forces. The operational protocol developed allowed for this occurrence to be avoided but it remained a design flaw. If it was desired to use centrifugal forces to perform mixing within the “mixing chamber” this design error would become fatal to the disc’s capabilities.

11.2 Biocompatibility of 3D printed disc

The biocompatibility of standard resin was a query that arose when considering a 3D printed microfluidic disc. This was tested during Test Six (**section 10.2**).

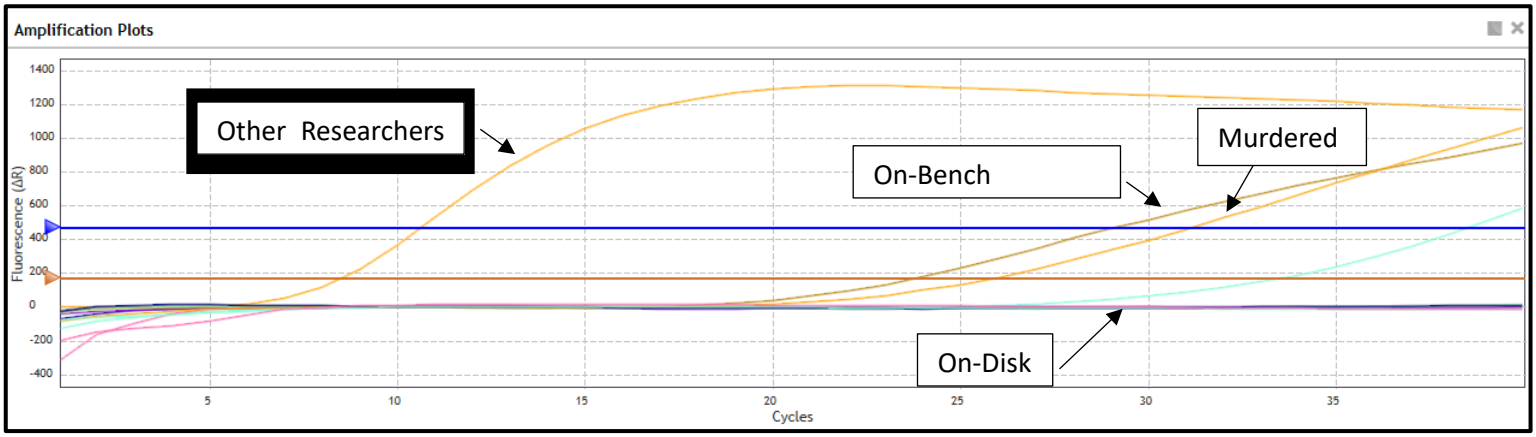


Figure 11.2.1: Image of q-PCR results from Test Six

This q-PCR plot shows that there is a minimal effect on the DNA sample when it is placed on the disk for 30 minutes. There was also the possibility that some of the DNA could have been lost to pipetting operations. Thus, it was determined that the biocompatibility of the disc was sufficient.

For the final tests where there was still a lack of any positive results, BSA was applied to the disc to ensure that biocompatibility was not an issue, but it did not appear to be the inhibiting factor.

11.3 DNA Isolation Analysis

Regrettably, it was not possible to successfully perform a DNA Isolation assay on the vLoaD disc as part of this thesis. At the conclusion of testing there were two potential areas of failure identified:

- Failure to successfully bond DNA to the bead due to lack of mixing action
- Possible contamination in the product that is inhibiting PCR

Towards the end of the masters project, the student had a consultation with Professor James P. Landers, the director of the RAPID institute where the project commenced. Potential flaws were identified within the design and James pointed towards a paper he had been a contributor to. [129]

Analysing this paper, made it clear that the challenges associated with performing a DNA Isolation assay had been previously overlooked. It made it apparent that if the vLoaD was to perform a DNA Isolation assay effectively, changes would need to be made to the design to improve the degree of mixing occurring. Diffusive mixing with slight agitation seems to not be enough to incur bonding of DNA to silica beads. An overhaul on the mixing chamber to allow the usage of rotational forces during mixing would be required to improve the DNA bonding process.

The DNA sample (**section 9.7.2**) is prepared using QIAGEN brand buffers, Buffer AL and Buffer ATL. Buffer AL contains Guanidine Hydrochloride, which is a strong PCR inhibitor. It denatures proteins such as the ones required for successful PCR operation. Meaning if the DNA product used for PCR has this chemical present, the sample is contaminated and PCR would not be possible. The continuous PCR results of zero-amplification lead the student and Nicklas Rondot to believe that this contamination was occurring.

While it is not possible to determine the exact source of contamination, it was speculated to be due to either the pipette being used for chemical handling or failure to wash away the Guanidine using wash buffers one and two. For future researchers, more experimentation with the DNA Isolation assay and tests to analyse sources of contamination would be required to remedy this issue.

12 Recommendations for Future Work

The vLoaD was successful at achieving its main goal of performing a DNA Isolation assay, however if I had the opportunity to work further on the device the following segments are ideas, I would wish to explore to further advance the microfluidic device.

12.1 Siphon Valves for Storage Chamber

If I had more time I would have liked to have tried to implement siphon valves into the storage chambers. If they worked as theorised it would be possible to fit more storage chambers onto the disc or reduce the size of the existing ones. I would have liked to place siphon valves at the outlet of the storage chambers such that the siphons can be primed one at a time with gravitational flow and then actuated with centrifugal flow.

This would allow for more reliable actuation of the storage chambers in theory, as well as the possibility of reducing the footprint of each storage chamber. Possibly allowing more than 4 storage chambers on the same size disc.

12.2 Remove Backflow from Mixing Chamber

The channels that connected the storage chambers to the mixing chamber were at a risk of backflow during operation. The outlets were modified during prototyping to prevent this during normal operations. However, it did make it that extra care had to be taken with the position of the disc during assay operations.

Usage of hydrophobic materials or redesign of capillary channels at the outlets could potentially create a one way flow system where liquids can enter the mixing chamber but not leave it. This would allow mixing in the mixing chamber via rotational forces, as opposed to purely diffusive operations. Although it would require precise tuning of the burst valve. This inclusion would also make the disc more reliable for operators.

12.3 Magnetic Bead Automation

For the final tests, the change to magnetic bead based DNA Isolation was made. Regrettably this reduced the autonomy of the system as the magnets had to be removed and added dependent on the step of the assay.

As a future development, an electromagnet or mechanism to move permanent magnets into place could be introduced to the rig, automating the process of magnetising the beads during the various assay steps.

12.4 Heat Activated Wax Plugs

To make the disc viable for consumer usage, wax plugs could be integrated into the outlets of the storage chambers, this way reagents can be stored on disc. Allowing for users to receive a disc and only need to add a DNA sample to run an assay.

The usage of wax plugs would require redesign of the disc and rig itself. A heat source would need to be integrated into the design such that plugs can be melted during operations. However, this would greatly advance the currently design vLoaD disc towards point of care testing applications.

12.5 Sensor Integration

The current rig is basic and simple in structure, only using a motor and microcontroller, all automation operations are done in a “blind” manner where the system assumes the disc is moving properly and operations are being conducted as intended. Integration of sensors to detect fluid levels within the storage could be utilized to ensure that the fluid is fully emptied from storage to mixing.

This would act as a failsafe where the system could adjust position appropriately to ensure correct actuation of storage chambers. In theory, this only requires a single sensor at the 12 o clock position as all storage chambers occupy this position when being actuated. Additionally, the same sort of sensor could be employed at the mixing chamber to ensure fluids are being pumped away successfully, choosing the spin the disk an additional time if required.

A capacitive sensor could be used for this, as these sensors can detect fluids without making direct contact, as they measure fluid using their surrounding electric field. With the current rig design sensor integration would be cumbersome, as the sensor may be in the pathway of disc insertion if placed in front of the disc. Experimentation with using one of these sensors behind the disc would be required as this appears to be a far more optimal location in terms of design philosophy.

12.6 Motor Adjustments

During Test Seven (**section 10.3**), it was determined that the RPM of the vLoaD rig motor was an inhibiting factor to the design. 1200 RPM as max speed is relatively low for a LoaD device, as such if another rig was designed, swapping out this stepper motor for a more powerful one would be a vital design consideration. Another option would be the introduction of a belt-drive or gearbox to increase the speed provided by the motor.

Additionally, the mounting mechanism for vLoaD disc is somewhat basic. The disc is held in place purely through the tolerance fit of the gear, while a Perspex shield was used during testing a locking mechanism should be introduced in future versions. The gear itself should also be made of a stronger

material and use a better mounting mechanism to couple to the motor shaft. Over time the gear would wear down, deteriorating the interference fit, this would lead to some “dead range” on the gear motor, reducing accuracy of angular position and erratic motion when vibrating the disc.

12.7 Downscaling / Theoretical Multi-Assay vLoaD

For the current vLoaD disc there was no issues with spacing in terms of fitting the DNA Isolation assay onto the disc, however a single assay taking up the entirety of the disc could lead to future issues. As one of the key aims of the vLoaD was to provide a more compact alternative to the traditional LoaD design.

With augmentation of the current design and introduction of some of the improvements proposed in this segment, in theory a more compact vLoaD disc where multiple assays can be “daisy-chained” together could be achieved on a single disc.

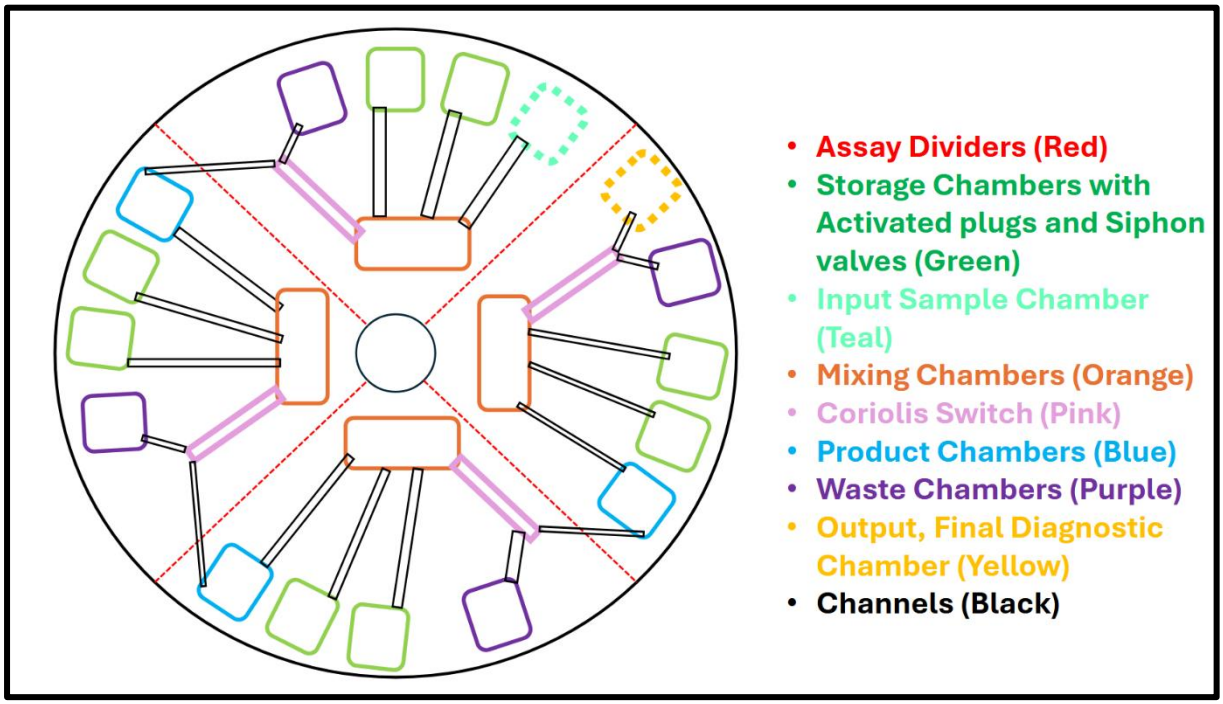


Figure 12.7.1: Illustration of Multi-Assay vLoaD disc

As seen in figure 12.7.1, usage of siphon valves and activated plugs could lead to a vLoaD design that can perform multiple assays at once. Note: The image seen in figure 12.7.1 is a loose depiction of what this design would look like, in reality the requirements for these chemical assays would lead to a far more complex design. Figure 12.7.2 is a comic strip showing a vague depiction of what the order of operations for such a disc could look like. If successful, this Multi-Assay design could be a huge breakthrough in point-of-care testing.

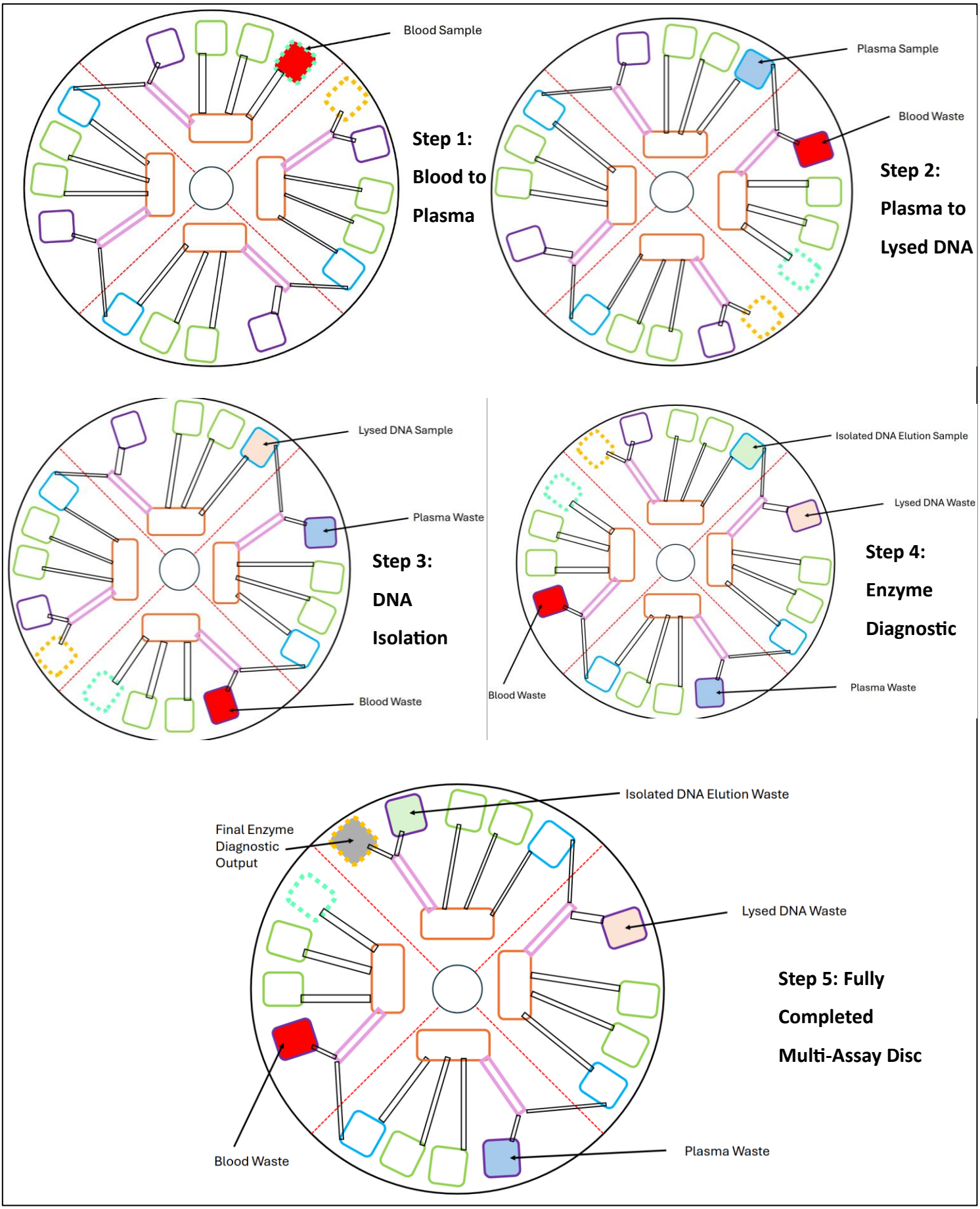


Figure 12.7.2: Comic Strip, theoretical operation of Multi-Assay vLoaD

13 Conclusions

The conclusions that can be drawn from the vLoaD project are as follows:

- Gravitational flow has been successfully utilized for a vertical configuration lab on a disc to achieve inward radial flow. The design challenge associated have been thoroughly reviewed.
- The vLoaD succeeded in utilizing gravitational and centrifugal flow in conjunction to perform tasks expected of a microfluidic device, such as actuation, mixing, valving and flow switching.
- Design calculations have been performed and analysed to experimental measured metrics to understand the variances from theory to real-world applications when using a vLoaD disc
- The vLoaD discs display what level of microfluidic disc fabrication is possible with a standard commercial 3D printer. This allows future researchers to analyse this paper to understand the opportunities and shortcomings of this fabrication method.
- The system is fully automated via software, allowing for the autonomous operation of assays. This improves the ease of usage for operators and minimizing training time.
- Protocols have been developed for both non-magnetic and magnetic silica bead based DNA Isolation on the vLoaD device
- The DNA Isolation assays were not successful, NanoDrop results hint at the possibility that DNA is present within the samples produced, however q-PCR testing shown no signs of amplification
- Testing and results were analysed, and two possible points of error were identified for future research
- DNA Isolation is a complex assay as it has three very different fluids present, guanidine salts and ethanol are both are more viscous than water making mixing difficult. Generally, it is deemed one of the more difficult assays to perform on a microfluidic disc
- While the vLoaD was unable to perform the assay, it was not far from a full implementation. With the identified changes and minimal troubleshooting, the device could work as intended.
- The work achieved did succeed in being a proof of concept for the vLoaD which was the main objective. The research completed opens the pathway for further developments that could lead to revolutionary changes within the microfluidic field.

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15 Appendices

15.1 Appendix 1: Work Breakdown Structure

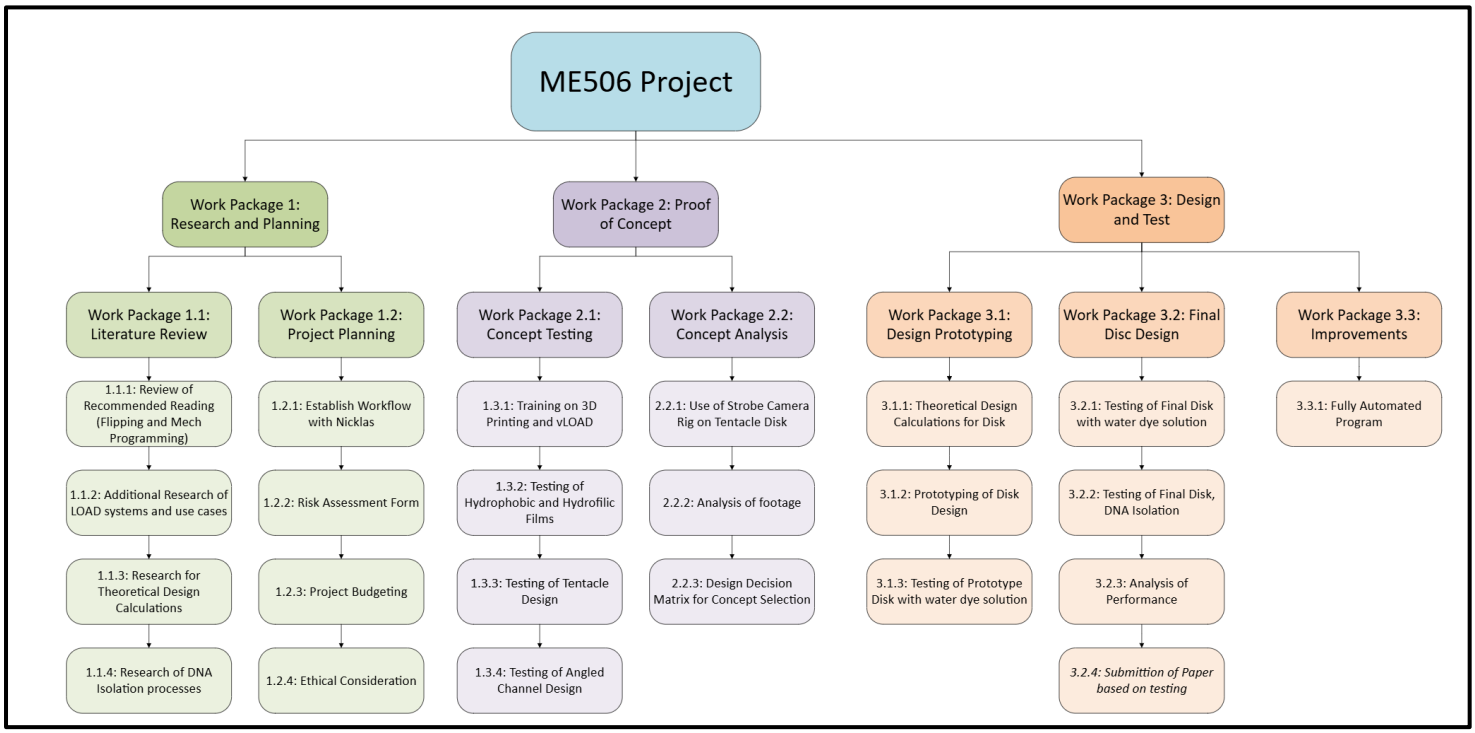


Figure 15.1.1: Work Breakdown Structure

15.2 Appendix 2: Gantt Chart (Semester 1 & 2)

15.2.1 Semester 1

				Sem 1											
Task Title	Status	Start	End	Week 1	Week 2	Week 3	Week 4	Week 5	Week 6	Week 7	Week 8	Week 9	Week 10	Week 11	Week 12
				08/09/25	15/09/25	22/09/25	29/09/25	06/10/25	13/10/25	20/10/25	27/10/25	03/11/25	10/11/25	17/11/25	24/11/25
Work Package 1.1: Literature Review															
Reading intitally recommended papers	Completed	15/09	22/09												
Reading on SLA Printing	Completed	29/09	01/10												
Independent research of LOAD system	Completed	08/10	3/11												
Additional research for Masters Status Report	Completed	27/10	11/11												
Additional for Theoretical Calculations	Completed	12/11	24/11												
Work Package 1.2: Project Planning															
Creation of Work Packages	Completed	22/09	28/09												
Creation of Gantt Chart	Completed	22/09	28/09												
Establishing Google Drive / Notion	Completed	22/09	28/09												
Risk assessment form	Completed	22/10	3/11												
Project Budgeting	Completed	22/10	24/11												
Masters Status Report	Completed	27/10	19/11												
Work Package 2.1: Concept Testing															
Training on film contact angle and disks	Completed	22/09	23/09												
Learning to use VLOAD	Completed	22/09	23/09												
Invesigating Contact Angles	Completed	03/11	03/11												
Learn to SLA Print	Completed	1/10	8/10												
Print Tentacle Disk v1	Completed	10/11	10/11												
Print Tentacle Disk v1_2	Completed	11/11	11/11												
Print Angled Channel disk	Completed	11/11	11/11												
Print Tentacle Disk v1_3	Completed	12/11	12/11												
Testing of angled disk	Completed	11/11	11/11												
Work Package 2.2: Concept Analysis															
Testing of Tentacle Disk w Strobe	Completed	12/11	12/11												
Analysis of Footage	Completed	13/11	16/11												
Design Concept Selection	Completed	13/11	16/11												
Work Package 3.1: Initial Design															
Definition of design parameters	Completed	13/11	14/11												
Design workbook calculations	Completed	14/11	24/11												

Figure 15.2.1: Gantt Chart for Semester 1

15.2.2 Semester 2

Task Title	Status	Start	End	Week 1	Week 2	Week 3	Week 4	Week 5	Week 6	Week 7	Week 8	Week 9	Week 10
				12/01/26	19/01/26	26/01/26	02/02/26	09/02/26	16/02/26	23/02/26	02/03/26	09/03/26	16/03/26
Work Package 3.1 Initial Design													
Disk Prototyping	Completed	12/01	16/02										
Work Package 3.2: Final Disc Design													
Testing with Water Dye Solution	Completed	23/02	23/02										
vLoaD Testing	Completed	24/02	18/03										
Analysis of Performance	Completed	18/03	19/03										
Work Package 3.3: Improvements													
Fully Automated Program	Completed	12/01	23/2										
PAPER DUE 20th of March	Completed	12/01	20/3										

Figure 15.2.2: Gantt Chart for Semester 2

15.3 Appendix 3: Hazard Assessment

DCU Health and Safety Office

Project Title:	An automated mini lab platform using gravitational and centrifugal flow	**Researcher Name:**	Tadhg Reid
Equipment/Process Name:	VLOAD Platform & DNA Isolation	**Preparation/Review Date:**	05/11/2025
Supervisor:	David Kinahan **Technical Staff:** Alan Meehan	**Location:**	S154

Who is affected by the Hazard: Staff, Research students, visitors

Hazards	Is the Hazard present? Y/N	What is the risk?	Likelihood	Severity	Risk Rating (L x S)	Risk Rating -prior to having controls in place H= high M=Medium L= Low	Controls - Controls in place to reduce risks	Is the control in place? Y/N	Likelihood	Severity	Risk Rating (L x S)	L/M/H	Control measure implementation - Person/Office/Unit/ Dept. responsible?	Approved by
Risk of exposure to chemicals during 3D printing Manufacturing process	Y	Exposure to Resin used for SLA Printing, Risk of Skin irritation, allergic recations and respiratory issues	3	2	6	M	Laboratory has appropriate PPE and tools for handling of 3D printed parts without exposure to resin, Student will only conduct experiments with appropriate PPE	Y	1	2	2	L	Tadhg Reid / Nicklas Rondot / Rapid Technical Staff	Dr. David Kinahan
							Standard Operating Procedures in place for 3D Resin printing operations	Y					Nicklas Rondot	Dr. David Kinahan
							Warning signs in laboratory to ensure PPE is worn and appropriate procedures are followed	Y					Nicklas Rondot	Dr. David Kinahan
							Student is aware where hand and eye wash stations are in S154						Tadhg Reid	Dr. David Kinahan
		Exposure to Isopropyl Alcohol, Risk of skin and eye irritation with risk of poisoning if swallowed	3	2	6	M	Laboratory has appropriate PPE and tools for handling of 3D printed parts without exposure to IPA	Y	1	2	2	L	Nicklas Rondot / Rapid Technical Staff	Dr. David Kinahan
							Student will only conduct experiments with appropraite PPE	Y					Tadhg Reid	Dr. David Kinahan
							Student is aware where hand and eye wash stations are in S154	Y					Tadhg Reid	Dr. David Kinahan
							Warning signs in laboratory to ensure PPE is worn and appropriate procedures are followed	Y					Nicklas Rondot	Dr. David Kinahan
Risk of burns due to using heat during manufacture of disks	Y	Burning hand when taking disk on and off of hotplate	2	3	6	M	Follow Safety Practices when using the hotplate, using metal plate to compress disc for film adhesion and using tweezers to remove disc from plate	Y	1	1	1	L	Tadhg Reid	Dr. David Kinahan
Risk of injury due to poor ergonomics	Y	Sitting at the PC too long, risk of eye strain and damage to posture	2	1	2	L	Follow Proper Egonomic practices, hands below elbow level, keeping head level with external monitor, taking frequent breaks to move and stretch, sitting in a chair with proper lumbar support	Y	1	1	1	L	Tadhg Reid	Dr. David Kinahan
Risk of Rotating platform at speed	Y	VLOAD Disc becomes loose during operations, Risk of Physical harm, such as abrasion, Risk of chemical exposure should liquids spill from holes in film	2	3	6	M	Chemicals used in DNA isolation process are pipetted into the chambers on the disc, at high speed there is a risk of these liquids leaking out of the disc, platform uses a perspex shield in front of the disc and appropraite PPE is worn during operation, should any chemicals come into contact with bare skin they can be removed by wiping with a cloth/tissue	Y	1	1	1	L	Tadhg Reid	Dr. David Kinahan
			1	3	3	L	VLOAD platform uses a properly designed gear spindle to mate the disk and driving motor, for each test the disc is mounted securely and checked for proper fit at low speed before using high speed operations (approximetly 1200 RPM), Rig is fitted with perspex shield infront of rotating disc	Y	1	1	1	L	Tadhg Reid	Dr. David Kinahan
Exposure to Biological Material Sample	Y	Exposure to biological material may cause risk of contracting virsus or diseases, with possibility of sever immune response	2	3	6	M	Laboratory has appropriate PPE with warning signs to ensure that is it used. Student is aware to not perform operations without appropraite PPE	Y	1	3	3	L	Tadhg Reid	Dr. David Kinahan
							Testing will occur using lab safe e-coli sample, student will be trained to safely carry out DNA isolation process							
							Laboratory has a wash station present for hands and eyes, should exposure occur							
Risk of exposure to chemicals during DNA Isolation Process	Y	Exposure to Wash solution (Ethanol), Risk of eye, skin and respiratory irritation, possibility of developing severe symptoms	2	2	4	M	Laboratory has appropriate PPE with warning signs to ensure that is it used. Student is aware to not perform operations without appropraite PPE	Y	1	2	2	L	Tadhg Reid	Dr. David Kinahan
		Exposure to Lysis Solution, Risk of chemical burn if skin contact occurs	2	3	6	M	Should chemicals come into contact with bare skin they can be removed by wiping with cloth/tissue	Y	1	2	2	L	Tadhg Reid	Dr. David Kinahan
		Exposure to Magnetic Bead Solution, Risk of skin and eye irritation, toxic material that can cause severe effects if consumed	2	2	4	M	Student is aware where hand and eye wash stations are in S154	Y	1	2	2	L	Tadhg Reid	Dr. David Kinahan

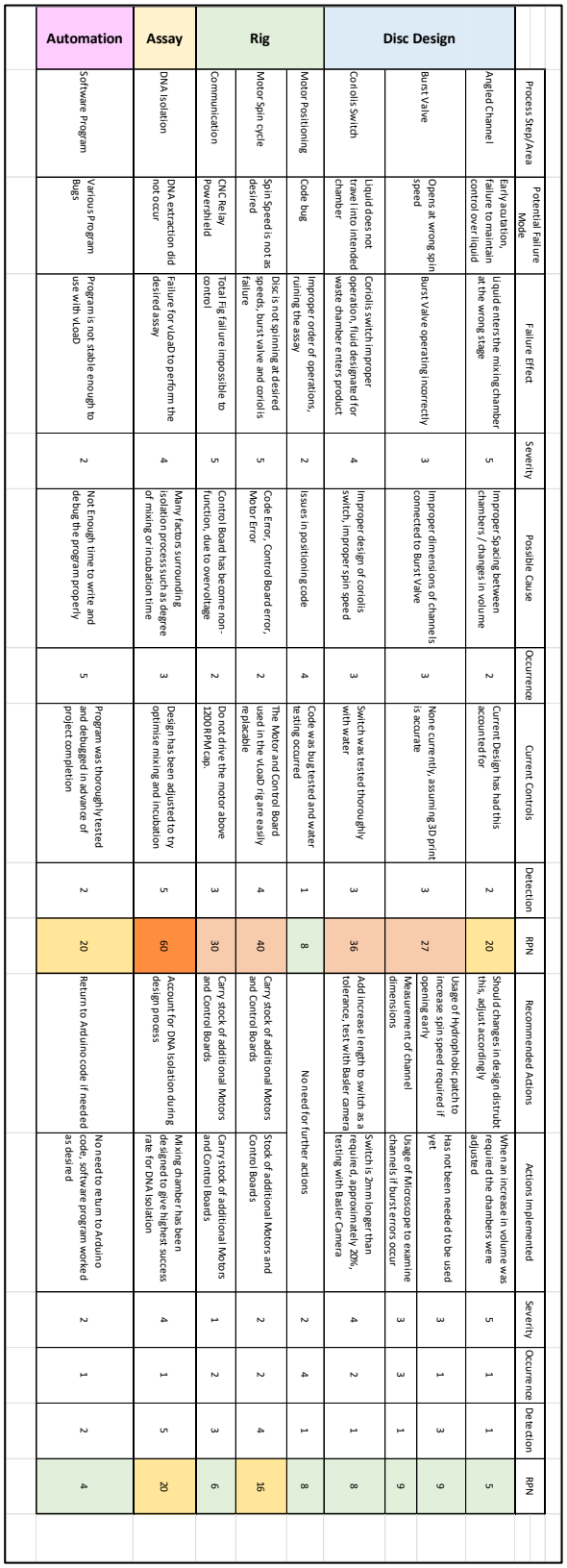
Researcher signature: Tadhg Reid **Technician signature(s):** _____ **Supervisor signature:** David Kinahan

Date: 06/11/2025 **Date:** _____ **Date:** _____ 06-Nov-25

A copy of this form when completed should be submitted through the FYP Loop page. A copy should also be available where project work is undertaken.

Figure 15.3.1: Risk Assessment for Project

15.4 Appendix 4: FMEA



	Process Step/Area	Potential Failure Mode	Failure Effect	Severity	Possible Cause	Occurrence	Current Controls	Detection	RPN	Recommended Actions	Actions Implemented	Severity	Occurrence	Detection	RPN
Disc Design	Angled Channel	Early actuation, failure to maintain control over liquid	Liquid enters the mixing chamber at the wrong stage	5	Improper Spacing between chambers / changes in volume	2	Current Design has had this accounted for	2	20	Should changes in design distrubt this, adjust accordingly	When an increase in volume was required the chambers were adjusted	5	1	1	5
	Burst Valve	Opens at wrong spin speed	Burst Valve operating incorrectly	3	Improper dimensions of channel is connected to Burst Valve	3	None currently, assuming 3D print is accurate	3	27	Usage of Hydrophobic patch to increase spin speed required if opening early Measurement of channel dimensions	Has not been needed to be used yet Usage of Microscope to examine channels if burst errors occur	3 3	1 3	3 1	9 9
	Coriolis Switch	Liquid does not travel into intended chamber	Coriolis switch improper operation, fluid designated for waste chamber enters product	4	Improper design of coriolis switch, improper spin speed	3	Switch was tested thoroughly with water	3	36	Add increase length to switch as a tolerance, test with Basler camera	Switch is 2mm longer than required, approximately 20%, testing with Basler Camera	4	2	1	8
Rig	Motor Positioning	Code bug	Improper order of operations, ruining the assay	2	Issues in positioning code	4	Code was bug tested and water testing occurred	1	8	No need for further actions		2	4	1	8
	Motor Spin cycle	Spin Speed is not as desired	Disc is not spinning at desired speeds, burst valve and coriolis failure	5	Code Error, Control Board error, Motor Error	2	The Motor and Control Board used in the vLoaD rig are easily replacable	4	40	Carry stock of additional Motors and Control Boards	Stock of additional Motors and Control Boards	2	2	4	16
	Communication	CNC Re-lay Powershield	Total Fig failure impossible to control	5	Control Board has become non-function, due to overvoltage	2	Do not drive the motor above 1200 RPM cap.	3	30	Carry stock of additional Motors and Control Boards	Carry stock of additional Motors and Control Boards	1	2	3	6
Assay	DNA Isolation	DNA extraction did not occur	Failure for vLoaD to perform the desired assay	4	Many factors surrounding isolation process such as degree of mixing or incubation time	3	Design has been adjusted to try optimise mixing and incubation	5	60	Account for DNA isolation during design process	Mixing chamber has been designed to give highest success rate for DNA isolation	4	1	5	20
Automation	Software Program	Various Program Bugs	Program is not stable enough to use with vLoaD	2	Not Enough time to write and debug the program properly	5	Program was thoroughly tested and debugged in advance of project completion	2	20	Return to Arduino code if needed	No need to return to Arduino code, software program worked as desired	2	1	2	4

Figure 15.4.1: FMEA for Project

15.5 Appendix 5: Results of Channel Film Testing

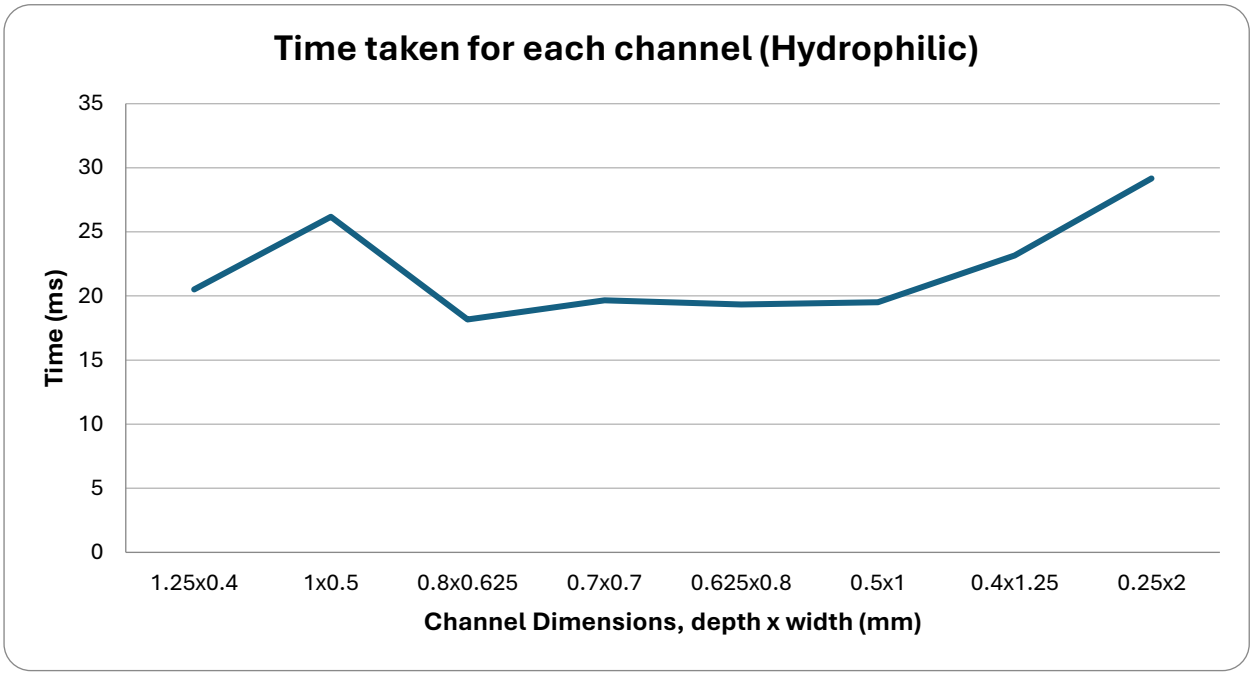


Figure 15.5.1: Line chart of times for each channel dimension, hydrophilic film

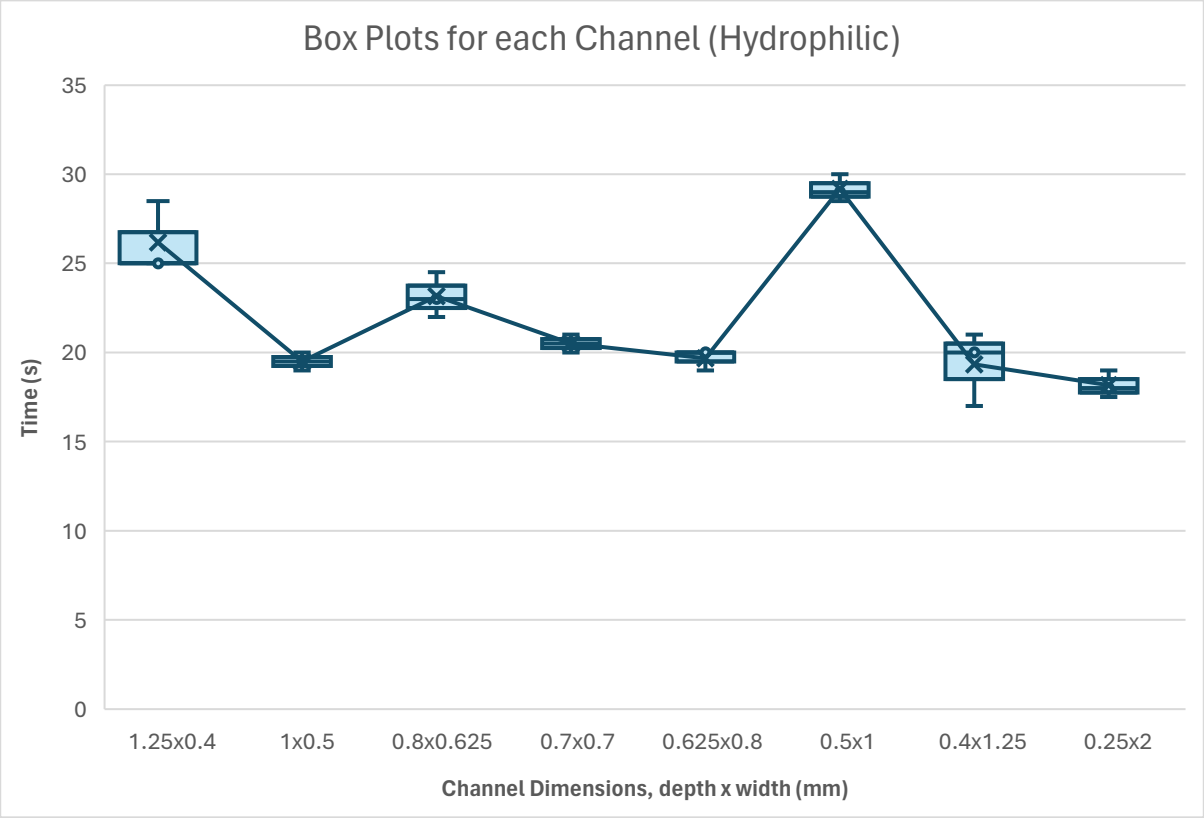


Figure 15.5.2: Boxplots of times for each channel dimension, hydrophilic film

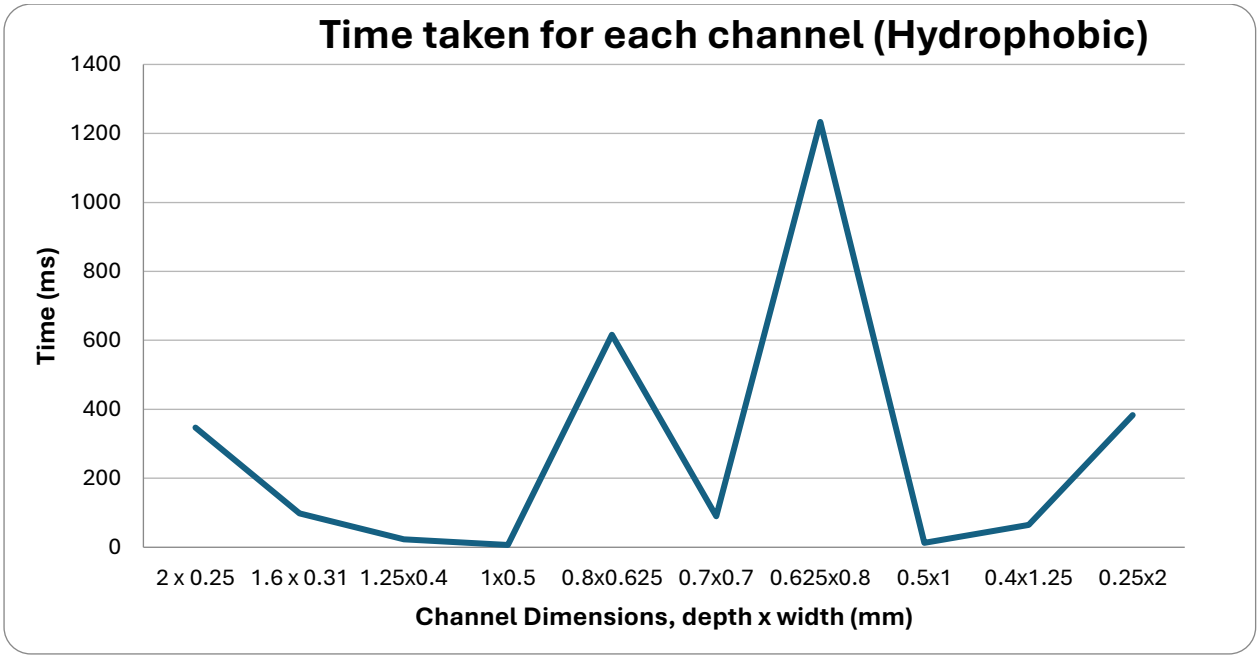


Figure 15.5.3: Line chart of times for each channel dimension, hydrophobic film

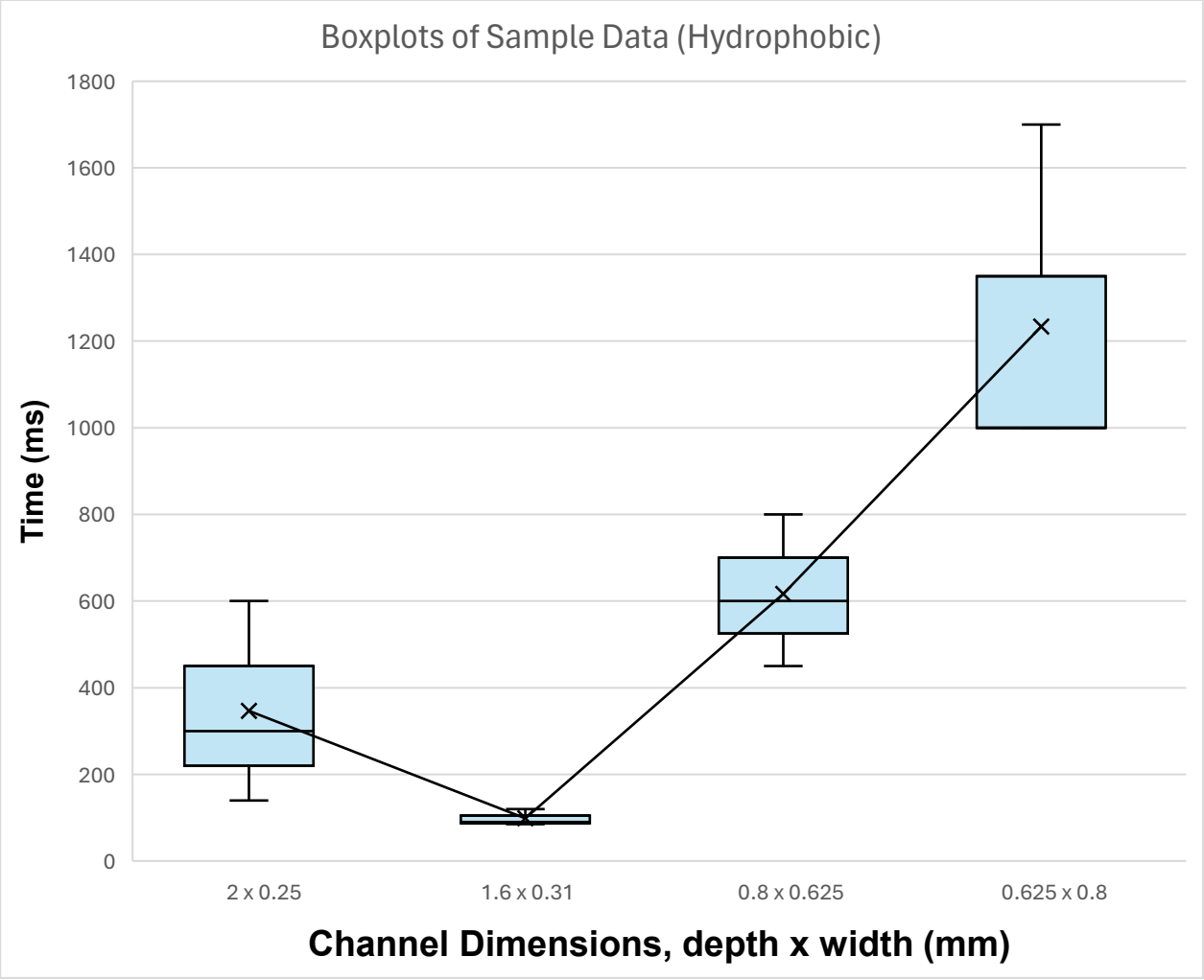


Figure 15.5.4: Boxplots of times for sample data of channel dimension, hydrophobic film

15.6 Appendix 6: Design Calculations, Testing Design 2-2

This segment covers the same theoretical calculations as used for Testing Design 2-1; the sole change is the numerical values applied.

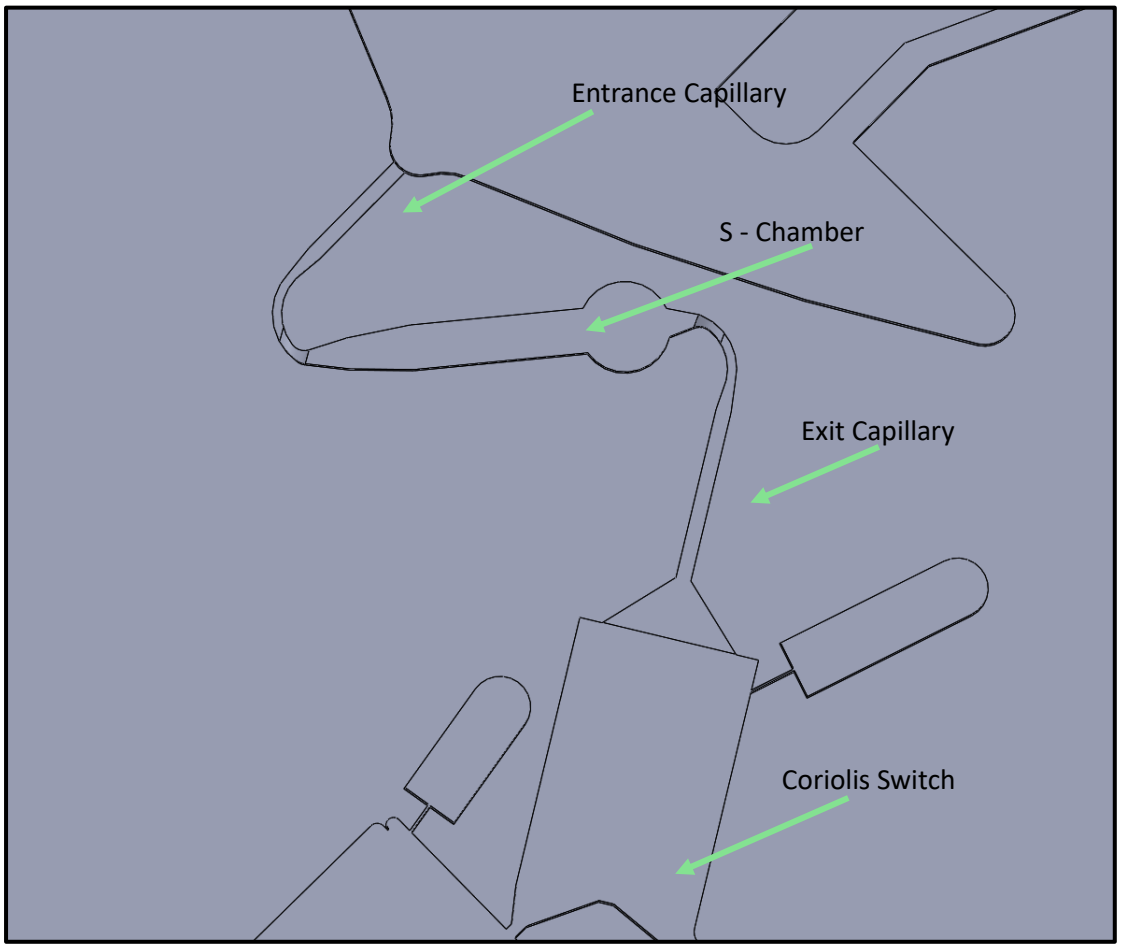


Figure 15.6.1: Image of vLoaD Revamp, highlighting design calculation relevant areas

15.6.1 Capillary Valve at entry to S-Chamber

Capillary Valve calculations were performed using the same method as seen in **section 8.2.3**. Adjusting radius values and capillary values to account for the new design changes. This valve is technically not a burst valve anymore due to its new geometry making it one long capillary, however burst pressure was still investigated for usage during operations.

Table 15.6.1.1: Calculation of max Gravitational Pressure

g	9.81	m/s
Δh	3	m
Fluid Density (ρ)	789	Kg/m^3
Gravitational Pressure	23.22	Pa

Table 15.6.1.2: Calculation of max Capillary Pressure

Contact Angle (Film)	14.2	Deg
Contact Angle (Resin)	22.905	Deg
Average Contact Angle	0.36178	rad
Surface Energy of Ethanol 96%	0.02239	J/m^2
Channel Width	0.5	Mm
Channel Depth	0.5	Mm
Capillary Pressure	167.525	Pa

Table 15.6.1.3: Calculation of Burst Frequency

Capillary Pressure	167.525	Pa
Gravitational Pressure	23.22	Pa
Δr	3	mm
$\bar{r}$	22.5	mm
rpm	497.07	

Table 15.6.1.4: Calculation of max Centrifugal Pressure

Angular Velocity	1200	rpm
r1	21	mm
r2	24	mm
r bar	22.5	mm
delta r	3	mm
Pressure	841	Pa

15.6.2 Capillary Valve at exit of S-Chamber

Capillary Valve calculations were performed using the same method as seen in **section 8.2.3**. Adjusting radius values and capillary values to account for the new design changes. This valve is technically not a burst valve anymore due to its new geometry making it one long capillary, however burst pressure was still investigated for usage during operations.

Table 15.6.2.1: Calculation of max Gravitational Pressure

g	9.81	m/s
Δh	9	mm
Fluid Density (ρ)	789	Kg/m^3
Gravitational Pressure	69.66081	Pa

Table 15.6.2.2: Calculation of max Capillary Pressure

Contact Angle (Film)	14.2	Deg
Contact Angle (Resin)	22.905	Deg
Average Contact Angle	0.36178	rad
Surface Energy of Ethanol 96%	0.02239	J/m^2
Channel Width	0.5	Mm
Channel Depth	0.5	Mm
Capillary Pressure	167.525	Pa

Table 15.6.2.3: Calculation of Burst Frequency

Capillary Pressure	167.525	Pa
Gravitational Pressure	69.66081	Pa
Δr	9	mm
$\bar{r}$	28.5	mm
rpm	209.99	

Table 15.6.2.4: Calculation of max Centrifugal Pressure

Angular Velocity (ω)	1200	rpm
r1	24	mm
r2	33	mm
r bar	28.5	mm
delta r	9	mm
Pressure	3195.833	Pa

The burst rpm, is well below the operational rpm of 1200, this exit valve should not inhibit the liquids entry toward the Coriolis switch.

15.6.3 Centrifugal Lift

Table 15.6.3.1: Calculation of Centrifugal Hydrostatic Pressure

Angular Velocity (ω)	1200	rpm
Fluid Density (ρ)	789	Kg/m^3
R Crest	22.75	mm
R Liquid	23.5	mm
Delta Pressure	-216.09	Pa

This pressure is a negative value which indicates that the liquid will not move over the bend itself, towards the exit of the S-Chamber. This means that other forces must be relied upon for the S-Chamber to operate as intended.

Table 15.6.3.2: Calculation of Chamber Capillary Pressure

Contact Angle (Film)	14.2	Deg
Surface Energy of Ethanol 96%	0.02239	J/m^2
Channel Width	2	Mm
Channel Depth	1	Mm
Capillary Pressure	50.64705	Pa

Table 15.6.3.3: Calculation of max Gravitational Pressure

g	9.81	m/s
Δh	7.66	mm
Fluid Density (ρ)	789	Kg/m^3
Gravitational Pressure	59.289	Pa

It was assumed that the liquid will take the path of least resistance and thus only the film contact angle was consulted. The gravitational pressure was calculated as it in theory would work against the fluid as it attempts to “climb” the slope. With the new design changes the capillary pressure is less than the gravitational pressure, the liquid will not wick itself up the slope. The vLoaD must be positioned such that gravitational flow can aid the siphon.

15.6.4 Bead Capillary Force

$$\kappa = \frac{d_p^2 \varepsilon^3}{150(1 - \varepsilon)^2} \quad (25)$$

Table 15.6.4.1: Calculation of Permeability

Porosity (ε)	0.40	
Mean Bead Diameter (d_p)	181	µm
Permeability (κ)	3.88E-11	m^2

Table 15.6.4.2: Calculation of Pressure Required to pump through beads

Permeability (κ)	3.88E-11	m^2
Channel Length (L)	32.19	mm
Target Velocity (v)	0.001	m/s
Fluid Viscosity (µ)	0.003	Pa.s
Delta Pressure	2489.13	Pa

15.6.5 Coriolis Switch

The Coriolis Switch design remained the same as previous versions, however radial positions of the disc affect the maximum flow velocity, so the calculations were run again.

Table 15.6.5.1: Calculations for Coriolis Switch

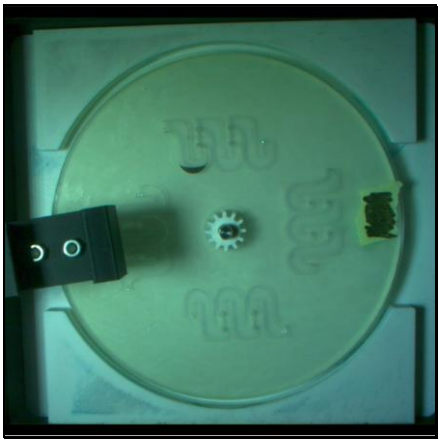
Angular Velocity (ω)	125.664	rad/s
r1	24	mm
r2	33	mm
r bar	28.5	mm
Density (ρ)	789	Kg/m^3
Viscosity (η) Ethanol 96%	1.2	mPa.s
Tangential Distance (x)	3	mm
Maximum Flow Velocity	1.3316	m/s
Time	4.234	ms
Length (L)	5.638	mm

The current length of the Coriolis switch is greater than this calculated value, meaning no design changes are required, the switch design is adequate.

15.7 Appendix 7: Comic Strip for Tentacle Strobe Test

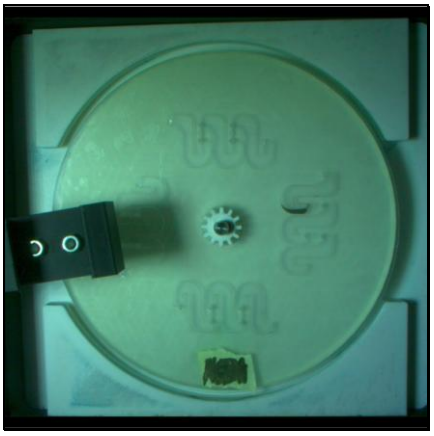
15.7.1Rotation performed at 180 degree position

Step 1



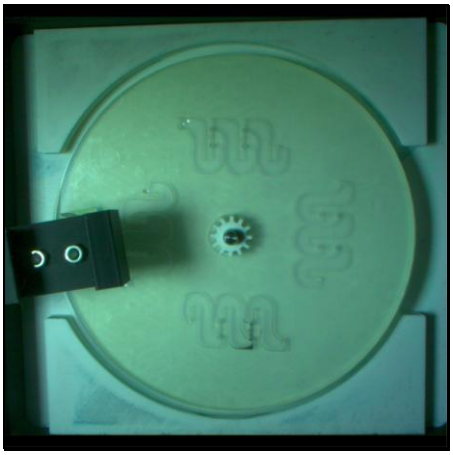
Liquid in First Chamber at 0 Degree Position

Step 2



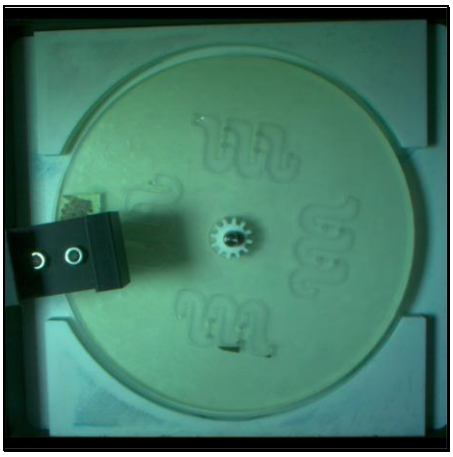
Rotate disc to plus 90 degree position

Step 3



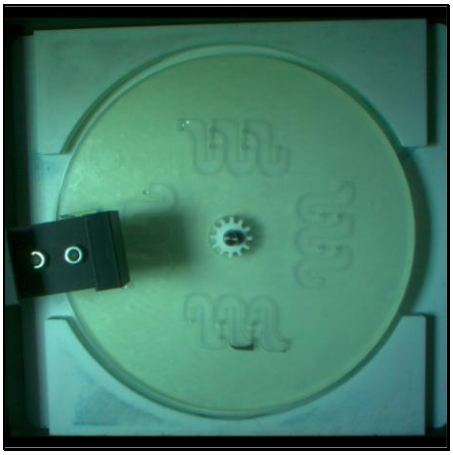
Rotate disc to 180 degree position, liquid is now in channel between two chambers

Step 4



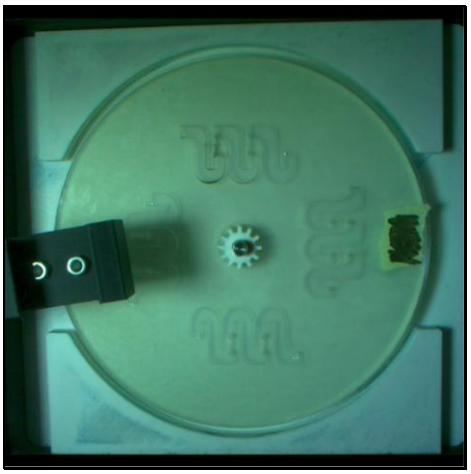
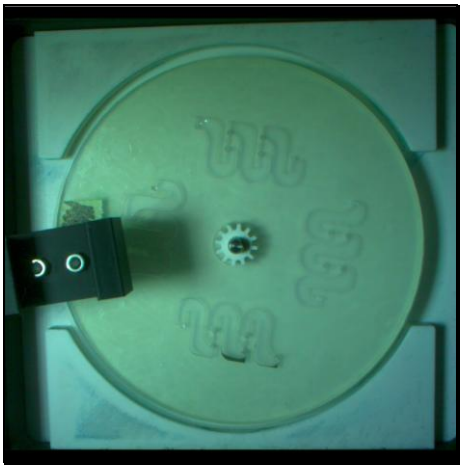
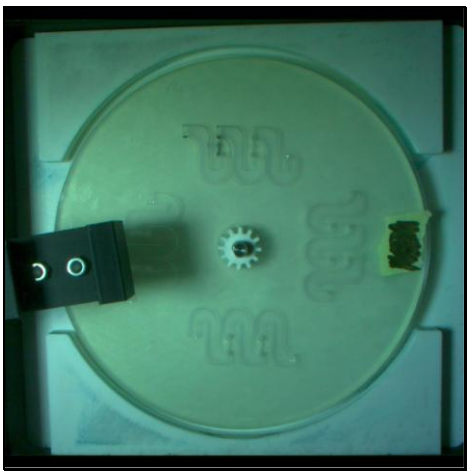
Disc during Rotation, Liquid is being transferred into second chamber

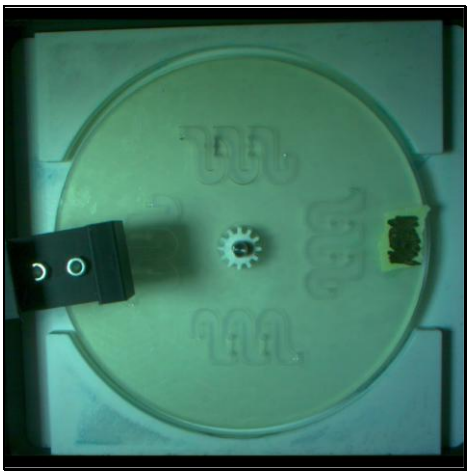
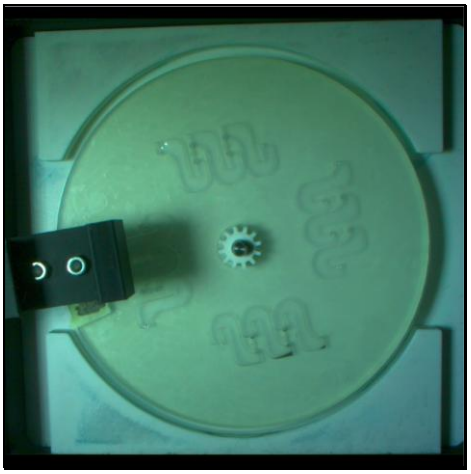
Step 5



Disc after rotation all liquid has been transferred from first chamber to second

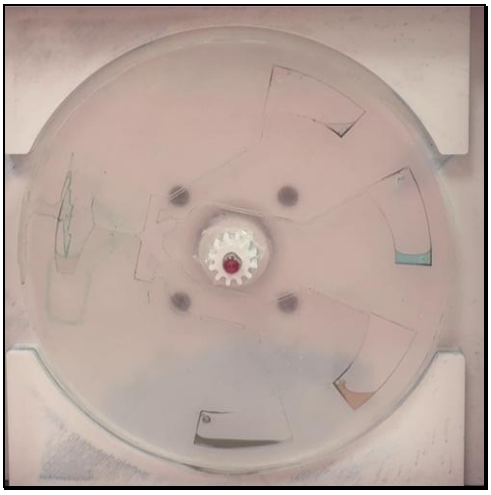
15.7.2 Rotation performed at 0 degree position

Clockwise Step 1		Liquid in channel between first and second chamber at 0 Degree Position
Clockwise Step 2		Disc being spun clockwise
Clockwise Step 3		End of clockwise spin liquid, disc returns to 0 degree position and liquid flows back into channel

Counter-Clockwise Step 1		Liquid in channel between first and second chamber at 0 Degree Position
Counter-Clockwise Step 2		Disc being spun counter-clockwise
Counter-Clockwise Step 3		End of counter-clockwise spin liquid, disc returns to 0 degree position and liquid flows around s-curve into second chamber

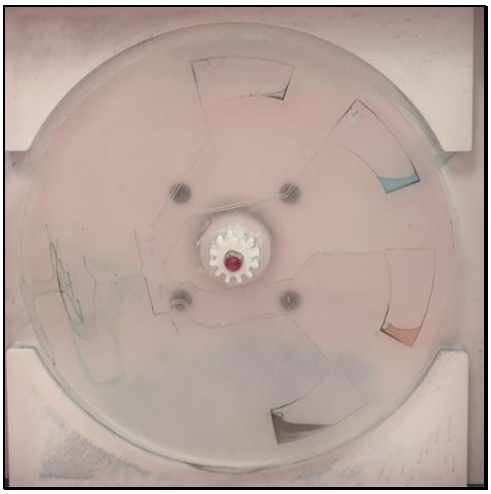
15.8 Appendix 8: Comic Strip for Angled Channel Multi-Colour Test

Step 1



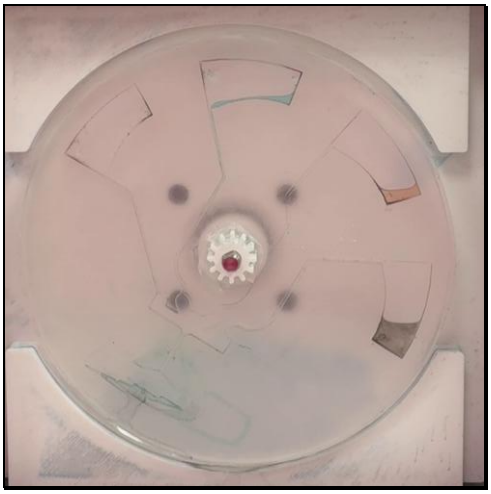
Start Position for Angled Channel Disc

Step 2



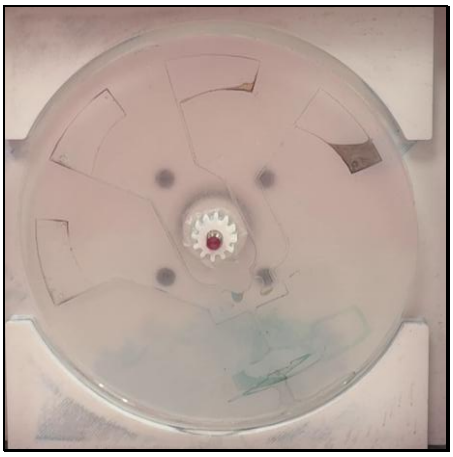
Release of liquid from chamber 1

Step 3



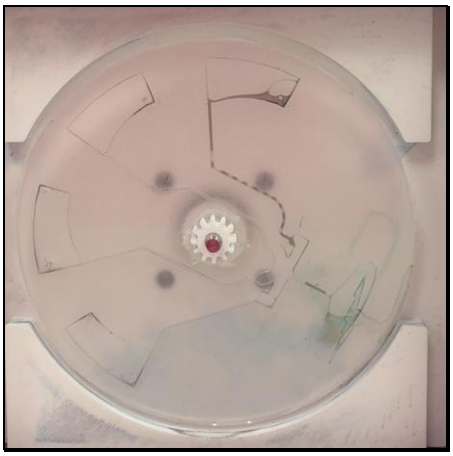
Release of liquid from chamber 2

Step 4



Release of liquid from chamber 3

Step 5



Release of liquid from chamber 4

15.9 Appendix 9: Prototype designs

15.9.1 Version 3

Differences from previous version:

- Modification of mixing chamber dimensions
- Introduction of “Stalagmite” style structures to try preventing liquid movement towards other outlets

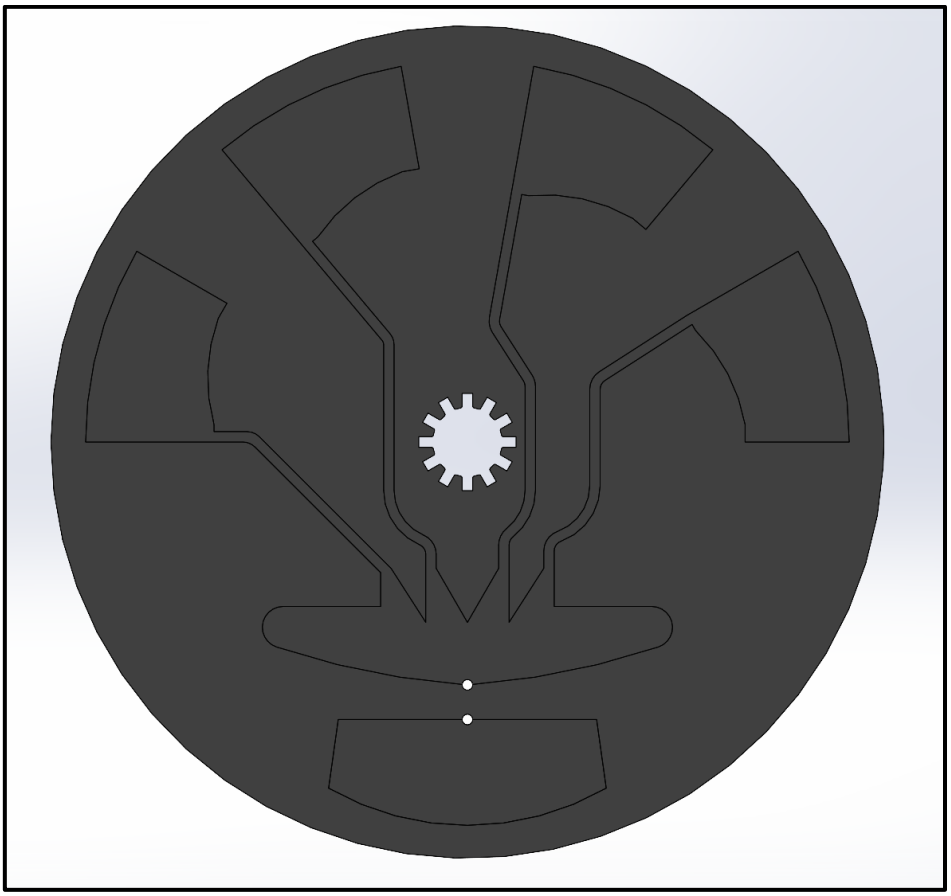


Figure 15.9.1: Image of Prototype Design V3

Testing Results:

Successes	Shortcomings
• None	• Stalagmite design did not work as intended; the issue was still not resolved

15.9.2 Version 4

Differences from previous version:

- New geometry to prevent liquid backflow, rounded corner approach

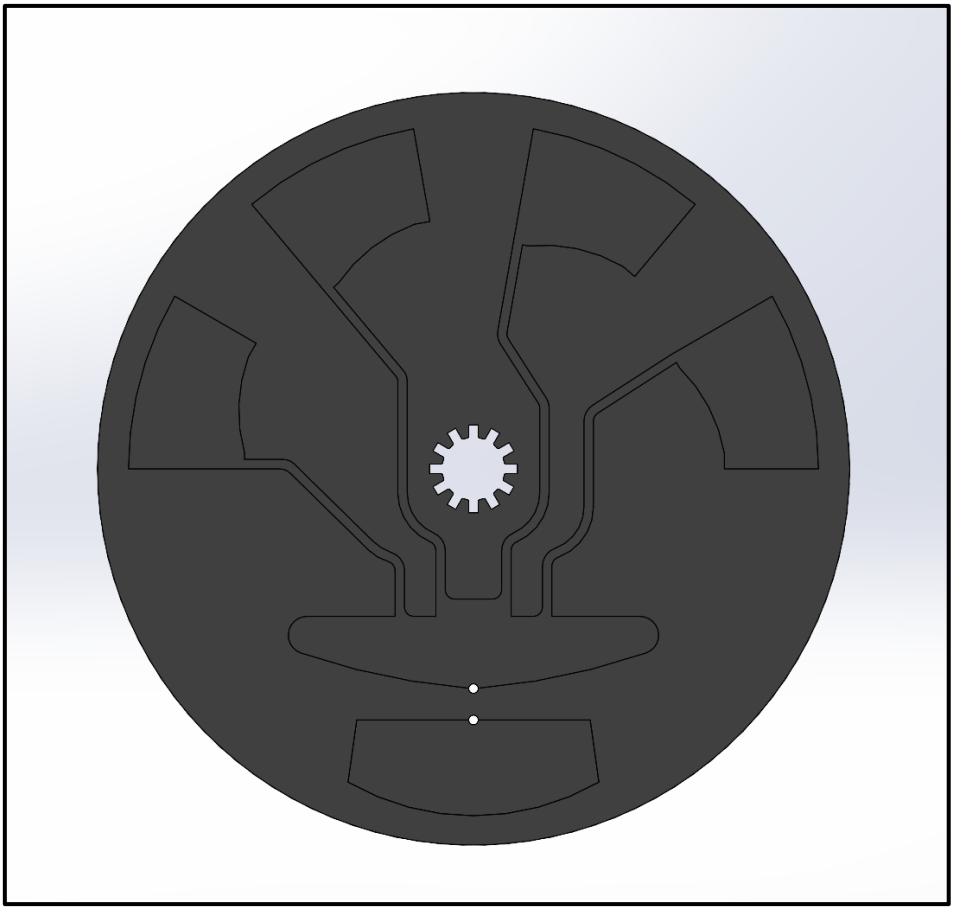


Figure 15.9.2: Image of Prototype Design V4

Testing Results:

Successes	Shortcomings
• Liquid would not roll over into other outlets, combating the previous problem	• Design changes to mixing chamber, "broke" the burst valve, the side wings were a further radial distance than the chamber exit, rendering centrifugal flow useless, as it will only pump to the furthest radial point

15.9.3 Version 7

Differences from previous version:

- Flipped waste and product chambers
- Introduction of vents throughout design

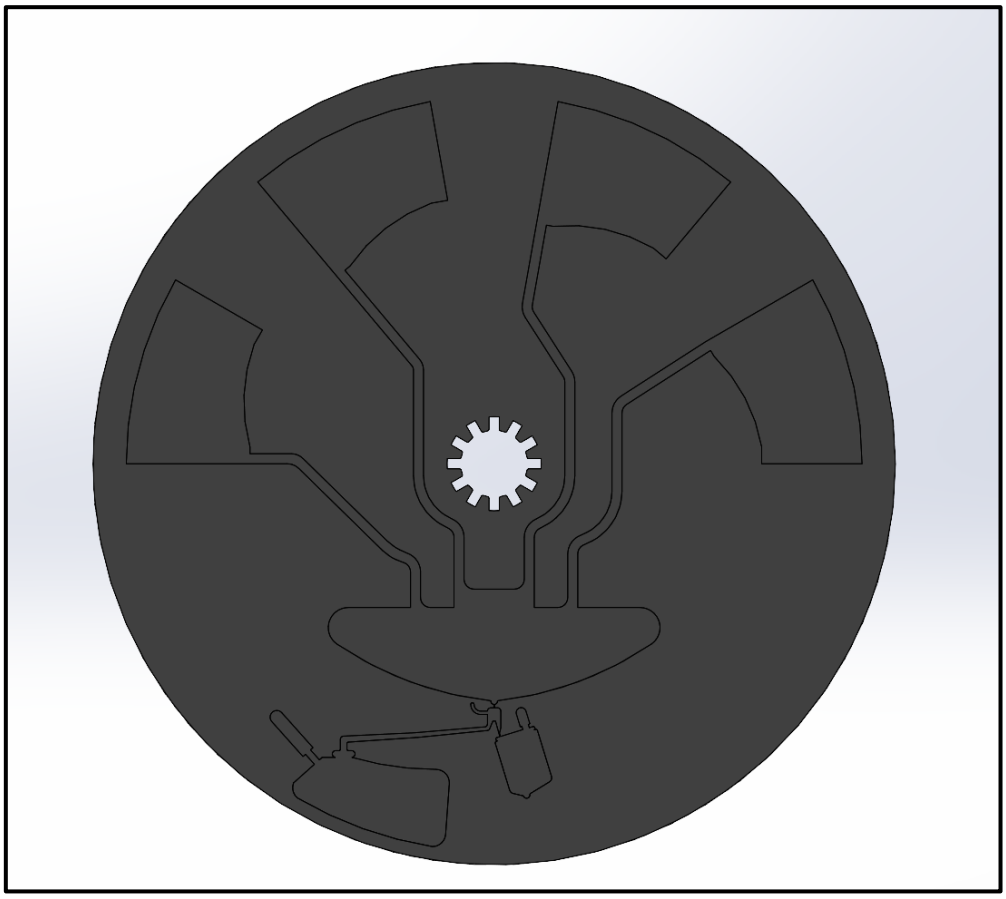


Figure 15.9.3: Image of Prototype Design V7

Testing Results:

Successes	Shortcomings
• Widened channel to the waste chamber was successful, liquid could now flow into the chamber	• Flipping the waste and product chambers positions was a mistake, the flow direction was correct initially

15.9.4 Version 8

Differences from previous version:

- Flipped waste and product chambers back to positions seen in version 6
- Moved mixing chamber closer to the centre of disc to leave more room
- Increased length of Coriolis switch, to improve flow trajectory
- Moved waste chamber further away from outer edge to minimize risk of print defects
- Adjustment of vent capillaries to prevent fluid flow

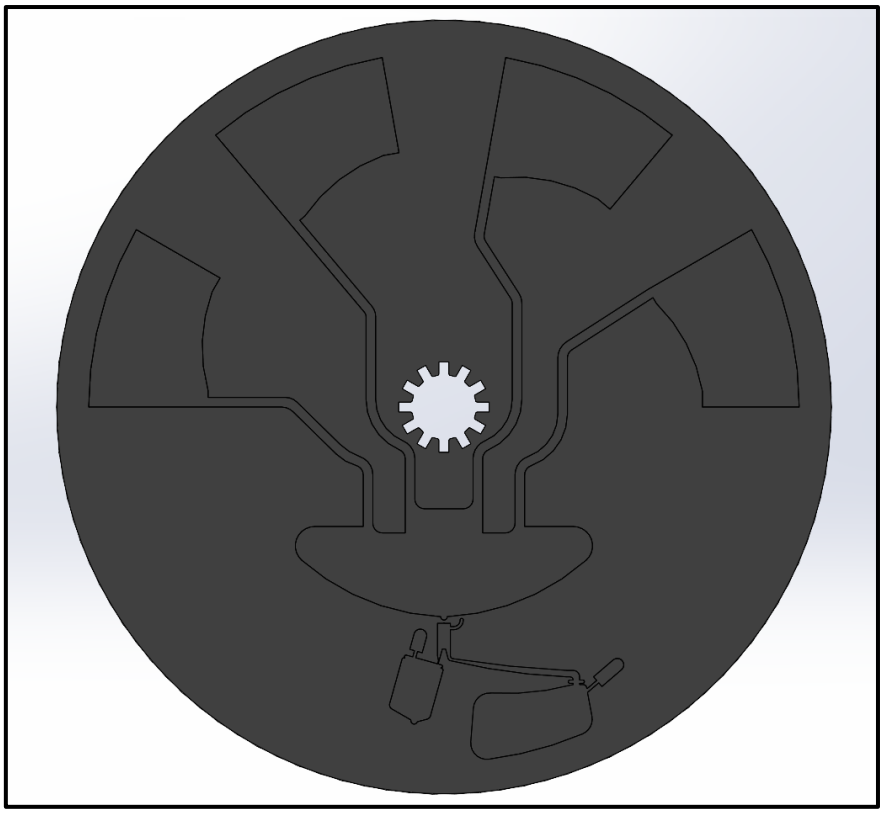


Figure 15.9.4: Image of Prototype Design V8

Testing Results:

Successes	Shortcomings
• Flow direction of Coriolis switch was corrected • Vents do not have the same leakage as seen in previous designs	• Flow into waste chamber was still difficult, channel needs to be widened • Coriolis switch still does not pump 100% of fluid into desired chamber

15.9.5 Version 9

Differences from previous version:

- Expanded size of channel leading to waste chamber
- Adjustment to size of wings in mixing chamber
- Augmentation of mixing chamber outlet to allow for beads to nestle there

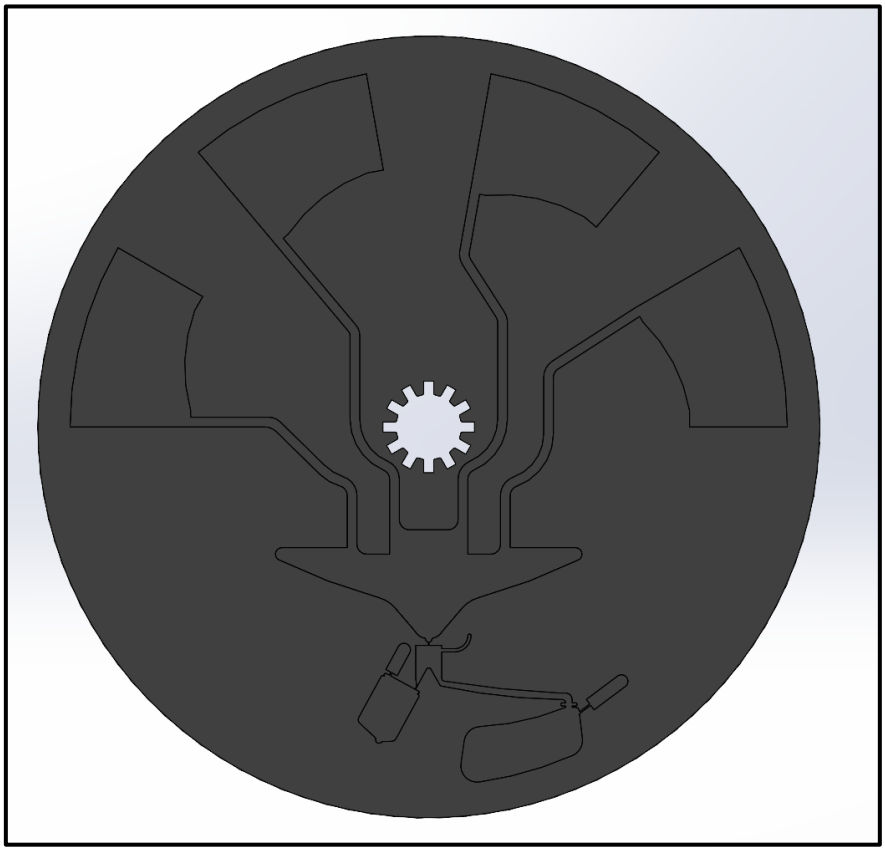


Figure 15.9.5: Image of Prototype Design V9

Testing Results:

Successes	Shortcomings
• Improved liquid flow to waste chamber with increase in channel width	• Coriolis switch still unsuccessful in moving 100% of fluid to desired chamber

15.9.6 Version 10

Differences from previous version:

- Cuts on the side of the disc, swapped to Elegoo Saturn 4 16K which has a smaller print bed, this change was necessary to fit disc on the bed
- This was the first test using the Basler camera integrated to the vLoaD rig

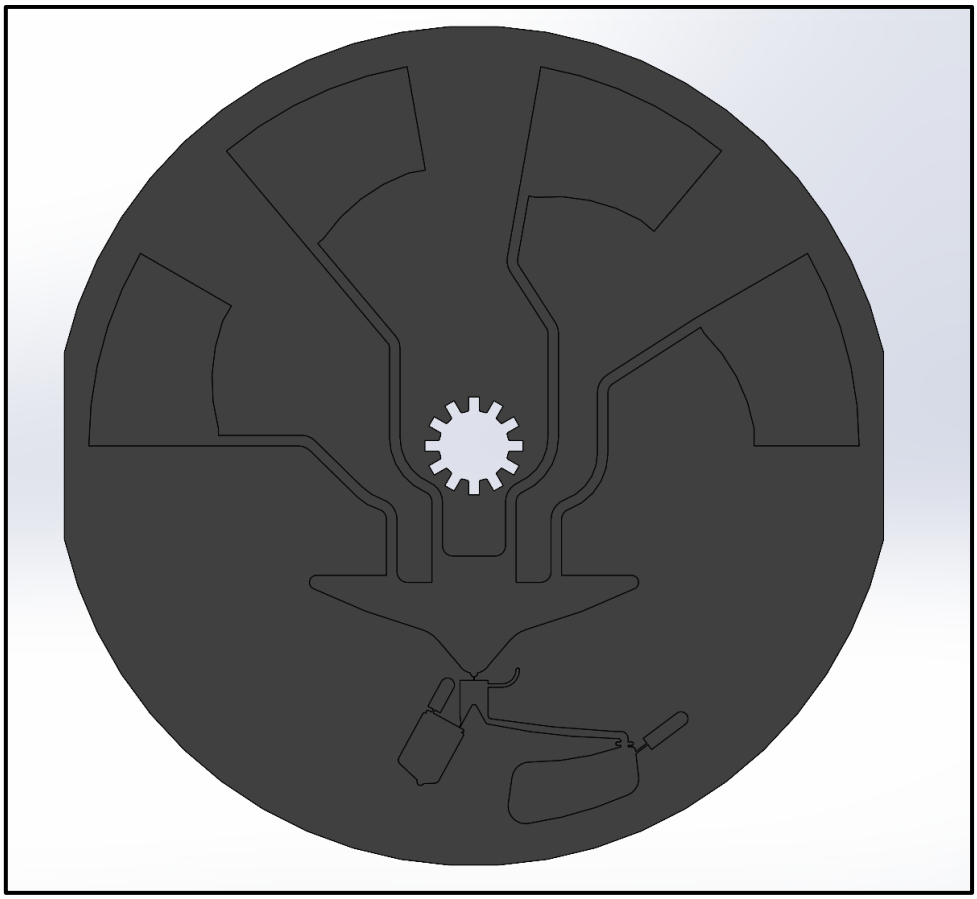


Figure 15.9.6: Image of Prototype Design V10

Testing Results:

Successes	Shortcomings
• Successful vacuum sealing on waste chamber thanks to fishbone valve element of the design	• Fluid loss from storage chambers upon conclusion of spinning. As liquid travels radially inwards, some leaves the chamber.

15.9.7 Version 11

Differences from previous version:

- Adjustment of vent capillary dimensions
- New storage chamber design to prevent fluid loss at the end of spinning
- Made Coriolis switch longer, to improve performance

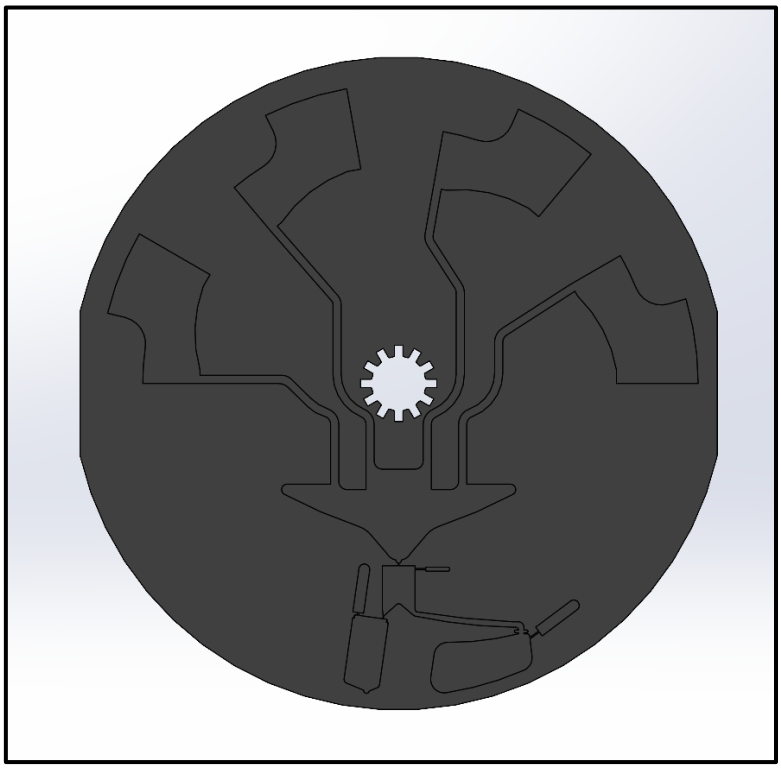


Figure 15.9.7: Image of Prototype Design V11

Testing Results:

Successes	Shortcomings
• No more fluid loss from storage chamber due to new design	• Issues noticed on Basler camera with liquid entering the product chamber • Vent performance is still not as desired, noticed failure in calculations to fully account for gravitational force

15.9.8 Version 12

Differences from previous version:

- Changed entrance to product chamber
- Final vent design calculations occurred here leading to the 0.1mm width
- Further increased length of Coriolis switch and adjustment of geometry

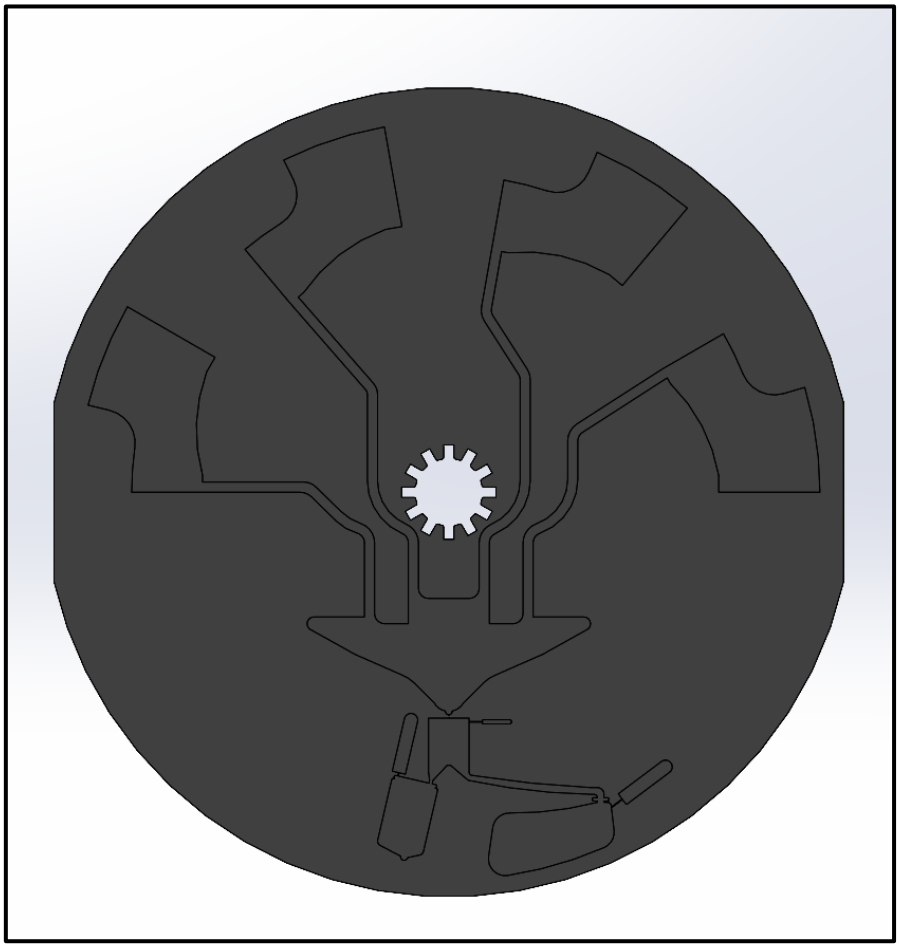


Figure 15.9.8: Image of Prototype Design V12

Testing Results:

Successes	Shortcomings
• Liquid was now able to enter the product chamber as desired • Vents sealed properly from fluids	• Coriolis switch still unsuccessful in moving 100% of fluid to desired chamber

15.10 Appendix 10: DNA Isolation protocol

15.10.1 For Testing Design 1

The basis for the DNA Isolation protocol comes from QIAGEN's DNA isolation kit. It follows on from the same protocol seen above, however the volumes used were adjusted for the vLoaD.

The volumes of each fluid used were (See **Appendix 14** for visual):

- 50 µL of sample mixture, see **section 9.6**.
- 50 µL of QIAGEN Buffer AW1 (Wash Buffer 1)
- 50 µL of QIAGEN Buffer AW2 (Wash Buffer 2)
- 50 µL of QIAGEN Buffer AE (Elution)

For the silica beads approximately 0.2 grams were used in the mixing chamber. This applies for test one through six.

Each fluid was handled as follows:

- Actuate Storage Chamber
- Allow fluid to flow into mixing chamber
- Reposition into diffusion position
- Allow Diffusion to Occur
- Spin to exit mixing chamber via Burst Valve at 1200 RPM
- Spin until fluid leaves Coriolis switch towards Product/Waste Chamber
- Notably for the elution solution the disk was rotated 180 degrees prior to spinning to ensure successful operation of the Coriolis switch, this was evaluated during testing
- After test four the Peclet numbers and diffusion times were calculated, see **section 8.2.8**. After this the diffusion times were adjusted for Test Five onwards, as seen below;

Table 15.10.1: Diffusion Times for each vLoaD Test

Test	Cell Sample	Wash Buffer 1	Wash Buffer 2	Elution Buffer
Test One --> Four	2 Minutes	2 Minutes	2 Minutes	3 Minutes
Test Five --> Six	10 Minutes	2 Minutes	2 Minutes	5 Minutes

Each liquid was dumped to waste, bar elution which went to product instead. Elution Buffer AE acts as a binding agent which makes the DNA bind together, hence it is flushed to the product chamber.

Table 15.10.2: Spin Cycles for each vLoaD Test

Test One --> Three	30 seconds for each step
Test Four	60 seconds for each step
Test Five --> Six	60 seconds for each step bar elution which spun for 120 seconds

15.10.2 For Testing Design 2

This segment covers the DNA Isolation protocol for the new vLoaD design

The volumes of each fluid used were:

- 50 µL of sample mixture, see **section 9.6**.
- 50 µL of QIAGEN Buffer AW1 (Wash Buffer 1)
- 50 µL of QIAGEN Buffer AW2 (Wash Buffer 2)
- 50 µL of QIAGEN Buffer AE (Elution)

For the silica beads approximately 0.5 grams were used in the mixing chamber.

There were no diffusion times for the vLoaD Revamp design.

Each fluid was handled as follows:

- Actuate Storage Chamber
- Allow fluid to flow into “mixing” chamber
- Spin to enter S-Chamber via Burst Valve
- Reposition to allow siphon effect to occur
- Spin to exit S-Chamber via Burst Valve at 1200 RPM
- Spin until fluid leaves Coriolis switch towards Product/Waste Chamber

Each liquid was dumped to waste, bar elution which went to product instead. Centrifugal force was always applied using 1200 RPM as it guaranteed successful operation of the burst valves and the Coriolis switch. Elution Buffer AE acts as a binding agent which makes the DNA bind together, hence it is flushed to the product chamber as the final DNA solution.

Table 15.10.3: Spin Cycles for vLoaD Testing Design 2

DNA Sample	120s
Wash Buffer 1	120s
Wash Buffer 2	120s
Elution Buffer	180s

15.10.3 For Testing Design 3

This segment covers the DNA Isolation protocol for the new vLoaD design

The volumes of each fluid used were:

- 50 µL of sample mixture, see **section 9.6**.
- 50 µL of QIAGEN Buffer AW1 (Wash Buffer 1)
- 50 µL of QIAGEN Buffer AW2 (Wash Buffer 2)
- 50 µL of QIAGEN Buffer AE (Elution)

For the silica beads approximately 0.5 grams were used in the mixing chamber.

There were no diffusion times for the vLoaD Revamp design.

Each fluid was handled as follows:

- Actuate Storage Chamber
- Allow fluid to flow into “mixing” chamber
- Spin to enter S-Chamber via Burst Valve
- Reposition to allow siphon effect to occur
- Spin to exit S-Chamber via Burst Valve at 1200 RPM
- Spin until fluid leaves Coriolis switch towards Product/Waste Chamber

Each liquid was dumped to waste, bar elution which went to product instead. Centrifugal force was always applied using 1200 RPM as it guaranteed successful operation of the burst valves and the Coriolis switch. Elution Buffer AE acts as a binding agent which makes the DNA bind together, hence it is flushed to the product chamber as the final DNA solution.

Table 15.10.4: Spin Cycles for vLoaD Testing Design 3

DNA Sample	120s
Wash Buffer 1	120s
Wash Buffer 2	120s
Elution Buffer	180s

15.10.4 For Testing Design 1 (Magnetic Bead Based Assay)

This segment covers the DNA Isolation protocol for the magnetic bead vLoaD design. The magnetic puck was placed on the disk anytime movement for the actuation of the storage chambers and for spin cycles. It was removed during diffusion times.

The volumes of each fluid used were:

- 50 µL of sample mixture, see **section 9.6**.
- 50 µL of QIAGEN Buffer AW1 (Wash Buffer 1)
- 50 µL of QIAGEN Buffer AW2 (Wash Buffer 2)
- 50 µL of QIAGEN Buffer AE (Elution)

Each fluid was handled as follows:

- Actuate Storage Chamber
- Allow fluid to flow into mixing chamber
- Reposition into diffusion position
- Allow Diffusion to Occur
- Agitate vLoaD via Vibrate Disk function while waiting
- Spin to exit mixing chamber via Burst Valve at 1200 RPM
- Spin until fluid leaves Coriolis switch towards Product/Waste Chamber
- Notably for the elution solution the disk was rotated 180 degrees prior to spinning to ensure successful operation of the Coriolis switch, this was discovered during testing

Each liquid was dumped to waste, bar elution which went to product instead. Centrifugal force was always applied using 1200 RPM as it guaranteed successful operation of the burst valves and the Coriolis switch. Elution Buffer AE acts as a binding agent which makes the DNA bind together, hence it is flushed to the product chamber as the final DNA solution. Spin times were for 2 minutes, except for once wash buffer 2 left the chamber, a 5 minute spin time was used and then the disk was allowed to rest for 2 minutes prior to the actuation of the elution storage chamber. This ensured that the wash buffers were fully removed from the beads.

Table 15.10.5: Diffusion Times for vLoaD Testing Design 1, Magnetic

DNA Sample	15 Minutes
Wash Buffer 1	3 Minutes
Wash Buffer 2	3 Minutes
Elution Buffer	20 Minutes

15.11 Appendix 11: Testing Results

15.11.1 Test One

15.11.1.1 vLoaD Test

This test was unsuccessful. Wash Buffer 2 leaked into the product chamber during its diffusion in the mixing chamber. For this test the beads were placed near the burst valve. As the liquid was held against the burst valve over time gravitational flow caused some fluid to leak. For future testing they were move to one of the wings of the mixing chamber to prevent this from occurring. During diffusion the disc will be held in a position such that this wing points downwards, keeping beads and fluid away from the burst valve.

15.11.1.2 DNA Test

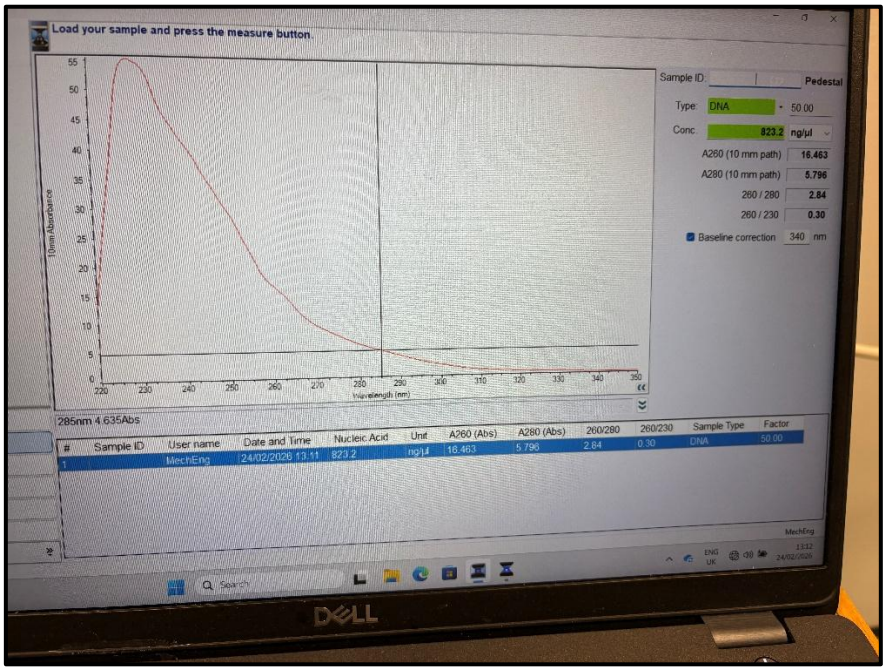


Figure 15.11.1: Image of NanoDrop Graph Test One

Table 15.11.1: NanoDrop Measurement Results for Test One

Measurement	ng/µL	A260	A280	260/280	260/230
1	823.2	16.463	5.796	2.84	0.30

The NanoDrop readings show signs of contaminants from the large nucleic acid readings and incorrect ratios. As the vLoaD Test was unsuccessful the extracted product was only tested via NanoDrop.

15.11.2 Test Two

15.11.2.1 vLoaD Test

This test was unsuccessful. The Elution mixture was first pumped to waster chamber as opposed to product. This is assumed to be due to start position of the spin cycle and the effects of gravity driven flow. For future tests, disc was repositioned such that the gravity-driven flow will point towards the product chamber.

15.11.2.2 DNA Test

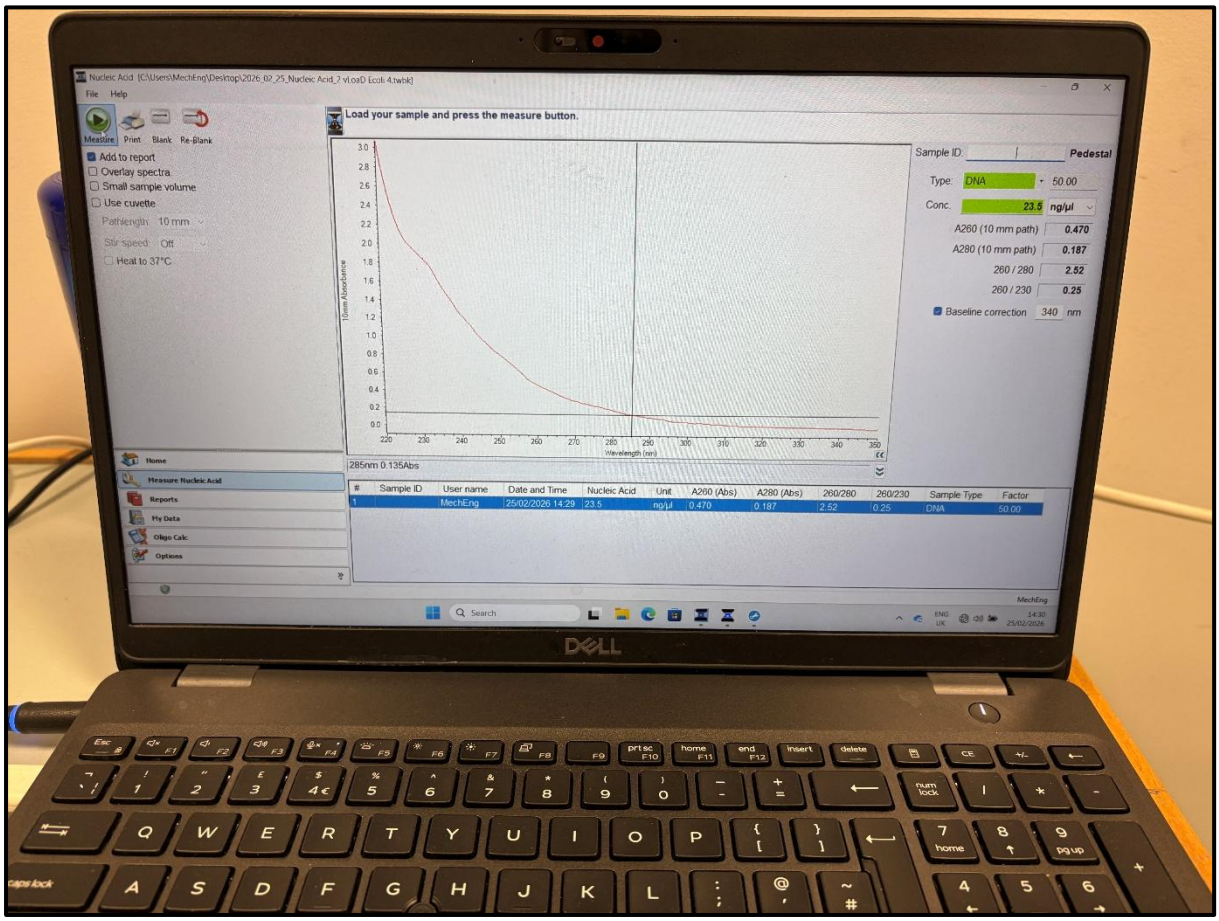


Figure 15.11.2: Image of NanoDrop Graph Test Two

Table 15.11.2: NanoDrop Measurement Results for Test Two

Measurement	ng/µL	A260	A280	260/280	260/230
1	108.2	2.164	0.817	2.65	0.28

Similar to Test one the nucleic acid reading and ratios indicate that contamination is present. As the vLoaD Test was unsuccessful the extracted product was only tested via NanoDrop.

After this first measurement, it was assumed that silica beads were present in the mixture, due to the tail at the end of the graph showing absorbance at 310nm forward. The sample was centrifuged at 10000 RPM for 60 seconds and a NanoDrop sample was taken from the top of the sample.

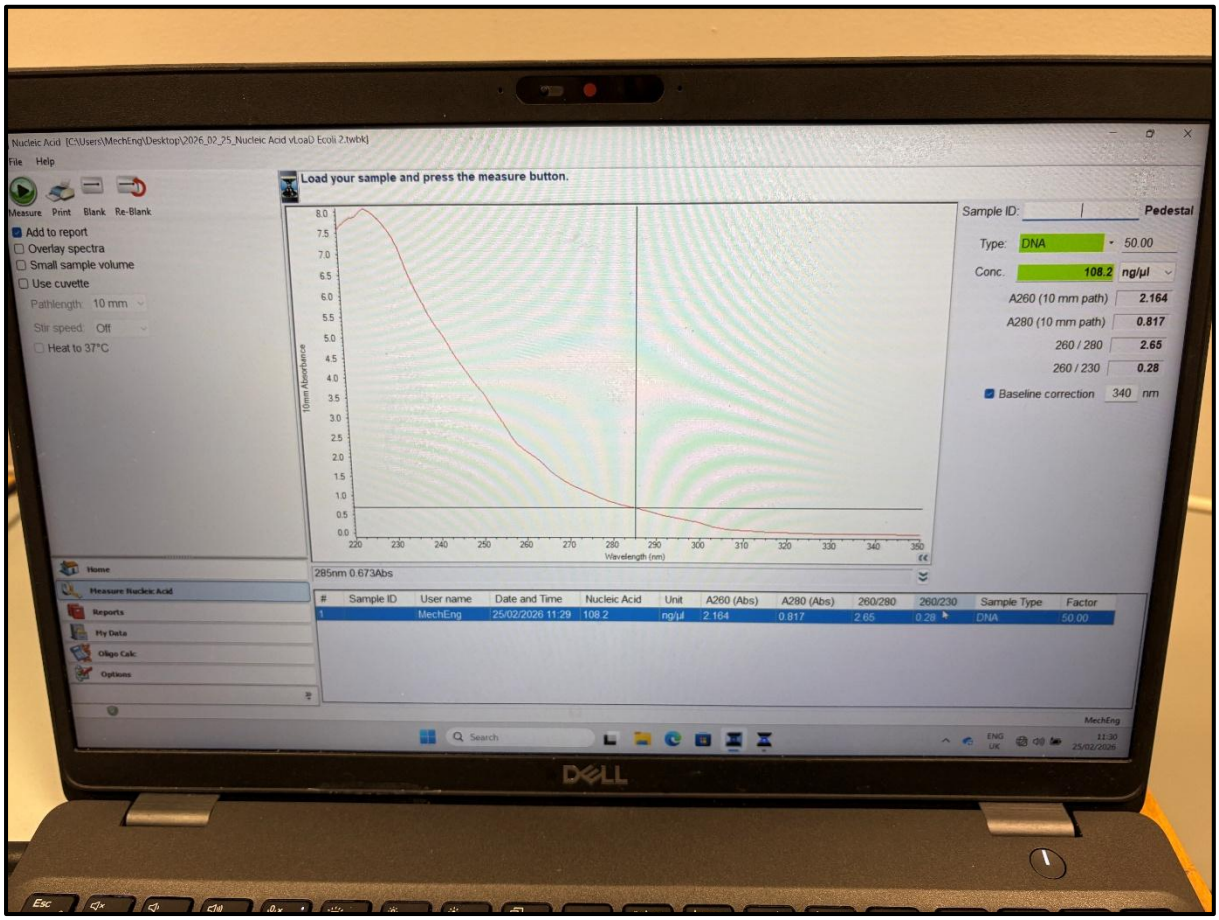


Figure 15.11.3: Image of NanoDrop Graph Test Two, Silica Removed

Table 15.11.3: NanoDrop Measurement Results for Test Two, Silica Removed

Measurement	ng/µL	A260	A280	260/280	260/230
1	23.5	0.470	0.187	2.52	0.25

NanoDrop readings did improve after “removal” of the silica beads from the sample, however the ratio and absolute value of absorbance at 260nm indicates that DNA is still not present in the sample.

15.11.3 Test Three

15.11.3.1 vLoaD Test

This test was unsuccessful. During operation fluid leaked from the disc, it is uncertain if it was due to an occlusion in the disc edge adjacent to the waste chamber or due to improper sealing of the mixing chamber with PSA. Analysis of the Basler camera footage seem to indicate both occurred. Fabrication based error, mistakes were noted for future tests.

15.11.4 Test Four

15.11.4.1 vLoaD Test

This test was conducted successfully, the only minor error was the storage chambers were difficult to actuate due to a vacuum sealing effect in the disc. Additional ventilation holes were added to the disc storage chambers. This vent position was select as during centrifugal spin these vents will not contact the fluid, preventing pumping of fluid to the outside of the disc.

DNA Test

NanoDrop

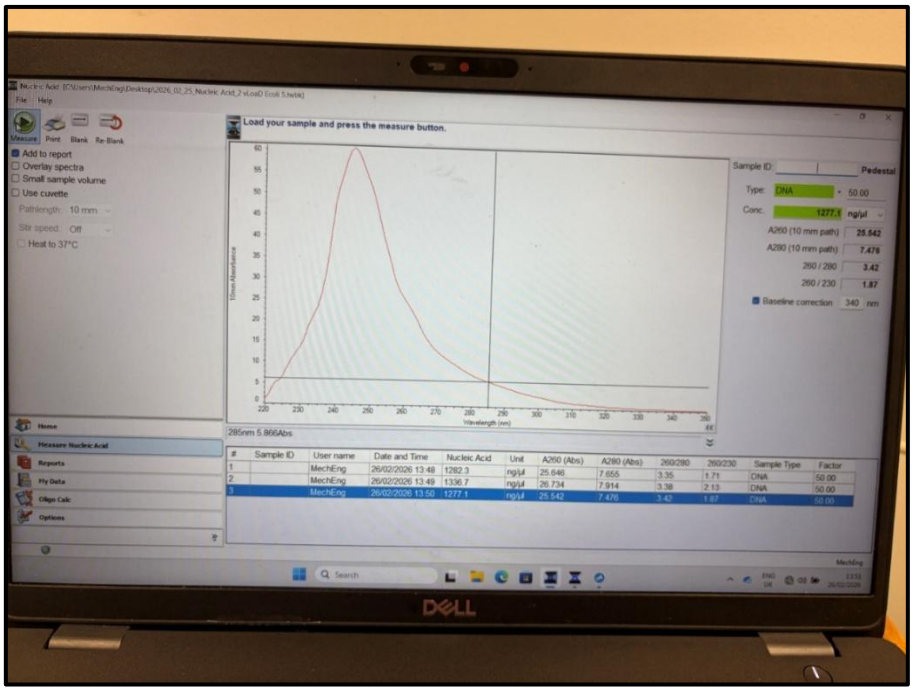


Figure 15.11.4: Image of NanoDrop Graph Test Four

Table 15.11.4: NanoDrop Measurement Results for Test Four

Measurement	ng/µL	A260	A280	260/280	260/230
1	1282.3	25.646	7.655	3.35	1.71
2	1336.7	26.734	7.914	3.38	2.13
3	1277.1	25.542	7.476	3.42	1.87

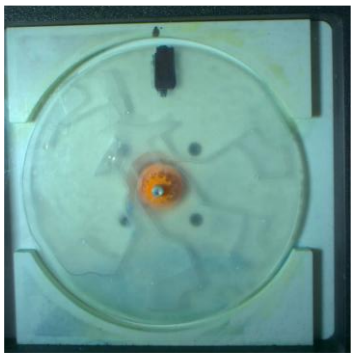
NanoDrop readings and the graph both indicate that contaminates are present however it was decided to run the q-PCR anyways as it is possible that while contaminated, the DNA can still be amplified.

q-PCR

For this q-PCR both the vLoaD product and an on-bench control were run. There was zero amplification across the forward and the reverse for both samples. It is believed that this occurred due an issue with the rotor-gene Q, and PCR protocol, swapped over to different method.

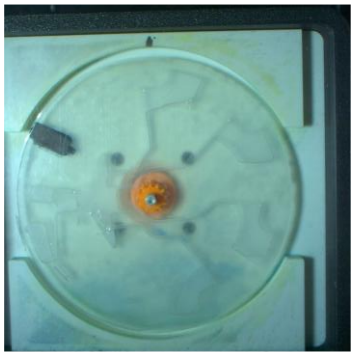
15.12 Appendix 12: Comic Strip depicting vLoaD Assay (Test Six)

Step 1



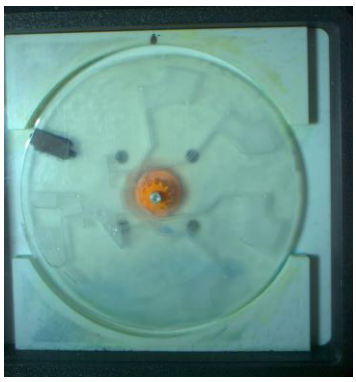
Start Position for vLoaD Disk

Step 2



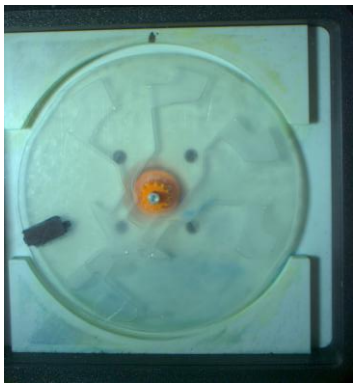
Release of DNA Sample from Chamber 1, Maintain Position for Diffusion

Step 3



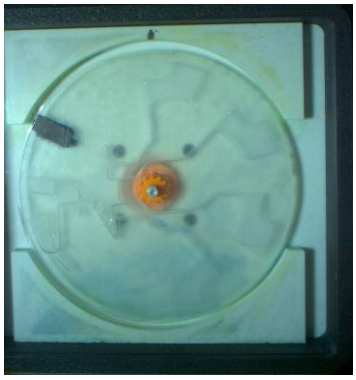
Pumped P DNA Sample to Waste Chamber

Step 4



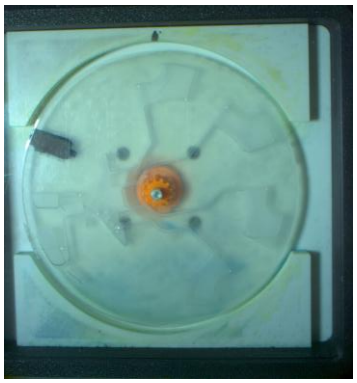
Release of Wash Buffer 1 from Chamber 2

Step 5



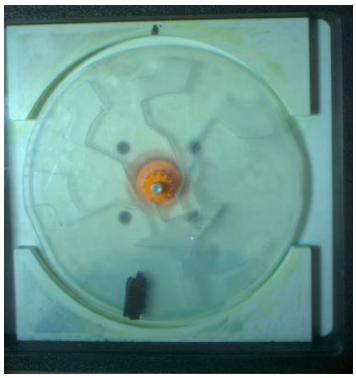
Diffusion Position for Wash Buffer 1

Step 6



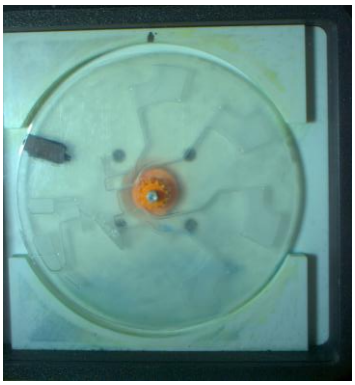
Pumped Wash Buffer 1 to Waste Chamber

Step 7



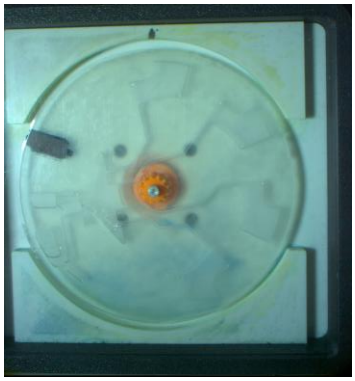
Release of Wash Buffer 2 from Chamber 3

Step 8



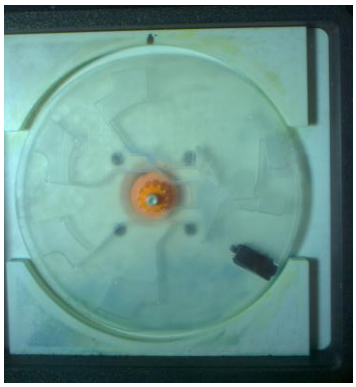
Diffusion Position for Wash Buffer 2

Step 9



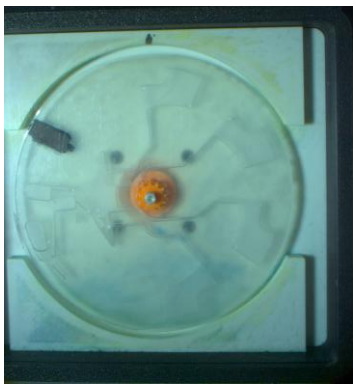
Pumped Wash Buffer 2 to Waste Chamber

Step 10



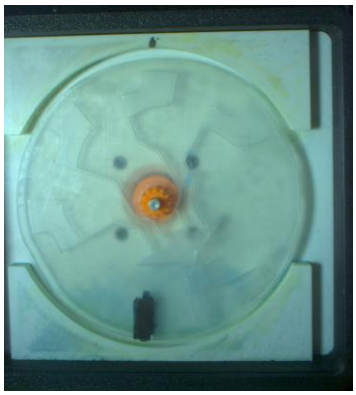
Release of Elution Buffer from Chamber 4

Step 11



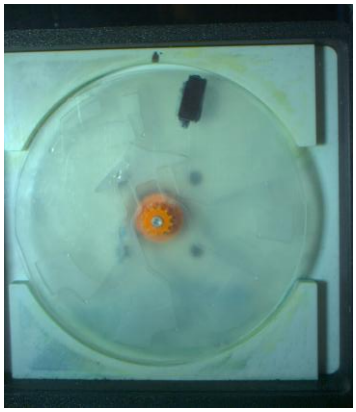
Diffusion Position for Elution Buffer

Step 12



Rotate Disk Prior to Pumping Elution Buffer to Product Chamber

Step 13



Pumped Elution Buffer to Product Chamber

15.13 Appendix 13: Microscope Images

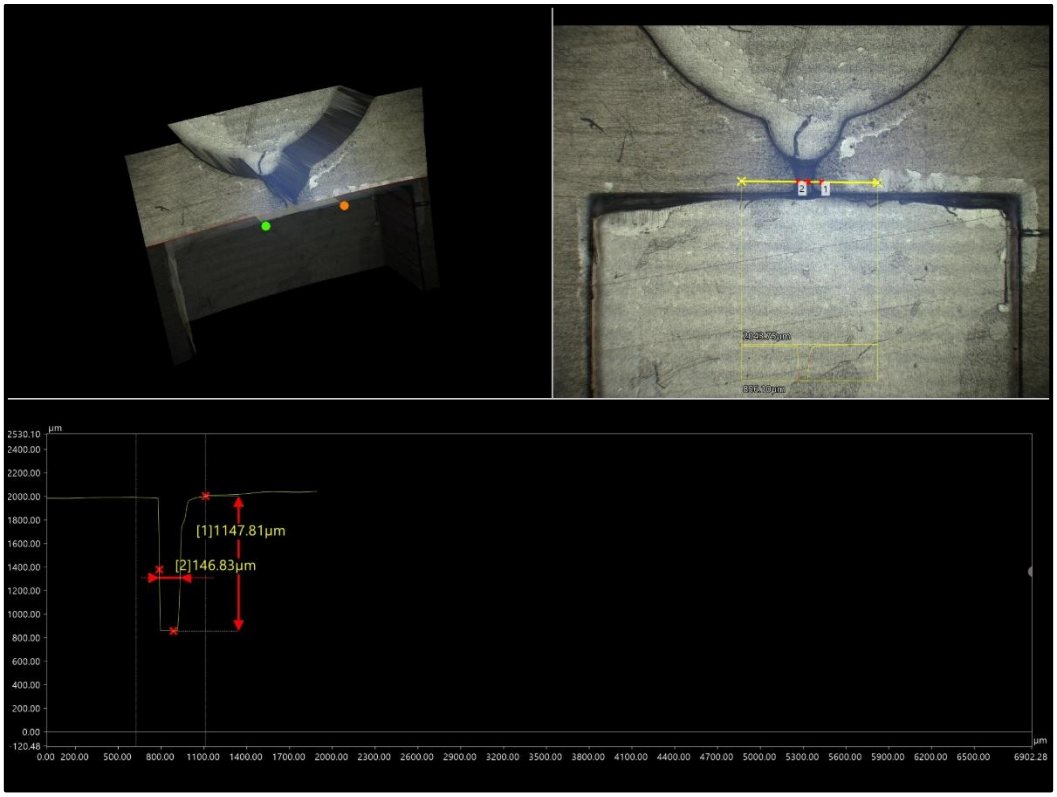


Figure 15.13.1: Microscope Dimension Image of Burst Valve

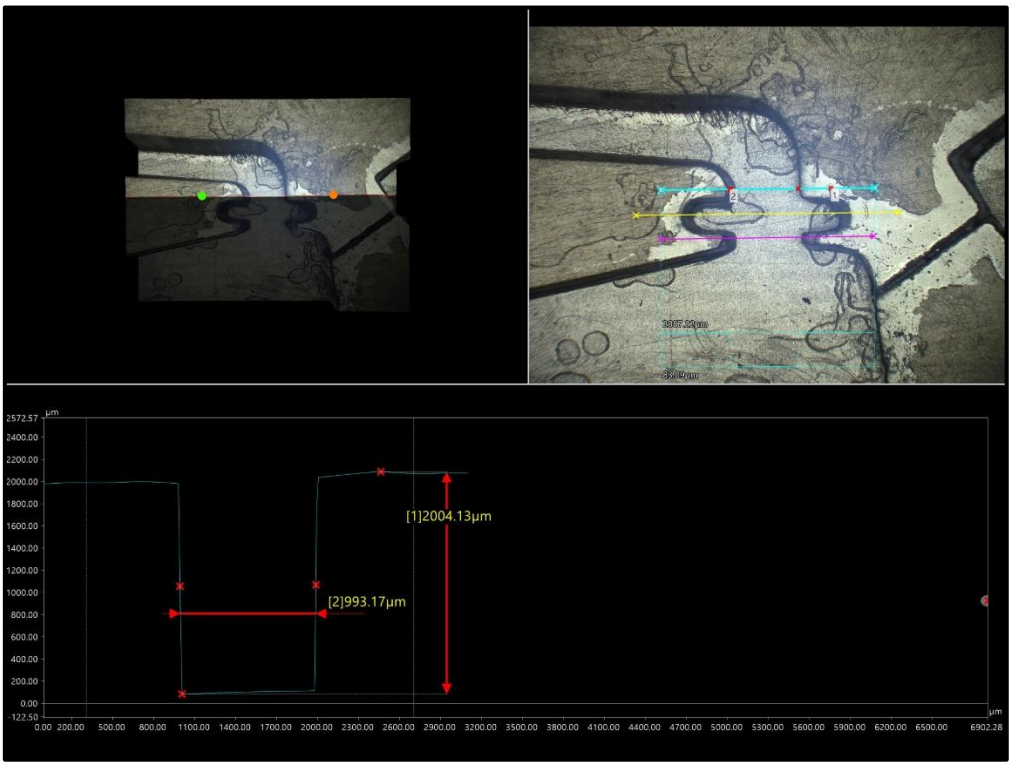


Figure 15.13.2: Microscope Dimension Image of Fishbone Valve, Entrance

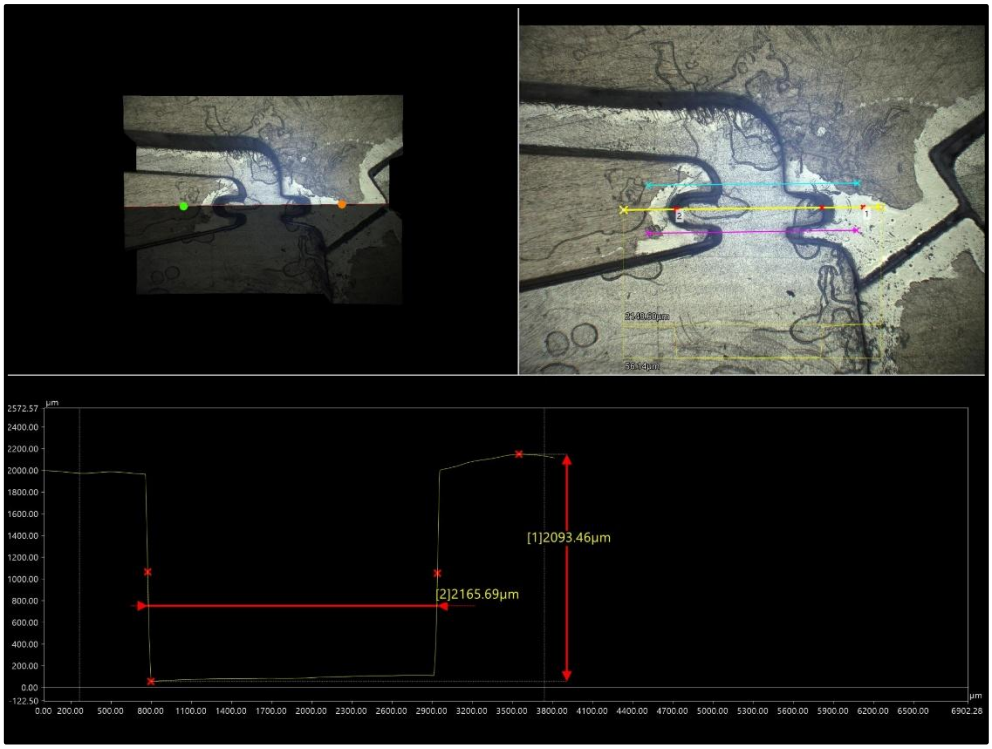


Figure 15.13.3: Microscope Dimension Image of Fishbone Valve, Midsection

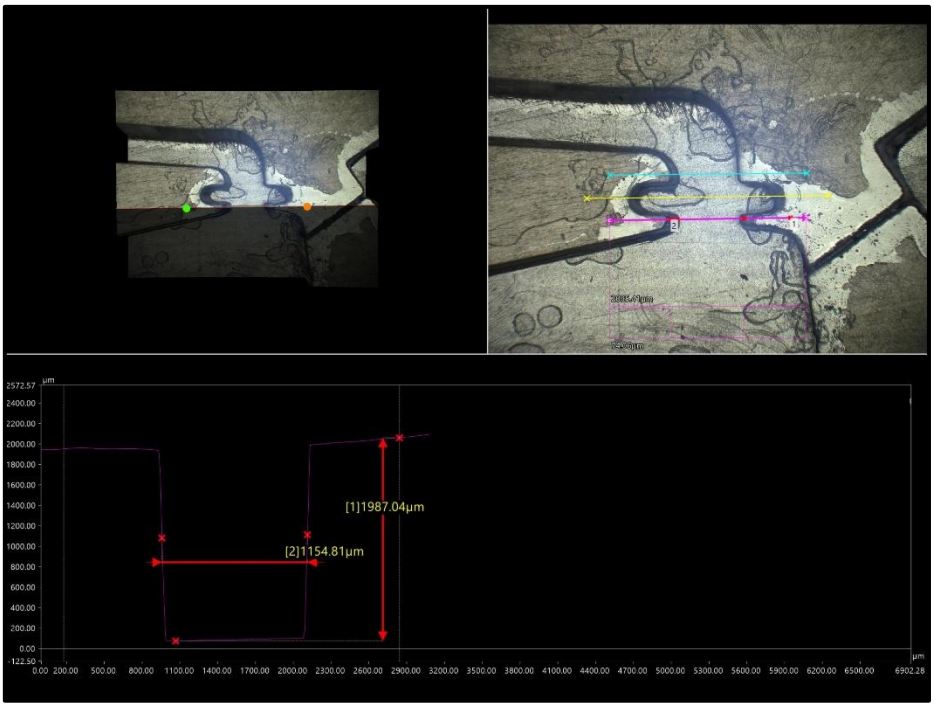


Figure 15.13.4: Microscope Dimension Image of Fishbone Valve, Exit

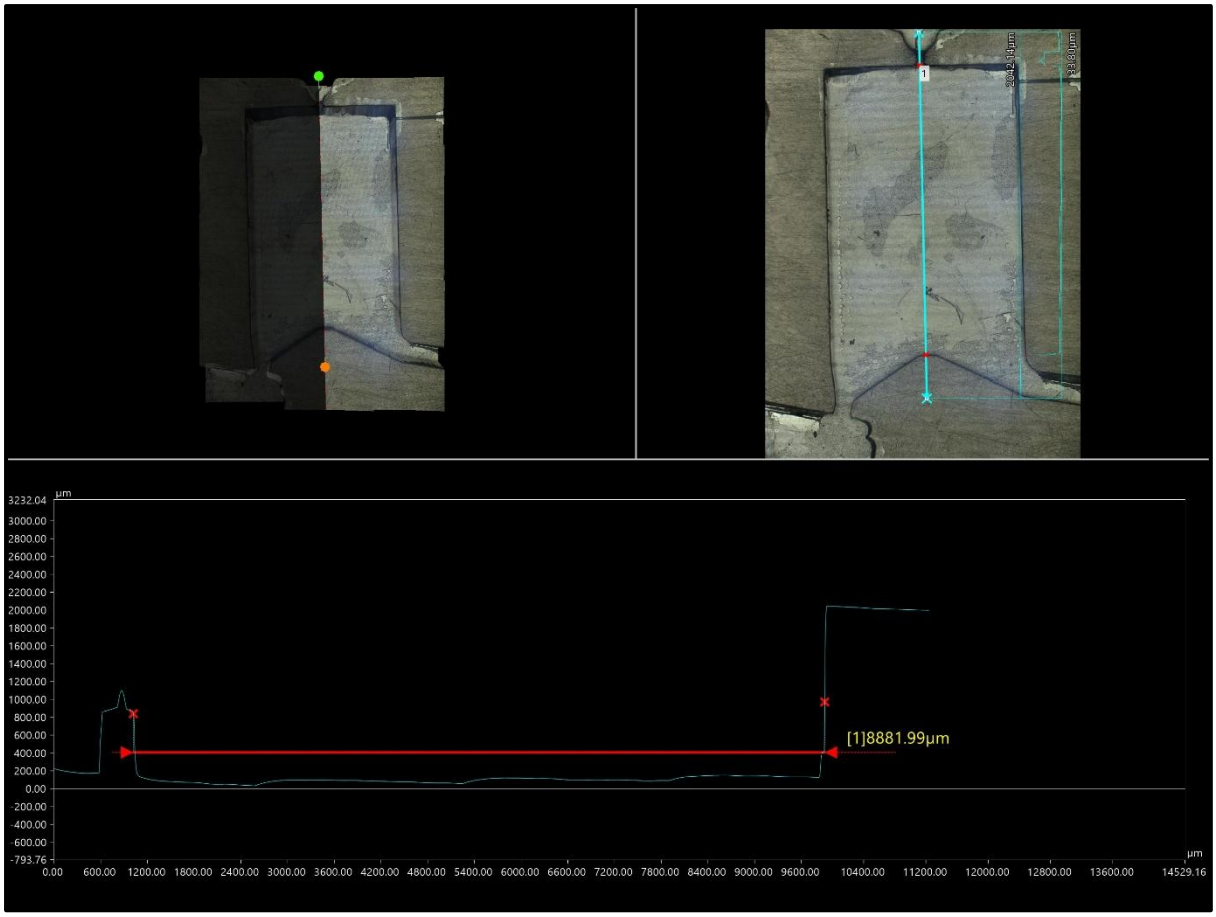


Figure 15.13.5: Microscope Dimension Image of Coriolis Switch

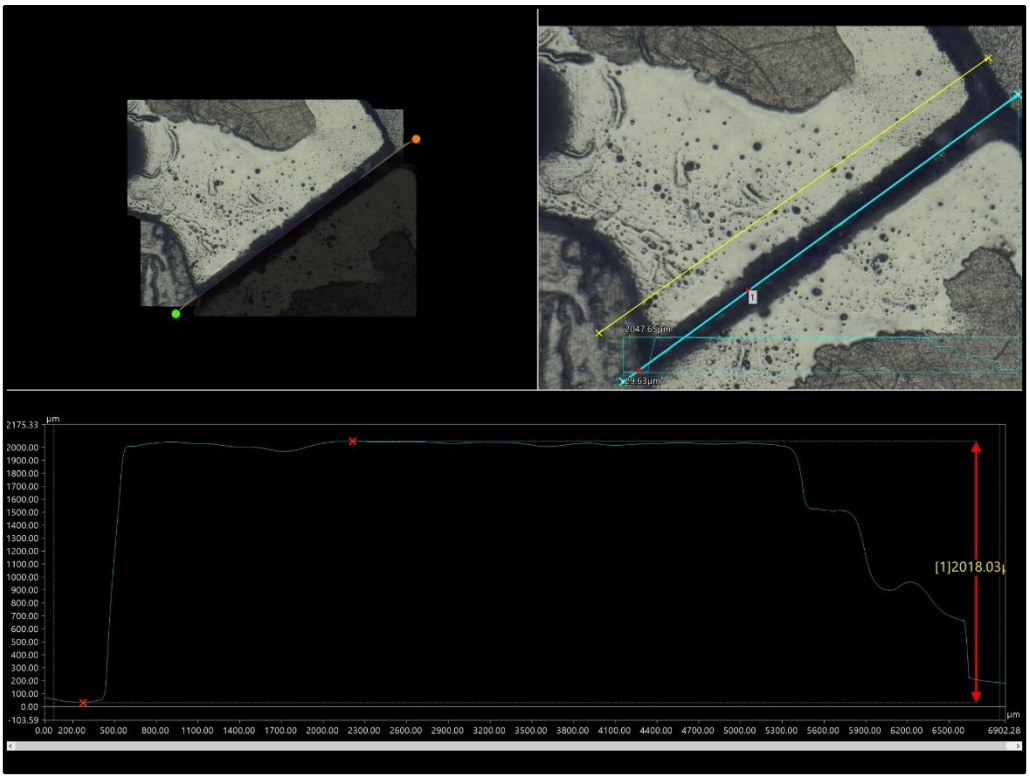


Figure 15.13.6: Microscope Dimension Image of Waste Chamber Vent

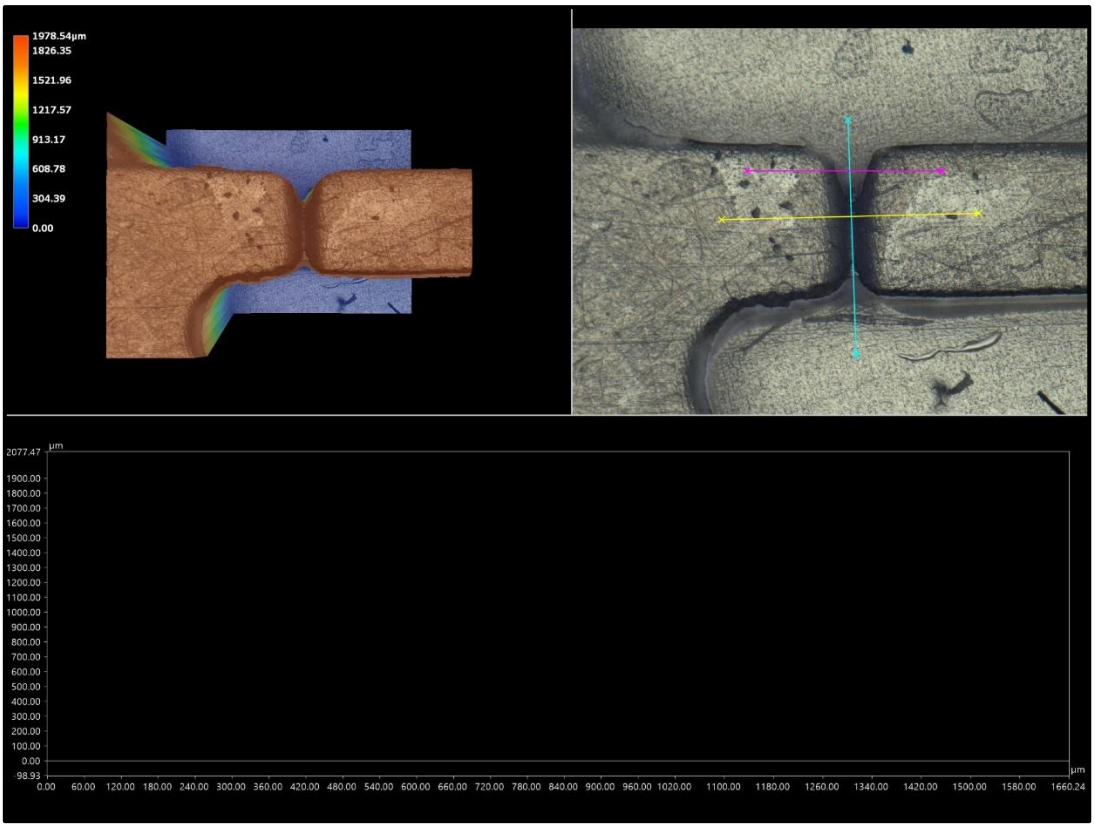


Figure 15.13.7: Microscope Topography Image of Product Chamber Vent

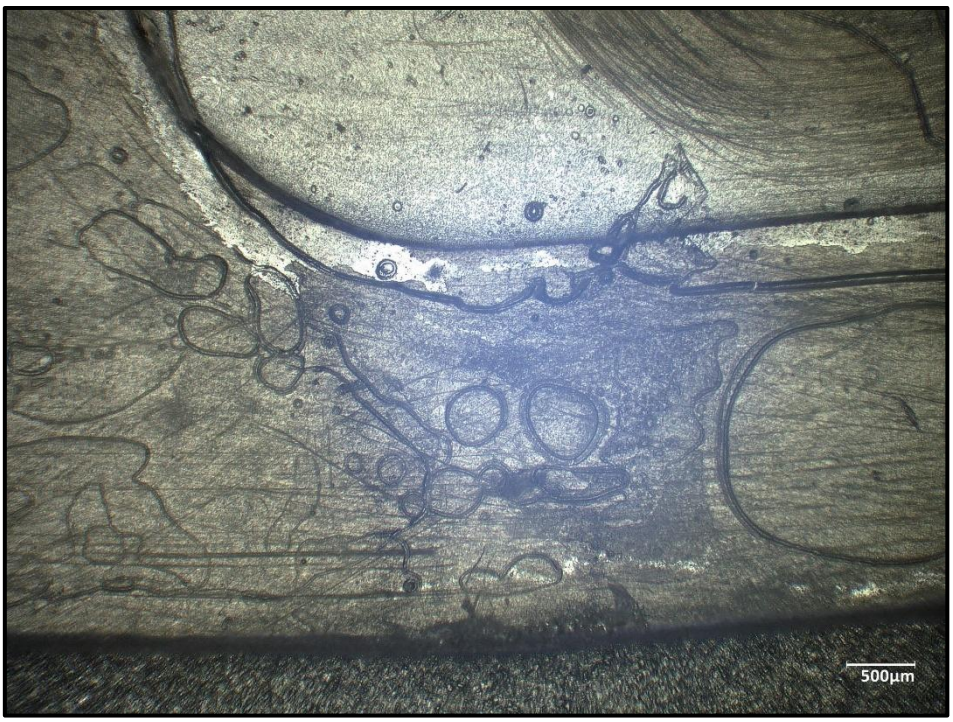


Figure 15.13.8: Microscope Image of Inclusion at base of Waste Chamber

15.14 Appendix 14: Arduino vLoaD Code

Listing 15.14.1: Arduino Code used to control vLoaD Rig

```
#include <AccelStepper.h>
// Define stepper motors
AccelStepper stepper(AccelStepper::DRIVER, 3, 6); // (Type: Driver, Step Pin, Dir Pin)
const int enPin = 8;   // Enable pin for motor driver
const float stepAngle = 1.8;
const int motorSteps = 360 / stepAngle;
float currentAngle = 0;
int acceleration = 2000; // Acceleration in steps per second^2
void setup() {
 Serial.begin(115200);
 pinMode(enPin, OUTPUT);
 digitalWrite(enPin, LOW); // Enable motor driver
 // Configure Stepper X (Main Motor)
 stepper.setMaxSpeed(rpmToSps(1200));     // Set max speed (steps/sec)
 stepper.setAcceleration(acceleration); // Set acceleration (steps/sec^2)
 //Serial.println(rpmToSps(1200));
}
void loop() {
 if (Serial.available()) {
  String input = Serial.readStringUntil('\n');  // Read input string
  input.trim();  // Remove leading/trailing whitespace

  if (input.length() > 0) {
   char commandType = input.charAt(0);  // Get the first character

   if (commandType == 'A' || commandType == 'S' || commandType == 'P') {
    // Parse the rest of the input
    int values[3] = {0, 0, 0};  // Array to hold extracted values
    int index = 0;

    char *token = strtok(&input[1], " ");  // Tokenize the input
    while (token != nullptr && index < 3) {
       values[index++] = atoi(token);
       token = strtok(nullptr, " ");
    }

    // Execute corresponding function
    if (commandType == 'A' && index == 2) {
       moveToAngle(rpmToSps(values[0]), values[1]);
    } else if (commandType == 'S' && index == 3) {
       spinToAngle(rpmToSps(values[0]), values[1], values[2]);
    } else if (commandType == 'P' && index == 3){
       persuade(rpmToSps(values[0]), values[1], values[2]);
    } else {
       Serial.println("Invalid command format.");
    }
   } else {
    Serial.println("Unknown command.");
```

```
    }
   }
  }
}

float rpmToSps(float rpm){
 return (rpm/60)*motorSteps;
}

float angleToSteps(float angle){
 return angle / stepAngle;
}

float spinToAngle(float speed, float runTime, float angle) {
 long int runSteps = speed * runTime;
 int displacementSteps = runSteps % motorSteps;
 float desiredDisplacementSteps = (angle-currentAngle) / stepAngle;
 float adjustedRunSteps = runSteps + desiredDisplacementSteps - displacementSteps;
 Serial.print("Current angle: ");
 Serial.print(currentAngle);
 Serial.print(", Moving by ");
 Serial.print(runSteps);
 Serial.print(" steps with a speed of ");
 Serial.println(speed);
 stepper.setMaxSpeed(speed);
 stepper.move(adjustedRunSteps); // Move by the specified steps
 while (stepper.distanceToGo() != 0) {
  stepper.run(); // Ensures acceleration is applied every time
 }
 currentAngle = angle;
 stepper.setCurrentPosition(desiredDisplacementSteps); // Set position to target angle
}

float moveToAngle(float speed, float angle){
 float runSteps;
 if(speed >= 0){ //positive direction
  if (angle - currentAngle >= 0){
   runSteps = angleToSteps((angle-currentAngle)); //calculate number of steps to move to new angle
  }
  else{
   runSteps = angleToSteps((angle-currentAngle)) + motorSteps; //if desired position minus current position is less than 0 then plus 360 due to the direction
  }

 }
 else{ //negative direction
  if (angle - currentAngle > 0){ //Changing this line from >= to only > fixed the error spotted
   runSteps = angleToSteps((angle-currentAngle)) - motorSteps; //minus 360 due to the directon
  }
  else{
   runSteps = angleToSteps((angle-currentAngle));
  }
 }
 Serial.print("Current angle: ");
 Serial.print(currentAngle);
```

```
  Serial.print(", Moving by ");
  Serial.print(runSteps);
  Serial.print(" steps with a speed of ");
  Serial.println(speed);
  stepper.setMaxSpeed(speed);
  stepper.move(runSteps);
  while (stepper.distanceToGo() != 0) {
    stepper.run(); // Ensures acceleration is applied every time
  }
  currentAngle = angle;
  stepper.setCurrentPosition(runSteps); // Set position to target angle
}

float moveToAnglePer(float speed, float angle){
  float runSteps;
  if(speed >= 0){ //positive direction
    if (angle - currentAngle >= 0){
      runSteps = angleToSteps((angle-currentAngle)); //calculate number of steps to move to new angle
    }
    else{
      runSteps = angleToSteps((angle-currentAngle)) + motorSteps; //if desired position minus current position is less than 0 then plus 360 due to the direction
    }

  }
  else{ //negative direction
    if (angle - currentAngle > 0){ //Changing this line from >= to only > fixed the error spotted
      runSteps = angleToSteps((angle-currentAngle)) - motorSteps; //minus 360 due to the directon
    }
    else{
      runSteps = angleToSteps((angle-currentAngle));
    }
  }
  //Serial.print("Current angle: ");
  //Serial.print(currentAngle);
  //Serial.print(", Moving by ");
  //Serial.print(runSteps);
  //Serial.print(" steps with a speed of ");
  //Serial.println(speed);
  stepper.setMaxSpeed(speed);
  stepper.move(runSteps);
  while (stepper.distanceToGo() != 0) {
    stepper.run(); // Ensures acceleration is applied every time
  }
  currentAngle = angle;
  stepper.setCurrentPosition(runSteps); // Set position to target angle
}
float persuade(float speed, float runCount, float angle){
//lock the current angle and memorise it
//create a function that moves the motor back and forth
//alternate until runtime is reached
  //might need to make something to force the angle and speed inputted to be positive
  float angleRange = angle/2;
  float resetAngle = 0;
  for (int i = 0; i <= runCount; i++){
```

```
    angleRange = abs(angleRange); //ensures positive numbers
    speed = abs(speed);
    int per1 = currentAngle + angleRange;
    moveToAnglePer(speed,per1); //positive range direction
    //delay(5000);
    //return to previous position
    resetAngle = currentAngle - angleRange; //undoing the previous movement
    speed = speed*(-1); //this may or may not be needed
    moveToAnglePer(speed, resetAngle);
    //delay(5000);
    angleRange = angleRange*(-1);
    int per2 = currentAngle + angleRange;
    moveToAnglePer(speed, per2);
    //delay(5000);
    resetAngle = currentAngle - angleRange; //may need a polarity change or plus 360
    speed = speed*(-1);
    moveToAnglePer(speed, resetAngle);
    //delay(5000);
    //Serial.println(i);
    //Serial.println(" loop completed");
  }
    Serial.println("Persuasion completed");

}
```

15.15 Appendix 15: vLoaD Automated Program Code

Listing 15.15.1: XAML File for Application

```
<Window x:Class="SerialGuiDemo.MainWindow"
    xmlns="http://schemas.microsoft.com/winfx/2006/xaml/presentation"
    xmlns:x="http://schemas.microsoft.com/winfx/2006/xaml"
    xmlns:local="clr-namespace:SerialGuiDemo"
    Title="vLoaD Control Program"
    Height="1200"
    Width="1000"
    Loaded="Window_Loaded"
    Closing="Window_Closing">


  <Window.Resources>
  <local:NegateDoubleConverter x:Key="NegateDouble"/>
  <local:HalverConverter x:Key="Halver"/>
  </Window.Resources>


  <Grid Margin="12">

    <Grid.RowDefinitions>
      <RowDefinition Height="Auto"/>
      <RowDefinition Height="Auto"/>
      <RowDefinition Height="Auto"/>
    </Grid.RowDefinitions>

    <!-- Connectivitiy Bar -->
    <StackPanel Grid.Row="0"
      Orientation="Horizontal"
      Margin="0,0,0,10">

      <TextBlock Text="Port:"
        VerticalAlignment="Center"
        Margin="0,0,8,0"/>

      <ComboBox x:Name="PortCombo"
       Width="160"
       Margin="0,0,8,0"/>

      <Button Content="Refresh"
      Width="80"
      Margin="0,0,8,0"
      Click="RefreshPorts_Click"/>

      <Button x:Name="ConnectBtn"
      Content="Connect"
      Width="110"
      Click="ConnectBtn_Click"/>

      <TextBlock x:Name="StatusText"
        Text="Disconnected"
        VerticalAlignment="Center"
        Margin="12,0,0,0"/>
    </StackPanel>

    <TabControl  Grid.Row="1">
      <!-- Manual mode tab -->
      <TabItem Header="Manual">
        <Grid>
          <Grid.RowDefinitions>
            <RowDefinition Height="Auto"/>
            <RowDefinition Height="Auto"/>
            <RowDefinition Height="Auto"/>
            <RowDefinition Height="Auto"/>
            <RowDefinition Height="Auto"/>
            <RowDefinition Height="*"/>
          </Grid.RowDefinitions>
```

```
        <Grid.ColumnDefinitions>
          <ColumnDefinition Width="600"/>
          <ColumnDefinition Width="Auto"/>
        </Grid.ColumnDefinitions>



        <!--  Angle Adjust Function -->
        <GroupBox Grid.Row="1"
                Grid.Column="0"
                Header="Angle Adjustment"
                Margin="0,0,0,10"
                Height="135"
                Foreground="Black"
                BorderBrush="White">

          <Grid Margin="10">
            <Grid.RowDefinitions>
              <RowDefinition Height="Auto"/>
              <RowDefinition Height="Auto"/>
              <RowDefinition Height="Auto"/>
            </Grid.RowDefinitions>

            <Grid.ColumnDefinitions>
              <ColumnDefinition Width="Auto"/>
              <ColumnDefinition Width="*"/>
              <ColumnDefinition Width="70"/>
            </Grid.ColumnDefinitions>

            <!-- Speed -->
            <TextBlock Grid.Row="0"
                    Grid.Column="0"
                    Text="Speed:"
                    VerticalAlignment="Center"
                    Margin="0,0,10,0"/>

            <Slider x:Name="A1Slider"
                  Grid.Row="0"
                  Grid.Column="1"
                  Minimum="-100"
                  Maximum="100"
                  TickFrequency="1"
                  IsSnapToTickEnabled="True"
                  ToolTip="Range: -100 –> 100"
                  ValueChanged="A1Slider_ValueChanged"/>

            <TextBox x:Name="A1Box"
                   Grid.Row="0"
                   Grid.Column="2"
                   Margin="10,0,0,0"
                   VerticalContentAlignment="Center"
                   TextChanged="A1Box_TextChanged"/>

            <!-- Angle -->
            <TextBlock Grid.Row="1"
                    Grid.Column="0"
                    Text="Angle:"
                    VerticalAlignment="Center"
                    Margin="0,8,10,0"/>

            <Slider x:Name="A2Slider"
                  Grid.Row="1"
                  Grid.Column="1"
                  Margin="0,8,0,0"
                  Minimum="-360"
                  Maximum="360"
                  TickFrequency="1"
                  IsSnapToTickEnabled="True"
                  ToolTip="Range: -360 –> 360"
                  ValueChanged="A2Slider_ValueChanged"/>

            <TextBox x:Name="A2Box"
                   Grid.Row="1"
                   Grid.Column="2"
                   Margin="10,8,0,0"
                   VerticalContentAlignment="Center"
```

```
                TextChanged="A2Box_TextChanged"/>

        <!-- Send -->
        <Button Grid.Row="2"
            Grid.Column="0"
            Grid.ColumnSpan="3"
            Height="32"
            Margin="0,12,0,0"
            Content="Update Angle"
            BorderBrush="Black"
            Click="SendA_Click"/>
      </Grid>
    </GroupBox>

    <!--Spin disk function -->
    <GroupBox Grid.Row="2"
          Grid.Column="0"
          Header="Spin Disk"
          Margin="0,0,0,10">

      <Grid Margin="10">
        <Grid.RowDefinitions>
          <RowDefinition Height="Auto"/>
          <RowDefinition Height="Auto"/>
          <RowDefinition Height="Auto"/>
          <RowDefinition Height="Auto"/>
        </Grid.RowDefinitions>

        <Grid.ColumnDefinitions>
          <ColumnDefinition Width="Auto"/>
          <ColumnDefinition Width="*"/>
          <ColumnDefinition Width="70"/>
        </Grid.ColumnDefinitions>

        <!-- Speed -->
        <TextBlock Grid.Row="0"
              Grid.Column="0"
              Text="Speed:"
              VerticalAlignment="Center"
              Margin="0,0,10,0"/>

        <Slider x:Name="S1Slider"
            Grid.Row="0"
            Grid.Column="1"
            Minimum="-1200"
            Maximum="1200"
            TickFrequency="1"
            IsSnapToTickEnabled="True"
            ToolTip="Range: -1200 –> 1200"
            ValueChanged="S1Slider_ValueChanged"/>

        <TextBox x:Name="S1Box"
              Grid.Row="0"
              Grid.Column="2"
              Margin="10,0,0,0"
              VerticalContentAlignment="Center"
              TextChanged="S1Box_TextChanged"/>

        <!-- Run Time -->
        <TextBlock Grid.Row="1"
              Grid.Column="0"
              Text="Run Time:"
              VerticalAlignment="Center"
              Margin="0,8,10,0"/>

        <Slider x:Name="S2Slider"
            Grid.Row="1"
            Grid.Column="1"
            Margin="0,8,0,0"
            Minimum="0"
            Maximum="30"
            TickFrequency="1"
            IsSnapToTickEnabled="True"
            ToolTip="Range: 0 –> 30"
            ValueChanged="S2Slider_ValueChanged"/>
```

```
            <TextBox x:Name="S2Box"
                  Grid.Row="1"
                  Grid.Column="2"
                  Margin="10,8,0,0"
                  VerticalContentAlignment="Center"
                  TextChanged="S2Box_TextChanged"/>

            <!-- Angle -->
            <TextBlock Grid.Row="2"
                   Grid.Column="0"
                   Text="Angle:"
                   VerticalAlignment="Center"
                   Margin="0,8,10,0"/>

            <Slider x:Name="S3Slider"
                  Grid.Row="2"
                  Grid.Column="1"
                  Margin="0,8,0,0"
                  Minimum="-360"
                  Maximum="360"
                  TickFrequency="1"
                  IsSnapToTickEnabled="True"
                  ToolTip="Range: -360 –> 360"
                  ValueChanged="S3Slider_ValueChanged"/>

            <TextBox x:Name="S3Box"
                   Grid.Row="2"
                   Grid.Column="2"
                   Margin="10,8,0,0"
                   VerticalContentAlignment="Center"
                   TextChanged="S3Box_TextChanged"/>

            <Button Grid.Row="3"
                  Grid.Column="0"
                  Grid.ColumnSpan="3"
                  Height="32"
                  Margin="0,12,0,0"
                  Content="Spin Disk"
                  Click="SendS_Click"/>
        </Grid>
    </GroupBox>
    <!-- Vibrate disk Function -->
    <GroupBox Grid.Row="3"
          Grid.Column="0"
      Header="Vibrate Disk"
      Margin="0,0,0,10">
        <Grid Margin="10">
            <Grid.RowDefinitions>
                <RowDefinition Height="Auto"/>
                <RowDefinition Height="Auto"/>
                <RowDefinition Height="Auto"/>
                <RowDefinition Height="Auto"/>
            </Grid.RowDefinitions>

            <Grid.ColumnDefinitions>
                <ColumnDefinition Width="Auto"/>
                <ColumnDefinition Width="*"/>
                <ColumnDefinition Width="70"/>
            </Grid.ColumnDefinitions>

            <!-- Speed -->
            <TextBlock Grid.Row="0"
              Grid.Column="0"
              Text="Speed:"
              VerticalAlignment="Center"
              Margin="0,0,10,0"/>

            <Slider x:Name="P1Slider"
            Grid.Row="0"
            Grid.Column="1"
            Minimum="0"
            Maximum="100"
            TickFrequency="1"
            IsSnapToTickEnabled="True"
            ToolTip="Range: 0 –> 100"
            ValueChanged="P1Slider_ValueChanged"/>
```

```
        <TextBox x:Name="P1Box"
         Grid.Row="0"
         Grid.Column="2"
         Margin="10,0,0,0"
         VerticalContentAlignment="Center"
         TextChanged="P1Box_TextChanged"/>

        <!-- Run Count -->
        <TextBlock Grid.Row="1"
          Grid.Column="0"
          Text="Run Count:"
          VerticalAlignment="Center"
          Margin="0,8,10,0"/>

        <Slider x:Name="P2Slider"
        Grid.Row="1"
        Grid.Column="1"
        Margin="0,8,0,0"
        Minimum="0"
        Maximum="30"
        TickFrequency="1"
        IsSnapToTickEnabled="True"
        ToolTip="Range: 0 –> 30"
        ValueChanged="P2Slider_ValueChanged"/>

        <TextBox x:Name="P2Box"
         Grid.Row="1"
         Grid.Column="2"
         Margin="10,8,0,0"
         VerticalContentAlignment="Center"
         TextChanged="P2Box_TextChanged"/>

        <!-- Range -->
        <TextBlock Grid.Row="2"
          Grid.Column="0"
          Text="Range:"
          VerticalAlignment="Center"
          Margin="0,8,10,0"/>

        <Slider x:Name="P3Slider"
        Grid.Row="2"
        Grid.Column="1"
        Margin="0,8,0,0"
        Minimum="0"
        Maximum="180"
        TickFrequency="1"
        IsSnapToTickEnabled="True"
        ToolTip="Range: 0 –> 180"
        ValueChanged="P3Slider_ValueChanged"/>

        <TextBox x:Name="P3Box"
         Grid.Row="2"
         Grid.Column="2"
         Margin="10,8,0,0"
         VerticalContentAlignment="Center"
         TextChanged="P3Box_TextChanged"/>

        <!-- Send -->
        <Button Grid.Row="3"
        Grid.Column="0"
        Grid.ColumnSpan="3"
        Height="32"
        Margin="0,12,0,0"
        Content="Vibrate Disk"
        Click="SendP_Click"/>
      </Grid>
    </GroupBox>


    <GroupBox Grid.Row="1"
           Grid.Column="1"
           Header="Angular Position"
           Margin="0,0,0,10">
```

```
<Grid Margin="10">
  <Viewbox Stretch="Uniform">
    <Grid Width="150" Height="150">
      <!-- Outer disk -->
      <Ellipse StrokeThickness="4" Stroke="Black" Fill="#FFEFEFEF"/>
      <!-- Rotating marker group around center -->
      <Grid x:Name="NeedleGroup"
      RenderTransformOrigin="0.5,0.5">
        <Grid.RenderTransform>
          <RotateTransform x:Name="NeedleRotate" Angle="{Binding ElementName=A2Slider,Path=Value}"/>
        </Grid.RenderTransform>

        <!-- Marker line pointing up initially -->
        <Line X1="75" Y1="75" X2="75" Y2="10"
         Stroke="Red"
         StrokeThickness="6"
         StrokeEndLineCap="Round"/>
      </Grid>

      <!-- Center dot -->
      <Ellipse Width="8" Height="8"
       Fill="Black"
       HorizontalAlignment="Center"
       VerticalAlignment="Center"/>

      <!-- Rim arrow: CW (clockwise) -->
      <Path x:Name="SpinArrowCW"
       Stroke="DodgerBlue"
       StrokeThickness="5"
       StrokeEndLineCap="Round"
       StrokeStartLineCap="Round"
       Visibility="Collapsed"
       Data="M 12.5,39.0 A 72,72 0 0 1 137,39.0"/>

      <Polygon x:Name="SpinArrowHeadCW"
       Fill="Purple"
       Visibility="Collapsed"
       Points="140,45.0 127,35.5 127,50">
        <Polygon.RenderTransform>
          <RotateTransform Angle="50"
           CenterX="140"
           CenterY="45.0"/>
        </Polygon.RenderTransform>
      </Polygon>

      <!-- Rim arrow: CCW (counter-clockwise) -->
      <Path x:Name="SpinArrowCCW"
       Stroke="DodgerBlue"
       StrokeThickness="5"
       StrokeEndLineCap="Round"
       StrokeStartLineCap="Round"
       Visibility="Collapsed"
       Data="M 137,39.0 A 72,72 0 0 0 12.5,39.0"/>
      <Polygon x:Name="SpinArrowHeadCCW"
       Fill="Purple"
       Visibility="Collapsed"
       Points="9,45 22.5,35.5 22.5,50">
        <Polygon.RenderTransform>
          <RotateTransform Angle="-50"
           CenterX="9"
           CenterY="45"/>
        </Polygon.RenderTransform>
      </Polygon>

    </Grid>
  </Viewbox>
 </Grid>
</GroupBox>
<GroupBox Grid.Row="2"
      Grid.Column="1"
      Header="Destination of Spin"
      Margin="0,0,0,10">
```

```
<Grid Margin="10">
  <Viewbox Stretch="Uniform">
    <Grid Width="150" Height="150">
      <!-- Outer disk -->
      <Ellipse StrokeThickness="4" Stroke="Black" Fill="#FFEFEFEF"/>
      <!-- Rotating marker group around center -->
      <Grid x:Name="NeedleGroup2"
      RenderTransformOrigin="0.5,0.5">
        <Grid.RenderTransform>
          <RotateTransform x:Name="NeedleRotate2" Angle="{Binding ElementName=S3Slider,Path=Value}"/>
        </Grid.RenderTransform>

        <!-- Marker line pointing up initially -->
        <Line X1="75" Y1="75" X2="75" Y2="10"
         Stroke="Blue"
         StrokeThickness="6"
         StrokeEndLineCap="Round"/>
      </Grid>

      <!-- Rim arrow: CW (clockwise) -->
      <Path x:Name="SpinArrow2CW"
       Stroke="DodgerBlue"
       StrokeThickness="5"
       StrokeEndLineCap="Round"
       StrokeStartLineCap="Round"
       Visibility="Collapsed"
       Data="M 12.5,39.0 A 72,72 0 0 1 137,39.0"/>

      <Polygon x:Name="SpinArrowHead2CW"
       Fill="Purple"
       Visibility="Collapsed"
       Points="140,45.0 127,35.5 127,50">
        <Polygon.RenderTransform>
          <RotateTransform Angle="50"
           CenterX="140"
           CenterY="45.0"/>
        </Polygon.RenderTransform>
      </Polygon>

      <!-- Rim arrow: CCW (counter-clockwise) -->
      <Path x:Name="SpinArrow2CCW"
       Stroke="DodgerBlue"
       StrokeThickness="5"
       StrokeEndLineCap="Round"
       StrokeStartLineCap="Round"
       Visibility="Collapsed"
       Data="M 137,39.0 A 72,72 0 0 0 12.5,39.0"/>
      <Polygon x:Name="SpinArrowHead2CCW"
       Fill="Purple"
       Visibility="Collapsed"
       Points="9,45 22.5,35.5 22.5,50">
        <Polygon.RenderTransform>
          <RotateTransform Angle="-50"
           CenterX="9"
           CenterY="45"/>
        </Polygon.RenderTransform>
      </Polygon>

      <!-- Center dot -->
      <Ellipse Width="8" Height="8"
       Fill="Black"
       HorizontalAlignment="Center"
       VerticalAlignment="Center"/>
     </Grid>


    </Viewbox>
  </Grid>
</GroupBox>

<GroupBox Grid.Row="3"
      Grid.Column="1"
      Header="Oscilation Range"
      Margin="0,0,0,10">
```

```
        <Grid Margin="10">
          <Viewbox Stretch="Uniform">
            <Grid Width="150" Height="150">
              <!-- Outer disk -->
              <Ellipse StrokeThickness="4" Stroke="Black" Fill="#FFEFEFEF"/>
              <!-- Rotating marker group around center -->
              <Grid x:Name="NeedleGroup31"
              RenderTransformOrigin="0.5,0.5">
                <Grid.RenderTransform>
                  <RotateTransform x:Name="NeedleRotate31" Angle="{Binding ElementName=P3Slider,Path=Value, Converter={StaticResource Halver}}"/>
                </Grid.RenderTransform>

                <!-- Marker line pointing up initially -->
                <Line X1="75" Y1="75" X2="75" Y2="10"
                  Stroke="Green"
                  StrokeThickness="6"
                  StrokeEndLineCap="Round"/>
              </Grid>
              <Grid x:Name="NeedleGroup32"
              RenderTransformOrigin="0.5,0.5">
                <Grid.RenderTransform>
                  <RotateTransform x:Name="NeedleRotate32" Angle="{Binding ElementName=P3Slider,Path=Value, Converter={StaticResource NegateDouble}}"/>
                </Grid.RenderTransform>

                <!-- Marker line pointing up initially -->
                <Line X1="75" Y1="75" X2="75" Y2="10"
                  Stroke="Green"
                  StrokeThickness="6"
                  StrokeEndLineCap="Round"/>
              </Grid>

              <!-- Center dot -->
              <Ellipse Width="8" Height="8"
               Fill="Black"
               HorizontalAlignment="Center"
               VerticalAlignment="Center"/>
            </Grid>
          </Viewbox>
        </Grid>
      </GroupBox>


    </Grid>
  </TabItem>

  <!-- Program Mode Tab -->
  <TabItem Header="Program">
    <Grid Margin="10">
      <Grid.RowDefinitions>
        <RowDefinition Height="*"/>
        <RowDefinition Height="*"/>
      </Grid.RowDefinitions>

      <Grid.ColumnDefinitions>
        <ColumnDefinition Width= "*"/>
      </Grid.ColumnDefinitions>

      <!-- Top Row buttons -->
      <StackPanel Orientation="Horizontal" Grid.Row="0" Margin="0,0,0,10">
        <Button Content="Add Step" Width="100" Margin="0,0,8,0" Click="AddStep_Click"/>
        <Button Content="Remove Step" Width="110" Margin="10,0,8,0" Click="RemoveStep_Click"/>
        <Button Content="Load" Width="80" Margin="10,0,8,0" Click="Load_Click"/>
        <Button Content="Save" Width="80" Margin="10,0,8,0"  Click="Save_Click"/>
        <Button Content="Paste" Width="80" Margin="10,0,8,0" Click="PasteProgram_Click"/>
        <Button x:Name="RunBtn" Content="Run" Width="80" Margin="10,0,8,0" Click="RunProgram_Click"/>
      </StackPanel>

      <!-- Program steps grid -->
      <DataGrid x:Name="StepsGrid"
                Grid.Row="1"
                VerticalAlignment="Stretch"
                HorizontalAlignment="Stretch"
```

```
                AutoGenerateColumns="False"
                CanUserAddRows="False"
                ItemsSource="{Binding Steps}"
                SelectionMode="Single"
                SelectionUnit="FullRow"
                IsReadOnly="False">
            <DataGrid.Columns>

              <DataGridTextColumn Width="70" Binding="{Binding CMD}" CanUserSort="False">
                <DataGridTextColumn.Header>
                  <TextBlock Text="CMD" ToolTip="A for Angle Adjustment, S for Spin Disk  P for Vibrate Disk"/>
                </DataGridTextColumn.Header>
              </DataGridTextColumn>

              <DataGridTextColumn Width="70" Binding="{Binding V1}" CanUserSort="False">
                <DataGridTextColumn.Header>
                  <TextBlock Text="V1" ToolTip="This controls speed for all functions" />
                </DataGridTextColumn.Header>
              </DataGridTextColumn>

              <DataGridTextColumn Width="70" Binding="{Binding V2}" CanUserSort="False">
                <DataGridTextColumn.Header>
                  <TextBlock Text="V2" ToolTip="This controls different variables for each function" />
                </DataGridTextColumn.Header>
              </DataGridTextColumn>

              <DataGridTextColumn Width="70" Binding="{Binding V3}" CanUserSort="False">
                <DataGridTextColumn.Header>
                  <TextBlock Text="V3" ToolTip="V3 only works for S or P; this will be ignored for A" />
                </DataGridTextColumn.Header>
              </DataGridTextColumn>

              <DataGridTextColumn Width="100" Binding="{Binding DelMS}" CanUserSort="False">
                <DataGridTextColumn.Header>
                  <TextBlock Text="Delay (s)" ToolTip="Experiment as necessary, do note that without delays some commands will overwrite another or cause errors"/>
                </DataGridTextColumn.Header>
              </DataGridTextColumn>

              <DataGridTextColumn Header="Comment" Binding="{Binding Cmnt}" CanUserSort="False" Width="*"/>
            </DataGrid.Columns>
          </DataGrid>
        </Grid>
      </TabItem>

    </TabControl>



    <GroupBox Grid.Row="2">
      <Grid>
        <Grid.RowDefinitions>
          <RowDefinition Height="Auto"/>
          <RowDefinition Height="Auto"/>
        </Grid.RowDefinitions>
        <!--  CMD history -->

        <GroupBox Grid.Row="0" Header="Command History" Margin="0,10,0,0">
          <Grid Margin="10">
            <Grid.RowDefinitions>
              <RowDefinition Height="Auto"/>
              <RowDefinition Height="*"/>
            </Grid.RowDefinitions>

            <StackPanel Orientation="Horizontal" Grid.Row="0" Margin="0,0,0,8">
              <Button Content="Clear" Width="80" Margin="0,0,8,0" Click="ClearHistory_Click"/>
              <Button Content="Copy" Width="80" Margin="0,0,8,0" Click="CopyHistory_Click"/>
              <CheckBox x:Name="AutoScrollHistoryCheck"
              Content="Auto-scroll"
              IsChecked="True"
              VerticalAlignment="Center"/>
            </StackPanel>

            <ListView x:Name="HistoryList"
              Grid.Row="1" Height="120"
              ItemsSource="{Binding CommandHistory}">
```

```
                    <ListView.View>
                      <GridView>
                        <GridViewColumn Header="Time" Width="150" DisplayMemberBinding="{Binding TimeText}"/>
                        <GridViewColumn Header="Command" Width="200" DisplayMemberBinding="{Binding Text}"/>
                        <GridViewColumn Header="Note" Width="150" DisplayMemberBinding="{Binding Note}"/>
                      </GridView>
                    </ListView.View>
                  </ListView>
                </Grid>
              </GroupBox>
              <!--  Message Recieve Box -->
              <GroupBox Grid.Row="1"
                 Header="Alert Center"
                 Margin="0,10,10,0">
                 <TextBox x:Name="ReceiveTextBox"
                Height="150"
                 Margin="10"
                AcceptsReturn="True"
                IsReadOnly="True"
                TextWrapping="NoWrap"
                VerticalScrollBarVisibility="Auto"
                HorizontalScrollBarVisibility="Auto"/>
              </GroupBox>
            </Grid>
        </GroupBox>


        <ContentControl x:Name="MainContent"/>
    </Grid>
</Window>
```

Listing 15.15.2: Cs File for Application

```
using Microsoft.Win32;
using System;
using System.Collections.ObjectModel;
using System.Globalization;
using System.IO;
using System.IO.Ports;
using System.Text;
using System.Text.Json;
using System.Windows;
using System.Windows.Controls;
using System.Windows.Data;
using System.Windows.Input;
using System.Windows.Shapes;


namespace SerialGuiDemo
{
    public partial class MainWindow : Window
    {
        private SerialPort? _port;
        public ObservableCollection<ProgramStep> Steps { get; } = new();

        public ObservableCollection<HistoryItem> CommandHistory { get; } = new();
        public MainWindow()
        {
            InitializeComponent();
            DataContext = this;
            Steps.Add(new ProgramStep
            {
                CMD = "A",
                V1 = 0,
                V2 = 0,
                V3 = 0,
                DelMS = 0,
                Cmnt = "Initialise System"
            });
        }

        private void AddHistory(string text, string? note = null)
        {
```

```
    var item = new HistoryItem
    {
        Text = text,
        Note = note
    };

    CommandHistory.Add(item);

    // Auto-scroll to newest
    if (AutoScrollHistoryCheck?.IsChecked == true)
    {
        HistoryList?.ScrollIntoView(item);
    }
}


public class HistoryItem
{
    public DateTime Time { get; set; } = DateTime.Now;
    public string TimeText => Time.ToString("HH:mm:ss");
    public string Text { get; set; } = "";
    public string? Note { get; set; }
}

private void ClearHistory_Click(object sender, RoutedEventArgs e)
{
    CommandHistory.Clear();
}

private void CopyHistory_Click(object sender, RoutedEventArgs e)
{
    // Simple TSV copy
    var sb = new System.Text.StringBuilder();
    foreach (var h in CommandHistory)
        sb.AppendLine($"{h.Text}\t;\t{h.Note}");

    Clipboard.SetText(sb.ToString());
}

private void Window_Loaded(object sender, RoutedEventArgs e)
{
    RefreshPorts();
    A1Slider.Value = 0;
    A2Slider.Value = 0;
    S1Slider.Value = 0;
    S2Slider.Value = 0;
    S3Slider.Value = 0;
    P1Slider.Value = 0;
    P2Slider.Value = 0;
    P3Slider.Value = 0;
}

private void Window_Closing(object sender, System.ComponentModel.CancelEventArgs e)
{
    Disconnect();
}

private void RefreshPorts_Click(object sender, RoutedEventArgs e)
{
    RefreshPorts();
}

private void RefreshPorts()
{
    var ports = SerialPort.GetPortNames();
    Array.Sort(ports, StringComparer.OrdinalIgnoreCase);

    PortCombo.ItemsSource = ports;

    // Keep selection if still present; otherwise select first
    if (ports.Length > 0)
    {
        if (PortCombo.SelectedItem == null)
            PortCombo.SelectedIndex = 0;
    }
    else
```

```
        {
            PortCombo.SelectedItem = null;
        }
    }

    private void ConnectBtn_Click(object sender, RoutedEventArgs e)
    {
        if (_port != null && _port.IsOpen)
        {
            Disconnect();
            return;
        }

        var selected = PortCombo.SelectedItem as string;
        if (string.IsNullOrWhiteSpace(selected))
        {
            MessageBox.Show("Please select a COM port.", "No port selected",
                MessageBoxButton.OK, MessageBoxImage.Warning);
            return;
        }

        try
        {
            // Adjust baud rate / settings to match your device
            _port = new SerialPort(selected, 115200, Parity.None, 8, StopBits.One)
            {
                Encoding = Encoding.ASCII,
                NewLine = "\n",
                ReadTimeout = 500,
                WriteTimeout = 500
            };

            _port.DataReceived += Port_DataReceived;
            _port.Open();

            ConnectBtn.Content = "Disconnect";
            StatusText.Text = $"Connected: {selected} @ 115200";
            AppendLine($"[INFO] Connected to {selected}");
        }
        catch (Exception ex)
        {
            MessageBox.Show($"Failed to connect:\n{ex.Message}", "Connection error",
                MessageBoxButton.OK, MessageBoxImage.Error);
            Disconnect();
        }
    }

    private void Disconnect()
    {
        try
        {
            if (_port != null)
            {
                _port.DataReceived -= Port_DataReceived;

                if (_port.IsOpen)
                    _port.Close();

                _port.Dispose();
                _port = null;
            }
        }
        catch
        {
            // Ignore cleanup errors
        }
        finally
        {
            Dispatcher.Invoke(() =>
            {
                ConnectBtn.Content = "Connect";
                StatusText.Text = "Disconnected";
            });
        }
    }
    private void Port_DataReceived(object sender, SerialDataReceivedEventArgs e)
```

```
    {
        if (_port == null) return;

        try
        {
            // Read whatever is available; often devices send newline-terminated lines
            string incoming = _port.ReadExisting();

            if (!string.IsNullOrEmpty(incoming))
            {
                // Must update UI on UI thread
                Dispatcher.Invoke(() =>
                {
                    ReceiveTextBox.AppendText(incoming);
                    ReceiveTextBox.ScrollToEnd();
                });
            }
        }
        catch (Exception ex)
        {
            Dispatcher.Invoke(() => AppendLine($"[ERROR] Read failed: {ex.Message}"));
        }
    }

    private void AppendLine(string line)
    {
        ReceiveTextBox.AppendText(line + Environment.NewLine);
        ReceiveTextBox.ScrollToEnd();
    }
    private bool _updatingA1;
    private bool _updatingA2;
    private bool _updatingS1;
    private bool _updatingS2;
    private bool _updatingS3;
    private bool _updatingP1;
    private bool _updatingP2;
    private bool _updatingP3;

    private void A1Slider_ValueChanged(object sender, RoutedPropertyChangedEventArgs<double> e)
    {
        if (_updatingA1) return;
        _updatingA1 = true;
        A1Box.Text = ((int)A1Slider.Value).ToString();
        _updatingA1 = false;
        UpdateSpinDirectionArrow(e.NewValue); //For Spin arrow visualisation
    }

    private void A2Slider_ValueChanged(object sender, RoutedPropertyChangedEventArgs<double> e)
    {
        if (_updatingA2) return;
        _updatingA2 = true;
        A2Box.Text = ((int)A2Slider.Value).ToString();
        _updatingA2 = false;
    }

    private void A1Box_TextChanged(object sender, System.Windows.Controls.TextChangedEventArgs e)
    {
        if (_updatingA1) return;

        if (int.TryParse(A1Box.Text, out int v))
        {
            v = Math.Clamp(v, -100, 100);
            _updatingA1 = true;
            A1Slider.Value = v;
            _updatingA1 = false;
        }
    }

    private void A2Box_TextChanged(object sender, System.Windows.Controls.TextChangedEventArgs e)
    {
        if (_updatingA2) return;

        if (int.TryParse(A2Box.Text, out int v))
        {
            v = Math.Clamp(v, -360, 360);
            _updatingA2 = true;
```

```
            A2Slider.Value = v;
            _updatingA2 = false;
        }
    }
    private void SendA_Click(object sender, RoutedEventArgs e)
    {
        if (_port == null || !_port.IsOpen)
        {
            AppendLine("[WARNING] Not connected.");
            return;
        }

        int v1 = (int)A1Slider.Value;
        int v2 = (int)A2Slider.Value;

        // Build Arduino Command Line: "A 50 50"
        string cmd = $"A {v1} {v2}";

        try
        {
            _port.WriteLine(cmd);   // adds NewLine at end
            AppendLine($"> {cmd}");

            AddHistory(cmd, "Angle Adjustment");
        }
        catch (Exception ex)
        {
            AppendLine($"[ERROR] Send failed: {ex.Message}");
        }
    }

    private void S1Slider_ValueChanged(object sender, RoutedPropertyChangedEventArgs<double> e)
    {
        if (_updatingS1) return;
        _updatingS1 = true;
        S1Box.Text = ((int)S1Slider.Value).ToString();
        _updatingS1 = false;
        UpdateSpinDirectionArrow2(e.NewValue); //For Spin arrow visualisation
    }

    private void S2Slider_ValueChanged(object sender, RoutedPropertyChangedEventArgs<double> e)
    {
        if (_updatingS2) return;
        _updatingS2 = true;
        S2Box.Text = ((int)S2Slider.Value).ToString();
        _updatingS2 = false;
    }

    private void S3Slider_ValueChanged(object sender, RoutedPropertyChangedEventArgs<double> e)
    {
        if (_updatingS3) return;
        _updatingS3 = true;
        S3Box.Text = ((int)S3Slider.Value).ToString();
        _updatingS3 = false;
    }

    private void S1Box_TextChanged(object sender, System.Windows.Controls.TextChangedEventArgs e)
    {
        if (_updatingS1) return;

        if (int.TryParse(S1Box.Text, out int v))
        {
            v = Math.Clamp(v, -1200, 1200);
            _updatingS1 = true;
            S1Slider.Value = v;
            _updatingS1 = false;
        }
    }

    private void S2Box_TextChanged(object sender, System.Windows.Controls.TextChangedEventArgs e)
    {
        if (_updatingS2) return;

        if (int.TryParse(S2Box.Text, out int v))
        {
            v = Math.Clamp(v, 0, 30);
```

```
        _updatingS2 = true;
        S2Slider.Value = v;
        _updatingS2 = false;
    }
}
private void S3Box_TextChanged(object sender, System.Windows.Controls.TextChangedEventArgs e)
{
    if (_updatingS3) return;

    if (int.TryParse(S3Box.Text, out int v))
    {
        v = Math.Clamp(v, -360, 360);
        _updatingS3 = true;
        S3Slider.Value = v;
        _updatingS3 = false;
    }
}
private void SendS_Click(object sender, RoutedEventArgs e)
{
    if (_port == null || !_port.IsOpen)
    {
        AppendLine("[WARNING] Not connected.");
        return;
    }

    int v1 = (int)S1Slider.Value;
    int v2 = (int)S2Slider.Value;
    int v3 = (int)S3Slider.Value;

    // Build Arduino Command Line: "S 50 50 50"
    string cmd = $"S {v1} {v2} {v3}";

    try
    {
        _port.WriteLine(cmd);   // adds NewLine at end
        AppendLine($"> {cmd}");
        AddHistory(cmd, "Spin Disk");
    }
    catch (Exception ex)
    {
        AppendLine($"[ERROR] Send failed: {ex.Message}");
    }
}


private void P1Slider_ValueChanged(object sender, RoutedPropertyChangedEventArgs<double> e)
{
    if (_updatingP1) return;
    _updatingP1 = true;
    P1Box.Text = ((int)P1Slider.Value).ToString();
    _updatingP1 = false;
}

private void P2Slider_ValueChanged(object sender, RoutedPropertyChangedEventArgs<double> e)
{
    if (_updatingP2) return;
    _updatingP2 = true;
    P2Box.Text = ((int)P2Slider.Value).ToString();
    _updatingP2 = false;
}

private void P3Slider_ValueChanged(object sender, RoutedPropertyChangedEventArgs<double> e)
{
    if (_updatingP3) return;
    _updatingP3 = true;
    P3Box.Text = ((int)P3Slider.Value).ToString();
    _updatingP3 = false;
}

private void P1Box_TextChanged(object sender, System.Windows.Controls.TextChangedEventArgs e)
{
    if (_updatingP1) return;

    if (int.TryParse(P1Box.Text, out int v))
    {
        v = Math.Clamp(v, 0, 100);
```

```
        _updatingP1 = true;
        P1Slider.Value = v;
        _updatingP1 = false;
    }
}


private void P2Box_TextChanged(object sender, System.Windows.Controls.TextChangedEventArgs e)
{
    if (_updatingP2) return;

    if (int.TryParse(P2Box.Text, out int v))
    {
        v = Math.Clamp(v, 0, 30);
        _updatingP2 = true;
        P2Slider.Value = v;
        _updatingP2 = false;
    }
}

private void P3Box_TextChanged(object sender, System.Windows.Controls.TextChangedEventArgs e)
{
    if (_updatingP3) return;

    if (int.TryParse(P3Box.Text, out int v))
    {
        v = Math.Clamp(v, 0, 180);
        _updatingP3 = true;
        P3Slider.Value = v;
        _updatingP3 = false;
    }
}

private void SendP_Click(object sender, RoutedEventArgs e)
{
    if (_port == null || !_port.IsOpen)
    {
        AppendLine("[WARNING] Not connected.");
        return;
    }

    int v1 = (int)P1Slider.Value;
    int v2 = (int)P2Slider.Value;
    int v3 = (int)P3Slider.Value;

    // Build Arduino Command Line: "P 50 50 50"
    string cmd = $"P {v1} {v2} {v3}";

    try
    {
        _port.WriteLine(cmd);   // adds NewLine at end
        AppendLine($"> {cmd}");
        AddHistory(cmd, "Vibrate Disk");
    }
    catch (Exception ex)
    {
        AppendLine($"[ERROR] Send failed: {ex.Message}");
    }
}

private void AddStep_Click(object sender, RoutedEventArgs e)
{
    Steps.Add(new ProgramStep { CMD = "A", V1 = 0, V2 = 0, V3 = 0, DelMS = 0 });
}

private void RemoveStep_Click(object sender, RoutedEventArgs e)
{
    if (StepsGrid.SelectedItem is ProgramStep step)
        Steps.Remove(step);
}

private void Save_Click(object sender, RoutedEventArgs e)
{
    var dlg = new SaveFileDialog
    {
        Filter = "Program Files (*.json)|*.json|All Files (*.*)|*.*",
```

```
            DefaultExt = "json"
        };
        if (dlg.ShowDialog() != true) return;

        var json = JsonSerializer.Serialize(Steps, new JsonSerializerOptions { WriteIndented = true });
        File.WriteAllText(dlg.FileName, json);
    }

    private void Load_Click(object sender, RoutedEventArgs e)
    {
        var dlg = new OpenFileDialog
        {
            Filter = "Program Files (*.json)|*.json|All Files (*.*)|*.*"
        };
        if (dlg.ShowDialog() != true) return;

        var json = File.ReadAllText(dlg.FileName);
        var loaded = JsonSerializer.Deserialize<ProgramStep[]>(json);

        Steps.Clear();
        if (loaded != null)
            foreach (var s in loaded)
                Steps.Add(s);
    }
    private void PasteProgram_Click(object sender, RoutedEventArgs e)
    {
        if (!Clipboard.ContainsText())
            return;

        var text = Clipboard.GetText();
        PasteProgramText(text);
    }

    private void PasteProgramText(string text)
    {
        var lines = text
            .Split(new[] { '\r', '\n' }, StringSplitOptions.RemoveEmptyEntries);

        int added = 0;

        foreach (var rawLine in lines)
        {
            var line = rawLine.Trim();

            // Skip empty or comment-only lines
            if (string.IsNullOrWhiteSpace(line))
                continue;

            // Split comment
            string? comment = null;
            int commentIdx = line.IndexOfAny(new[] { ';', '#' });
            if (commentIdx >= 0)
            {
                comment = line[(commentIdx + 1)..].Trim();
                line = line[..commentIdx].Trim();
            }

            if (string.IsNullOrWhiteSpace(line))
                continue;

            var parts = line.Split(new[] { ' ', '\t', ',' },
                           StringSplitOptions.RemoveEmptyEntries);

            if (parts.Length == 0)
                continue;

            var cmd = parts[0].ToUpperInvariant();

            if (cmd != "A" && cmd != "S" && cmd != "P")
            {
                AppendLine($"Invalid command skipped: {rawLine}");
                continue;
            }

            int v1 = 0, v2 = 0, v3 = 0, delay = 0;
```

```
            TryParse(parts, 1, ref v1);
            TryParse(parts, 2, ref v2);
            TryParse(parts, 3, ref v3);
            TryParse(parts, 4, ref delay);

            Steps.Add(new ProgramStep
            {
                CMD = cmd,
                V1 = v1,
                V2 = v2,
                V3 = v3,
                DelMS = delay,
                Cmnt = comment
            });

            added++;
        }

        if (added > 0)
            AppendLine($"Pasted {added} program step(s)");
    }

    private static void TryParse(string[] parts, int index, ref int value)
    {
        if (index < parts.Length && int.TryParse(parts[index], out var v))
            value = v;
    }

    private void StepsGrid_PreviewKeyDown(object sender, KeyEventArgs e)
    {
        if (e.Key == Key.V && Keyboard.Modifiers.HasFlag(ModifierKeys.Control))
        {
            PasteProgram_Click(sender, e);
            e.Handled = true;
        }
    }
    private async void RunProgram_Click(object sender, RoutedEventArgs e)
    {
        if (_port == null || !_port.IsOpen)
        {
            AppendLine("[WARNING] Not connected.");
            return;
        }
        for (int i = 0; i < Steps.Count; i++)
        {
            var step = Steps[i];
            string CMD = step.CMD;
            if (CMD == "A")
            {
                int V1 = step.V1;
                int V2 = step.V2;
                string msg = $"{CMD} {V1} {V2}";
                try
                {
                    _port.WriteLine(msg);   // adds NewLine at end
                    AppendLine($"> {msg}");
                }
                catch (Exception ex)
                {
                    AppendLine($"[ERROR] Send failed: {ex.Message}");
                }
                await Task.Delay(step.DelMS * 1000);
            }
            else
            {
                int V1 = step.V1;
                int V2 = step.V2;
                int V3 = step.V3;
                string msg = $"{CMD} {V1} {V2} {V3}";
                try
                {
                    _port.WriteLine(msg);   // adds NewLine at end
                    AppendLine($"> {msg}");
                }
                catch (Exception ex)
                {
```

```
                    AppendLine($"[ERROR] Send failed: {ex.Message}");
                }
                await Task.Delay(step.DelMS * 1000);
            }
        }
    }

    private void UpdateSpinDirectionArrow(double speed)
    {
        // speed > 0 => Clock, speed < 0 => Counter, speed == 0 => hide both
        bool cw = speed > 0;
        bool ccw = speed < 0;

        SpinArrowCW.Visibility = cw ? Visibility.Visible : Visibility.Collapsed;
        SpinArrowHeadCW.Visibility = cw ? Visibility.Visible : Visibility.Collapsed;

        SpinArrowCCW.Visibility = ccw ? Visibility.Visible : Visibility.Collapsed;
        SpinArrowHeadCCW.Visibility = ccw ? Visibility.Visible : Visibility.Collapsed;
    }
    private void UpdateSpinDirectionArrow2(double speed)
    {
        // speed > 0 => Clock, speed < 0 => Counter, speed == 0 => hide both
        bool cw = speed > 0;
        bool ccw = speed < 0;

        SpinArrow2CW.Visibility = cw ? Visibility.Visible : Visibility.Collapsed;
        SpinArrowHead2CW.Visibility = cw ? Visibility.Visible : Visibility.Collapsed;

        SpinArrow2CCW.Visibility = ccw ? Visibility.Visible : Visibility.Collapsed;
        SpinArrowHead2CCW.Visibility = ccw ? Visibility.Visible : Visibility.Collapsed;
    }



  }
  public class ProgramStep
  {
    public string CMD { get; set; } = "A";
    public int V1 { get; set; }
    public int V2 { get; set; }
    public int V3 { get; set; }
    public int DelMS { get; set; }
    public string? Cmnt { get; set; }

    public string ToArduino() => $"{CMD} {V1} {V2} {V3}";

    // Useful for UI preview binding (won't throw if SelectedItem is null)
    public string SerialLinePreview => ToArduino();



  }


}
```

Listing 15.15.3: HalverConverter File for Application

```
using System;
using System.Globalization;
using System.Windows.Data;

namespace SerialGuiDemo
{
   public class HalverConverter : IValueConverter
   {
      public object Convert(object value, Type targetType, object parameter, CultureInfo culture)
      {
         if (value is double d)
            return d/2;

         return 0.0;
      }

      public object ConvertBack(object value, Type targetType, object parameter, CultureInfo culture)
      {
         if (value is double d)
            return d/2;

         return 0.0;
      }
   }
}
```

Listing 15.15.4: NegateDoubleConverter File for Application

```
using System;
using System.Globalization;
using System.Windows.Data;

namespace SerialGuiDemo
{
   public class NegateDoubleConverter : IValueConverter
   {
      public object Convert(object value, Type targetType, object parameter, CultureInfo culture)
      {
         if (value is double d)
            return -d/2;

         return 0.0;
      }

      public object ConvertBack(object value, Type targetType, object parameter, CultureInfo culture)
      {
         if (value is double d)
            return -d/2;

         return 0.0;
      }
   }
}
```